

**BEAR VALLEY SCHOOL REHABILITATION PHASE 1
PINNACLES NATIONAL PARK, CALIFORNIA**

PMIS NO. 331043

PROJECT SPECIFICATIONS



NATIONAL PARK SERVICE

100% CONSTRUCTION DOCUMENTS

JANUARY 16, 2026

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Nicole Christ
Landmark Environmental, Inc.

SECTION 01 11 00 - SUMMARY OF WORK

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes the following:

1. Work Covered by Contract Documents
2. Work Phases
3. Work Under Other Contracts
4. Government-Furnished Materials
5. Construction Contractor Use of Site
6. Public Use of Site
7. Occupancy Requirements for Buildings
8. Conduct of Operations
9. Work Restrictions
10. Special Construction Requirements
11. Soils Investigation Report
12. Additional Reports

1.2 WORK COVERED BY CONTRACT DOCUMENTS

A. Project Location: Pinnacles National Park, San Benito County, CA

B. The Work consists of:

1. Removal of existing foundation around building, including front porch framing and installing new poured concrete continuous strip footing under building and front porch. Coordination with existing historic tree, buried sanitary lines, and historic concrete rear porch. Crawl space vents to be included in new concrete foundation.
2. Bear Valley School was completed in 1903 and was an active rural school until June 1950, after which it served as a community hall until 2002 when it was shuttered. Ownership of the property was conveyed to the NPS in 2011 and it was listed in the National Register of Historic Places in 2014. Its period of significance is understood to be 1902–1950.
 - a. Character defining features include:
 - 1) Site near center of parcel in a grove of mature valley oaks
 - 2) East-west orientation
 - 3) Rectangular one-story form with central schoolroom and half-height open porch
 - 4) With coatrooms at the north and south sides
 - 5) Moderately pitched front-gabled roof with shallow overhangs and hipped roof porch
 - 6) Double-hung, four-over-four wood windows
 - 7) Four-paneled wood entry doors
 - 8) Interior millwork and finishes including extant original hardwood flooring
 - 9) Interior features related to the building's use as a school including a decorative

- 10) Brass plate at the west schoolroom ceiling that once admitted the belfry cord and
- 11) Chalkboards from the period of significance

C. Project will be constructed under a single prime contract.

1.3 WORK PHASES

A. The Work shall be conducted in one phase.

1.4 WORK UNDER OTHER CONTRACTS

A. General: Cooperate with separate Construction Contractors so work on those contracts may be carried out smoothly, without interfering with or delaying work under this Contract. Coordinate Work of this Contract with work performed under separate contracts.

B. Future Work: Government will award separate contract(s) for the following additional work to be performed at site after Substantial Completion. Completion of that work will depend on successful completion of preparatory work under this Contract.

1. Bear Valley Schoolhouse Rehabilitation: A separate contract will be awarded to NPS Historic Preservation Treatment Construction Crew for full building rehabilitation including but not limited to window and door rehabilitation, exterior wood siding repair, full building cleaning, finishes rehabilitation, installation of new metal roof, and reconstruction of historic belfry.

1.5 CONSTRUCTION CONTRACTOR USE OF SITE

A. General: Construction Contractor shall have full use of the site for construction operations during the construction period. Construction Contractor's use of the site is limited only by the Government's right to perform work or to retain other Construction Contractors on portions of Project.

B. General: Construction Contractor shall have limited use of site for construction operations. Limit use of premises to areas within the Construction limits indicated on drawings. Do not disturb portions of Project site beyond areas in which the Work is indicated.

C. Storage of Materials: Confine storage of materials to staging area shown on drawings.

D. Parking: Confine parking to staging area as shown on drawings.

E. Stockpiling: Confine stockpiling to staging area as shown on drawings.

F. Preservation of Natural Features:

1. Prevent damage to natural surroundings. Restore damaged areas, repairing or replacing damaged trees and plants, at no additional expense to the Government.
2. Provide temporary barriers to protect existing trees and plants and root zones.

3. Do not remove, injure, or destroy trees or other plants without prior approval. Consult with Contracting Officer (CO) and remove agreed-on roots and branches that interfere with construction.
 4. Do not fasten ropes, cables, or guys to existing trees.
 5. Carefully supervise excavating, grading, filling, and other construction operations near trees to prevent damage.
- G. Driveways and Entrances: Keep driveways and entrances serving premises clear and available to Government employees, and emergency vehicles at all times. Do not use for parking or storage of materials.
1. Schedule deliveries to minimize use of driveways and entrances.
 2. Schedule deliveries to minimize space and time requirements for storage of materials and equipment on-site.
- H. Construction Camp: Establishment of a camp within park will not be permitted.
- I. Hauling Restrictions: Comply with legal load restrictions in hauling of materials. Load restrictions on park roads are identical to state load restrictions with such additional regulations as may be imposed by the Park Superintendent. Information regarding rules and regulations for vehicular traffic on park roads may be obtained from the Office of the Park Superintendent. A special permit will not relieve Construction Contractor of liability for damage which may result from moving of equipment.
- J. Bridge Restrictions: Identify jurisdictions, load restrictions, permit requirements, time and calendar restrictions as outlined.

1.6 PUBLIC USE OF SITE

- A. The building will be closed to the public during construction.
- B. Construction Contractor shall conduct his operations to ensure the least inconvenience to public. The access road from Hwy 25 to the Bear Valley School will be closed to the public during construction. NPS has no authority over State Hwy 25, and no road closures will be permitted other than temporary traffic calming measures, such as flaggers, and coordinated with California Department of Transportation. Contractor will be responsible for any traffic control that may be necessary to safely access the Bear Valley School site from Hwy 25.

1.7 OCCUPANCY REQUIREMENTS FOR BUILDINGS

- A. Existing Buildings
 1. Government will not occupy building under construction during the entire contract period. Maintain existing building in weather tight condition throughout construction period. Repair damage caused by construction operations. Protect building during construction period.

1.8 CONDUCT OF OPERATIONS

- A. Construction Contractor shall conduct his operations in conformance with rules and regulations promulgated by the Secretary of the Interior for the National Park Service, and applicable park rules and regulations prescribed by Park Superintendent.
- B. Work on Saturdays, Sundays, Federal holidays or at night may not be performed unless stated in the Work Restrictions below or without prior consent from the Contracting Officer. Submit requests 3 business days in advance of the work to the Contracting Officer for approval.
- C. No signs or advertisements (except those specified herein) shall be displayed on the construction site or within the park unless approved by the Contracting Officer.

1.9 WORK RESTRICTIONS

- A. On-Site Work Hours: Work shall be generally performed during normal business working hours of 8 a.m. to 5 p.m., Monday through Friday, except when otherwise indicated.
 - 1. Weekend Hours: 9a.m. to 5 p.m.
 - 2. Early Morning Hours: Not permitted unless approved by the Contracting Officer in writing.
 - 3. Hours for noisy activity is limited to permitted working hours.
- B. Existing Utilities
 - 1. Existing Utilities: Notify Contracting Officer and utility companies of proposed locations and times for excavation.
 - 2. Construction Contractor shall be responsible for locating and preventing damage to known utilities. If damage occurs, repair utility at no additional expense to the Government.
 - 3. If damage occurs to an unknown utility, repair utility. An equitable adjustment will be made in accordance with the Changes clause of the contract.
- C. Nonsmoking Building/Tobacco Use/Vaping: Smoking is not permitted within building or within 50 feet of entrances, operable windows, or outdoor air intakes. Tobacco use and vaping is restricted to area at least 50 feet away from building.
- D. All vehicles, tools, or other equipment that will be used off pavement or that will come into contact with the ground or with vegetation must be properly cleaned before entering the park. This equipment must be free of dirt, plant parts, and other debris that might harbor diseases or plant propagules. NPS recommends carefully pressure washing equipment to thoroughly remove these materials, paying extra attention to crevices, undercarriages, tire treads, decks, and other horizontal surfaces where material may accumulate. At least three business days (72 hours) prior to arrival, contact Contracting Officer to arrange an inspection by a resources management specialist at the park entrance upon arrival. Any vehicles, equipment, or tools not meeting the above standards will not be allowed to enter the park until it has been properly cleaned outside the park.

1.10 SPECIAL CONSTRUCTION REQUIREMENTS

- A. Project Management and Communication Software: Software administered by NPS shall be used to manage communication and document sharing during construction.
 - 1. See Section 01 31 00 "Project Management and Coordination" for requirements on using NPS project management and communication software.
- B. Coordination with the future rehabilitation phase of the project will be critical in understand items to be salvaged and stored for the park or reinstalled in order to maintain a weathertight condition between phases of work.
- C. Extreme amounts of animal excrement is present in all areas of the building. Additional PPE may be desired by contractors.

1.11 SOILS INVESTIGATION REPORT

- A. The report "Geotechnical Report" dated January 2025 and prepared by Ninyo & Moore is Appendix B with this package.
- B. In case of conflict between report and drawings or specifications, the drawings and specifications govern.

1.12 ADDITIONAL REPORTS

- A. The hazardous materials report prepared by Landmark Environmental is Appendix A with this package.
- B. In case of conflict between report and drawings or specifications, drawings and specifications govern.

PART 2 - PRODUCTS (Not Used)

PART 3 - EXECUTION (Not Used)

END OF SECTION 01 11 00

SECTION 01 26 01 – CONTRACT MODIFICATION PROCEDURES

PART 1 - GENERAL

1.1 SUMMARY

- A. This section consists of administrative and procedural requirements for contract modifications.

1.2 DEFINITIONS AND ALLOWANCES

- A. Home Office Overhead: Costs incurred in support of all a contractor's projects and not attributable to a specific job. The cost for home office overhead is only allowed as a percentage of all direct work excluding profit. The following items represent allowable home office overhead costs identified in Part 31 of the Federal Acquisition Regulation (FAR):

1. Rent
2. Utilities
3. Furnishings
4. Office equipment
5. Executive and management staff not exclusively assigned to the project
6. Support, accounting, and administrative staff
7. Preparation of cost proposals, estimating, and schedule analyses connected with Modifications
8. Estimating and preconstruction services
9. Mortgage costs
10. Real estate and corporate taxes
11. Automobile maintenance and travel costs for home office personnel
12. Home office insurances i.e. structure, automotive, umbrella, flood, etc.
13. Depreciation of equipment and other assets
14. Home office supplies (paper, staples, etc.)
15. Legal services
16. Accounting and data processing
17. Professional fees/registration

- B. General Conditions (Field Office Overhead): Management and administrative costs incurred on site for the designated project. Costs associated with preparation of modifications will not be allowed. Costs for these items are to be included only in the general conditions of the modification estimate. Only in the case of a contract time extension are additional general conditions included in modifications. The following items, if applicable, are considered allowable costs for calculating General Conditions:

1. Project Manager (PM), Assistant Project Manager
2. Superintendent, Assistant Superintendent
3. Quality Control, Safety Officer, Environmental Manager, etc.
4. Engineers
5. Travel, lodging, and per diem (as established by Federal Travel Regulations)
6. Scheduling

7. Field Office Trailers and associated temporary utilities
8. Field office supplies
 - a. Mailing and couriers
 - b. Reproduction costs
 - c. Storage
 - d. Phones
 - e. Computers
 - f. Copiers
9. Personal vehicles i.e. Superintendent Pickup trucks

C. General Requirements: Costs directly associated with the project and are necessary to perform the actual work of the modification. These costs shall be shown as direct costs in the estimate. The following items, if applicable, are considered allowable costs for calculating General Requirements:

1. Hoisting
2. Material handling
3. Temporary fencing
4. Port-a-lets
5. Trash removal, dumpsters
6. Barricades
7. Small tools
8. Safety supplies
9. Scaffolding
10. Daily cleaning
11. Traffic control
12. Temporary signage
13. Temporary heating and power

D. Personnel Costs: Costs included in the modification must only be for General Conditions staff and workers present and working on the project site. Modification costs for salaried workers are only allowed within the structure of a 40-hour week, and no overtime or holiday pay will be allowed.

1. Worker Hourly Rates are costs directly associated with the individual worker and consist of the following:
 - a. Base Rate: The hourly rate paid directly to the worker
 - b. Labor Burden: Employer payments of all applicable burdens include insurance and taxes the business must pay on behalf of the worker to government entities and educational forums, such as:
 - 1) Social Security
 - 2) Medicare
 - 3) Workers Compensation – Policy and company calculation to be made available.
 - 4) Federal Unemployment Tax Act (FUTA) - Cap Rate and percentage to be proportionally allocated over one year.
 - 5) State Unemployment Tax Act (SUTA) - Cap Rate and percentage to be proportionally allocated over one year.

- 6) Union agreement costs - Other costs required under an enforceable collective bargaining agreement.
- c. Fringe Benefits: Various non-wage compensations provided to employees such as:
 - 1) Health Care Insurance Premiums
 - 2) Cell Phone
 - 3) Clothing
 - 4) 401K and Pensions
 - 5) Vehicle allowances
 - 6) Gas allowance
 - 7) Life insurance premiums
 - 8) Disability insurance
 - 9) Other Fringe Benefits required under an enforceable collective bargaining agreement
- E. Bonuses or Deferred Compensation: No Bonus or Deferred Compensation will be allowed within any components of pricing including Home Office Overhead, General Conditions, General Requirements, Hourly Worker Rates, or the direct costs of work.
- F. General Liability Insurance: An insurance policy that protects Contractor from claims resulting from bodily injury or property damage to a third party. Include as a separate line item within all modification proposals and provide a current insurance quote upon request.
- G. Consent of Surety:
 - 1. When any contract is modified, the Contractor shall provide the consent of surety if-
 - a. An additional bond is obtained from other than the original surety;
 - b. No additional bond is required and
 - 1) The modification is for new work beyond the scope of the original contract; or
 - 2) The modification does not change the contract scope but changes the contract price (upward or downward) by more than 25 percent or \$50,000; or
 - c. Consent of surety is required for a novation agreement.
 - 2. When a contract for which performance or payment is secured by any of the types of security listed in FAR 28.204 is modified as described in paragraph (a) of this subsection, no consent of surety is required.
 - 3. Agencies shall use Standard Form 1414, Consent of Surety, for all types of contracts.
- H. Builder's Risk Insurance: Covers the contractor's loss due to fire, high winds, or other natural forces. Not reimbursed by the National Park Service (NPS) and shall not be included in modification proposals.

1.3 MODIFICATION PROPOSAL PRICING REQUIREMENTS

A. General:

1. Proposal be received in the format and within the time frame specified in the Request for Proposal (RFP) letter. Costs or delays resulting from failure of contractor to submit within the time frame specified will not be compensable.
2. The proposal shall be detailed with itemized lists of equipment, materials, labor, production rates, overhead, profit, and bond markup for each item. Labor costs must be itemized by craft and hourly rate, including Fringe Benefits and Labor Burden. If the costs of Fringe Benefits and Labor Burden are not itemized, it is assumed they are included in the hourly rate shown, or the contractor is not requesting reimbursement. Contractor may utilize the government provided [Contractor Estimate Form](#), or their own form, provided that it contains the same information and level of detail as the Government's form.
3. Requests for extensions of contract time as a result of change must be justified with a Time Impact Analysis (TIA). Refer to Section 01 32 16 "Construction Schedule", for time impact analysis requirements. TIA and associated costs shall be received with the proposal by the date shown within the Request for Proposal letter. Contractor's failure to submit within the specified time frame will be construed as the Contractor waiving right for additional time and no time extension will be allowed.
4. All supporting documentation used to justify the proposed modification will be made available to the Contracting Officer (CO) upon request.
5. Contractor shall review and approve all subcontractor/supplier pricing in detail for proper format, scope, production rates, and pricing prior to submission to NPS. All delay costs associated with not reviewing and approving subcontractor/supplier pricing will be borne by the Contractor.
6. All pricing and production rates within the estimate must be based on fair and reasonable pricing and cannot include built-in contingency.

B. Labor:

1. Contractor shall estimate the cost of labor by itemizing each craft involved, indicating worker hourly rate (base rate + labor burden + fringe benefits) for each and itemizing hours required for each craft directly engaged in modification work. Any work proposed requiring overtime work or premium pay shall be itemized separately. Rates shall be in accordance with the Davis-Bacon Act as incorporated herein. Labor Burden may include payroll taxes, Social Security, unemployment insurance, workers compensation insurance, Federal Insurance Contributions Act (FICA), FUTA, and other direct costs resulting from Federal, State or local laws.
2. Itemize labor costs for equipment operators separate from equipment costs.
3. Labor cost for foremen shall only be costs for related work required for the modification.

C. Materials:

1. Estimated cost for materials shall include quotes from multiple sources. Material prices shall include applicable fees and credits, including but not limited to, sales tax, freight and delivery charges, and tax rebates.
2. No markup shall be applied to any material provided by NPS.

D. Equipment:

1. The equipment used for the project must be appropriately sized for work being performed.
2. Do not include costs for "miscellaneous tools and equipment", in your proposal for a replacement value of \$500 or less. Costs shown in excess of \$500 shall be broken out separately.
3. Regardless of ownership, rates to be used in determining equipment rental costs shall be the lowest cost from one of the following sources:
 - a. United States (U.S.) Army Corps of Engineers, Ownership and Operating Expense Schedule (use latest edition and applicable region)
 - b. Construction Blue Book
 - c. Local equipment rental rates, documented by actual invoice charges, or itemized vendor quotes.
4. Estimated equipment rates shall include operating costs of all fuel, oil, lubrication, supplies, small tools, necessary attachments, ground engaging components, tires and tracks, routine repairs and maintenance (cost of major repair and overhaul is not allowed per Federal Acquisition Regulation (FAR) 31.105(d)(2)), depreciation, storage, insurance, and all incidentals. Mobilization, if applicable, may be included for equipment solely used on the modification work but must be listed separately.
5. Estimate full rate for equipment only for duration that equipment will be utilized to accomplish work of the modification.
6. Standby unit rates used in accordance with paragraph 1.3, D, 2, above. If the U.S. Army Corp of Engineers is utilized, then their standby rates prevail. If Bluebook or local equipment pricing is accepted, then 1/2 of equipment costs minus any operating costs, major repair and overhaul will be accepted.
7. If equipment is in standby mode due solely to a documented NPS delay, established standby rate shall apply from the first day of the delay.
8. Equipment not used and on the job site for up to five consecutive days may be classified at standby rates, provided the equipment is or has been used solely to perform work on the modification and will be necessary to complete additional modification work. Equipment still on the jobsite but not in use after five consecutive days will not be considered in the modification pricing.
9. Requests for compensation for equipment stand by time must be justified, documented and itemized separately.
10. The estimated timeframe (daily, weekly, monthly) for use of the equipment must reflect the lowest cost to the Government.

E. Establishment and Application of Overhead and Profit Percentages:

1. Home Office Overhead and Profit (OH&P) shall be applied to direct costs only. Profit shall not be applied to overhead amounts; and overhead shall not be applied to profit. Home office overhead shall contain only allowable, allocable, and reasonable costs per the contract documents and FAR Part 31. Profit percentages are based on risk factors found in FAR Part 31 which have been applied to the specific type of work included in this project. Negotiated rates shall not exceed the following percentages for OH&P for contractor self-performed work:

Overhead.....	10%
Profit.....	6%

2. Total aggregate limit of markup (OH&P) for Contractor and Subcontractors on modification work shall not exceed 25%. The NPS will not be responsible for allocation of percentages between contractor and subcontractors at any tier.
3. If Contractors form a partnership, partnership may only receive home office overhead and profit in same amount as an individual Contractor (refer to paragraph 1.3,E,1 above). It is the responsibility of the partners to decide on division of revenue.
4. Combined Increases and Decreases: On proposals involving both increases and decreases in the Contract Price, overhead and profit mark-ups are required on net increases and deducted on net decreases.
5. At no time can profit be calculated on Overhead or itself, it must be calculated on direct costs of work only.

PART 2 - PRODUCTS

PART 3 - EXECUTION

END OF SECTION 012601

SECTION 01 27 00 – DEFINITION OF CONTRACT LINE ITEMS

PART 1 - GENERAL

1.1 SUMMARY

- A. Section explains in general, what is and is not included in a contract line item, and limits or cut-off points where one item ends and another begins.
- B. If no contract line item exists for a portion of work, include costs in a related item.

PART 2 - PRODUCTS (Not Used)

PART 3 - EXECUTION

3.1 LIST OF CONTRACT LINE ITEMS

- A. Contract Line Item Number 01: Bear Valley Schoolhouse, Phase 1 – Foundation Replacement
 - 1. This item consists of removal of existing foundation around building, including front porch and installing new poured concrete continuous strip footing under building and front porch. Coordination with existing historic tree, buried sanitary lines, and historic concrete rear porch. Crawl space vents to be included in new concrete foundation.
 - 2. Measurement for payment will be by percentage of work completed in the previous month.
 - 3. Payment will be made at the contract lump sum price.

END OF SECTION 01 27 00

SECTION 01 31 00 - PROJECT MANAGEMENT AND COORDINATION

PART 1 - GENERAL

1.1 SUMMARY

- A. Section includes administrative provisions for coordinating construction operations on Project including, but not limited to, the following:
 - 1. Definitions
 - 2. Construction Coordination
 - 3. Submittals
 - 4. Coordination Drawings
 - 5. Requests for Information (RFIs)
 - 6. NPS Project Management and Communication Software
 - 7. Project Meetings
 - 8. Environmental Coordination
- B. Related Requirements:
 - 1. Section 01 32 16 "Construction Schedule" for preparing and submitting Construction Contractor's construction schedule.
 - 2. Section 01 73 40 "Execution" for procedures for coordinating general installation and field-engineering services, including establishment of benchmarks and control points.
 - 3. Section 01 77 00 "Closeout Procedures" for coordinating closeout of the Contract.

1.2 DEFINITIONS

- A. [Agency with Jurisdiction](#)

1.3 CONSTRUCTION COORDINATION

- A. Coordination: Coordinate construction operations included in different Sections of Specifications to ensure efficient and orderly installation of each part of the Work. Coordinate construction operations, included in different Sections, which depend on each other for proper installation, connection, and operation.
 - 1. Schedule construction operations in sequence required to obtain best results where installation of one part of the Work depends on installation of other components, before or after its own installation.
 - 2. Coordinate installation of different components with other Construction Contractors to ensure maximum accessibility for required maintenance, service, and repair.
 - 3. Make provisions to accommodate items scheduled for later installation.
 - 4. Where availability of space is limited, coordinate installation of different components to ensure maximum performance and accessibility for required maintenance, service, and repair of components, including mechanical and electrical.

- B. Prepare memoranda for distribution to each party involved, outlining special procedures required for coordination. Include such items as required notices, reports, and list of attendees at meetings.
- C. Administrative Procedures: Coordinate scheduling and timing of required administrative procedures with other construction activities and activities of other Construction Contractors to avoid conflicts and to ensure orderly progress of the Work. Such administrative activities include, but are not limited to:
 - 1. Preparation of Construction Contractor Construction Schedule
 - 2. Preparation of the Schedule of Values
 - 3. Installation and removal of temporary facilities and controls
 - 4. Delivery and processing of submittals
 - 5. Progress meetings
 - 6. Pre-installation conferences
 - 7. Project closeout activities

1.4 SUBMITTALS

- A. Coordination Drawings: Prepare Coordination Drawings if limited space availability necessitates maximum utilization of space for efficient installation of different components or if coordination is required for installation of products and materials fabricated by separate entities.
 - 1. Content: Project-specific information, drawn accurately to scale. Do not base Coordination Drawings on reproductions of Contract Documents or standard printed data. Include following information, as applicable:
 - a. Indicate functional and spatial relationships of components of architectural, structural, and civil.
 - b. Indicate dimensions shown on Contract Drawings and make specific note of dimensions that appear to be in conflict with submitted equipment and minimum clearance requirements. Provide alternate sketches to Contracting Officer (CO) for resolution of such conflicts.
 - c. Indicate space requirements for routine maintenance and anticipated replacement of components during the life of the installation.
 - d. Indicate required installation sequences.
 - 2. Refer to individual Sections for Coordination Drawing requirements for Work in those Sections.
 - 3. Follow **NPS DSC Electronic Drawing Standards**. See [CAD/BIM \(Computer Aided Design/Building Information Modeling\) & Drafting Standards](#) page > **DSC Requirements** section.
- B. Coordination Drawing Organization: Organize coordination drawings as follows:
 - 1. Floor Plans: Show architectural, structural elements. Supplement plan drawings with section drawings where required to adequately represent the Work.
 - 2. Structural Penetrations: Indicate penetrations and openings.
 - 3. Embedded Items: Indicate sizes and locations of embedded items for anchor bolts, bearing plates, angles, and similar items.

4. Review: Contracting Officer will review coordination drawings to confirm Work is being coordinated; details of coordination are Construction Contractor's responsibility. If the Contracting Officer determines coordination drawings are not prepared in scope or detail, or are otherwise deficient, the Contracting Officer will inform Construction Contractor, who shall make changes and resubmit.
5. Coordination Drawing Prints: Prepare coordination drawing prints according to requirements in Section 01 33 23 "Submittal Procedures."

C. Coordination Digital Data Files: Prepare coordination digital data files according to:

1. File Preparation Format: Same digital data software program, version, and operating system as original Drawings.
2. File Submittal Format: Submit or post coordination drawing files using Portable Document Format (PDF) file format.
3. Contracting Officer will furnish Construction Contractor one set of digital data files (AutoCad.dwg) of Drawings for use in preparing coordination digital data files.
 - a. Contracting Officer makes no representations as to accuracy or completeness of digital data files as they relate to Drawings.
 - b. Digital Data Software Program: Drawings are available in AutoCad.dwg.

D. Division 1 submittals required prior to mobilization:

1. Letter designating Project Superintendent
2. Construction Baseline Schedule
3. A Comprehensive Schedule of Values
4. Accident Prevention Plan
5. A List of Subcontractors for this project
6. Written statements from Subcontractors certifying compliance with applicable labor standard clauses.
7. Certificates of Insurance and Standard Form SF1413 for Contactor and all Subcontractors
8. Waste Management Plan
9. Quality Control Plan
10. Under-an-Acre Pollution Prevention Plan (UPPP)
11. Historic Preservation Treatment Plan (HPTP)

1.5 REQUESTS FOR INFORMATION (RFIs)

- A. General: Immediately on discovery of the need for additional information or interpretation of Contract Documents, Construction Contractor shall prepare and submit an RFI utilizing form created on NPS project management and communication software.
1. Contracting Officer will not respond to RFIs submitted by other entities controlled by Construction Contractor.
 2. Coordinate and submit RFIs in a prompt manner to avoid delays in the work.
- B. Content of RFI: Include detailed, legible description of item needing information or interpretation and the following:
1. RFI number, numbered sequentially

2. Date
 3. RFI subject
 4. Specification Section number and title and related paragraphs, as appropriate.
 5. Drawing number and detail references, as appropriate.
 6. Field dimensions and conditions, as appropriate.
 7. Construction Contractor's suggested resolution: If suggested resolution impacts Contract Time or Contract Sum, Construction Contractor shall state impact in RFI.
 8. Construction Contractor's signature
 9. Requested date for response
 10. Attachments: Include sketches, descriptions, measurements, photos, Product Data, Shop Drawings, coordination drawings, and other information necessary to fully describe items needing interpretation.
 - a. Include dimensions, thicknesses, structural grid references, and details of affected materials, assemblies, and attachments on attached sketches.
- C. RFI Form: Complete and submit on NPS project management and communication software.
- D. Contracting Officer's Action: Contracting Officer will review each RFI, determine action required, and respond. The Contracting Officer will determine the critical nature of each RFI and issue response accordingly.
1. The following are not considered to be RFIs and will receive no action:
 - a. Requests for approval of submittals.
 - b. Requests for approval of substitutions.
 - c. Requests for approval of Construction Contractor's means and methods.
 - d. Requests for coordination information already indicated in Contract Documents.
 - e. Requests for adjustments in Contract Time or Contract Sum.
 - f. Requests for interpretation of Architect's actions on submittals.
 - g. Incomplete RFIs or inaccurately prepared RFIs.
 2. Contracting Officer's action may include a request for additional information; time for response will date from time of receipt of additional information.
 3. Contracting Officer's action on RFIs may result in need for a change to Contract Time or Contract Sum. All contract changes will be processed according to the terms and conditions of the contract.

1.6 PROJECT MANAGEMENT AND COMMUNICATION SOFTWARE

- A. Use to manage project communication and document sharing during contract period on:
1. Project directory
 2. Project correspondence
 3. Meeting agendas and minutes
 4. Contract modifications forms and logs
 5. RFI form and processing
 6. Task and issue management
 7. Photo documentation
 8. Baseline schedule, schedule updates and calendar management

9. Submittal form and processing
10. Payment coordination documentation
11. Reminder and tracking functions
12. Archiving functions
13. Notification of submittal and RFI statuses and current responsible party
14. Addendums

- B. Some documents are not suitable to be shared using NPS project management and communication software. Documents containing Personal Identifying Information (PII) (i.e. certified payrolls) shall not be shared and shall be coordinated with the project team as appropriate.

1.7 PROJECT MEETINGS

- A. Preconstruction Conference: Before the start of construction, the Contracting Officer will arrange an on-site meeting with Construction Contractor. Meeting agenda will include the following as a minimum:

1. Roles & Responsibilities / Lines of Authority
2. Park rules and regulations
3. Jobsite Safety
4. Resolution of comments on required Division 1 documents
5. Historic Preservation Treatment Procedures
6. Coordination of Subcontractors
7. Labor law application
8. Modifications
9. Payments to Construction Contractor
10. Payroll reports
11. Contract time
12. Liquidated damages
13. Construction Contractor Performance Evaluation
14. Display of Hotline posters
15. Notice to proceed
16. Correspondence procedures
17. NPS project management and communication software
18. Acceptance/rejection of work
19. Progress meetings
20. Submittal procedures
21. Environmental requirements
22. As-constructed drawings
23. Saturday, Sunday, holiday and night work.
24. Reference materials
25. Value engineering
26. Schedule of Values

- B. Progress Meetings: Contracting Officer will schedule weekly meetings with Construction Contractor.

1. Attendees: In addition to Government Representatives, each Construction Contractor, Subcontractor, supplier, and other entity concerned with current progress or involved in planning, coordination, or performance of future activities shall be represented.

Participants at meetings shall be familiar with the Project and authorized to conclude matters relating to the Work.

2. Agenda: Review and correct or approve minutes of previous progress meeting. Review other items of significance that could affect progress. Meeting agenda will include:

- a. Approval of minutes of previous meetings
- b. Submittal status
- c. Review of off-site fabrication and delivery schedules.
- d. Requests for information (RFI) and other issues.
- e. Modifications
- f. Work in progress and projected.
 - 1) Status of required inspections (Special Inspections, Accessibility, etc.)
- g. Inspections of work in progress and projected (Special inspections, Accessibility, etc.)
- h. Construction Schedule update (provide updated Critical Path Method (CPM)).
- i. Status of Project Record Drawings and O&M manuals.
- j. Other business relating to work.

- C. Preinstallation Conferences: Conduct at Project site before each construction activity that requires coordination with other construction.

1. Attendees: Installer and representatives of manufacturers and fabricators involved in or affected by installation and its coordination or integration with other materials and installations that have preceded or will follow shall attend meetings. Advise the Contracting Officer of scheduled meeting dates.
2. Agenda: Review progress of other construction activities and preparations for particular activity under consideration, including requirements for:

- a. Contract Documents
- b. Options
- c. Related RFIs
- d. Related Change Orders
- e. Purchases
- f. Deliveries
- g. Submittals
- h. Review of mockups
- i. Possible conflicts
- j. Compatibility requirements
- k. Time schedules
- l. Weather limitations
- m. Manufacturer's written instructions
- n. Warranty requirements
- o. Compatibility of materials
- p. Acceptability of substrates
- q. Temporary facilities and controls
- r. Space and access limitations
- s. Regulations of agency(ies) with jurisdiction
- t. Testing and inspecting requirements
- u. Installation procedures
- v. Coordination with other work
- w. Required performance results

- x. Protection of adjacent work
 - y. Protection of construction and personnel
3. Record significant conference discussions, agreements, and disagreements, including required corrective measures and actions.
 4. Reporting: Distribute minutes of meetings to each party present and to other parties requiring information.
 5. Do not proceed with installation if the conference cannot be successfully concluded. Initiate whatever actions are necessary to resolve impediments to the performance of Work and reconvene the conference at the earliest feasible date.

1.8 ENVIRONMENTAL COORDINATION

- A. Construction Contractor's Environmental Manager: Designate on-site party responsible for overseeing Construction Contractor's conformance to environmental goals for project and implementing procedures for environmental protection.
 1. Qualifications: Minimum 3 years Construction experience on projects of similar size and scope; with environmental procedures similar to this project; familiarity with Environmental Management Systems (EMSs) such as International Organization for Standardization (ISO) 14001; familiar with environmental regulations applicable to construction operations.
 2. Responsibilities: Responsibilities shall include:
 - a. Compliance with applicable Federal, State, and local environmental regulations, including maintaining required documentation.
 - b. Implementation of Waste Management Plan (WMP).
 - c. Present overview of environmental issues and summarize site specific procedures relating to management plans at Preconstruction conference.
 - d. Training for Construction Contractor personnel in accordance with position requirements.
 - e. Monitoring and documentation of environmental procedures.
- B. Perform project quality control in accordance with requirements specified in Related Sections, including:
 1. Quality Requirements
 2. Regulatory Requirements
 3. Noise and Acoustics Management
 4. Construction Waste Management
- C. Construction Contractor's Environmental Training Program: Construction Contractor shall provide environmental training for workers performing work on project site. Training shall include:
 1. Overview of environmental issues related to building industry.
 2. Overview of environmental issues related to Project.
 3. Review of site-specific procedures and management plans:
 - a. Construction Waste Management

- b. Noise and Acoustics Management
- c. Temporary Storm Water Pollution Prevention
- 4. Pollution Prevention (P2) practices: Submit evidence of familiarity with P2 practices.
- 5. Compliance with environmental regulations: As specified in Regulatory Requirements. Submit Construction Contractor 40 CFR (Code of Federal Regulations) employee training records upon request of the Contracting Officer.
- D. Provide documentation for environmental procedures as specified herein and in accordance with the approved Waste Management Plan and Under-an-Acre Pollution Prevention Plan.

PART 2 - PRODUCTS (Not Used)

PART 3 - EXECUTION (Not Used)

END OF SECTION 01 31 00

SECTION 01 32 16 – CONSTRUCTION SCHEDULE

PART 1 - GENERAL

1.1 SUMMARY

- A. Section consists of Construction Schedule requirements including:
 - 1. Schedule of Values
 - 2. Construction Schedule Requirements.
 - 3. Construction Schedule Updates.
 - 4. Time Impact Analysis.
- B. Purpose: The Construction Schedule ensures adequate planning, coordination, scheduling, and reporting during execution of the work by the Construction Contractor. It shall assist the Construction Contractor and Contracting Officer (CO) in monitoring the progress of the work, evaluating proposed changes, and processing Construction Contractor's monthly progress payments. It shall include the dates in the contract, phases, milestones, occupancies, holidays, weather consideration, a critical path, and the requirements of this section.

1.2 DEFINITIONS

- A. Activity: A discrete part of a project identified for planning, scheduling, monitoring, and controlling the construction project. Activities included in a construction schedule consume time and resources.
 - 1. Critical activities are activities on the critical path. They start and finish on the planned early start and finish times.
 - 2. Predecessor Activity: An activity that precedes another activity in the network.
 - 3. Successor Activity: An activity that follows another activity in the network.
- B. Cost Loading: Allocation of the Schedule of Values for completion of an activity as scheduled. The sum of costs for all activities must equal the total Contract Sum, unless otherwise approved by the Contracting Officer.
- C. Critical Path Method (CPM): Method of planning and scheduling a construction project where activities are arranged based on activity relationships. Network calculations determine when activities can be performed and the critical path of Project.
- D. Critical Path: Longest connected chain of interdependent activities through the network schedule that establishes minimum overall Project duration and contains no float.
- E. Float: Measure of leeway in starting and completing an activity.
 - 1. Float: Not for the exclusive use or benefit of the Government or Construction Contractor but is jointly owned.
 - 2. Free Float: Amount of time an activity can be delayed without adversely affecting the early start of the successor activity.

3. Total Float: Measure of leeway in starting or completing an activity without adversely affecting planned Project completion date.
- F. Resource Loading: Allocation of manpower and equipment necessary for completion of an activity as scheduled.
 - G. Fragnet: A partial or fragmentary network that breaks down activities into smaller activities for greater detail.

1.3 SUBMITTALS

- A. Electronic Copies: Schedules and reports submitted shall be posted on the NPS project management and communication software as PDF (portable document format) files. The intent of the Government is to limit the number of printed reports to those determined by the project team as essential.
- B. Schedule of Values: After contract award and before Pre-Construction conference, submit schedule of dollar values based on Contract Price Schedule.
- C. Construction Baseline Schedule: After contract award and before Pre-Construction conference, submit two paper copies of baseline schedule, large enough to show entire schedule for entire construction period. Utilize Schedule of Values in preparation of Construction Baseline Schedule.
- D. Critical Path Method (CPM) Reports: Concurrent with CPM schedule, submit three paper copies of the following computer-generated reports. For each activity, include activity number, activity description, cost and resource loading, original duration, remaining duration, early start date, early finish date, late start date, late finish date, and total float in calendar days.
 1. Activity Report: List of activities sorted by activity number and then early start date, or actual start date if known.
 2. Logic Report: List of predecessor and successor tasks for activities sorted in ascending order by activity number and then early start date, or actual start date if known.
 3. Total Float Report: List of activities sorted in ascending order of total float.
- E. Construction Schedule Updates: On or before 7th day preceding progress payment request date, submit estimates of percent completion of each schedule activity and necessary supporting data. Provide two paper copies.
- F. Construction Schedule Revisions and Time Impact Analysis: For each Construction Schedule revision, submit two paper copies of a Time Impact Analysis. Incorporate a Fragmentary Network (Fragnet) into currently accepted Construction Schedule demonstrating proposal to incorporate a modification, change, delay, or Construction Contractor request.

1.4 QUALITY ASSURANCE

- A. Construction Contractor shall meet with Contracting Officer on day of the preconstruction conference to go over:
 1. Review software limitations, content and format for reports.

2. Verify availability of qualified personnel needed to develop and update schedule.
3. Discuss constraints, including historic tree protection and coordination with future construction phase.
4. Review delivery dates for Government-furnished products.
5. Review schedule for work of separate Government contracts.
6. Review time required for review of submittals and re-submittals.
7. Review requirements for tests and inspections by independent testing and inspecting agencies.
8. Review time required for completion and startup procedures.
9. Review time required for obtaining and activating permits.
10. Review and finalize list of construction activities to be included in schedule.
11. Review baseline schedule comments, resolve issues and progress on incorporating them
12. Review procedures for updating schedule.
13. Discuss reporting requirements and establish protocol for naming and transmitting electronic schedules.

- B. Construction Contractor's Schedule Representative: Before the preconstruction conference, designate an authorized representative to be responsible for preparing and maintaining the Construction Schedule. Submit resume outlining qualifications of Scheduler to Contracting Officer for acceptance. Scheduler shall have prepared and maintained at least 5 previous schedules of similar size and complexity similar to this Contract, demonstrating proficiency of using scheduling software. Authorized representative will be responsible for preparing the Baseline Schedule, required updates, revisions, Time Impact Analyses, and reports.

1.5 COORDINATION

- A. Coordinate preparation and processing of schedules and reports with performance of construction activities and with scheduling and reporting of separate Construction Contractors.
- B. Coordinate Construction Baseline Schedule with Schedule of Values, list of subcontracts, Submittals Schedule, progress reports, payment requests, and other required schedules and reports.
1. In developing Construction Baseline Schedule, ensure Subcontractor's work at all tiers, and prime Construction Contractor's work, is included and coordinated.
 2. Secure time commitments for performing critical elements of work from parties involved.
 3. Coordinate each construction activity in network with other activities and schedule in proper sequence.

PART 2 - PRODUCTS

2.1 SCHEDULE OF VALUES

- A. Breakdown each lump-sum item into component work activities used in the schedule for which progress payments may be requested. Work activities broken out within schedule of values shall be integrated into and made a logical part of the construction baseline schedule. Total costs for the component work activities shall equal contract price for that lump-sum item. Contracting Officer may request data to verify accuracy of dollar values. Include mobilization, general

condition costs, overhead and profit in the total dollar value of unit price items and in the component work activities for each lump-sum item. Do not include mobilization, general condition costs, overhead or profit as a separate item.

- B. Do not break down unit price items. Use only the contract price for unit price items.
- C. Total cost of all items shall equal the contract price. The Schedule of Values will form the basis for progress payments and the Construction Schedule.

2.2 CONSTRUCTION SCHEDULE REQUIREMENTS

- A. Construction Baseline Schedule: Prepare Construction Baseline Schedule using a computerized, cost and resource-based, time-scaled Critical Path Method network analysis diagram for the Work.
 - 1. Develop and finalize Construction Baseline Schedule so it can be accepted for use no later than 30 days after date established for the Notice of Award.
 - a. Failure to include any work item required for performance of this Contract shall not excuse Construction Contractor from completing work within applicable completion dates, regardless of Governments acceptance of schedule.
 - 2. Establish procedures for monitoring and updating Construction Baseline Schedule and for reporting progress. Coordinate procedures with progress meeting and payment request dates.
- B. Construction Baseline Schedule Preparation: Prepare a list of all activities required to complete the Work. Using preliminary Critical Path Method network diagram, prepare a skeleton network to identify probable critical paths.
 - 1. Activities: Indicate estimated duration, sequence requirements, and relationship of each activity in relation to other activities.
 - 2. Critical Path Activities: Identify critical path activities, including those for interim completion dates. Scheduled start and completion dates shall be consistent with Contract milestone dates.
 - 3. Processing: Process data to produce output data on a computer-drawn, time-scaled network. Revise data, reorganize activity sequences, and reproduce as often as necessary to produce the Critical Path Method schedule within the limitations of the Contract Time.
 - 4. Show sequence and interdependence of activities required for completion of work. Ensure work sequences are logical and Construction Baseline Schedule shows a coordinated plan of the work.
 - 5. Resource loading of each activity shall include personnel by labor category and equipment type and capacity proposed to complete the activity in duration shown.
 - 6. Consider seasonal weather conditions in planning and scheduling work influenced by high and low ambient temperatures, wind, or precipitation to ensure completion of work within contract time.
 - 7. Time Frame: Proposed duration assigned to each activity shall be Construction Contractor's best estimate of time required to complete activity considering the scope and resources planned for activity.

- a. An early finish date may be shown but the late finish date shall be same date as last day of contract period. An early completion schedule shall contain:
 - 1) Insert an activity titled "Project Float" as a successor to last activity in early project completion schedule network.
 - 2) Add a milestone titled "Contract End Date" as a successor to the activity "Project Float".
 - 3) Add duration to the activity "Project Float" as required so the milestone "Contract End Date" equals the last day of Contract Period.
 - b. Contract completion date shall not be changed by submission of a schedule that shows an early completion date.
 - c. Construction Contractor shall limit use of lead or lag duration's between schedule activities.
 - d. Project Calendars: Develop and incorporate the following calendars:
 - 1) Administrative Calendar: Include calendar based on a 7-day week to be used on activities based on calendar days. Apply this calendar to administrative tasks or other tasks not affected by non-working days (Federal Holidays, weather, etc.).
 - 2) Project Calendar: Include calendar based on planned work week for the project. Include Federal Holidays, weekends, and non-workdays indicated in contract documents. Apply this calendar to activities not anticipated to be affected by weather. Be clear when identifying number of work days in work week.
 - 3) Weather Calendar: Utilize Project Calendar and show anticipated normal downtime related to weather as non-working time. Weather days shall be based on data for local area from a reliable source like the National Oceanic and Atmospheric Administration (NOAA), National Park Service records, or source acceptable to Contracting Officer. Apply this calendar to activities anticipated to be affected by weather.
 - e. Activity Duration: Define so no activity is longer than 14 days, except for non-construction activities including mobilization, shop drawings and submittals, fabrication and delivery of materials and equipment.
 - f. Procurement Activities: Include procurement process activities for long lead items and major items, requiring a cycle of more than 60 calendar days, as separate activities in the schedule. Procurement cycle activities can include submittals, approvals, purchasing, fabrication, and delivery.
 - g. Submittal Review Time: Include review and re-submittal times indicated. Coordinate submittal review times in Construction Baseline Schedule.
 - h. Substantial Completion: Allow time for Government administrative procedures necessary for certification of Substantial Completion. (For more information, refer to Specification 01 77 00 "Closeout Procedures.")
8. Constraints: Include constraints and work restrictions indicated in Contract Documents and as follows in schedule and show how the sequence of Work is affected.
- a. Work Restrictions: Show effect of the following on the schedule:
 - 1) Coordination with existing construction

- 2) Protection of adjacent trees
- 3) Use of premises restrictions
- 4) Provisions for future construction
- 5) Seasonal variations
- 6) Environmental control

b. Work Stages: Indicate important stages of construction for each major portion of the Work.

- 1) Subcontract awards
- 2) Submittals
- 3) Purchases
- 4) Mockups
- 5) Fabrication
- 6) Sample testing
- 7) Deliveries
- 8) Installation
- 9) Tests and inspections
- 10) Adjusting
- 11) Curing

9. Milestones: Include milestones indicated in Contract Documents in schedule, including, but not limited to Notice to Proceed, Substantial Completion.

C. Joint Review, Revision, and Acceptance:

1. Within seven calendar days of receiving Construction Contractor's proposed Construction Baseline Schedule, Contracting Officer shall review initial Construction Baseline Schedule.
2. Within seven calendar days after review, Construction Contractor shall revise and resubmit Construction Baseline Schedule in accordance with comments presented from review.
3. In the event the Construction Contractor fails to define any element of work, activity, or logic and the Contracting Officer review does not detect this omission or error, such omission or error, when discovered by Construction Contractor or Contracting Officer, shall be corrected by Construction Contractor within seven calendar days and shall not affect contract period.
4. Upon acceptance of the Construction Baseline Schedule, Contracting Officer saves schedule as a baseline and updates on a monthly basis. Construction schedule update will be used to evaluate Construction Contractor's monthly applications for payment based upon information developed at monthly Construction Schedule update meeting.

D. Cost Correlation: In the heading of the schedule, provide a cost correlation line, indicating planned and actual costs. On the line, show dollar volume of Work performed as of dates used to prepare payment requests.

1. Construction Contractor shall assign cost to construction activities on Construction Baseline Schedule. Costs shall not be assigned to submittal activities unless specified otherwise but may, with Contracting Officer's approval, be assigned to fabrication and delivery activities. Costs shall be included for testing and commissioning activities, operation and maintenance manuals, punch list activities, Project Record Documents, and demonstration and training (if applicable).

2. Each activity cost shall reflect an accurate value based on the Contract Price Schedule.
 3. Total cost assigned to activities shall equal total Contract Price.
- E. Recovery Schedule: When periodic schedule update indicates Work is 14 or more calendar days behind current accepted schedule, a separate recovery schedule indicating means by which Construction Contractor intends to regain compliance with the schedule shall also be submitted. Indicate changes to working hours, working days, crew sizes, and equipment required to achieve compliance, and date by which recovery shall be accomplished.
- F. Computer Software: Prepare schedules using a program developed specifically to manage construction schedules.
1. Use Microsoft Project or Primavera, for Windows 7 and newer.

PART 3 - EXECUTION

3.1 CONSTRUCTION SCHEDULE UPDATES

- A. Progress Meeting Updates: Provide a 2 week look-ahead schedule, derived from the currently accepted schedule, before each weekly progress meeting. Utilize look-ahead schedule to facilitate and take notes on discussions held during progress meeting.
- B. Monthly Schedule Updates:
1. General: Update Construction Schedule on monthly basis to reflect construction progress and activities throughout entire contract period and until project substantial completion. The status date of each schedule update shall be the 7th day preceding the progress payment request date.
 2. Procedure: Construction Contractor shall meet with Contracting Officer each month at Construction Schedule update meeting to review progress made through the status date of the Construction Schedule update, including dates activities were started or completed and percentage of work completed on each activity started or completed.
 3. Reports: Concurrent schedule revisions, prepare tabulated reports showing:
 - a. Identification of activities that have changed
 - b. Changes in early and late start dates
 - c. Changes in early and late finish dates
 - d. Changes in activity durations in workdays
 - e. Changes in the critical path
 - f. Changes in total float or slack time
 - g. Changes in the Contract Time
 4. Narrative: Report shall include a brief description of actual progress made during update period; actual and potential delaying activities; impediments to progress; issues related to inclement weather; progress toward established milestones and project float. Report shall include a brief description of work anticipated to be performed in the next month. Minor revisions to the schedule should be identified for evaluation and acceptance or rejection.
 5. As Work progresses, indicate Actual Completion percentage for each activity.
 6. If schedule update shows a late finish date after contract completion date, include:

- a. Known delays
 - b. Actions to get back on schedule
 - c. Pending modifications
 - d. Impediments or constraints affecting progress
- 7. Progress Payments: Monthly updating of the currently accepted Construction Schedule shall be an integral part of the process upon which progress payments will be made. If Construction Contractor fails to provide schedule updates or revisions, a portion of the monthly payment may be retained until corrections have been made.
- C. Distribution: Distribute copies of accepted schedule to Contracting Officer, Contracting Officers Representative, Construction Management Representative, Subcontractors, testing and inspecting agencies, and other parties identified by Construction Contractor with a need-to-know schedule responsibility.
 - 1. When revisions are made, distribute updated schedules to same parties and post in same locations. Delete parties from distribution when they have completed their assigned portion of the Work.
- D. Construction Schedule Revisions:
 - 1. Required Revisions: If, as a result of the monthly schedule update, it appears the currently accepted Construction Schedule no longer represents actual prosecution and progress of the work, Contracting Officer will request, and Construction Contractor shall submit, a revision to the Construction Schedule. Construction Contractor may also request reasonable revisions to currently accepted Construction Schedule in event the Construction Contractor's planning for the work is revised. If Construction Contractor desires to make changes, Construction Contractor shall notify Contracting Officer in writing, stating reason for proposed revision. Accepted revisions shall be incorporated into currently accepted Construction Schedule for next monthly schedule update.
 - 2. Procedure: If revision to currently accepted Construction Schedule is contemplated, Construction Contractor or Contracting Officer shall advise the other in writing at least seven calendar days prior to next monthly schedule update meeting, describing revision and reasons for the revision. Government-requested revisions will be presented in writing to the Construction Contractor, who shall respond in writing within seven calendar days.
 - 3. Reports: Concurrent with making revisions to schedule, prepare tabulated reports showing:
 - a. Identification of activities changed
 - b. Changes in early and late start dates
 - c. Changes in early and late finish dates
 - d. Changes in activity durations in workdays
 - e. Changes in critical path
 - f. Changes in total float or slack time

3.2 TIME IMPACT ANALYSIS FOR CONTRACT MODIFICATIONS CHANGES DELAYS AND CONSTRUCTION CONTRACTOR REQUESTS:

- 1. Requirements: When contract modifications or changes are initiated, delays experienced, or Construction Contractor desires to revise currently accepted Construction Schedule, Construction Contractor shall submit to Contracting Officer a written time impact analysis

illustrating the influence of modification, change, delay, or Construction Contractor request on contract time.

2. Time Extensions: Activity delays, resulting in a late completion date projection, shall not automatically mean an extension of contract time is warranted or due to Construction Contractor. It is possible a modification, change, or delay will not affect existing critical path activities or cause non-critical activities to become critical. A modification, change, or delay may result in absorbing a part of available total float that may exist within an activity chain of the Schedule, not causing any effect on contract time. Time extensions will be granted in accordance with terms of contract.
3. Extension of contract time will be granted only to the extent the equitable time adjustments to activity or activities affected by modification, change, or delay exceeds total (positive or zero) float available on a particular activity.
4. Procedure: Each time impact analysis shall be submitted within time period stated in a request for proposal, or time period designated under the clauses entitled Changes or Default. In cases where Construction Contractor does not submit a written request for extension of time and a time impact analysis within the designated time, it is mutually agreed that the particular modification, change, delay, or Construction Contractor request does not require an extension of the contract time. Upon acceptance, time impact analysis shall be incorporated into currently accepted Construction Schedule at next monthly schedule update.
5. Contract Modifications: Prepare time-impact analysis using fragnets to demonstrate effect of proposed change on overall Construction Schedule for each proposed contract modification concurrent with submission.

END OF SECTION 01 32 16

SECTION 01 32 33 – PHOTO DOCUMENTATION

PART 1 - GENERAL

1.1 SUMMARY

- A. Section includes administrative and procedural requirements for:
 - 1. Existing condition images
 - 2. Periodic construction images
- B. See Section 01 77 00 "Closeout Procedures" for a complete listing of closeout documents.

1.2 SUBMITTALS

- A. Construction Images: Submit images electronically within seven days of taking the image. Include:
 - 1. Date, time and number (sequentially number all images) in filename.
 - 2. Description of vantage point, indicating location, direction (by compass point), and elevation or story of construction.
 - 3. Submit digital images exactly as originally recorded in digital camera, without alteration, or modifications using image-editing software.
- B. Closeout: Submit complete set of digital image electronic files as a Project Record Document. Submit on Digital Video Disc (DVD).
 - 1. Provide index as separate file on Disc. List each image as a file name with number, date, and time. Include description and or vantage point image was taken.
 - 2. Submit images that have the same aspect ratio as the sensor, un-cropped.

PART 2 - PRODUCTS

2.1 FORMAT REQUIREMENTS

- A. Media: DVD-R Archival Gold
- B. Media Labels: Archival DVD labeling markers, archival labels, or direct print.
- C. Images: Provide sRGB (standard Red Green Blue) color images in JPEG (Joint Photographic Experts Group) format. Minimum sensor size of 8 megapixels, and at image resolution of not less than 3200 by 2400 and 300 dpi (dots per inch).

PART 3 - EXECUTION

3.1 CONSTRUCTION IMAGES

- A. General: Take digital images using the maximum range of depth of field, in-focus, to clearly show the Work. No blurry or out-of-focus areas accepted.
 - 1. Maintain index with each set of Construction images and identify the number, date, time, and description for each.
 - 2. Maintain one set of images accessible in field office at Project site available for reference.
- B. Existing Condition Images: Before commencement of excavation, commencement of demolition, starting construction, take color digital images of Project site and surrounding properties, including existing items to remain during construction, from different vantage points.
 - 1. Flag construction limits before recording construction images.
 - 2. Take eight separate images to show existing conditions adjacent to property before starting Work.
- C. Periodic Construction Images: Take 12 color, digital images weekly, with timing each month adjusted to coincide with cutoff date associated with each Application for Payment. Select vantage points to show status of construction and progress since last images were taken.
- D. Additional Images: Contracting Officer may issue requests for additional images.
 - 1. Three days advance, where feasible.
 - 2. In emergency situations, take additional images within 24 hours of request.
 - 3. Additional images include, but are not limited to:
 - a. Immediate follow-up when on-site events result in construction damage or losses.
 - b. Fabrication locations away from Project site.
 - c. Substantial Completion of a major phase or component of Work.
 - d. Extra record images at time of final acceptance.

END OF SECTION 01 32 33

SECTION 01 33 23 - SUBMITTAL PROCEDURES

PART 1 - GENERAL

1.1 SUMMARY

- A. Section includes administrative and procedural requirements for submitting Shop Drawings, Product Data, Samples, and other submittals.

1.2 DEFINITIONS

- A. Action Submittals: Written, graphic information, and physical samples that require Government's responsive action.
- B. Informational Submittals: Written information that does not require Government's responsive action. Submittals may be rejected for not complying with requirements.
- C. File Transfer Protocol (FTP): Communications protocol that enables transfer of files to and from another computer over a network and that serves as the basis for standard Internet protocols. An FTP site is a portion of a network located outside of network firewalls within which internal and external users are able to access files.
- D. Portable Document Format (PDF): An open standard file format licensed by Adobe Systems used for representing documents in a device-independent and display resolution-independent fixed-layout document format.

1.3 GENERAL SUBMITTAL PROCEDURES

- A. General: Prepare and submit submittals required by individual Specification Sections and in some cases as requested in drawings. Types of submittals are indicated in individual specific sections.
 - 1. Contracting Officer (CO) reserves right to require submittals in addition to those called for in individual sections.
- B. Coordination: Coordinate preparation and processing of submittals with performance of construction activities. Review for legibility, accuracy, completeness, and compliance with Contract Documents.
 - 1. Coordinate submittal with fabrication, purchasing, testing, delivery, other submittals, and related activities that require sequential activity.
 - 2. Coordinate transmittal of different types of submittals for related parts of Work so processing will not be delayed because of need for concurrent review coordination.
 - a. Contracting Officer reserves right to withhold action on a submittal requiring coordination with other submittals until related submittals are received.

- C. Submittal List: Submittal list is attached to the end of this Specification Section. The intent is to provide an overall summary of submittal requirements. The requirements of individual Specification Sections and terms and conditions of the Contract still apply regardless of what is shown on submittal list.
- D. Processing Time: Allow enough time for submittal review and re-submittals. Contract time extensions will not be authorized if submittals are not transmitted with enough time to permit processing, including re-submittals.
 - 1. Action Submittals
 - a. Initial Review: Allow 30 days for initial review of each submittal. Allow additional time if coordination with subsequent submittals is required.
 - b. Re-submittal Review: Allow 10 days for review of each re-submittal.
 - 2. Informational submittals
 - a. Review: Allow 10 days for review of each submittal.
- E. Approved Equal Submittal Requirements:
 - 1. For each item proposed, submit supporting data, including:
 - a. Drawings and samples as appropriate.
 - b. Comparison of the salient characteristics of the proposed item with that specified.
 - c. Changes required in other elements of the work because of the substitution.
 - d. Name, address, and telephone number of vendor.
 - e. Manufacturer's literature regarding installation, operation, and maintenance, including schematics for electrical and hydraulic systems, lubrication requirements, and parts lists. Describe availability of maintenance service, and state source of replacement materials.
 - 2. A request for approval constitutes a representation that Construction Contractor:
 - a. Has investigated the proposed item and determined that it is equal or superior in all respects to that specified.
 - b. Will provide the same warranties for the proposed item as for the item specified.
 - c. Has determined that the proposed item is compatible with interfacing items.
 - d. Will coordinate installation of an approved item and make changes required in other elements of the work because of the substitution.
 - e. Waives claims for additional expenses that may be incurred as a result of the substitution.
- F. Submittals: Identify and incorporate information in each submittal file as follows:
 - 1. Transmittal Form (CM-16): Utilize [NPS Transmittal Form](#) for all submittals. No action will be taken on a submittal item unless accompanied by this Transmittal Form.
 - 2. Physical samples: Complete Transmittal Form and deliver physical sample to the Contracting Officer (or designee) on site for processing. All comments and actions will be documented on the Transmittal Form.

- G. Identification: Submittal number or other unique identifier, including revision identifier.
 - 1. Submittal number shall use a sequential number (e.g. .001). Re-submittals shall include alphabetic suffix after another decimal point (e.g. .001.A).
- H. Re-submittals: Make re-submittals using same process used with initial submittal.
 - 1. Note date and content of previous submittal.
 - 2. Note date and content of revision in the title block on the Transmittal Form (CM-16) and clearly indicate extent of revision.
 - 3. Re-submit submittals until they are marked "No Exceptions Taken" or "Reviewed with Notations".
- I. Distribution: Furnish copies of final submittals to manufacturers, subcontractors, suppliers, fabricators, installers, and others as necessary for performance of construction activities.
- J. Use for Construction: Use only final submittals with mark indicating "No Exceptions Taken" or "Reviewed with Notations". Ensure notations have been incorporated and, at a minimum, keep one copy of final approved submittal on site for use during construction.

1.4 CONSTRUCTION CONTRACTOR'S USE OF CAD/BIM FILES

- A. General: At Construction Contractor's written request, copies of CAD/BIM (Computer Aided Design/Building Information Modeling) files will be provided to Construction Contractor for use in connection with Project, subject to:
 - 1. Files provided as is; no format or other changes to files or changes to objects in the drawing will be done by the Government.

PART 2 - PRODUCTS

2.1 ACTION SUBMITTALS

- A. Product Data: Collect information into a single submittal for each element of construction and type of product or equipment.
 - 1. If information must be specially prepared for submittal because standard printed data are not suitable for use, submit as Shop Drawings, not as Product Data.
 - 2. Mark each submittal to show which products and options are applicable.
 - 3. As applicable, include:
 - a. Manufacturer's product specifications.
 - b. Manufacturer's installation instructions: When Contract Documents require compliance with manufacturer's printed instructions, provide one complete set of instructions to Contracting Officer and keep another complete set of instructions at the project site until substantial completion.
 - c. Manufacturer's catalog cuts: Submit only pertinent pages; mark each page of standard printed data to identify specific products proposed for use.
 - d. Wiring diagrams showing factory-installed wiring.

- e. Printed performance curves.
 - f. Operational range diagrams.
 - g. Compliance with specified referenced standards.
 - h. Testing by recognized testing agency.
- 4. Submit product data in PDF (portable document format) file format before or concurrent with samples.
- B. Shop Drawings: Prepare Project-specific information, drawn accurately to scale. Do not base Shop Drawings on reproductions of the Contract Documents or standard printed data , unless submittal of CAD/BIM Drawings is otherwise permitted.
 - 1. Preparation: Fully illustrate requirements in Contract Documents. As applicable, include:
 - a. Dimensions
 - b. Identification of products
 - c. Fabrication and installation drawings
 - d. Roughing-in and setting diagrams
 - e. Shopwork manufacturing instructions
 - f. Templates and patterns
 - g. Schedules
 - h. Notation of coordination requirements
 - i. Notation of dimensions established by field measurement
 - j. Relationship to adjoining construction clearly indicated
 - k. Seal and signature of professional engineer if specified
 - 2. Submit shop drawings as PDF electronic file
- C. Samples: Submit Samples for review of kind, color, pattern, and texture for a check of these characteristics with other elements and for a comparison of these characteristics between submittal and actual component as delivered and installed.
 - 1. Transmit Samples that contain multiple, related components such as accessories together in one submittal package.
 - 2. Complete and post the Transmittal Form (CM-16) on the NPS project management and communication software website for processing and documentation of action on submitted samples.
 - 3. Identification: Attach label on unexposed side of Samples that includes:
 - a. Generic description of Sample
 - b. Product name and name of manufacturer
 - c. Sample source
 - d. Submittal Number and title of appropriate Specification Section
 - 4. Disposition: Maintain sets of approved Samples at Project site, available for quality-control comparisons throughout course of construction activity. Sample sets may be used to determine final acceptance of construction associated with each set.
 - 5. Samples for Initial Selection: Submit manufacturer's color charts consisting of units or sections of units showing full range of colors, textures, and patterns available.

- a. Number of Samples: Submit **two** full set(s) of available choices where color, pattern, texture, or similar characteristics are required to be selected from manufacturer's product line. Contracting Officer will return with options selected.
- 6. Samples for Verification: Submit full-size units or Samples of size indicated, prepared from same material to be used for the Work, cured and finished in manner specified, and physically identical with material or product proposed for use, and that show full range of color and texture variations expected. Samples include: partial sections of manufactured or fabricated components; small cuts or containers of materials; complete units of repetitively used materials; swatches showing color, texture, and pattern; color range sets; and components used for independent testing and inspection.
 - a. Number of Samples: Submit four sets of Samples. Contracting Officer will retain three Sample sets; remainder will be returned. Retain Sample set as a Project Record Sample.
- D. Construction Materials: Construction Contractor is encouraged to submit products made out of recycled or environmentally responsible material. Every effort will be made by NPS to approve these materials.

2.2 INFORMATIONAL SUBMITTALS

- A. General: Prepare and submit Informational Submittals required by individual Specification Sections.
 - 1. Post informational submittals as PDF electronic files directly to the NPS project management and communication software.
 - 2. Certificates and Certifications: Provide a notarized statement with signature of entity responsible for preparing certification. Certificates and certifications shall be signed by officer or other individual authorized to sign documents on behalf of that entity.
 - 3. Informational submittals that do not comply with requirements specified in Contract Documents will be rejected and one copy will be returned.
- B. Coordination Drawings: Comply with requirements specified in Section 01 31 00 "Project Management and Coordination."
- C. Construction Contractors Construction Schedule: Comply with requirements specified in Section 01 32 16 "Construction Schedule."
- D. Accident Prevention Plan: Comply with requirements specified in Section 01 35 23 "Safety Requirements."
- E. Schedule of Values: Comply with requirements specified in Section 01 32 16 "Construction Schedule."
- F. Waste Recycling Plan: Comply with requirements specified in Section 01 74 19 "Construction Waste Management and Disposal."
- G. Quality Control Plan: Comply with requirements specified in Section 01 40 00 "Quality Requirements."

- H. Storm Water Pollution Prevention Plan: Comply with requirements specified in Section 01 57 23 "Temporary Storm Water Pollution Prevention" and storm water permit requirements identified in Section 01 31 00 "Project Management and Coordination."
- I. Qualification Data: Prepare written information demonstrating capabilities and experience of firm or person. Include lists of completed projects with project names and addresses, names and addresses of architects and owners, and other information specified.
- J. Installer Certificates: Prepare written statements on manufacturer's letterhead certifying Installer complies with Contract Documents and, where required, is authorized by manufacturer for this specific Project.
- K. Manufacturer Certificates: Prepare written statements on manufacturer's letterhead certifying manufacturer complies with Contract Documents. Include evidence of manufacturing experience where required.
- L. Product Certificates: Prepare written statements on manufacturer's letterhead certifying product complies with Contract Documents.
- M. Material Certificates: Prepare written statements on manufacturer's letterhead certifying material complies with Contract Documents.
- N. Material Test Reports: Prepare reports written by a qualified testing agency, on testing agency's standard form, indicating and interpreting test results of material for compliance with requirements in Contract Documents.
- O. Product Test Reports: Prepare written reports indicating current product produced by manufacturer complies with requirements in Contract Documents. Base reports on evaluation of tests performed by manufacturer and witnessed by a qualified testing agency, or on comprehensive tests performed by a qualified testing agency.
- P. Research/Evaluation Reports: Prepare written evidence, from a model code organization acceptable to authorities having jurisdiction, that product complies with building code in effect for Project.
- Q. Preconstruction Test Reports: Prepare reports written by a qualified testing agency, on testing agency's standard form, indicating and interpreting results of tests performed before installation of product, for compliance with performance requirements in Contract Documents.
- R. Compatibility Test Reports: Prepare reports written by a qualified testing agency, on testing agency's standard form, indicating and interpreting results of compatibility tests performed before installation of product. Include written recommendations for primers and substrate preparation needed for adhesion.
- S. Field Test Reports: Prepare reports written by a qualified testing agency, on testing agency's standard form, indicating and interpreting results of field tests performed either during installation of product or after product is installed in its final location, for compliance with requirements in Contract Documents.

- T. Maintenance Data: Prepare written and graphic instructions and procedures for operation and normal maintenance of products and equipment. Comply with requirements specified in Section 01 78 23 "Operation and Maintenance Data."
- U. Design Data: Prepare written and graphic information, including: performance and design criteria, list of applicable codes and regulations, and calculations. Include list of assumptions and other performance and design criteria and a summary of loads. Include load diagrams if applicable. Provide name and version of software, if any, used for calculations. Include page numbers.
- V. Manufacturer's Instructions: Prepare written or published information documenting manufacturer's recommendations, guidelines, and procedures for installing or operating a product or equipment. Include name of product and name, address, and telephone number of manufacturer.
- W. Manufacturer's Field Reports: Prepare written information documenting factory-authorized service representative's tests and inspections. As applicable, include:
 - 1. Statement on condition of substrates and their acceptability for installation of product.
 - 2. Summary of installation procedures being followed, compliance with requirements and, if not, what corrective action was taken.
 - 3. Results of operational and other tests and a statement of whether observed performance complies with the requirements.

PART 3 - EXECUTION

3.1 CONSTRUCTION CONTRACTOR'S REVIEW

- A. Review each submittal and check for coordination with other Work of Contract and for compliance with Contract Documents. Note corrections and field dimensions.

3.2 CONTRACTING OFFICER'S ACTION

- A. General: Submittals will be disapproved without technical review if identification information is missing, not filled in, or if placed on back of submittal; an incorrect format of submittals is provided; transmittal form is incorrectly filled out; submittals are not coordinated; or submittals do not show evidence of Construction Contractor's approval.
 - 1. Any work done or orders for materials or services placed before approval shall be at Construction Contractor's own risk.
- B. Action Submittals: Contracting Officer will review each submittal, generate comments on corrections or modifications required, and indicate appropriate action on the Transmittal Form (CM-16). Submittal will be marked as defined below:
 - 1. NO EXCEPTIONS TAKEN: Submittal appears to confirm to design requirements, but contractor is still responsible for all aspects of construction.

2. REVIEWED WITH NOTATIONS: Minor corrections or clarifications required. Comments are clear and no further review is required. Construction Contractor shall address review comments when proceeding with the work.
 3. DISAPPROVED - RESUBMIT: Rejected as not in accordance with the contract or as requiring major corrections or clarifications. Contracting Officer will identify reasons for disapproval. Construction Contractor shall revise and resubmit with changes clearly identified.
- C. Informational Submittals: Contracting Officer will review each submittal and will either accept or reject it.
- D. Partial submittals are not acceptable, will be considered non-responsive, and will be returned without review.

END OF SECTION 01 33 23

Submittal List with Review Estimate Template

National Park Service (NPS) - Denver Service Center (DSC) | 1-27-21

SUBMITTAL LIST

Park Acronym/Project Management Information System (PMIS) Number:

PINN 331043

Project Title **Bear Valley School Rehabilitation Phase 1**

SUBMITTAL			REQUIREMENTS (Indicate with X)							
SPECIFICATION SECTION	PARAGRAPH NUMBER	DESCRIPTION	INFORMATIONAL				ACTION			
			CERTIFICATIONS OR LABORATORY TESTS	REPORTS OR CALCULATIONS OR PLAN	MANUFACTURER DATA AND INSTRUCTIONS	OTHER	SAMPLES	SHOP DRAWINGS	MANUFACTURER DATA AND INSTRUCTIONS	OTHER
01 31 00		Project Management and Coordination								
	1.4.A	Coordination Drawings								X
	1.4.B	Coordination Drawing Organization								X
	1.4.C	Coordination Digital Data Files				X				
	1.4.D	Division 1 documents								X
01 33 23		Submittal Procedures								
	2.1.A	Submittal Schedule								X
	2.1.B	Product Data							X	
	2.1.C	Shop Drawings						X		
	2.1.D	Samples					X			
	2.1.E	Construction Materials								X
	2.1.F	Coordination Drawings								X
	2.1.G	Quality Control Plan								X
	2.1.H	Qualification Data								X
	2.2.B	Construction Contractors Construction Schedule		X						

SUBMITTAL LIST										
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	2.2.C	Accident Prevention Plan		X						
	2.2.D	Schedule of Values		X						
	2.2.E	Waste Recycling Plan		X						
	2.2.F	Storm Water Pollution Prevention Plan		X						
	2.2.G	Indoor Air Quality Management Plan		X						
	2.2.H	Welding Certificates	X							
	2.2.I	Installer Certificates	X							
	2.2.J	Manufacturer Certificates	X							
	2.2.K	Product Certificates	X							
	2.2.L	Material Certificates	X							
	2.2.M	Material Test Reports		X						
	2.2.N	Product Test Reports		X						
	2.2.O	Research/Evaluation Reports		X						
	2.2.P	Preconstruction Test Reports		X						
	2.2.Q	Compatibility Test Reports		X						
	2.2.R	Field Test Reports		X						
	2.2.S	Maintenance Data			X					

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	2.2.T	Design Data			X					
	2.2.U	Manufacturer's Instructions			X					
	2.2.V	Manufacturer's Field Reports		X						
	2.2.W	Permit Compliance Products				X				
01 32 16		Construction Schedule								
	1.3.A	Electronic Copies				X				
	1.3.B	Schedule of Values:				X				
	1.3.C	Construction Baseline Schedule				X				
	1.3.D	Critical Path Method (CPM) Reports				X				
	1.3.E	Construction Schedule Updates				X				
	1.3.F	Construction Schedule Revisions and Time Impact Analysis				X				
01 32 33		Photographic Documentation								
	1.2.A	Construction Images				X				
	1.2.B	Closeout				X				
01 35 91		Historic Preservation Treatment Procedures								
	1.3.A	Historic Preservation Treatment Plan								X
	1.3.B	Alternative Methods and Materials								X
	1.3.C	Photographs or Videotape								X
01 40 00		Quality Requirements								

SUBMITTAL LIST										
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	1.4.A	Quality Control Plan								X
	1.4.B	Qualification Data								X
	1.4.C	Contractor Quality Control (CQC) Daily Reports		X						
	1.4.D	Test Reports		X						
	1.4.E	Accessibility Inspection Report		X						
	1.4.F	Off-Site Inspection Reports: Submit prior to shipment		X						
	1.4.H	Permits, Licenses, and Certificates	X							
01 57 23		Under-an-Acre Pollution Prevention								
	1.3.A	UPPP		x						
	1.3.B	Inspection Schedule				x				
	1.3.C	Erosion Control Products			x					
01 67 00		Product Requirements								
	1.3	Record Submittals-Sustainable Design Close-out Documentation			X					
01 73 40		Execution								
	1.2.A	Certificates	X							
	1.2.B	Landfill Receipts				X				
	1.2.C	Certified Surveys	X							
	1.2.D	Quantity Surveys		X						
01 74 19		Construction Waste Management and Disposal								
	1.4.A	Waste Management Plan		X						
	1.4.B	Progress Documentation		X						

SUBMITTAL LIST										
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			CERTIFICATIONS OR LABORATORY TESTS	REPORTS OR CALCULATIONS OR PLAN	MANUFACTURER DATA AND INSTRUCTIONS	OTHER	SAMPLES	SHOP DRAWINGS	MANUFACTURER DATA AND INSTRUCTIONS	OTHER
	1.4.C	Waste Reduction Calculations		X						
	1.4.D	Records of Donations				X				
	1.4.E	Records of Sales				X				
	1.4.F	Recycling and Processing Facility Records				X				
	1.4.G	Landfill and Incinerator Disposal Records				X				
	1.4.H	Qualification Data			X					
	1.4.I	Statement of Refrigerant Recovery				X				
	1.4.J	Progress payment requirements		X						
	1.4.K	Closeout Submittals				X				
02 43 96		Historic Removal and Dismantling								
	1.4.A	Qualification Statements								X
	1.4.C	Schedule								X
	1.4.D	Shop Drawings						X		
	1.4.E	Preconstruction Documentation								
	1.4.F	Removal and Dismantling Historic Treatment Program								
	1.4.G	List of Items to be Salvaged								
03 20 00		Concrete Reinforcing								
	1.2.A	Product data							X	
	1.2.B	Shop Drawings					X			
	1.3.A	Qualification data				X				
	1.3.B	Material Test Reports		X						

SUBMITTAL LIST

Park Acronym/Project Management Information System (PMIS) Number:

PINN 331043

Project Title **Bear Valley School Rehabilitation Phase 1**

SUBMITTAL			REQUIREMENTS (Indicate with X)							
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			CERTIFICATIONS OR LABORATORY TESTS	REPORTS OR CALCULATIONS OR PLAN	MANUFACTURER DATA AND INSTRUCTIONS	OTHER	SAMPLES	SHOP DRAWINGS	MANUFACTURER DATA AND INSTRUCTIONS	OTHER
03 30 00	1.3.C	Field quality-control reports		X						
		Cast-in-Place Concrete								
	1.3.A	Product data for each type of product							X	
	1.3.B	Design Mixtures						X		
	1.3.C	Concrete Schedule								X
	1.4.A	Qualification data				X				
	1.4.B	Material Certificates	X							
	1.4.C	Material Test Reports		X						
	1.4.D	Research Reports		X						
	1.4.E	Field quality-control reports		X						
06 10 00		Rough Carpentry								
	1.3.A	Product data for each type of product							X	
	1.4.A	Material Certificates		X						
06 16 00		Sheathing								
	1.2.A	Product data for each type of product							X	
	1.2.B	Sustainable Design Submittals							X	
07 16 00		Cementitious Dampproofing								
	1.2.A	Product Data							X	
07 21 00		Thermal Insulation								
	1.2.A	Product Data							X	
	1.2.B	Installer Qualifications						X		

SUBMITTAL LIST										
Park Acronym/Project Management Information System (PMIS) Number:						PINN 331043				
Project Title Bear Valley School Rehabilitation Phase 1										
SUBMITTAL			REQUIREMENTS (Indicate with X)							
SPECIFICATION SECTION	PARAGRAPH NUMBER	DESCRIPTION	INFORMATIONAL				ACTION			
			CERTIFICATIONS OR LABORATORY TESTS	REPORTS OR CALCULATIONS OR PLAN	MANUFACTURER DATA AND INSTRUCTIONS	OTHER	SAMPLES	SHOP DRAWINGS	MANUFACTURER DATA AND INSTRUCTIONS	OTHER
	1.2.C	Post installation inspection report data						X		
07 26 00		Vapor Retarder								
	1.2.A	Product Data								X
07 62 00		Sheet Metal Flashing and Trim								
	1.2.A	Product Data								X
	1.2.B	Shop Drawings							X	
	1.2.C	Samples						X		
	1.2.D	Samples for Initial Selection						X		
	1.2.E	Samples for Verification						X		
	1.3.A	Product Test Reports		X						X
	1.3.B	Qualification Statements				X				
	1.3.C	Sample warranties				X				
08 95 16		Wall Vents								
	1.2.A	Product Data								X
	1.2.B	Samples						X		

SECTION 01 35 13.22 – ARCHEOLOGICAL PROTECTION

PART 1 - GENERAL

1.1 SUMMARY

- A. The work of this section consists of protecting archeological and historical resources, and within and/or around the limits of disturbance limits.
- B. The area of work has the potential to contain precontact and historically significant buried features and artifacts. The Contractor shall strictly limit and control the depth and extent of excavations and disturbance, minimizing them to the greatest extent possible in order to avoid impacts within the project work area.
- C. The contractor shall provide professional archeological services that includes archeological monitoring to observe and document all excavation and ground disturbing activities. Archeological monitor requirements are specified in this Section.
- D. The Contractor shall provide maximum protection of government-provided data and information about precontact, historic resources, and other sensitive areas within the park. The Contractor may be required to sign the park's Sensitive Archeological and Ethnographic Data Confidentiality Agreement. Upon the completion of the project, such data and information will either be (1) returned to the government or (2) destroyed in an appropriate manner approved by the Contracting Officer.
- E. The contractor may not publish articles based on the research covered under this contract, without prior written consent of the park. The NPS will have unlimited rights to all records, images, maps, and notes produced in relation to this project. The completed documentation may be used and reproduced in any fashion that the NPS chooses, with appropriate credit given to the contractor.
- F. Any consultations and/or coordination with the office of the State Historic Preservation Officer or American Indian tribes traditionally associated with park lands will be the responsibility of the NPS and is not required of the contractor.

1.2 DEFINITIONS

- A. Archeological Resources: Archeological resources are physical evidences of past human activity, including evidences of effects of that activity on the environment. Archeological resources represent both prehistoric and historic time periods, found above and below ground and under water.
- B. Archeologically Sensitive Areas: Areas having potential to contain significant (National Register eligible) archeological resources. If National Register eligible or listed archeological resources could not be avoided, an appropriate mitigation strategy would be developed in consultation with state historic preservation officer and, if necessary, associated American Indian tribes.
- C. Cultural Resource Specialist: A person employed by the park that meets the Secretary of the Interior's Professional Qualification Standards.

- D. Archeological Monitor: A person employed by the Contractor that meets the Secretary of Interior's Professional Qualification Standards for Archeology which can be found at [Professional Qualifications Standards \(U.S. National Park Service\) \(nps.gov\) \(https://www.nps.gov/articles/sec-standards-prof-quals.htm\)](https://www.nps.gov/articles/sec-standards-prof-quals.htm) and meets the Archaeological Resource Protection Act (ARPA) standards for undertaking archeological investigations (43 CFR 7.8).
- A. Archeological Resources Protection Act (ARPA) of 1979 (Public Law (P.L.) 96-95; 93 United States Statutes at Large (Stat.) 712): defines archeological resources as any material remains of past human life or activities that are of archeological interest and at least 100 years old; Section 4 of the statute describes the requirements that must be met before Federal authorities can issue a permit to excavate or remove any archeological resource on Federal or Indian lands; the curatorial requirements of artifacts, and other materials excavated or removed.
- B. Non-sensitive Areas: Areas with little potential of containing significant (National Register eligible) archeological resources. Section 106 of the National Historic Preservation Act (54 U.S.C. 306108): requires federal agencies to take into account the effects of their undertakings on historic properties either listed in or eligible to be listed in the National Register of Historic Places and provide consulting parties a reasonable opportunity to comment before making decisions.
- C. [NPS Management Policies \(http://www.nps.gov/subjects/policy/management-policies.htm\)](http://www.nps.gov/subjects/policy/management-policies.htm)
- D. [NPS Director's Order No. 28 and 28A \(Archeology\) \(see http://www.nps.gov/subjects/policy/directors-orders.htm\)](http://www.nps.gov/subjects/policy/directors-orders.htm)
- E. Secretary of the Interior's Standards and Guidelines for Archeology and Historic Preservation. [Archeology and Historic Preservation; Secretary of the Interior's Standards and Guidelines \(nps.gov\) \(https://www.nps.gov/subjects/historicpreservation/upload/standards-guidelines-archeology-historic-preservation.pdf\)](https://www.nps.gov/subjects/historicpreservation/upload/standards-guidelines-archeology-historic-preservation.pdf)
- F. Secretary of the Interior's Standards for the Treatment of Historic Properties: common sense historic preservation principles in non-technical language. They promote historic preservation best practices and can be found at <https://www.nps.gov/tps/standards.htm>

1.3 SUBMITTALS

- A. Archaeological Resources Protection Act (ARPA) permit application for work performed by a contracted archeological monitor. ARPA permit application shall be submitted at least 75 days before Contractor mobilization unless approved by Contracting Officer.
- B. Resume and qualifications of proposed Archeological Monitor(s) for the project.
- C. Preliminary Report
 - 1. Draft report - Due 30 days after completion of the field component.
 - a. Draft report
 - b. Draft site records
 - c. Geodatabase
 - d. Photographs and photo log
 - 2. Revised Draft report – Due 21 days after receipt of NPS comments on draft report
 - a. Revised draft report

- b. Revised draft site records
 - c. Geodatabase
 - d. Photographs and photo log
- D. Final Report/Final Project Deliverables - Due 15 days after receiving Revised Report comments.
 - 1. Final Report
 - 2. Final Site Records
 - 3. Geodatabase
 - 4. Photographs (all project photographs) and photo log
 - 5. All field records, including field notes
 - 6. Any archeological resources collected during the project (if applicable)

1.4 QUALITY ASSURANCE

- A. At least one week before on-site work begins, Contractor shall meet with Contracting Officer and Archeological Monitor to discuss Daily Work Schedule, equipment, and special methods used in archeologically sensitive areas. Contractor shall ensure approved Daily Work Schedule is followed throughout construction.
- B. Weekly progress meetings, as specified in Section 01 31 00 "Project Management and Coordination" shall include Archeological, and Historical Resource Protection as an agenda item to address:
 - 1. Identification of significant features within the work area and a protection plan for features that will be:
 - a. Protected and avoided.
 - b. Protected with minimal impacts (features that will have minimal destruction, damage, or alteration and features that will be temporarily removed, stored, and reinstalled).
 - c. Removed and replaced with materials in-kind following the Secretary of the Interior's Standards for the Treatment of Historic Properties.
- C. The contractor is responsible for quality control, which includes ensuring the technical accuracy and completeness of the deliverables prior to submission to the park. The contractor's quality control efforts shall ensure that all draft and final deliverables have been reviewed for consistency, professional and copy quality, technical adequacy, and readability. Quality control review shall ensure appropriate analysis with sufficient support and rationale for National Register of Historic Places (NRHP) eligibility recommendations.
- D. Archaeological Resources Protection Act (ARPA) Permit as specified in the Section.
- E. Report data, GPS information, photographs and final report shall meet the electronic format requirements specified in this Section.

1.5 ARCHEOLOGICAL INVESTIGATION BY NON-NPS PERSONNEL

- A. An Archaeological Resources Protection Act (ARPA) permit is required for any archeological work that is performed by a subcontractor.

- B. Permits are issued by the NPS under the Archaeological Resources Protection Act of 1979 (16 U.S.C. 470aa-mm) and its regulations (43 CFR 7). The NPS does not issue a permit for archeological investigations carried out by NPS archeologists, or to archeologists working on NPS archeological projects under a contract or cooperative agreement.
- C. Permit form and guidance can be found at <https://home.nps.gov/orgs/1187/issue-arpa-permits.htm> and <https://www.nps.gov/articles/series.htm>
 - 1. Applicants should submit Permit Application (DI Form 1926). Submit permit application in Word (.docx) format to the Contracting Officer.
 - 2. The ARPA permit process typically takes 75 days.

PART 2 - PRODUCTS

2.1 DAILY WORK SCHEDULE

- A. Daily Work Schedule is required for work occurring within archeologically sensitive areas. Include all work that is to occur within the area and key the schedule to the drawings to include:
 - 1. Starting and ending dates of ground-disturbing construction.
 - 2. Locations of temporary facilities, such as barriers, field offices, staging areas, sanitary facilities, borrow pits, and haul and access roads.
 - 3. Types of construction, such as clearing, topsoil stripping, structure or trench excavation, landscaping, and post construction clean-up.
 - 4. Methods and equipment used for each type of construction.
 - 5. Plan for relocating work in the event of temporary work stoppages at each archeologically sensitive area

2.2 ARCHEOLOGICAL SERVICES

- B. Documentation and evaluation of archeological and historic resources shall be conducted by historic preservation professionals meeting the Secretary of the Interior's Standards.
- C. Data Transfer: All sensitive archeological site location information will be submitted to the park electronically via the MS Teams project link to the park's cultural resource specialist. If submittals cannot be made via the MS Teams, submittals shall be made to the park via the Contractor's FTP transfer or encrypted USB Drives.
- D. Photographs: The Contractor will follow the NRHP Submittal Standards (see link). A photo log will be completed and include the view, location, direction, and description of each image. Photography will follow the standards set out in the National Register of Historic Places and National Historic Landmarks Consolidated and Updated Photograph Policy 2024 (NRHP/NHL Consolidated Photograph Policy 2024): <https://www.nps.gov/subjects/nationalregister/upload/NR-NHL-photo-policy-2024-01-02.pdf>
- E. Artifact Collection Guidelines. It is not anticipated that any artifacts or other archeological materials will be collected during the monitoring. If unusual or particularly significant diagnostic

artifacts are discovered, the Park Archeologist shall be consulted, and their approval secured prior to collection.

2.3 ARCHEOLOGICAL MONITORING

- A. The Contractor Shall perform archeological monitoring to identify and document any archeological deposits or features that may be revealed during ground disturbing construction activities.
- B. The methodology shall consist of visual observation and monitoring of project activities with the potential to cause disturbance to surface/subsurface soils. The on-site monitor must be always present when ground disturbing project activities are being conducted.
- C. On-site monitoring shall be documented using park specific field forms and a daily field log, supplemented by photography and scaled drawings as required. GPS data collection shall include the location of identified archeological deposits, artifacts, and features.
- D. As ground disturbance occurs, the archeological monitor shall examine excavated materials, using construction layout centerline and perimeter staking as a reference point to record locations of findings in addition to GPS data.
- E. If the archeological monitor discovers potentially significant archeological resources or human remains, the monitor shall stop work in the area of the discovery and make a reasonable effort to protect the area from further disturbance and then immediately notify the CO, COR, and park archeologist.
- F. The archeological monitor shall be empowered to stop and/or redirect ground disturbing activities in the event that a discovery of potentially significant archeological resources or human remains is made. If human remains are identified, the archeological monitor shall confirm the discovery in writing to the CO and the Cultural Resources Program Manager/Archeologist and the NPS will then develop and implement an appropriate course of action.
- G. As directed by Cultural Resources Program Manager/Archeologist, the archeological monitor shall record, document, and assess the discovery using the appropriate park Cultural Resource Documentation Form.
- H. The archeological monitor shall fully photo-document such discoveries and record: measurements, including dimensions of discoveries and depth at which they were identified; geospatial locations of discoveries using GPS with sub-meter accuracy; sediment types and Munsell colors observed for all identified strata; and excavation profiles through drawing and digital photography.
- I. Artifacts shall undergo in-field analysis and be counted, measured, and photographed before being left at the site
- J. Daily field notes shall be kept to document time on site, NPS and other contacts, and any other information relevant to describing project activities and supporting invoices.
- K. Photographs shall be taken of each day's activities, and any potentially significant archeological deposits or features discovered. A photo log will be completed with information on each photo,

including view, location, direction, and description of each image. Select photographs shall be included in the monitoring report. All photographs taken in the course of the project will be submitted to the NPS with the final deliverables. Photographic documentation of features or artifacts encountered shall include but not be limited to an appropriate scale, north arrow, and signboard describing the date, location, trinomial site number if associated with a recorded site and soil horizon if known. Specifications for taking (TIFF or RAW format) the photographs and labeling the files and disks shall meet the NRHP/NHL Consolidated Photograph Policy (2024), <https://www.nps.gov/subjects/nationalregister/upload/NR-NHL-photo-policy-2024-01-02.pdf>.

- L. Artifact collection is not anticipated. If unusual or particularly significant diagnostic artifacts are discovered, the park archeologist shall be consulted, and their approval secured prior to collection. If artifact collection is authorized, project documentation shall require preparation of artifacts for curation in consultation with the park archeologist/curator.
- M. All final digital files pertaining to the project, including the final report in MS Word and PDF, and all graphics, maps plans, photographs, and supporting materials will be provided electronically.

2.4 REPORTING

A. Preliminary Report

1. The monitoring report shall include a summary of the undertaking and description of the project area. It shall include detailed documentation of archeological monitoring, investigations, and discoveries of archeological resources within the areas within the project area. This shall include high-resolution digital photography, and a narrative description of the field methods, and results. The report shall include the steps to taken to protect the resources and documentation of resource impacts.
2. The report should be organized by Description of Project/Undertaking, Environmental Setting, Cultural Historical Context, Class I File Search, Monitoring Methods, Results, Discussion and Conclusion.
3. Daily field notes and park Cultural Resource Documentation Forms shall be included as appendices.
4. The contractor shall submit the geodatabase (zipped), compatible with ArcGIS Pro version 3.x
5. Existing cartographic standards for the NPS shall be incorporated into all GIS mapping deliverables. The following elements shall be incorporated into all GIS generated maps:
 - a. All layers represented in the geographic extent must be identified and clearly labeled in the legend.
 - b. Symbolology in legend should match that in the geographic extent.
 - c. Where applicable, include a set of reference maps (e.g., overview maps with plot locations, cities, roads).
 - d. Include reference citations for all data layers reproduced in the map.
 - e. Map will contain relative path name location of the document within the Project folder.

6. The draft report and documentation forms shall be complete, and reviewed by archeological contractor's subject matter expert, technical editor, and copy editor before submission to NPS. If the park begins review and finds any of the following the report and/or documentation forms will not be accepted:
 - a. Three or more instances where NRHP eligibility recommendations are not consistent between the report and corresponding documentation form.
 - b. Report sections are incomplete or missing.
 - c. Report and/or documentation forms shall be free of typos, grammatical errors, or technical errors.
 - d. Documentation forms are not properly formatted and/or missing photos/map.
 - e. Two or more documentation forms are missing values for two or more data fields.
 7. Draft report: (one copy (1) in MS Word and one (1) copy in Adobe Acrobat PDF with lines numbered on numbered pages. Draft site records: one (1) copy in MS Word and one (1) copy in Adobe Acrobat PDF of all site forms. Geodatabase: one (1) geodatabase (zipped), compatible with ArcGIS Pro version 3.x. Photographs: one (1) copy of all photographs following guidance in NRHP/NHL Consolidated Photograph Policy 2024 and one (1) copy of project photo log.
 - a. The NPS will have thirty (30) calendar days to review the first draft report submittal and provide consolidated comments to the Contractor.
 8. Revised draft report: (one copy (1) in MS Word and one (1) copy in Adobe Acrobat PDF with lines numbered on numbered pages of both copies. Revised draft site records: one (1) copy in MS Word and one (1) copy in Adobe Acrobat PDF of all site forms. Geodatabase: one (1) geodatabase (zipped), compatible with ArcGIS Pro version 3.x. Photographs: one (1) copy of all photographs following guidance in NRHP/NHL Consolidated Photograph Policy 2024 and one (1) copy of project photo log.
 - a. Within twenty-one (21) calendar days of receipt of the NPS comments on the draft report, the Contractor shall revise the draft report and submit the revised draft report to the NPS for review. The NPS will have thirty (30) calendar days to review the revised draft report submittal and provide consolidated comments to the Contractor.
- B. Final Report/Project Deliverables
1. Within fifteen (15) calendar days of receipt of the NPS comments on the revised draft report, the Contractor shall submit the final report and all associated documentation.
 2. Deliverables:
 - a. Final report: one copy (1) in MS Word and one (1) copy in Adobe Acrobat PDF. Five hard copies, spiral bound.
 - b. Final site records: one (1) copy in MS Word and one (1) copy in Adobe Acrobat PDF of all site forms.
 - c. Geodatabase: one (1) geodatabase (zipped), compatible with ArcMap or ArcPro.
 - d. Photographs: one (1) copy of all project photographs in following guidance in NRHP/NHL Consolidated Photograph Policy 2024 and one (1) copy of project photo log.
 - e. All field records - originals or copies of material associated with the project - including field notes.
 - f. If applicable, archeological resources collected during the project shall be returned to Cultural Resources Program Manager/Archeologist for NPS cataloging, accessioning, and curation.

PART 3 - EXECUTION

3.1 BARRICADES

- A. Comply with requirements specified in Section 01 50 00 "Temporary Facilities and Controls."

3.2 OBSERVATION

- A. Archeological Monitor will observe ground-disturbing site work, including construction of temporary facilities, at archeologically sensitive areas, from a safe location mutually agreed on by Contractor and Monitor. As new ground is broken, Monitor will examine excavated materials, using construction layout centerline and perimeter staking as a reference point to record locations of findings.

3.3 DISCOVERY OF RESOURCES

- A. If Archeological Monitor discovers resources, immediate relocation of work to a non-sensitive area may be required for Monitor to identify and document resources and, if necessary, develop appropriate mitigation plan. While Archeological Monitor is documenting resources in sensitive areas, Contractor shall relocate work to non-sensitive areas where monitoring is not normally required.
- B. If resources are discovered while Archeological Monitor is absent, stop work immediately and report the discovery to the Contracting Officer.

3.4 WORK STOPPAGE

- A. Contractor shall plan, schedule, and execute work to prevent stoppages at one area from stopping all work at construction site.

END OF SECTION 01 35 13.22

SECTION 01 35 23 - SAFETY REQUIREMENTS

PART 1 - GENERAL

1.1 SUMMARY

- A. Section includes establishing an effective accident prevention program and providing a safe working environment for personnel and visitors.

1.2 CONDITIONS PRESENT FOR PROJECT

- A. Confined spaces, trenching, scaffolding, crane and lifting equipment procedures, wildlife, weather, hazardous materials, flooding, fatigue, heat, cold, among many other requirements are present. Wildlife include but are not limited to deer, bat, raccoon, squirrel, and bird.
 - 1. Wildfire issues may result in local fire restrictions.
 - 2. Wildlife interactions shall be minimized for everyone's safety.
 - a. No feeding the wildlife.
 - b. Food storage regulations shall be followed.
 - c. Bear, mountain lion, and any unusual sightings shall be reported to the Contracting Officer.
 - 3. The contractor is recommended to include these subjects in the weekly safety meeting on an ongoing basis.

1.3 SUBMITTALS

- A. Accident Prevention Plan (APP): Submit APP after contract award and before Pre-Construction conference. Contracting Officer (CO) will review proposed APP. If APP requires any revisions or corrections, Contractor shall resubmit Plan within 10 days. No progress payments will be made until the APP is accepted.

1.4 QUALITY ASSURANCE

- A. Comply with contract clauses "Accident Prevention" and "Permits and Responsibilities." In case of conflicts between Federal, State, and local safety and health requirements, the most stringent shall apply. Onsite equipment shall meet 29 CFR 1926 (Code of Federal Regulations) (Occupational Safety and Health Administration (OSHA)) requirements. Failure to comply with requirements of this section and related sections may result in suspension of work.
- B. Site Safety Supervisor:
 - 1. Designate authorized onsite representative for preparation and maintenance of the APP.
 - 2. Shall be responsible for:
 - a. Implementation and enforcement of the APP
 - b. Daily safety inspections
 - c. Conducting and documenting weekly and monthly safety meetings

- d. Review of safety requirements at progress meetings
- e. Compilation and maintenance of Safety Data Sheets (SDS) and safety reference materials
- f. Tracking and resolution of safety violations
- g. Site personnel and visitor compliance with site safety and health requirements and APP
- h. Investigation and reporting of accidents and injuries

C. Qualifications of Employees:

- 1. Physically and able to perform their assigned duties in a safe manner.
- 2. Do not allow employees whose ability or alertness is impaired because of prescription or illegal drug use, fatigue, illness, intoxication, or other conditions that may expose themselves or others to injury to perform work.
- 3. Provide operating instructions for equipment. Operators of vehicles, hoisting equipment, and hazardous plant equipment shall be able to understand signs, signals, operating instructions, and be fully capable of operating such equipment. Retain copies of operator licenses and certifications onsite.

1.5 ACCIDENT REPORTING

- A. Reportable Accidents: Defined as: death, occupational disease, and/or traumatic injury to employees or the public; fires; and/or property damage by accident in excess of \$100.
 - 1. Notify Contracting Officer immediately in the event of a reportable accident.
 - 2. Fill out and forward an Accident/Property Damage Report Form (CM-22) to Contracting Officer within 7 days of a reportable accident. Obtain form from Contracting Officer.

PART 2 - PRODUCTS

2.1 ACCIDENT PREVENTION PLAN (APP)

- A. APP shall be written to comply with OSHA and project requirements (generic plan is not acceptable) including but not limited to:
 - 1. Name and qualifications of supervisor responsible to carry out program.
 - 2. Weekly and monthly safety meetings shall be documented with topics and attendees.
 - 3. First aid and rescue procedures.
 - 4. Job Hazard Analysis (JHA) for each major phase. List of hazards associated and methods proposed to provide for property protection and safety of the public, National Park Service personnel, and Contractor's employees. Include initial and continuing training.
 - 5. Planning for possible emergency situations, as detailed in Article 1.2. Such planning shall take nature of construction, site conditions, and degree of exposure of persons and property into consideration.

2.2 FIRST AID FACILITIES

- A. Provide adequate facilities for number of employees and appropriate to construction hazards.

2.3 PERSONNEL PROTECTIVE EQUIPMENT (PPE)

- A. Selection shall conform to OSHA Subpart E.

PART 3 - EXECUTION

3.1 DAILY SAFETY INSPECTIONS

- A. Conduct daily safety inspections and maintain daily safety reports which include:
 - 1. Area/operation inspected
 - 2. Date of inspection
 - 3. Identified hazards
 - 4. Corrective actions taken

3.2 EMERGENCY INSTRUCTIONS

- A. Post telephone numbers and reporting instructions for ambulance, physician, hospital, fire department, and police in conspicuous locations at work site.

3.3 FIRE AND LIFE SAFETY

- A. Comply with requirements of 2024 International Fire Code Chapter 33 Fire Safety During Construction and Demolition.

3.4 HAZARDOUS MATERIALS

- A. Hazardous materials: Explosive, flammable, poisonous, corrosive, oxidizing, irritating, or otherwise harmful substances that could cause death or injury.
- B. Store hazardous materials in accordance with manufacturer's and OSHA Subpart D requirements. Maintain Safety Data Sheets (SDS) for each chemical readily available on site.
 - 1. Immediately report spills of hazardous materials to the Park.
 - 2. Maintain a spill emergency response kit.
 - 3. Train employees how to respond to a spill and use emergency response kit.

3.5 PROTECTIVE EQUIPMENT

- A. Inspect personal protective equipment daily and maintain in a serviceable condition. Clean, sanitize, and repair personal items as appropriate before issuing to another individual.

3.6 SAFETY MEETINGS

- A. As a minimum, conduct one weekly 15-minute "toolbox" safety meeting conducted by a foreman or supervisor and attended by construction personnel at worksite. Topics shall coincide with work scheduled for following week. Document and submit meeting minutes to Contracting Officer within one day after meeting.
- B. Conduct monthly safety meetings for personnel, contractors, and subcontractors performing work on the site. Notify Contracting Officer of meeting dates and times. Meetings shall be used to: review effectiveness of Contractor's safety effort; resolve current health and safety problems; provide a forum for planning safe construction activities, and for updating Accident Prevention Plan. Contracting Officers Representative will attend meetings and enter results of meetings into the daily log.

3.7 HARD HATS AND PROTECTIVE EQUIPMENT AREAS

- A. A hard hat use area shall be designated by Contractor. Hard hat area shall be posted by Contractor in a manner satisfactory to Contracting Officer.
- B. It is Contractor's responsibility to require persons working on or visiting site to wear hard hats and PPE in good repair at all times. As a minimum, maintain two hard hats and other APP required equipment.

3.8 TRAINING

- A. First Aid: Provide training to personnel to ensure prompt and efficient first aid.
- B. Hazardous Material: Train and instruct each employee exposed to hazardous material in safe and approved methods of handling and storage.

END OF SECTION 01 35 23

SECTION 01 35 91 - HISTORIC PRESERVATION TREATMENT PROCEDURES

PART 1 - GENERAL

1.1 SUMMARY

- A. Section includes special procedures for historic treatment on Project including, but not limited to:
 - 1. Definitions
 - 2. Submittals
 - 3. Quality Assurance
 - 4. Storage and protection of existing historic materials
 - 5. Project site conditions
 - 6. Historic Preservation Treatment Plan
 - 7. Protection, General
 - 8. Protection during application of chemicals
 - 9. Protection during use of heat-generating equipment
 - 10. Historic preservation treatment procedures

1.2 DEFINITIONS

- A. "Preservation" - To apply measures necessary to sustain existing form, integrity, and materials of historic property. Work may include preliminary measures to protect and stabilize the property.
- B. "Rehabilitation" - To make possible a compatible use for property through repair, alterations, and additions while preserving those portions or features that convey its historical, cultural, or architectural values.
- C. "Restoration" - To accurately return form, features, and character of a property to its appearance at a particular period of time by means of removal of features from other periods in its history and repair and reconstruction of missing and deteriorated features from the restoration period.
- D. "Reconstruction" - To reproduce in exact form and detail, a building, structure, or artifact as it appeared at a specific period in time. Reconstructed elements do not possess historic integrity in their own right since they are-not original fabric.
- E. "Stabilize" - To apply measures designed to reestablish a weather-resistant enclosure and structural reinforcement of an item or portion of the building while maintaining essential form as it exists at present. This level of intervention is aimed at retarding or arresting adverse impacts to structures.
- F. "Protect and Maintain" - To remove deteriorating corrosion, reapply protective coatings, and install protective measures such as temporary guards; to provide the least degree of intervention.
- G. "Repair" - To stabilize, consolidate, or conserve; to retain existing materials and features while employing as little new material as possible. Repair includes patching, piecing-in, splicing, consolidating, or otherwise reinforcing or upgrading materials. Within restoration, repair also includes limited replacement in kind, rehabilitation, and reconstruction, with compatible

substitute materials for deteriorated or missing parts of features when there are surviving prototypes.

- H. "Replace" - To duplicate in its entirety, a historic element or feature by matching its historic pattern, detail and appearance. Replacement is justified when original or historic elements are damaged beyond repair or are missing. Replacement conditions and methods include:
 - 1. Replacement with Original or Historic Fabric: Includes fabric salvaged from other locations or projects having identical architectural qualities. Duplication of appearance using identical material possessing historical significance.
 - 2. Replacement with New Materials: Includes replacement with new material of like kind (custom fabricated or manufactured). Duplication of appearance using like material.
 - 3. Replacement with Substitute Materials: Includes replacement with a compatible substitute that is frequently contemporary and unlike the historic fabric. Duplication of appearance using modern (non-traditional) material. Use of substitute materials is not approved unless matching materials are not available.
- I. "Remove" - To demolish or detach items from existing construction and legally dispose of them off-site unless indicated to be removed and salvaged or removed and reinstalled.
- J. "Remove and Salvage" - To detach items from existing construction and deliver them to Contracting Officer (CO) or designee.
- K. "Remove and Reinstall" - To detach items from existing construction, repair and prepare for reuse, and reinstall where indicated.
- L. "Existing to Remain" or "Retain" - Existing items of construction not to be removed and not otherwise indicated to be removed and salvaged or removed and reinstalled.
- M. "Material in Kind" - Material that closely matches existing materials through comparison of architectural qualities and salient characteristic such as species, cut, color, grain, dimension, profile, thickness, and finish.

1.3 SUBMITTALS

- A. Historic Preservation Treatment Plan:
 - 1. After contract award and before Pre-Construction conference, submit for approval a written Historic Preservation Treatment Plan (HPTP).
 - 2. If the plan requires revisions or corrections, Contractor shall resubmit plan within 10 days.
 - 3. No change in approved plan may be made without written concurrence by Contracting Officer.
- B. Alternative Methods and Materials: If alternative methods and materials to those indicated are proposed for any phase of work, provide a written description including evidence of successful use on other, comparable projects, and program of testing to demonstrate effectiveness for use on this Project.
- C. Photographs or Videotape: In accordance with Section 01 32 33 "Photographic Documentation," show existing conditions of adjoining construction and site improvements including finish

surfaces that might be misconstrued as damage caused by historic treatment operations. Submit before work begins.

1.4 QUALITY ASSURANCE

- A. Historic Preservation Treatment Specialist Qualifications: Experienced firm with required certifications and training able to demonstrate through past performance they are qualified to perform this work.
- B. Supervisory Personnel: Must have at least 10 years' experience in the field of historic preservation treatment.

1.5 STORAGE AND PROTECTION OF HISTORIC MATERIALS

- A. Removed and Salvaged Historic Materials:
 - 1. Clean salvaged historic items.
 - 2. Pack or crate items after cleaning. Identify contents of containers.
 - 3. Store items in secure area until delivery to Contracting Officer.
 - 4. Transport items to storage area indicated by Contracting Officer.
 - 5. Protect items from damage during transport and storage.
 - 6. Do not dispose of items removed from existing construction without prior written consent of Contracting Officer.
- B. Removed and Reinstalled Historic Materials:
 - 1. Clean and repair historic items to functional condition adequate for intended reuse.
 - 2. Pack or crate items after cleaning and repairing. Identify contents of containers.
 - 3. Protect items from damage during transport and storage.
 - 4. Reinstall items in locations indicated. Comply with installation requirements for new materials and equipment. Provide connections, supports, and miscellaneous materials necessary to make item functional for use as designed.
- C. Existing Historic Materials to Remain: Protect construction indicated to remain against damage and soiling during historic treatment. When permitted by Contracting Officer, items may be removed to suitable, protected storage location during historic treatment and reinstalled in their original locations after historic treatment operations are complete.
- D. Storage and Protection: When removed from existing location, store historic materials within weather-tight enclosure protected from wetting by rain, snow, or ground water, and temperature variations. Secure stored materials to protect from theft.
 - 1. Identify removed items with an inconspicuous mark indicating original location.
 - 2. Develop key plan when many similar items are scheduled for removal and reinstallation.
 - 3. Ensure that the containers used for storage are pest free, including rodent and insects.

1.6 PROJECT-SITE CONDITIONS

- A. Exterior Cleaning and Repairing:

1. Proceed with work only when forecasted weather conditions are favorable.
 - a. Wet Weather: Do not attempt repairs during rainy or foggy weather. Do not apply primer, paint, putty, or epoxy when relative humidity is above 80 percent. Do not remove exterior elements of structures when rain is forecast or in progress.
 - b. Do not perform exterior wet work when air temperature is below 40 degrees Fahrenheit (5 degrees Celsius).
 - c. Do not begin cleaning, patching, or repairing given likelihood of frost or freezing.
 - d. Do not begin cleaning when either air or surface temperature is below 45 degrees Fahrenheit (7 degrees Celsius) unless approved means are provided for maintaining 45 degrees Fahrenheit (7 degrees Celsius) temperature of air and materials during, and for 48 hours subsequent to, cleaning.
2. Perform cleaning and rinsing of the exterior only during daylight hours.

PART 2 - PRODUCTS (Not Used)

PART 3 - EXECUTION

3.1 HISTORIC PRESERVATION TREATMENT PLAN

- A. Prepare written plan for preservation work covering preservation components of project. Plan shall verify construction strategy and intent is compatible with Department of the Interior's (DOI) standards for Treatment of Historic Properties, guidelines for Treatment of Cultural Landscapes, and National Park Service management policies for cultural resources. Plan shall satisfy both project scope and resource protection requirements. Plan shall include:
 1. Organized list of preservation components of project, systems, and tasks
 2. Staging and sequence of work
 3. Disassembly and reassembly techniques and steps
 4. Equipment and tools required
 5. Supplies and materials with manufacturer or supplier identified including specific clean up/storage procedure including frequency and documentation of such.
 6. Skilled trades and crafts required
 7. Anticipated testing and analysis of fabric
 8. Additional investigations for extents or magnitude of treatments needed
 9. Protective measures
 10. Seasonal limitations on work
 11. Alternative means if primary treatment method is unfeasible
 12. Work conducted off-site (Approval from Contracting Officer required prior to taking resources off-site).
- B. The Historic Preservation Treatment Plan shall cover the following topics, at a minimum:
 1. Worker training regarding the site's history, cultural sensitivity and Secretary of the Interior's guidance with the regard to the treatment of historic properties.
 2. Protection of existing historical elements to remain during the removal of adjacent material systems, elements, etc.

- C. Review Preservation Plan and historic preservation priorities in detail with park at a pre-construction meeting prior to the start of any demolition. May be combined with pre-construction meeting detailed in Section 01 31 00 "Project Management" with the approval of the Contracting Officer.
 - 1. Review extent hazardous material abatement
 - 2. Review removed devices at the exterior to facilitate foundation work
 - 3. Review method anticipated for conducting foundation work under the historic building

3.2 HISTORIC PRESERVATION TREATMENT PROCEDURES

- A. Refer to material specific rehabilitation specifications for material specific procedures and evaluation criteria.
- B. The principal aim of preservation work is to halt the process of deterioration and stabilize the item's condition to sustain the integrity of the historic element, feature or structure being preserved. Cyclic maintenance is often required as well as repair work. Repair is required where specifically indicated. The following procedures shall be followed:
 - 1. Retain as much existing material as possible; repair and consolidate rather than replace.
 - 2. Use additional material or structure to reinforce, strengthen, prop, tie, and support existing material or structure.
 - 3. Use reversible processes wherever possible.
 - 4. Use traditional replacement materials and techniques if possible. New work shall be distinguishable from old work and original materials and techniques.
 - 5. Record the existing condition before commencing with repair work; document with preconstruction photos, sketches and field notes. Record repair work during construction with periodic construction photos and daily inspection reporting. Photo documentation is specified in Division 01 Section "Photo Documentation".
- C. Smoking is prohibited in and around the building. It shall be permitted in the fenced staging area only, provided that it is greater than 50 feet from the building.
- D. Notify Contracting Officer of visible changes in integrity of material or components due to environmental causes including biological attack, UV degradation, freezing, or thawing, or due to structural defects including cracks, movement, or distortion.
 - 1. Do not proceed with work in question until directed by Contracting Officer.
- E. Where Work requires existing features to be removed, cleaned, and reinstalled, perform these operations without damage to the material itself, to adjacent materials, or to the substrate. In some cases, materials are to be salvaged and returned to the park for archiving. Refer to Section 02 42 96 "Historic Removal and Dismantling" for more specific information.
- F. Identify new or replacement materials and features with inconspicuous, permanent marks (including the date of installation, if possible) on areas not observable by the public to distinguish them from original materials. Record the legend of identification marks and the locations of these marks on Record Drawings.
- G. When cleaning, match samples of existing materials that have been cleaned and identified for acceptable cleaning levels. Avoid over-cleaning to prevent damage to existing materials during

cleaning. Only the gentlest methods available should be attempted. Initiate cleaning using hand cleaning methods before introducing power cleaning methods and equipment.

3.3 PROTECTION, GENERAL

- A. Comply with manufacturer's written instructions for precautions and effects of products and procedures on adjacent building materials, components, and vegetation.
- B. Ensure supervisory personnel are present when work begins and during progress.
- C. Temporary Protection of Historic Materials during Construction:
 - 1. Protect existing materials during installation of temporary protections and construction. Do not deface or remove existing materials.
 - 2. Attachments of temporary protection to existing construction shall be approved by Contracting Officer prior to installation.
- D. Protect landscape work adjacent to or within work areas as follows:
 - 1. Provide barriers to protect tree trunks.
 - 2. Bind spreading shrubs.
 - 3. Coverings shall allow plants to breathe. Remove coverings at end of day. Do not cover plant material with waterproof membrane more than 8 hours at a time.
 - 4. Set scaffolding and ladder legs away from plants.

3.4 PROTECTION DURING APPLICATION OF CHEMICALS

- A. Protect persons, motor vehicles, surrounding surfaces of building being restored, building site, plants, and surrounding buildings from harm or damage resulting from applications of chemical cleaners and paint removers.
- B. Comply with requirements in Section 01 50 00 "Temporary Facilities and Controls."
- C. Cover adjacent surfaces with materials proven to resist chemical cleaners selected for Project unless chemicals being used will not damage adjacent surfaces. Use covering materials containing only waterproof, UV (ultraviolet)-resistant adhesives. Apply masking agents to comply with manufacturer's written instructions. Do not apply liquid masking agent to painted or porous surfaces. Promptly remove masking to prevent adhesive staining on completion.
- D. Do not clean surfaces during winds of sufficient force to spread cleaning solutions to unprotected surfaces.
- E. Neutralize and collect alkaline and acid wastes and dispose of outside park boundaries.
- F. Dispose of runoff from chemical operations by legal means and in a manner preventing soil erosion, undermining of paving and foundations, damage to landscaping, and water penetration into building interiors.
- G. Document daily clean-up, proper storage, and disposition of any flammable/combustible materials with daily reports.

3.5 PROTECTION DURING USE OF HEAT-GENERATING EQUIPMENT

- A. Comply with following procedures while performing work with heat-generating equipment, including welding, cutting, soldering, brazing, paint removal with heat, and other operations where open flames or implements utilizing heat are used:
1. Obtain Contracting Officer's approval for operations involving use of open-flame or welding equipment.
 - a. Notification shall be given for each occurrence and location of work with heat-generating equipment.
 - b. Obtain appropriate permit from the park as required.
 2. As far as practical, use heat-generating equipment in shop areas or outside building.
 3. Before work with heat-generating equipment commences, furnish fire watch (or watches) for location(s) where work is to be performed.
 4. Do not perform work with heat-generating equipment in or near rooms or in areas where flammable liquids or explosive vapors are present or thought to be present. Use combustible gas indicator test to ensure area is safe.
 5. Remove and keep area free of combustibles, including, rubbish, paper, waste, etc., within area of operations.
 - a. If combustible material cannot be removed, provide fireproof blankets to cover such materials.
 6. Where possible, furnish and use baffles of metal or gypsum board to prevent spraying of sparks or hot slag into surrounding combustible material.
 7. Prevent extension of sparks and particles of hot metal through open windows, doors, holes, and cracks in floors, walls, ceilings, roofs, and other openings.
 8. Inspect each location of day's work not sooner than 30 minutes after completion of operations to detect hidden or smoldering fires; ensure proper housekeeping is maintained.

END OF SECTION 01 35 91

SECTION 01 40 00 - QUALITY REQUIREMENTS

PART 1 - GENERAL

1.1 SUMMARY

- A. Section includes administrative and procedural requirements for quality assurance and quality control.
- B. Testing and inspecting services are required to verify compliance with requirements specified or indicated. These services do not relieve Contractor of responsibility for compliance with Contract Document requirements. Quality of work shall be responsibility of the Contractor.
 - 1. Specified tests, inspections, and related actions do not limit Contractor's other quality assurance and control procedures that facilitate compliance with Contract Document requirements.
- C. See Divisions 2 through 49 Sections for specific test and inspection requirements.

1.2 DEFINITIONS

- A. Quality Assurance Services: Activities, actions, and procedures performed before and during execution of the work to guard against defects and deficiencies and substantiate that proposed construction will comply with requirements.
- B. Quality Control Services: Tests, inspections, procedures, and related actions during and after execution of the work to evaluate actual products incorporated into the work and completed construction comply with requirements.
- C. Preconstruction Testing: Tests and inspections performed specifically for project before products and materials are incorporated into work to verify performance or compliance with specified criteria.
- D. Product Testing: Tests and inspections performed by a Nationally Recognized Testing Laboratory (NRTL), a National Voluntary Laboratory Accreditation Program (NVLAP), or a testing agency qualified to conduct product testing, to establish product performance and compliance with industry standards.
- E. Source Quality Control Testing: Tests and inspections performed at the source, i.e., plant, mill, factory, or shop.
- F. Field Quality Control Testing: Tests and inspections performed on-site for installation of work and for completed work.
- G. Testing Agency or Laboratory: Entity engaged to perform specific tests, inspections, or both. Testing laboratory shall mean the same as testing agency.

- H. Installer/Applicator/Erector: Contractor or another entity engaged by Contractor as an employee, Subcontractor, or Sub-subcontractor, to perform a particular construction operation, including installation, erection, application, and similar operations.
 - 1. Using a term such as "carpentry" does not imply certain construction activities must be performed by accredited or unionized individuals of a corresponding generic name, such as "carpenter." It also does not imply that requirements specified apply exclusively to trades people of corresponding generic name.

1.3 CONFLICTING REQUIREMENTS

- A. Reference Standards: If compliance with two or more standards is specified and standards establish different or conflicting requirements for minimum quality levels, comply with most stringent requirement. Refer uncertainties and requirements that are different, but apparently equal, to Contracting Officer before proceeding.
- B. Minimum Quality Levels: Quality level shown or specified shall be minimum provided or performed. Actual installation may comply exactly with minimum quality specified, or it may exceed minimum within reasonable limits. To comply with requirements, indicated numeric values are minimum or maximum, as appropriate, for the context of requirements. Refer uncertainties to Contracting Officer before proceeding.

1.4 SUBMITTALS

- A. Quality Control Plan:
 - 1. After contract award and before Pre-Construction conference, submit a written Contractor Quality Control (CQC) plan.
 - 2. If plan requires revisions or corrections, Contractor shall resubmit plan within 10 days.
 - 3. Government reserves the right to require changes in plan during contract period as necessary to obtain the quality specified.
 - 4. No change in the approved plan may be made without written concurrence by Contracting Officer.
- B. Qualification Data: For testing agencies specified in "Quality Assurance" Article to demonstrate their capabilities and experience. Include proof of qualifications in form of a recent report on inspection of testing agency by a recognized authority.
- C. Contractor Quality Control (CQC) Daily Reports: Submit showing inspections and tests on first workday following date covered by report. Quality Control Supervisor shall utilize [DSC Forms](#).
 - 1. Review Construction Management Representative (CMR) Daily report if applicable and reconcile any differences prior to posting.
- D. Test Reports
 - 1. Test reports shall be completed by person performing test.
 - 2. Submit Daily Test Information Sheets with Quality Control Daily Reports.
 - 3. Submit failing test results and proposed remedial actions within four hours of noted deficiency.

4. Submit three copies of complete test results no later than one calendar day after test was performed.
- E. Off-Site Inspection Reports: Submit prior to shipment.
- F. If Contractor Quality Control plan and Quality Control Daily Reports are not submitted as specified, Contracting Officer may retain payments until such time plan(s) is/are accepted and implemented, or may retain payments for work completed on days with no Quality Control Daily Reports.
- G. Permits, Licenses, and Certificates: For National Park Service (NPS) records, submit copies of permits, licenses, certifications, inspection reports, releases, jurisdictional settlements, notices, receipts for fee payments, judgments, correspondence, records, and similar documents, established for compliance with standards and regulations bearing on performance of work.

1.5 QUALITY ASSURANCE

- A. General: Qualifications paragraphs in this Article establish minimum qualification levels required; individual Specification Sections specify additional requirements.
- B. Contractors Quality Control Staff:
 1. Contractor's Quality Control Supervisor may also perform other duties.
 2. Contractor's Quality Control Supervisor shall be assigned no other duties.
 3. Contractor's designated Quality Control Supervisor shall be on the project site whenever contract work is in progress.
 4. Contractor's job supervisory staff may be used to assist Quality Control Supervisor supplemented, as necessary, by additional certified testing technicians.
- C. Installer Qualifications: Firm or individual experienced in installing, erecting, or assembling work similar in material, design, and extent indicated for Project, whose work has resulted in construction with a record of successful in-service performance.
- D. Manufacturer Qualifications: Firm experienced in manufacturing products or systems similar to those indicated for Project and with a record of successful in-service performance, as well as sufficient production capacity to produce required units.
- E. Fabricator Qualifications: Firm experienced in producing products similar to those indicated for Project and with a record of successful in-service performance, as well as sufficient production capacity to produce required units.
- F. Professional Engineer Qualifications: Professional engineer legally qualified to practice in jurisdiction where Project is located and is experienced in providing engineering services of kind indicated (including Structural Tests and Special Inspections (STSI)). Engineering services are defined as those performed for installations of system, assembly, or products similar to those indicated for Project in material, design, and extent.
- G. Specialists: Certain sections of the Specifications require that specific construction activities shall be performed by entities who are recognized experts in those operations. Specialists shall satisfy qualification requirements indicated and shall be engaged for activities indicated.

1. Requirement for specialists shall not supersede building codes and regulations governing Work.
- H. Testing Agency Qualifications: A Nationally Recognized Testing Laboratory (NRTL), a National Voluntary Laboratory Accreditation Program (NVLAP), or an independent agency with experience and capability to conduct testing and inspecting indicated, according to ASTM E 329; and with additional qualifications specified in individual Sections; and where required by Contract, is acceptable to Contracting Officer.
1. Nationally Recognized Testing Laboratory (NRTL): Nationally recognized testing laboratory according to 29 CFR 1910.7 (Code of Federal Regulations).
 2. National Voluntary Laboratory Accreditation Program (NVLAP): Testing agency accredited according to National Institute of Standards and Technology's (NIST) National Voluntary Laboratory Accreditation Program.
 3. Measuring devices, laboratory equipment, and instruments shall be calibrated at established intervals against certified standards in accordance with NIST requirements. Measuring and testing devices shall be made available for use by Government for verification tests.
- I. Factory-Authorized Service Representative Qualifications: Authorized representative of manufacturer who is trained and approved by manufacturer to inspect installation of manufacturer's products similar in material, design, and extent to those indicated for Project.

1.6 QUALITY CONTROL

- A. Contractor is responsible for testing and inspections, including Structural Tests and Special Inspections (STSI), as identified in attached STSI. Inspect and test work as needed to ensure quality of materials, workmanship, construction, finish, and functional performance are in compliance with applicable specifications, drawings, and those required by the Building Code.
1. Engage qualified testing agency to perform quality-control services.
 2. Submit appropriate report for each quality-control service.
 3. Testing and inspecting requested by Contractor and not required by Contract Documents are Contractor's responsibility.
 4. Contracting Officer may designate test locations.
- B. Manufacturer's Field Services: Where indicated, engage factory-authorized service representative to inspect field-assembled components and equipment installation, including service connections. Report results in writing.
- C. Re-testing/Re-inspecting: Regardless of whether original tests or inspections were Contractor's responsibility, provide quality-control services, including retesting and re-inspecting, for construction of replaced work that failed to comply with Contract Documents.
- D. Testing Agency Responsibilities: Cooperate with NPS and Contractor in performance of duties. Provide qualified personnel to perform required tests and inspections.
1. Notify Contracting Officer and Contractor promptly of irregularities or deficiencies observed in work during performance of services.
 2. Determine location from which test samples will be taken and in which in-situ tests are conducted.

3. Conduct and interpret tests and inspections, State in each report whether tested and inspected work complies with or deviates from requirements.
 4. Submit 3 copies of certified written report of each test, inspection, and similar quality-control service through Contractor.
 5. Do not release, revoke, alter, or increase Contract Document requirements or approve or accept any portion of Work.
- E. Associated Services: Cooperate with agencies performing required tests, inspections, and similar quality-control services, and provide reasonable auxiliary services as requested. Notify agency sufficiently in advance of operations to permit assignment of personnel. Provide:
1. Access to Work.
 2. Incidental labor and facilities necessary to facilitate tests and inspections.
 3. Adequate quantities of representative samples of materials that require testing and inspecting. Assist agency in obtaining samples.
 4. Facilities for storage and field curing of test samples.
 5. Delivery of samples to testing agencies.
 6. Preliminary design mix proposed for material mixes that require control by testing agency.
 7. Security and protection for samples and testing and inspecting equipment at Project site.
- F. Coordination: Coordinate sequence of activities to accommodate required quality assurance and control services with minimum delay and to avoid removing and replacing construction to accommodate testing and inspecting.
1. Schedule times for tests, inspections, obtaining samples, and similar activities.

PART 2 - PRODUCTS

2.1 QUALITY CONTROL PLAN

- A. Quality Control Plan shall include:
1. List of personnel responsible for quality control and assigned duties. Include each person's qualifications. Include alternate(s) and qualifications.
 2. Copy of letter of direction to Contractor's Quality Control Supervisor(s) outlining assigned duties and authorities designated by principal or owner.
 3. Names, qualifications / accreditations, and descriptions of laboratories to perform sampling and testing, and samples of proposed report forms from laboratories.
 4. Methods of performing, documenting, and enforcing quality control of work including Contractor report forms and acknowledgment of NPS forms.
 5. Methods of monitoring and controlling environmental pollution and contamination as required by regulations and laws.
 6. Specific discussion regarding mockups, off-site visits, receiving inspections, manufacturers representation, startup requirements, and other aspects of performance specific to Project.
 7. Provisions for substantial completion(s) and final inspection(s) per Contract.

PART 3 - EXECUTION

3.1 OFF-SITE CONTROL

- A. Items fabricated or assembled off-site shall be inspected for quality control at place of fabrication.

3.2 ON-SITE CONTROL

- A. Notification:

- 1. Notify Contracting Officer at least 48 hours in advance of preparatory phase meeting.
 - 2. Notify Contracting Officer at least 24 hours in advance of initial and follow-up phases.

- B. Preparatory Phase: Perform before beginning each feature of work.

- 1. Review control submittal requirements with personnel directly responsible for quality assurance and quantity control of the work. As a minimum, Contractor's Quality Control Supervisor and foreman responsible for feature of work shall be in attendance.
 - 2. Review applicable specifications sections and drawings related to feature of work.
 - 3. Ensure copies of referenced standards related to sampling, testing, and execution for feature of work are available on site.
 - 4. Ensure provisions have been made for field control testing.
 - 5. Examine work area to ensure preliminary work has been completed.
 - 6. Verify field dimensions and advise Contracting Officer of discrepancies with contract documents.
 - 7. Ensure necessary equipment and materials are at project site and they comply with approved shop drawings and submittals.
 - 8. Document preparatory phase activities and discussions on Contractor's Quality Control Daily Report.

- C. Initial Phase:

- 1. As soon as work begins, inspect and test representative portion of particular feature of work for quality of workmanship.
 - 2. Review control testing procedures to ensure compliance with contract requirements.
 - 3. Document initial phase activities and discussions on Contractor's Quality Control Daily Report. Exact location of initial phase shall be indicated for future reference and comparison with follow-up phases.

- D. Follow-Up Phase: Inspect and test as work progresses to ensure compliance with contract requirements until completion of work.

- E. Additional Preparatory and Initial Phases: Additional preparatory and initial phases may be required on same feature of work for following reasons:

- 1. Quality of on-going work is unacceptable.
 - 2. Changes in quality control staff, on-site production supervision, or work crew.
 - 3. Work on particular feature of work is resumed after substantial period of inactivity.

3.3 DOCUMENTATION

- A. Maintain Quality Control Daily Reports, Daily Test Report Information Sheets, and Accessibility Inspection Reports of quality control activities and tests. (Download from DSC Workflows website > Forms/Templates/Samples/Guidelines page > [Construction Forms](#) section.)
- B. Quality Control Daily Reports shall not be substituted for other written reports required under clauses of contract, such as Disputes, Differing Site Conditions, or Changes.
- C. Quality Control Daily Reports shall accurately portray all work and materials. The materials shall match the materials as approved from the Construction Documents (CD) requirements. Combustible/flammable materials shall be documented as well.

3.4 ENFORCEMENT

- A. Contractor shall stop work on any item or feature pending satisfactory correction of deficiency noted by quality control staff or Contracting Officer.

3.5 REPAIR AND PROTECTION

- A. General: On completion of testing, inspecting, sample taking, and similar services, repair damaged construction and restore substrates and finishes.
 - 1. Provide materials and comply with installation requirements specified in other Specification Sections. Restore patched areas and extend restoration into adjoining areas with durable seams as invisible as possible.
 - 2. Comply with Contract Document requirements for Section 01 73 29 "Cutting and Patching."
- B. Protect construction exposed by or for quality-control service activities.
- C. Repair and protection are Contractor's responsibility, regardless of the assignment of responsibility for quality-control services.

END OF SECTION 01 40 00

Construction Management Representative (CMR) & Contractor Quality Control (CQC) Daily Report

National Park Service (NPS) - Denver Service Center (DSC) | 2-6-18

REPORT NUMBER		PROJECT MANAGEMENT INFORMATION SYSTEM NUMBER (PMIS No.):					
PROJECT				CONTRACT NUMBER		DATE	
PARK				CONTRACTOR		CMR REPORTING	
WEATHER (Rain, Snow, Cloudy, Windy)		MOISTURE AMOUNT (INCHES)	TEMPERATURE		WEATHER DELAY Yes ☐ No ☐	DESCRIBE WEATHER DELAYED WORK	GROUND CONDITIONS (Dry, Damp, Wet, Frozen)
			MAXIMUM	MINIMUM			
PRIME CONTRACTOR:							
EMPLOYEES BY JOB CATEGORIES	NUMBER	HOURS	EQUIPMENT ON JOB (Include model, manufacture, size, year.)	NUMBER OF UNITS	EQUIPMENT HOURS WORKING		
					YES	STANDBY	MOBILIZED/ DEMOBILIZED DATE
Project Manager (PM)							
Superintendent							
Quality Control Officer							
Safety Officer							
Laborers							
MATERIALS DELIVERED				QUANTITY		OFFICIAL VISITORS	
WORK PERFORMED BY PRIME: (Include detail description per each activity including location, quantities, and production.)							
SPECIFIC INSPECTIONS: (Inspections performed, results, and corrective actions, and Primes and Subprimes.)							
TESTING:							
Was any testing performed today YES / NO. (Complete and attach Test Report Information Sheets.)							
☐ ☐							
Type and Location of Testing:							

SUBCONTRACTOR(S):			PMIS Number:		Date:			
SUBCONTRACTOR NAME								
EMPLOYEES BY JOB CATEGORIES	NUMBER	HOURS	EQUIPMENT ON JOB (Include model, manufacturer, size, and year.)	NUMBER OF UNITS	EQUIPMENT HOURS WORKING			
					YES	STANDBY	MOBILIZED/ DEMOBILIZED DATE	
MATERIALS DELIVERED			QUANTITY		EQUIPMENT MOBILIZED/DEMOBILIZED			
WORK PERFORMED BY SUBCONTRACTOR: Include detail description per each activity. Include location, quantity, and production.								
SUBCONTRACTOR NAME								
EMPLOYEES BY JOB CATEGORIES	NUMBER	HOURS	EQUIPMENT ON JOB (Include model, manufacturer, size, and year.)	NUMBER OF UNITS	EQUIPMENT HOURS WORKING			
					YES	STANDBY	MOBILIZED/ DEMOBILIZED DATE	
MATERIALS DELIVERED			QUANTITY		EQUIPMENT MOBILIZED/DEMOBILIZED			
WORK PERFORMED BY SUBCONTRACTOR: Include detail description per each activity. Include location, quantity, and production.								
SUBCONTRACTOR NAME								
EMPLOYEES BY JOB CATEGORIES	NUMBER	HOURS	EQUIPMENT ON JOB (Include model, manufacturer, size, and year.)	NUMBER OF UNITS	EQUIPMENT HOURS WORKING			
					YES	STANDBY	MOBILIZED/ DEMOBILIZED DATE	
MATERIALS DELIVERED			QUANTITY		EQUIPMENT MOBILIZED/DEMOBILIZED			
WORK PERFORMED BY SUBCONTRACTOR: Include detail description per each activity. Include location, quantity, and production.								

SUBCONTRACTOR(S):			PMIS Number:		Date:			
SUBCONTRACTOR NAME								
EMPLOYEES BY JOB CATEGORIES	NUMBER	HOURS	EQUIPMENT ON JOB (Include model, manufacturer, size, and year.)	NUMBER OF UNITS	EQUIPMENT HOURS WORKING			
					YES	STANDBY	MOBILIZED/ DEMOLIZED DATE	
MATERIALS DELIVERED			QUANTITY		EQUIPMENT MOBILIZED/DEMOLIZED			
WORK PERFORMED BY SUBCONTRACTOR: Include detail description per each activity. Include location, quantity, and production.								
MATERIALS DELIVERED			QUANTITY		EQUIPMENT MOBILIZED/DEMOLIZED			
WORK PERFORMED BY SUBCONTRACTOR: Include detail description per each activity. Include location, quantity, and production.								
MATERIALS DELIVERED			QUANTITY		EQUIPMENT MOBILIZED/DEMOLIZED			
WORK PERFORMED BY SUBCONTRACTOR: Include detail description per each activity. Include location, quantity, and production.								
MATERIALS DELIVERED			QUANTITY		EQUIPMENT MOBILIZED/DEMOLIZED			
WORK PERFORMED BY SUBCONTRACTOR: Include detail description per each activity. Include location, quantity, and production.								

PMIS Number:

Date:

CRITICAL POINT/MILESTONE OBSERVATIONS *(Describe item inspected, observations, problems, and action items.)*

DESCRIPTION OF WORK BEING MONITORED ON A TIME AND MATERIAL BASIS AND WHY

BREAKDOWN OF TIME AND MATERIAL WORK PERFORMED

Location:

Labor:

Equipment:

Material:

Production Rates:

STATUS OF GOVERNMENT FURNISHED SERVICES AND/OR SUPPLIES

SAFETY COMMENTS

Accidents / Lost Time:

Incidents:

First Aid Administered:

Other:

DIFFICULTIES WITH CONSTRUCTION CONTRACTOR

UNFORESEEN DEVELOPMENTS *(Describe conditions, action taken, person contacted, and recommended actions.)*

CONSTRUCTION DEFICIENCIES OR RE-TESTING REQUIRED

OTHER COMMENTS

CERTIFICATION

For Construction Management Representative (CMR): I certify the above report is complete and correct and that I, or my authorized representative, have inspected all work performed this day by the Prime Contractor and each Subcontractor and determined all materials, equipment, and workmanship are in strict compliance with plans and specifications except as may be noted above.

I certify that I or my authorized representative have reviewed and reconciled all differences between this daily report and the Construction Contractors. In the event a discrepancy is found, the Construction Contractor was notified and an acceptable revision was completed and resubmitted by either or both parties within this document. If unable to agree with the Construction Contractor, explain nature of disagreement in "Other Comments" above as needed.

For Contractor's Representative: I certify that the above report is complete and correct and that I, or my authorized representative, have inspected all work performed this day by the Prime Contractor and each Subcontractor and determined that all materials, equipment, and workmanship are in strict compliance with the plans and specifications except as may be noted above. I have reconciled all differences found when comparing this CQC daily diary with the CMR's daily diary. All unresolved issues are described under "Other Comments" above.

SIGNATURE

TITLE:

PMIS Number:		Date:	
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ADDITIONAL COMMENTS OR INFORMATION *(Describe item inspected, observations, problems, and action items.)*

PMIS Number:		Date:	
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IMAGES IN SUPPORT OF REPORT INFORMATION

INSPECTION CHECKLIST

National Park Service
Parks Planning Facilities and Lands, Building Code Official
Building_code@nps.gov

Review	Required	Signature of Inspector	Date Approved
Footing & Foundation	X		
Concrete Slab, Foundation Wall & Under Floor	X		
Lowest Floor Elevation (in flood areas)			
Underground			
Roof Framing			
Wall Framing			
Wall Close-In			
Ceiling Close-In			
Life Safety			
Fire & Smoke Resistant Penetrations			
Fire Protection			
Fire Alarm			
Lath & Gypsum Board			
Accessibility			
Underground Plumbing			
Rough Plumbing			
Underground Electrical			
Rough Electrical			
Mechanical			
Energy Efficiency			
Elevator			
Waterproofing	X		
Other:			
Other:			
Other:			
Other:			
Other:			
Other:			
Final Inspection:			
Fire/Life Safety			
Electrical			
Mechanical			
Plumbing			
Accessibility			

Statement of Structural Tests and Special Inspections

National Park Service (NPS) - Denver Service Center (DSC) | 8-22-24

Park: Pinnacles National Park

Project Management Information System (PMIS) Number: 331043

Project Name: Building Bear Valley School Rehabilitation, Phase 1

Structural Engineering Firm: JVA, Inc

This Statement of Structural Tests and Special Inspections is being submitted as required by Chapter 17 of the **2024 International Building Code** (IBC 2024). It includes the following:

1. Seismic requirements
2. Wind requirements
3. Qualification Requirements for Inspectors and Testing Technicians
4. Listing of Required Structural Tests and Special Inspections

The Construction Contractor's Quality Control Supervisor will provide copies of all special inspection reports and associated documentation to the Contracting Officer (CO). The Construction Contractor will be required to correct all deficiencies discovered in the Special Inspection and Structural Testing program.

Prepared By: Jeffrey Schalk, SE

(Type or print name)

Signature: _____



Date: 01/16/2026



Stamped by Professional Engineer (PE) or Structural Engineer (SE)

Seismic and Wind Requirements

Seismic Requirements, IBC 2024 Section 1704.3

Description of seismic-force-resisting system and designated seismic systems subject to special inspections:

Seismic Design Category E:

IBC 1705.13.2, Structural Wood; Building Frame System, Light-Frame (Wood) Walls with Wood Structural Panels Rated for Shear; Periodic Special Inspection for bolting and anchoring where fastener spacing is less than 6 inches on center, all hold downs and adhesive anchors.

Wind Requirements, IBC 2024 Section 1704.3

Description of wind-force-resisting system and designated wind systems subject to special inspections:

None. 3-second gust wind speed = 92 mph (Basic Design) < 140 mph (Ultimate), Exposure C; therefore, no special inspections of wind-force resisting systems are required (IBC 1705.12)

Instructions:

1. Under [Listing of Required Structural Tests and Special Inspections](#):
 - a. **Required?** - Place an **X** for all Special Inspections and Tests required for this project.
 - b. **Required Qualifications** - Provide qualifications for the special inspector, using the [Qualification Requirements for Inspector and Testing Technicians](#) list below, for all required Structural Tests and Special Inspections.
 - c. **Continuous** - If marked with an **X**, continuous special inspection shall be as defined in Chapter 2, IBC 2024.
 - d. **Periodic** - If marked with an **X**, provide the minimum number of tests, i.e. 20% of all field welds, or the amount of work to be inspected (e.g. 10% of all wall surfaces).
2. Attach completed Statement of Structural Tests and Special Inspections to the end of NPS DSC Division 1 Specifications **Section 01 40 00 Quality Requirements**.

Qualification Requirements for Inspectors and Testing Technicians

PE/SE	Structural Engineer – licensed PE or SE specializing in the design of buildings and structures
PE/GE	Geotechnical Engineer – licensed PE specializing in soil mechanics and foundations
EIT	Engineer-In-Training – graduate engineer who has passed the Fundamentals of engineering examination

American Concrete Institute (ACI) Certification

ACI-CCSI	Concrete Construction Special Inspector
ACI-LTT	Concrete Laboratory Testing Technician Level 1 or 2
ACI-STT	Concrete Strength Testing Technician
ACI-FTT	Concrete Field Testing Technician – Grade I

American Society of Non-Destructive Testing (ASNT) Certification

Non-Destructive Testing Technician – Level II or III

American Welding Society (AWS) Certification

AWS-CWI	Certified Welding Inspector
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Exterior Design Institute (EDI) Certification

EDI-EIFS	Certified EIFS inspector
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International Code Council (ICC) Certification

ICC-PCSI	Prestressed Concrete Special Inspector
ICC-RCSI	Reinforced Concrete Special Inspector
ICC-SSI	Soils Special Inspector
ICC-SFSI	Spray-applied Fireproofing Special Inspector
ICC-SMSI	Structural Masonry Special Inspector
ICC-SSBSI	Structural Steel and Bolting Special Inspector
ICC-SWSI	Structural Welding Special Inspector

National Institute for Certification in Engineering Technologies (NICET) Certification

NICET-CT	Concrete Technician – Levels I, II, III and IV
NICET-GET	Geotechnical Engineering Technician - Levels I, II, III and IV
NICET-ST	Soils Technician - Levels I, II, III and IV

Other

Listing of Required Structural Tests and Special Inspections

Required?	Structural Test or Special Inspection	Required Qualifications	Continuous	Periodic	Frequency of Periodic Test or Inspection
	Concrete Construction (IBC 2024 Section 1705.3)				
X	1. Inspect reinforcing steel, including prestressing tendons, and verify placement.	ACI-CCSI ICC-RCSI		X	Prior to each pour
	2. Inspection of reinforcing steel welding				
	a. Verify weldability of reinforcing bars other than ASTM A706.			X	
	b. Inspect welding of reinforcement for special moment frames, boundary elements of special structural walls and coupling beams.		X		
	c. Inspect welded reinforcement splices.		X		
	d. Inspect welding of primary tension reinforcement in corbels.		X		
	e. Inspect single pass fillet welds, maximum 5/16 inch.			X	
	f. Inspect all other welds.			X	
X	3. Inspection of anchors cast in concrete.	ACI-CCSI ICC-RCSI		X	Prior to each pour
	4. Inspection of anchors post-installed in hardened concrete members.	ACI-CCSI ICC-RCSI			
	a. Adhesive anchors installed in horizontally or upwardly inclined orientations to resist sustained tension load.		X		
X	b. Mechanical anchors and adhesive anchors not defined in 4a.			X	50% of anchors
X	5. Verify use of approved design mix.	ACI-CCSI ICC-RCSI		X	
X	6. Prior to placement fabricate specimens for strength tests, perform slump and air content tests, and determine the temperature of the concrete.	ACI-CCSI ICC-RCSI	X		
X	7. Inspect concrete and shotcrete placement for proper application techniques.	ACI-CCSI	X		

Required?	Structural Test or Special Inspection	Required Qualifications	Continuous	Periodic	Frequency of Periodic Test or Inspection
		ICC-RCSI			
X	8. Inspect for maintenance of specified curing temperature and techniques.	ACI-CCSI ICC-RCSI		X	After each pour
	9. Inspection of prestressed concrete:				
	a. Application of prestressing forces.		X		
	b. Grouting of bonded prestressing tendons in the seismic-force-resisting system.		X		
	10. Erection of precast structural members.			X	
	11. For precast concrete diaphragm connections or reinforcement at joints classified as moderate or high deformability elements (MDE or HDE) in structures assigned to Seismic Design Category C, D, E, or F, inspect such connections and reinforcement in the field for:				
	a. Installation of the embedded parts.		X		
	b. Completion of the continuity of reinforcement across joints.		X		
	c. Completion of connections in the field.		X		
	12. Inspect installation tolerances of precast concrete diaphragm connections for compliance with ACI 550.5 (American Concrete Institute).			X	
	13. Verification of in-situ concrete strength, prior to stressing of tendons in post-tensioned concrete and prior to removal of shores and forms from beams and structural slabs.			X	
X	14. Inspection formwork for shape, location and dimensions of the concrete member being formed.	ACI-CCSI ICC-RCSI		X	Prior to each pour
Wood Construction (IBC 2024 Section 1705.5)					
	1. Inspect prefabricated wood structural elements in accordance with Section 1704.2.5.			X	
	2. High load diaphragms:				
	a. Verify sheathing grade and thickness.			X	

Required?	Structural Test or Special Inspection	Required Qualifications	Continuous	Periodic	Frequency of Periodic Test or Inspection
	b. Verify nominal size of framing members at adjoining panel edges.			X	
	c. Verify nail or staple diameter and length.			X	
	d. Verify number of fastener lines.			X	
	e. Verify spacing between fasteners in each line and at panel edges.			X	
	3. Shearwalls:	PE/SE			
	a. Verify sheathing grade and thickness.			X	
	b. Verify nominal size of framing members at adjoining panel edges.			X	
	c. Verify nail or staple diameter and length.			X	
	d. Verify number of fastener lines.			X	
	e. Verify spacing between fasteners in each line and at panel edges.			X	
X	f. Location and size of holdowns.			X	50% and one of each unique condition
	4. Verify nailing, bolting, anchoring and fastening of:	PE/SE			
	a. Drag struts and collectors.			X	
	b. Braces.			X	
X	c. Hold-downs.			X	50% and one of each unique condition
	5. Metal-plate-connected wood trusses spanning 60 feet or greater:				
	a. Verify temporary installation restraint/bracing installed in accordance with the approved shop drawings.			X	
	b. Verify permanent individual truss member restraint/bracing installed in accordance with the approved shop drawings.			X	

Required?	Structural Test or Special Inspection	Required Qualifications	Continuous	Periodic	Frequency of Periodic Test or Inspection
Soils (IBC 2024 Section 1705.6)					
X	1. Verify materials below shallow foundations are adequate to achieve the required bearing capacity.	PE/GE		X	Once after excavation
X	2. Verify excavations are extended to proper depth and have reached proper material.	PE/GE		X	Prior to each pour
	3. Perform classification and testing of compacted fill materials.			X	
	4. During fill placement, verify use of proper materials and procedure in accordance with the provisions of the approved geotechnical report. Verify densities and lift thicknesses during placement and compaction of compacted fill.		X		
	5. Prior to placement of compacted fill, inspect subgrade and verify that site has been prepared properly.			X	
Wind Resistance (IBC 2024 Section 1705.12)					
	1. Provide inspections when required by Section 1705.12.				
Seismic Resistance (IBC 2024 Section 1705.13)					
X	1. Provide inspections when required by Section 1705.13.	PE/SE			
Testing and Qualification for Seismic Resistance (IBC 2024 Section 1705.14)					
	1. Test and qualify seismic resistance in accordance with IBC 2024 Section 1705.14 and the project specifications.				

SECTION 01 42 00 – REFERENCE STANDARDS

PART 1 - GENERAL

1.1 ENVIRONMENTAL DEFINITIONS

- A. Definitions pertaining to sustainable development: As defined in ASTM E2114 and as specified herein.
- B. Chain-of-Custody: Process whereby a product or material is maintained under physical possession or control during its entire life cycle.
- C. Deconstruction: Disassembly of buildings for purpose of recovering materials.
- D. DFE (Design for the Environment): A technique that includes elements of resource conservation and pollution prevention as applied in various product sectors. A technique that incorporates approaches which are part of product (or assembly) concept, need and design. Considerations involve material selection, material and energy efficiency, reuse, maintainability and design for disassembly and recyclability. Refer to International Organization for Standardization (ISO) Guide 64 for additional clarification.
- E. Environmentally preferable products: Products and services that have a lesser or reduced effect on the environment in comparison to conventional products and services. Refer to EPA's Final Guidance on [Environmentally Preferable Purchasing Program](#).
- F. Non-Renewable Resource: A resource that exists in a fixed amount that cannot be replenished on a human time scale. Non-renewable resources have potential for renewal only by geological, physical, and chemical processes taking place over of millions of years. Examples include iron ore, coal, and oil.
- G. Perpetual Resource: A resource that is virtually inexhaustible on a human time scale. Examples include solar energy, tidal energy, and wind energy.
- H. Recycled Content Materials: Products that contain pre-consumer or post-consumer materials as all or part of their feedstock. Recycled content claim shall be consistent Federal Trade Commission (FTC) Guide for Use of Environmental Marketing Claims.
- I. Renewable Resource: A resource that is grown, naturally replenished, or cleansed, at a rate which exceeds depletion of the usable supply of that resource. A renewable resource can be exhausted if improperly managed. However, a renewable resource can last indefinitely with proper stewardship. Examples include trees in forests, grasses in grasslands, and fertile soil.

1.2 QUALITY ASSURANCE

- A. Applicability of Standards: Unless the Contract Documents include more stringent requirements, applicable construction industry standards have the same force and effect as if bound or copied directly into Contract Documents to the extent referenced. Such standards are made a part of Contract Documents by reference.

- B. Publication Dates: Comply with standards in effect as of date of Contract Documents, unless otherwise indicated.
- C. Conflicting Requirements: Where compliance with two or more standards is specified, and standards may establish different or conflicting requirements for minimum quantities or quality levels, comply with most stringent requirement. Refer uncertainties and requirements that are different, but apparently equal, to Contracting Officer (CO) for decision before proceeding.

1.3 INDUSTRY STANDARDS

- A. Applicability of Standards: Unless Contract Documents include more stringent requirements, applicable construction industry standards have same force and effect as if bound or copied directly into Contract Documents to the extent referenced. Such standards are made a part of Contract Documents by reference.
- B. Publication Dates: Comply with standards in effect as of date of Contract Documents unless otherwise indicated.
- C. Copies of Standards: Each entity engaged in construction on Project should be familiar with industry standards applicable to its construction activity. Copies of applicable standards are not bound with Contract Documents.
 - 1. Where copies of standards are needed to perform a required construction activity, obtain copies directly from publication source.

1.4 ABBREVIATIONS AND ACRONYMS

- A. Industry Organizations: Where abbreviations and acronyms are used in Specifications or other Contract Documents, they shall mean the recognized name of the entities found in Section 01 42 00 Sources for Reference Publications, [Unified Facilities Guide Specifications](#) (UFGS) (accessible via [Masters](#) website > Downloads section > click on UFGS Master (WBDG Website). Names, telephone numbers, and websites are subject to change and are believed to be accurate and up-to-date as of date of Contract Documents.

XX EXAMPLE Association (The)
 [www.EXAMPLE.org](#)

- B. Code Agencies: Where abbreviations and acronyms are used in Specifications or other Contract Documents, they shall mean the recognized name of the entities in following list. Names, telephone numbers, and websites are subject to change and are believed to be accurate and up-to-date as of date of Contract Documents.

DIN Deutsches Institut fur Normung e.V. 49 30 2601-3003
 [www.din.de](#)

IAPMO International Association of Plumbing and Mechanical Officials (909) 472-4100
 [www.iapmo.org](#)

ICC	International Code Council www.iccsafe.org	(888) 422-7233
ICC-ES	ICC Evaluation Service, Inc. icc-es.org	(800) 423-6587 (562) 699-0543

- C. Federal Government Agencies: Where abbreviations and acronyms are used in Specifications or other Contract Documents, they shall mean the recognized name of the entities in following list. Names, and websites are subject to change and are believed to be accurate and up-to-date as of date of Contract Documents.

ABA & ABAAS United States Access Board	Architectural Barriers Act (ABA) Architectural Barriers Act Accessibility Standards (ABAAS) www.access-board.gov	
CoE	Army Corps of Engineers www.usace.army.mil	
CPSC	Consumer Product Safety Commission www.cpsc.gov	
DOC	Department of Commerce www.commerce.gov	
DOD	Department of Defense www.defense.gov	
DOJ	Department of Justice www.justice.gov	
DOE	Department of Energy www.energy.gov	
EPA	Environmental Protection Agency www.epa.gov	
FAA	Federal Aviation Administration www.faa.gov	
FCC	Federal Communications Commission www.fcc.gov	
FDA	Food and Drug Administration www.fda.gov	
GSA	General Services Administration www.gsa.gov	

HUD	Department of Housing and Urban Development www.hud.gov
LBL	Lawrence Berkeley National Laboratory www.lbl.gov
NCHRP	National Cooperative Highway Research Program (See TRB (Transportation Resource Board))
NIST	National Institute of Standards and Technology www.nist.gov
OSHA	Occupational Safety & Health Administration www.osha.gov
PHS	U.S. Department of Health and Human Services www.hhs.gov
RUS	Rural Utilities Service (See USDA (Department of Agriculture))
SD	State Department www.state.gov
TRB	Transportation Research Board www.nationalacademies.org/trb/transportation-research-board
USDA	Department of Agriculture www.usda.gov
USP	U.S. Pharmacopeia www.usp.org
USPS	Postal Service www.usps.com

- D. Standards and Regulations: Where abbreviations and acronyms are used in Specifications or other Contract Documents, they shall mean the recognized name of the standards and regulations in following list. Names, telephone numbers, and websites are subject to change and are believed to be accurate and up-to-date as of date of Contract Documents.

ABAAS	Architectural Barriers Act Accessibility Standards www.access-board.gov
CFR	Code of Federal Regulations Available from Government Printing Office www.govinfo.gov/app/collection/cfr
DOD	Department of Defense Military Specifications and Standards Available from Department of Defense Single Stock Point www.dsp.dla.mil/Specs-Standards/

DSCC	Defense Supply Center Columbus (See FS (Federal Specification))
FED-STD	Federal Standard (See FS (Federal Specification))
FS	Federal Specification Available from Department of Defense Single Stock Point www.dsp.dla.mil/Specs-Standards/ Available from General Services Administration www.gsa.gov Available from National Institute of Building Sciences www.nibs.org
FTMS	Federal Test Method Standard (See FS (Federal Specification))
MIL	(See MILSPEC (Military Specification and Standards))
MIL-STD	(See MILSPEC (Military Specification and Standards))
MILSPEC	Military Specification and Standards Available from Department of Defense Single Stock Point www.dsp.dla.mil/Specs-Standards/
UFAS	Uniform Federal Accessibility Standards Available from Access Board www.access-board.gov/guidelines-and-standards/buildings-and-sites/about-the-aba-standards/ufas <i>(UFAS is only for housing projects per Fair Housing Act. See also the Fair Housing Act Design Manual, www.huduser.gov/portal/publications/destech/fairhousing)</i>

1.5 ENVIRONMENTAL REFERENCE STANDARDS

A. American Forest and Paper Association:

1. Sustainable Forestry Initiative

B. American Society for Testing and Materials International (ASTM):

- A653/A653M Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process
- C1240 Standard Specification for Silica Fume Used in Cementitious Mixtures
- C128 Standard Test Method for Density, Relative Density (Specific Gravity), and Absorption of Fine Aggregate
- C131 Standard Test Method for Resistance to Degradation of Small-Size Coarse Aggregate by Abrasion and Impact in the Los Angeles Machine
- C136 Standard Test Method for Sieve Analysis of Fine and Coarse Aggregates

- C1601 Standard Test Method for Field Determination of Water Penetration of Masonry Wall Surfaces
- C289 Standard Test Method for Potential Alkali-Silica Reactivity of Aggregates (Chemical Method)
- C311 Test Methods for Sampling and Testing Fly Ash or Natural Pozzolans for Use as a Mineral Admixture in Portland-Cement Concrete
- C33 Standard Specification for Concrete Aggregates
- C593 Standard Specification for Fly Ash and Other Pozzolans for Use With Lime
- C595 Standard Specification for Blended Hydraulic Cements
- C618 Standard Specification for Coal Fly Ash and Raw or Calcined Natural Pozzolan for Use as a Mineral Admixture in Concrete
- C989 Standard Specification for Ground Granulated Blast-Furnace Slag for Use in Concrete and Mortars
- D198 Standard Test Methods of Static Tests of Lumber in Structural Sizes
- D3864 Standard Guide for Continual On-Line Monitoring Systems for Water Analysis
- D4444 Standard Test Methods for Use and Calibration of Hand-Held Moisture Meters
- D4552 Standard Practice for Classifying Hot-Mix Recycling Agents
- D4840 Standard Guide for Sampling Chain-of-Custody Procedures
- D5505 Standard Practice for Classifying Emulsified Recycling Agents
- D5539 Standard Specification for Seed Starter Mix
- D5663 Standard Guide for Validating Recycled Content in Packaging Paper and Paperboard
- D5792 Standard Practice for Generation of Environmental Data Related to Waste Management Activities: Development of Data Quality Objectives
- D5834 Standard Guide for Source Reduction Reuse, Recycling, and Disposal of Solid and Corrugated Fiberboard (Cardboard)
- D5851 Standard Guide for Planning and Implementing a Water Monitoring Program
- D5852 Standard Test Method for Erodibility Determination of Soil in the Field or in the Laboratory by the Jet Index Method
- D6629 Standard Guide for Selection of Methods for Estimating Soil Loss by Erosion
- D6886 Standard Test Method for Speciation of the Volatile Organic Compounds (VOCs) in Low VOC Content Waterborne Air-Dry Coatings by Gas Chromatography
- E1609 Standard Guide for Development and Implementation of a Pollution Prevention Program
- E1686 Standard Guide for Selection of Environmental Noise Measurements and Criteria
- E1780 Standard Guide for Measuring Outdoor Sound Received from a Nearby Fixed Source
- E1971 Standard Guide for Stewardship for the Cleaning of Commercial and Institutional Buildings

C. Bat Conservation International:

- Bat Approved Bat Houses

D. Center for Resource Solutions

- Green-e program

E. Environmental Protection Agency (EPA):

- Comprehensive Procurement Guidelines
- ENERGY STAR

- Environmentally Preferable Purchasing Program Final Guidance
 - GreenScapes program
 - Heat Island Initiative
 - Indoor Air Quality Building Education and Assessment Model (I-BEAM)
 - National Environmental Performance Track
 - Pollution Prevention (P2)
 - Product Stewardship Program
 - Significant New Alternatives Policy (SNAP) Program
- F. Federal Trade Commission:
- Guide for the Use of Environmental Marketing Claims
- G. Forest Stewardship Council:
- Chain-Of-Custody
 - Forest Management
- H. International Organization of Standardization:
- Guide 64; Guide for Inclusion of Environmental Aspects in Product Standards
 - 9660 Information processing -- Volume and file structure of CD-ROM for information interchange
 - 14001 Environmental management systems – Specification with guidance for use
 - 14004 Environmental Management Systems – General Guidelines on Principles, Systems and Supporting Techniques
 - 14020 Environmental labels and declarations – General principles
 - 14024 Environmental labels and declarations – Type I environmental labelling - Principles and procedures
 - 14040 Environmental management – Life cycle assessment – Principles and framework
- I. National Institute of Building Sciences:
- MOIST program for transfer of heat and moisture
 - Whole Building Design Guide
- J. National Institute of Standards and Technology:
- BEES (Building for Environmental and Economic Sustainability) Lifecycle Decision Support Tool
- K. Sheet Metal and Air Conditioning Contractors' National Association (SMACNA):
- IAQ Guidelines for Occupied Buildings Under Construction
- L. Southcoast Air Quality Management District:
- 1168 Adhesive And Sealant Applications
- M. US Composting Council:

- Seal of Testing Assurance Program

PART 2 - PRODUCTS (Not Used)

PART 3 - EXECUTION (Not Used)

END OF SECTION 01 42 00

SECTION 01 50 00 - TEMPORARY FACILITIES AND CONTROLS

PART 1 - GENERAL

1.1 SUMMARY

- A. Section includes requirements for temporary utilities, support facilities, and security and protection facilities.

1.2 DEFINITIONS

- A. Permanent Enclosure: As determined by Contracting Officer (CO), permanent or temporary roofing is complete, insulated, and weathertight; exterior walls are insulated and weathertight; and openings are closed with permanent construction or substantial temporary closures.

1.3 USE CHARGES

- A. General: Cost or use charges for temporary facilities shall be included in Contract Sum as required.
- B. Water Service: Nonpotable Water from existing water system is available for use without metering and without payment of use charges. Provide connections and extensions of services as required for construction operations without any additional permit costs.
- C. Electric Power Service: Electric power from existing system is not available for use.

1.4 QUALITY ASSURANCE

- A. Environmental Protection: Provide environmental protection as required by agency(ies) with jurisdiction and as indicated in Contract Documents. Coordinate with requirements of the following:
 - 1. Regulatory Requirements
 - 2. Noise and Acoustics Management
 - 3. Environmental Management
 - 4. Construction Waste Management

1.5 PROJECT CONDITIONS

- A. Temporary Use of Permanent Facilities: Installer of each permanent service shall assume responsibility for operation, maintenance, and protection of each permanent service during its use as a construction facility before NPS acceptance, regardless of previously assigned responsibilities.

PART 2 - PRODUCTS

2.1 MATERIALS

- A. Temporary materials may be new or used, but must be adequate in capacity for required usage, must not create unsafe conditions, and must not violate requirements of applicable codes and standards.
- B. Chain-Link Fencing: Minimum 2 inch (50 millimeters), 0.148 inch (3.76 millimeters) thick, galvanized steel, chain-link fabric fencing; minimum 6 feet (1.8 millimeter) high with galvanized steel pipe posts; minimum 2-3/8 inch (60 millimeters) OD (outside diameter) line posts and 2-7/8 inch (73 millimeters) OD corner and pull posts, with 1-5/8 inch (42 millimeters) OD top rails
- C. Portable Chain-Link Fencing: Minimum 2 inch (50 millimeters), 9-gage, galvanized steel, chain-link fabric fencing; minimum 6 feet (1.8 millimeters) high with galvanized steel pipe posts; minimum 2-3/8 inch (60 millimeters) OD line posts and 2-7/8 inch (73 millimeters) OD corner and pull posts, with 1-5/8 inch (42 millimeters) OD top and bottom rails. Provide concrete or galvanized steel bases for supporting posts.
- D. Safety Barrier Fence: Orange plastic fence, minimum height, 4 feet.
- E. Lumber and Plywood: Comply with requirements in Division 06 Section "Rough Carpentry".
- F. Insulation: Unfaced mineral-fiber blanket, manufactured from glass, slag wool, or rock wool; with maximum flame-spread and smoke-developed indexes of 25 and 50, respectively.
- G. Polyethylene Sheet: Reinforced, fire-resistive sheet, 10 mil (0.25-millimeter) minimum thickness, with flame-spread rating of 15 or less per ASTM E 84 and passing NFPA 701 Test Method 2.

2.2 TEMPORARY FACILITIES

- A. Field Offices, General: Prefabricated or mobile units with serviceable finishes, temperature controls, and foundations adequate for normal loading.
- B. Contracting Officers Field Office: Field office shall be a separate structure from Contractor's office.
 - 1. Prefabricated, job built, or a mobile unit; excellent condition, structurally sound, non-flammable exterior construction, weather tight, minimum 300 square feet.
 - 2. Operable windows and security screens, adjustable ventilation.
 - 3. Restroom (minimum of lavatory and toilet, with exhaust fan if room is windowless) or accessible Contractor facilities if available.
 - 4. Air conditioner and heater.
 - 5. Interior partition with lockable door to divide office
 - 6. Paneling or freshly painted walls, acoustical tile or painted ceilings, and resilient flooring.
 - 7. Two exterior doors with dead bolts keyed from outside,
 - 8. Minimum 20-square-foot landing and steps at each exterior door.

- C. Storage and Fabrication Sheds: Temporary weather tight sheds or other covered facilities for storage of materials subject to weather damage. Number and size of structures shall be subject to Contracting Officer's approval.
- D. Toilets: Sufficiently lighted and ventilated toilet facilities in weatherproof, sight proof, accessible, sturdy enclosures with privacy locks.
 - 1. Provide separate toilet facilities for men and women.

2.3 EQUIPMENT

- A. Fire Extinguishers: Portable, UL rated; with class and extinguishing agent as required by locations and classes of fire exposures.
- B. Direct Digital Control (DDC) Internet Connection: Furnish, install and maintain a high-speed connection (Digital Subscriber Line (DSL) or similar) between the project's DDC system and the internet through a Contractor furnished internet service provider. Contractor is responsible for maintenance of this connection and costs associated with internet service provider through warranty period of this contract. Upon completion of warranty period, service shall be transferable to the Government at which time future costs for connection will no longer be the responsibility of Contractor.
- C. Contracting Officers Field Office
 - 1. Outlets: Minimum of two quad outlets with surge protection.
 - 2. Refrigerator: Under counter, 3.2-cubic-foot volume with 0.8-cubic-foot freezer with dedicated power receptacle.
 - 3. Fire Extinguisher: UL listed and FM (Fire Pump Motors) approved, minimum rating of 2-A:10-B:C, dry chemical.
 - 4. First-Aid Kit: General office/light industrial kit.
 - 5. Two desks with five drawers each and two chairs with casters; two drafting tables (minimum 40 inches wide by 5 feet long) and two stools; drawing rack; two 2-drawer and one 4-drawer legal size locking filing cabinets with keys; 8 feet of 12 inch deep shelving; coat rack; two additional guest chairs; two desk lamps; two drafting table lamps; and a maximum/minimum thermometer.
 - 6. Manufactured computer work station, capable of containing CPU (central processing unit), monitor, keyboard, printer; work station chair.
 - 7. Additional tables necessary for FAX machine and copier.
 - 8. Two 5-gallon trash cans and one 30-gallon trash can with lid.
 - 9. Ceiling mounted general lighting fixtures.

PART 3 - EXECUTION

3.1 INSTALLATION, GENERAL

- A. Locate facilities where they will serve Project adequately and result in minimum interference with performance of the Work. Relocate and modify facilities as required by progress of the Work.

- B. Provide each facility ready for use when needed to avoid delay. Do not remove until facilities are no longer needed or are replaced by authorized use of completed permanent facilities.

3.2 TEMPORARY UTILITY INSTALLATION

- A. General: Install temporary service or connect to existing service.
 - 1. Arrange with utility company, NPS, and existing users for time when service can be interrupted, if necessary, to make connections for temporary services. Acquire necessary permits.
- B. Storm Sewers and Drainage: Provide temporary utilities to remove effluent lawfully.
 - 1. Connect temporary sewers to private system indicated as directed by the agency(ies) with jurisdiction.
- C. Non-potable water for construction is not available within the park boundaries. Contractor shall furnish non-potable water from a source outside park boundary.
- D. Potable water is not available on site. Furnish cool, potable water for construction personnel in locations convenient to work stations.
- E. Sanitary Facilities: Provide temporary toilets, and wash facilities for use by construction personnel.
 - 1. Place in approved locations secluded from public observation and convenient to work stations. Relocate as work progress requires.
 - 2. Maintain and clean toilet facilities at least weekly.
 - 3. Completely remove sanitary facilities on completion of work.
- F. Electric Power Service: Provide electric power service and distribution system of sufficient size, capacity, and power characteristics required for construction operations.
- G. Lighting: Provide temporary lighting with local switching that provides adequate illumination for construction operations, observations, and inspections.
 - 1. Install and operate temporary lighting that fulfills security and protection requirements without operating entire system.
- H. Telephone Service: No telephone service is available on site for Contractor's use. Make arrangements with Telephone Company and pay costs.

3.3 SUPPORT FACILITIES INSTALLATION

- A. General: Comply with the following:
 - 1. Provide incombustible construction for offices, shops, and sheds located within construction area or within 50 feet of building lines. Comply with NFPA 241.
 - 2. Maintain support facilities until near Substantial Completion. Remove structures, equipment, and furnishings, and terminate services after punch list is 100 percent

completed or when directed by Contracting Officer. Personnel remaining after Substantial Completion will be permitted to use permanent facilities, under conditions acceptable to Contracting Officer.

B. Contracting Officers Field Office:

1. Provide Heat, lights, power, air conditioning, temporary water pressure and sewage holding tanks.
2. Provide office, furnishings, and utility connections no later than 7 days after date of Notice to Proceed. Exact location will be determined by Contracting Officer.
3. Maintain equipment, furnishings, and structures. Provide equipment replacement elements as needed. Provide weekly cleaning services and trash disposal. Maintain and service water and sewer holding tanks as required.

C. Traffic Controls: Erect and maintain barricades, lights, danger signals, and warning signs in accordance with Manual on Uniform Traffic Control Devices (MUTCD), Part 6, latest edition.

1. Protect existing site improvements to remain including curbs, pavement, and utilities.
2. Maintain access for fire-fighting equipment.
3. Illuminate barricades and obstructions at night; keep safety lights burning from sunset to sunrise.
4. Adequately barricade and post open cuts in or adjacent to thoroughfares.
5. Install Barrier Tape where directed by Contracting Officer. Keep a minimum of two rolls on site.

D. Parking: Use designated areas of existing shown on drawings parking areas for construction personnel.

E. Dewatering Facilities and Drains: Comply with requirements of the agency(ies) with jurisdiction. Maintain Project site, excavations, and construction free of water.

1. Dispose of rainwater in lawful manner that will not result in flooding Project or adjoining properties nor endanger permanent Work or temporary facilities.
2. Remove snow and ice as required to minimize accumulations.

F. Project Identification and Temporary Signs: Provide Project identification and other signs. Fence, barricade, or otherwise block off the immediate work area to prevent unauthorized entry.

1. Provide temporary, directional signs for construction personnel and visitors.
2. Maintain and touchup signs so they are legible at all times.
3. Erect and maintain sufficient detour signs at road closures and along detour routes.

G. Waste Disposal Facilities: Provide waste-collection containers in sizes adequate to handle waste from construction operations. Comply with requirements of agency(ies) with jurisdiction.

H. Lifts and Hoists: Provide facilities necessary for hoisting materials and personnel.

1. Truck cranes and similar devices used for hoisting materials are considered "tools and equipment" and not temporary facilities.

3.4 SECURITY AND PROTECTION FACILITIES INSTALLATION

- A. Environmental Protection: Provide protection, operate temporary facilities, and conduct construction in ways and by methods that comply with environmental regulations and minimize possible air, waterway, and subsoil contamination or pollution or other undesirable effects.
- B. Cleaning of Equipment: Contractor shall ensure prior to moving on to Project Area, equipment, is free of soil, seeds, vegetative matter, or other debris that could contain or hold seeds. Ensure equipment has been pressure washed and is free of exotic species. Equipment shall be considered free of soil, seeds, and other debris when visual inspection does not disclose such material. Disassembly of equipment components or specialized inspection tools are not required.
- C. Temporary Erosion and Sedimentation Control: Refer to Section 01 57 23 "Temporary Storm Water Pollution Prevention".
- D. Tree and Plant Protection: Refer to Section 01 11 00 "Summary of Work".
- E. Pest Control: Follow NPS requirements to minimize attraction and harboring of rodents, roaches, and other pests and perform extermination and control procedures at regular intervals so Project will be free of pests and residues at Substantial Completion. Perform control operations lawfully, using environmentally safe materials.
- F. Site Enclosure Fence: Before construction operations begin, furnish and install chain link fencing to prevent people and animals from easily entering site except by entrance gates.
 - 1. Extent of Fence: As required to enclose entire Project site or portion determined sufficient to accommodate construction operations.
 - 2. Locate vehicular gates to avoid interference with traffic on public thoroughfares.
 - 3. Locate pedestrian entrance gates as required to provide controlled personnel entry.
 - 4. Maintain security by limiting number of keys and restricting distribution to authorized personnel. Provide Contracting Officer with one set of keys.
- G. Security Enclosure and Lockup: Install substantial temporary enclosure around partially completed areas of construction. Provide lockable entrances to prevent unauthorized entrance, vandalism, theft, and similar violations of security.
- H. Barricades, Warning Signs, and Lights: Comply with requirements of agency(ies) with jurisdiction for erecting structurally adequate barricades, including warning signs and lighting.
- I. Temporary Fire Protection: Install and maintain temporary fire-protection facilities of types needed to protect against reasonably predictable and controllable fire losses. Comply with NFPA 241.
 - 1. Responsible Person: Capable and qualified person shall be placed in charge of fire protection. Responsibilities shall include locating and maintaining fire protective equipment and establishing and maintaining safe torch cutting and welding procedures.
 - 2. Tobacco Use, Smoking, and Vaping: Smoking within buildings or temporary storage sheds is prohibited.

3. Supervise welding operations, combustion-type temporary heating units, and similar sources of fire ignition according to requirements of Agency(ies) with Jurisdiction. Check with Park; many require "burn permits" for welding.
4. Develop and supervise overall fire-prevention and -protection program for personnel at Project Site. Review needs with local fire department and establish procedures to be followed. Instruct personnel in methods and procedures. Post warnings and information.
5. Provide temporary standpipes and hoses for fire protection. Hang hoses with warning sign stating hoses are for fire-protection purposes only and are not to be removed. Match hose size with outlet size and equip with suitable nozzles.
6. Hazard Control: Take necessary precautions to prevent fire during construction. Do not store flammable or combustible liquids in historic structures. Provide adequate ventilation during use of volatile or noxious substances. Ensure cleanup procedures and storage requirements are followed at close of every work session.
7. Spark Arresters: Equip gasoline or diesel powered equipment used during periods of potential fire hazards or in potential forest and grass fire locations with spark arresters approved by United States Department of Agriculture (USDA) Forest Service.
 - a. Written determinations of periods and areas of potential fire hazard will be issued by Contracting Officer.
8. Buildings: Furnish a minimum of one extinguisher for each 1,500 square feet of area or major fraction thereof.
 - a. Travel distance from any work station to nearest extinguisher shall not exceed 75 feet.
9. Vehicles and Equipment: Provide one extinguisher on each vehicle or piece of equipment.
10. Service and Refueling Areas: Locate areas a minimum of 50 feet from buildings. Shut down equipment before refueling.

3.5 OPERATION, TERMINATION, AND REMOVAL

- A. Supervision: Enforce strict discipline in use of temporary facilities. Minimize waste and abuse, limit availability of temporary facilities to essential and intended uses.
- B. Maintenance: Maintain facilities in good operating condition until removal.
 1. Maintain operation of temporary enclosures, heating, cooling, humidity control, ventilation, and similar facilities on 24-hour basis where required to achieve indicated results and avoid possibility of damage.
- C. Temporary Facility Changeover: Do not change over from using temporary security and protection facilities to permanent facilities until Substantial Completion.
- D. Termination and Removal: Remove each temporary facility when need for its service has ended, when it has been replaced by authorized use of a permanent facility, or no later than Substantial Completion. Complete or, if necessary, restore permanent construction that may have been delayed because of interference with temporary facility. Repair damaged Work, clean exposed surfaces, and replace construction that cannot be satisfactorily repaired.

1. Materials and facilities that constitute temporary facilities are property of Contractor. NPS reserves right to take possession of Project identification signs.
2. At Substantial Completion, clean and renovate permanent facilities used during construction period.

END OF SECTION 01 50 00

SECTION 01 57 19.12 – NOISE AND ACCOUSTICS MANAGEMENT

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes:

1. Special requirements for noise and acoustics management during construction operations.

1.2 DEFINITIONS

- A. Ambient noise level: The total noise associated with a given environment, being usually a composite of normal or existing sounds from all sources near and far, excluding the noise source at issue.
- B. Daytime: The hours from 7 A.M. to 9 P.M. on weekdays and 9 A.M. to 9 P.M. on weekends and holidays.
- C. Nighttime: All non-daytime hours.
- D. Property line: The real or imaginary line along the ground surface and its vertical extension, which separates real property owned or controlled by one person from contiguous real property owned or controlled by another person or from any public right-of-way or from any public space.
- E. Receiving noise area: Any real property where people live or work and where noise is heard, excluding the project or source area.

PART 2 - PRODUCTS (Not Used)

PART 3 - EXECUTION

3.1 NOISE MANGEMENT

- A. Noise Control: Perform construction operations to minimize noise. Perform noise-producing work in less sensitive hours of the day or week as directed by the Contracting Officer CO).
- B. Repetitive and/or intermittent, high-level noise: Permitted only during Daytime.
1. Do not exceed the following dB(A) limitations at 50 feet:

<u>Sound Level in dB(A)</u>	<u>Time Duration of Impact Noise</u>
70	More than 12 minutes in any hour
80	More than 3 minutes in any hour
 2. Maximum permissible construction equipment noise levels at 50 feet:

<u>EARTHMOVING</u>	<u>dB(A)</u>	<u>MATERIALS HANDLING</u>	<u>dB(A)</u>
Front Loaders	75	Concrete Mixers	75
Backhoes	75	Concrete Pumps	75
Dozers	75	Cranes	75
Tractors	75	Derricks Impact	75
Scrapers	80	Pile Drivers	95
Graders	75	Jack Hammers	75
Trucks	75	Rock Drills	80
Pavers, Stationary	80	Pneumatic Tools	80
Pumps	75	Saws	75
Generators	75	Vibrators	75
Compressors	75		

C. Ambient Noise:

1. Maximum noise levels (dB (decibel)) for receiving noise area at property line shall be as follows:
 - a. Residential receiving area
 - Daytime: 65 dB
 - Nighttime: 45 dB
 - b. Commercial/Industrial receiving area
 - Daytime: 67 dB
 - Nighttime: 65 dB
 - c. In the event the existing local ambient noise level exceeds the maximum allowable receiving noise level (dB), the receiving noise level maximum for construction operations shall be adjusted as follows:
 - d. Residential receiving area: Maximum 3 additional dB above the local ambient as measured at property line.
 - e. Commercial/Industrial receiving area: Maximum 5 additional dB above the local ambient as measured at the property line.

3.2 FIELD QUALITY CONTROL

- A. Assess potential effects of construction noise on adjacent neighbors in accordance with ASTM E1686 and as follows:
 1. Ambient noise measurement: Measure at property line at a height of at least four (4) feet above the immediate surrounding surface. Average the ambient noise level over a period of at least 15 minutes.
 2. Ambient noise measurement at urban sites: Conduct during morning peak traffic hour between 7 A.M. and 9 A.M. and afternoon peak traffic hour between 4 P.M. and 6 P.M. In addition, conduct a 24-hour measurement at the proposed project site to document the noise pattern throughout the day. Adjust and weight for seasonal and climatic variations.
- B. Monitor noise produced from construction operations in accordance with ASTM E1780.

END OF SECTION 01 57 19.12

SECTION 01 57 23 – UNDER-AN-ACRE POLLUTION PREVENTION

PART 1 - GENERAL

1.1 SUMMARY

- A. National Park Service (NPS) Standards and Guidelines require water quality be protected to ensure compliance with Organic Act. Contractor shall prepare an Under-An-Acre Pollution Prevention Plan (UPPP) for each project resulting in less than 1 acre of soil disturbance or not otherwise subject to requirements of NPDES program. ([UPPP Guideline](#))
- B. The work of this section consists of implementing measures to prevent discharges of pollutants, including temporary storm water pollution during construction activities, either through compliance with NPDES permit program, or in conformance with NPS guidance for UPPPs.
- C. Work of this section consists of implementing measures to Temporary Storm Water Pollution during construction activities, either through compliance with NPDES permit program; or in conformance with NPS guidance for UPPPs.

1.2 DEFINITIONS

- A. Definitions pertaining to sustainable development: As defined in ASTM E2114.
- B. Environmental Pollution and Damage: The presence of chemical, physical, or biological elements or agents which adversely affect human health or welfare; unfavorably alter ecological balances; or degrade utility of the environment for aesthetic, cultural, or historical purposes.
- C. UPPP: Developed and implemented pollution prevention plan (including stormwater management measures, if needed) to protect environment from pollutants on construction projects with less than one acre of disturbance in conformance with NPS guidelines.

1.3 SUBMITTALS

- A. After contract award and before pre-construction conference, prepare and submit:
 - 1. A UPPP in conformance with NPS guidelines and adherence to applicable construction storm water management practices.
- B. Inspection Schedule: Submit schedule for inspection and monitoring of storm water pollution prevention measures.
- C. Erosion Control Products: Submit manufacturer's product information and installation recommendations for silt fence, filter fabric, erosion control blanket, ~~straw bales~~, and other materials proposed for use on this project.

1.4 QUALITY ASSURANCE

- A. Contractor shall prepare and submit a plan to Contracting Officer (CO) for review and concurrence.
- B. Orientation Meeting: Contractor shall arrange and conduct an Erosion and Sediment Control meeting/briefing to inform parties, scheduled to be on-site during project, of measures to be implemented for proper erosion and sediment control (may be included as part of Pre-Construction Meeting).
 - 1. Installation of silt fences, storm drain protection, and other forms of erosion and sediment control shall not begin until after this meeting has occurred.
- C. Pollution Prevention Manager: Contractor shall designate Pollution Prevention Manager who will be responsible for implementation, inspection, maintenance, and amendments to approved plan.
 - 1. Pollution Prevention and Erosion Control Manager shall be familiar with UPPP procedures and Best Management Practices (BMPs) and shall ensure emergency procedures and plan are updated as needed and available for inspection.
 - 2. When changes in approved plan are required, Pollution Prevention and Erosion Control Manager shall prepare and certify an amendment and submit to Contracting Officer for review and concurrence.

PART 2 - PRODUCTS

2.1 UNDER-AN-ACRE POLLUTION PREVENTION PLAN

- A. Provide UPPP which conforms to NPS requirements (utilize [UPPP template](#)) and include:
 - 1. Responsible Parties
 - 2. General Information: Project Scope, Project Details, Site Information, and Spill Prevention
 - 3. Standards and Constraints
 - 4. Project Scheduling
 - 5. Known Data on Soil and Fill
 - 6. Activities with Potential to Generate Sediment
 - 7. Activities and Materials with Potential to Pollute Storm Water
 - 8. Management and Reporting BMPs
 - 9. Waste Management BMPs
 - 10. Non-Storm Water Pollution Control BMPs
 - 11. Soil Stabilization BMPs
 - 12. Sediment Control BMPs
 - 13. Other Pollution Control BMPs
 - 14. References
 - 15. Preparer's Certification
 - 16. Appendices: Contact Information, Pollution Prevention Control Map or Sheet(s), Standard Installation Specifications for each BMP, and Blank forms.

PART 3 - EXECUTION

3.1 ENVIRONMENTAL PROTECTION

- A. Protection of Natural Resources: Comply with applicable regulations and these specifications. Preserve natural resources within project boundaries and outside limits of work performed under this Contract in their existing condition or restore to an equivalent or improved condition as approved by Contracting Officer.
- B. Construction Zone: Arrange construction activities to minimize pollution (i.e., erosion, trash, etc.) to maximum practical extent.
 - 1. Clearing, excavation, and grading shall be limited to those areas of project site necessary for construction. Minimize area exposed and unprotected.
 - 2. Clearly mark and delineate limits of work activities.
 - 3. Clearly delineate the sensitive resource area. Double fence the sensitive resource area with ESC and chain link fencing to prevent impacts/disturbance to the sensitive areas.
 - 4. Equipment shall not be allowed to operate outside limits of work or to disturb existing vegetation.
 - 5. Excavation and grading shall be completed during dry season to maximum extent possible.
 - 6. Material shall be stored away from locations where water is present to greatest extent practicable.

3.2 UNDER-AN_ACRE POLLUTION PREVENTION PLAN

- A. Review and Acceptance: Contractor and Contracting Officer will jointly review draft Plan and agree to needed revisions. Contractor shall incorporate revisions, sign, and submit final Plan to Contracting Officer. Final Plan will be the document enforced on the project.
 - 1. Accepted Plan will describe and ensure implementation of practices to be used to reduce pollutants in storm water discharges.
 - 2. Contractor shall maintain current copy of Plan and associated records and forms at jobsite throughout duration of project.
 - 3. Plan shall be available for public inspection and inspection and use of Contracting Officer.
 - 4. Approval of Contractor's Plan will not relieve Contractor of responsibility for compliance with applicable environmental regulations.
- B. Implementation: Implement Plan as required throughout construction period and maintain erosion control elements in proper working order.
 - 1. Do not perform clearing and grubbing or earthwork until Plan has been implemented.

3.3 SITE INSPECTIONS AND PLAN REVISIONS

- A. Inspections: Contractor and Contracting Officer will perform a weekly inspection onsite.

1. Inspection shall include disturbed areas not completely stabilized, areas used for storage of materials, locations where vehicles enter or exit site, and other erosion and sediment controls included in the Plan.
 2. Inspections shall be documented.
 3. Inspection forms shall be retained onsite in Plan notebook throughout construction period.
- B. Plan Revisions: It may be necessary to revise Plan during construction to make necessary improvements, revisions, or to respond to unforeseen conditions noted during construction or site inspections.
1. Plan shall specify mechanism whereby revisions may be proposed by Contractor or Contracting Officer.
 2. Contractor and Contracting Officer will jointly review each revision to Plan before changes incorporated and implemented. Contractor will then provide revised copy of Plan to Contracting Officer.
 3. Accepted modifications will be implemented within 7 calendar days following date of inspection when deficiencies or necessary corrections are first noted.
- C. Negligence: Provide additional temporary erosion and pollution controls made necessary by Contractor's errors or negligence at no additional cost to Government.

3.4 HOUSEKEEPING AND SITE MANAGEMENT

- A. Store materials onsite in conformance to Federal, state, local, and manufacturer's regulations and specifications. Use Best Management Practices to minimize risk of materials coming into contact with environmental conditions (i.e. water and wind) that could disperse them.
- B. Manage solid waste in conformance to Federal, state, and local regulations. Best Management Practices should be used to minimize risk of materials coming into contact with environmental conditions (i.e. water and wind) that could disperse them.
- C. Include a spill prevention and control plan with provisions placed in UPPP.

3.5 EROSION CONTROL MEASURES

- A. Erosion control measures shall consist of Best Management Practices for storm water discharges, including silt fencing, barrier protectors, temporary soil retention blankets, excelsior drainage filters, sediment traps and berms.
- B. Berms and excelsior drainage filters shall be used to form sediment traps and control run-on and run-off into other areas, including creeks, streams, marshes, access roads, well areas, and staging areas.
- C. Erosion control measures shall be used to contain only direct precipitation in construction zone. Contained water shall be allowed to percolate into ground or drain slowly through drainage filter sediment traps.
- D. Earthen sediment traps or holding ponds shall not be used unless accepted by Contracting Officer.

- E. Reduce runoff velocity and direct surface runoff around and away from fuel containment, storage, and borrow areas.
- F. Divert surface runoff around and away from cut and fill slopes.
- G. Excess water used for dust control shall be contained within demolition areas by erosion control measures.
- H. Contractor shall prevent deposition of materials onto paved areas. Contractor shall inspect paved areas for deposited materials weekly and remove materials immediately.
- I. Furnish, install, maintain, and operate necessary control measures and other equipment necessary to prevent erosion as described in approved UPPP.
- J. Before work begins, sufficient equipment shall be available on site to assure operation and adequacy of erosion control system can be maintained.

3.6 MAINTENANCE OF TEMPORARY FACILITIES

- A. Ensure erosion and sediment control structures remain effective throughout excavation and grading operations. Relocate structures as necessary.
- B. Inspect control structures after each significant rainfall. Promptly repair breaches which occur.
- C. Contractor shall remove entrapped sediment from behind excelsior drainage filter after each storm.

3.7 REPORTING

- A. If a discharge occurs or if project receives written notice or order from regulatory agency, Contractor shall immediately notify Contracting Officer and shall file written report to Agency(ies) with Jurisdiction within 7 days of discharge event, notice, or order. Corrective measures shall be implemented immediately following discharge, notice, or order. The report to the Agency(ies) with Jurisdiction shall contain:
 - 1. Date, time, location, nature of operation, and type of discharge, including cause or nature of notice or order.
 - 2. Best Management Practices deployed before discharge event, or prior to receiving notice or order.
 - 3. Date of deployment and type of Best Management Practices deployed after discharge event, or after receiving notice or order, including additional Best Management Practices installed or planned to reduce or prevent re-occurrence.
 - 4. An implementation and maintenance schedule for affected Best Management Practices.

3.8 SEDIMENT DISPOSAL

- A. Sediment excavated from temporary sediment control structures shall be disposed on site with general fill, or with topsoil. Sediment shall be allowed to dry out as required before reuse.

- B. Contractor shall place sediment removed from traps and other structures where it will not enter a storm drain or watercourse and where it will not immediately reenter the basin.

3.9 REMOVAL OF TEMPORARY POLLUTION CONTROL MEASURES

- A. Temporary control measures shall be removed with permission of Contracting Officer within 20 working days after final acceptance of project, and/or once grading is complete and slopes have stabilized.

END OF SECTION 01 57 23

SECTION 01 67 00 - PRODUCT REQUIREMENTS

PART 1 - GENERAL

1.1 SUMMARY

- A. Section includes administrative and procedural requirements for selection of products for use in Project; product delivery, storage, and handling; manufacturers' standard warranties on products; special warranties; and environmental requirements.

1.2 DEFINITIONS

- A. Products: Items purchased for incorporating into Work, whether purchased for Project or taken from previously purchased stock. The term "product" includes "material," "equipment," "system," and terms of similar intent.
 - 1. Named Products: Items identified by manufacturer's product name, including make or model number and other designation shown or listed in manufacturer's published product literature, current as of date of Contract Documents.
 - 2. New Products: Items that have not previously been incorporated into another project or facility, except that products consisting of recycled-content materials are allowed, unless explicitly stated otherwise. Products salvaged or recycled from other projects are not considered new products.
 - 3. Approved Equals: Product demonstrated and approved through submittal process, or where indicated as a product substitution, to have indicated qualities related to type, function, dimension, in-service performance, physical properties, appearance, and other characteristics that equal or exceed those of specified product.
- B. Definitions pertaining to sustainable development: As defined in ASTM E2114.
- C. Biobased Materials: As defined in Farm Security and Rural Investment Act, for purposes of Federal procurement of biobased products, "biobased" means a "commercial or industrial product (other than food or feed) composed, in whole or in significant part, of biological products or renewable domestic agricultural materials (including plant, animal, and marine materials) or forestry materials." Biobased materials also include fuels, chemicals, building materials, or electric power or heat produced from biomass as defined by The Biomass Research and Development Act of 2000.
 - 1. Biobased content: Amount of biobased carbon in material or product as a percentage of weight (mass) of total organic carbon in material or product.
- D. Chain-of-Custody: Process whereby a product or material is maintained under physical possession or control during its entire life cycle.
- E. Environmentally preferable products: Products and services with lesser or reduced effect on the environment in comparison to conventional products and services. Refer to Environmental

Protection Agency's (EPA) [Environmentally Preferable Purchasing Program](#) for more information.

- F. Stewardship: Responsible use and management of resources.
- G. Maintenance of ecosystem components and functions for future generations.
 - 1. Recycled Content Materials: Products containing pre-consumer or post-consumer materials as all or part of their feedstock. Recycled content claim shall be consistent with International Organization for Standardization (ISO) 140001 Standard for the Use of Environmental Marketing Claims.
 - 2. Rapidly Renewable Material: Material made from plants typically harvested within a ten-year cycle.
 - 3. Regional Materials: Materials manufactured and extracted, harvested, or recovered within a radius of 500 miles from Project location.

1.3 SUBMITTALS

- A. Record Submittals and submit:
 - 1. Affirmative Procurement Reporting Form. Submit on form in Appendix A of this Section, or similar form as approved by Contracting Officer (CO).
 - 2. Submit environmental data in accordance with Table 1 of ASTM E2129 for these products:
 - a. Masonry
 - b. Joint Sealers

.Material Safety Data Sheets (MSDS): For each product required by OSHA to have a MSDS, submit an MSDS. MSDS shall be prepared within the previous five years. Include information for MSDS Sections 1 to 16 in accordance with ANSI Z400.1 and as follows:

- a. Section 1: Chemical Product and Company Identification
- b. Section 2: Composition/Information on Ingredients
- c. Section 3: Hazards Identification
- d. Section 4: First Aid Measures
- e. Section 5: Fire Fighting Measures
- f. Section 6: Accidental Release Measures
- g. Section 7: Handling and Storage
- h. Section 8: Exposure Controls/Person Protection
- i. Section 9: Physical and Chemical Properties
- j. Section 10: Stability and Reactivity Data
- k. Section 11: Toxicological Information. Include data used to determine the hazards cited in Section 3. Identify acute data, carcinogenicity, reproductive effects, and target organ effects. Provide written description of the process used in evaluating chemical hazards relative to preparation of the MSDS.
- l. Section 12: Ecological Information. Include data regarding environmental impacts during raw materials acquisition, manufacture, and use. Include data regarding environmental impacts in event of accidental release.
- m. Section 13: Disposal Considerations. Include data regarding proper disposal of the chemical. Include information regarding recycling and reuse. Indicate whether or

- not product is considered to be "hazardous waste" according to United States EPA Hazardous Waste Regulations 40 CFR 261 (Code of Federal Regulations).
- n. Section 14: Transportation Information. Identify hazard class for shipping.
- o. Section 15: Regulatory Information. Identify federal, state, and local regulations applicable to the material.
- p. Section 16: Other Information. Include additional information relative to recycled content, biobased content, and other information regarding environmental and health impacts. Identify the date MSDS was prepared.
- 4. Chain of Custody: Submit chain-of-custody documentation for sustainable forestry for these products:
 - a. Rough Carpentry

1.4 QUALITY ASSURANCE

- A. Compatibility of Options: If Contractor is given option of selecting between two or more products for use on Project, product selected shall be compatible with products previously selected, even if previously selected products were also options.

1.5 PRODUCT DELIVERY, STORAGE, AND HANDLING

- A. Deliver, store, and handle products using means and methods that prevent damage, deterioration, and loss, including theft. Comply with manufacturer's written instructions.
- B. Delivery and Handling:
 - 1. Schedule delivery to minimize long-term storage at Project site and prevent overcrowding of construction spaces.
 - 2. Coordinate delivery with installation time to ensure minimum holding time for items that are flammable, hazardous, easily damaged, or sensitive to deterioration, theft, and other losses.
 - 3. Deliver products to Project site in undamaged condition; in manufacturer's original sealed container or other packaging system; complete with labels and instructions for handling, storing, unpacking, protecting, and installing.
 - 4. Inspect products on delivery to ensure compliance with Contract Documents. Ensure products are undamaged and properly protected.
 - 5. Obtain materials in biodegradable or recyclable/reusable packaging which uses minimum amount of packaging possible.
- C. Storage:
 - 1. Allow for inspection and measurement of quantity or counting of units.
 - 2. Store materials in manner to not endanger Project structure.
 - 3. Store products subject to damage by the elements, under cover in weather tight enclosure above ground, with ventilation adequate to prevent condensation.
 - 4. Comply with product manufacturer's written instructions for temperature, humidity, ventilation, and weather-protection requirements for storage.
 - 5. Protect stored products from damage and liquids from freezing.

6. Store loose granular materials in well-drained area on solid surfaces to prevent mixing with foreign matter.

1.6 PACKAGING

- A. Where Contractor has option to provide one of listed products or equal, preference shall be given to products with minimal packaging and easily recyclable packaging as defined in ASTM D5834.
- B. Maximize use of source reduction and recycling procedures outlined in ASTM D5834.
- C. Provide minimum 45 percent post-consumer recycled content and minimum 100 percent recovered fiber content of industrial paperboard in accordance with EPA's Comprehensive Procurement Guidelines and ASTM D5663.
- D. Provide minimum 10 percent post-consumer recycled content and minimum 10 percent recovered fiber content of carrier board in accordance with EPA's Comprehensive Procurement Guidelines and ASTM D5663.
- E. Provide minimum 5 percent post-consumer recycled content and minimum 5 percent recovered fiber content of brown papers (e.g., wrapping papers and bags) in accordance with EPA's Comprehensive Procurement Guidelines and ASTM D5663.

1.7 ENVIRONMENTALLY PREFERABLE PRODUCTS

- A. Provide environmentally preferable products to greatest extent possible.
 1. To greatest extent possible, provide products and materials with a lesser or reduced effect on the environment considering raw materials acquisition, production, manufacturing, packaging, distribution, reuse, operation, maintenance, and/or disposal of the product.
 2. Eliminate use of ozone depleting compounds during and after construction where alternative environmentally preferable products are available, consistent with either Montreal Protocol and Title VI or Clean Air Act Amendments of 1990, or equivalent overall air quality benefits that take into account life cycle impacts.
 3. Use products meeting or exceeding EPA's recycled content recommendations for EPA-designated products. Use materials with recycled content such that the sum of post-consumer recycled content plus one-half of the pre-consumer content constitutes at least 10% (based on cost) of total value of the materials in project.

1.8 PRODUCT WARRANTIES

- A. Warranties specified in other Sections shall be in addition to, and run concurrent with, other warranties required by Contract Documents. Manufacturer's disclaimers and limitations on product warranties do not relieve Contractor of obligations under requirements of Contract Documents.
 1. Manufacturer's Warranty: Preprinted written warranty published by individual manufacturer for product specifically endorsed by manufacturer to Owner.

2. Special Warranty: Written warranty required by or incorporated into Contract Documents, either to extend time limit provided by manufacturer's warranty or to provide more rights for Owner.
- B. Special Warranties: Prepare written document containing appropriate terms and identification, ready for execution. Submit draft for approval before final execution.
1. Manufacturer's Standard Form: Modified to include Project-specific information and properly executed.
 2. Specified Form: When specified forms are included with Specifications, prepare written document using appropriate form properly executed.
 3. Refer to Divisions 2 through 49 Sections for specific content requirements and particular requirements for submitting special warranties.
- C. Submittal Time: Comply with requirements in Section 01 77 00 "Closeout Procedures."

PART 2 - PRODUCTS

2.1 PRODUCT SELECTION PROCEDURES

- A. General Product Requirements: Provide products to comply with Contract Documents, undamaged and, unless otherwise indicated, new at time of installation.
1. Provide products complete with accessories, trim, finish, fasteners, and other items needed for a complete installation and indicated use and effect.
 2. Standard Products: If available, and unless custom products or nonstandard options are specified, provide standard products of types produced and used successfully in similar situations on other projects.
 3. Government reserves right to limit selection to products with warranties not in conflict with requirements of Contract Documents.
 4. Where products are accompanied by term "as selected," Contracting Officer will make selection.
 5. Where products are accompanied by term "match sample," sample to be matched is Governments.
 6. Descriptive, performance, and reference standard requirements in Specifications establish "salient characteristics" of products.
- B. Product Selection Procedures:
1. Product: Where Specifications name single product and manufacturer, provide named product that complies with requirements or approved equal.
 2. Manufacturer/Source: Where Specifications name single manufacturer or source, provide product by named manufacturer or source that complies with requirements or approved equal.
 3. Products: Where Specifications include list of names of both products and manufacturers, provide one of the products listed that complies with requirements or approved equal.
 4. Manufacturers: Where Specifications include list of manufacturers' names, provide a product by one of the manufacturers listed that complies with requirements or approved equal.

5. Available Products: Where Specifications include list of names of both products and manufacturers, provide one of the products listed, or an unnamed product, that complies with requirements. Comply with provisions in Part 2 "Products" Articles for consideration of an unnamed product.
6. Available Manufacturers: Where Specifications include list of manufacturers, provide a product by one of the manufacturers listed, or an unnamed manufacturer, that complies with requirements. Comply with provisions in Part 2 "Products" Articles for consideration of an unnamed product.
7. Product Options: Where Specifications indicate sizes, profiles, and dimensional requirements on Drawings are based on a specific product or system, provide specified product, system, or approved equal.
8. Visual Matching Specification: Where Specifications require matching an established Sample, select product that complies with requirements and matches Architect's sample. Contracting Officers decision will be final on whether a proposed product matches.
 - a. If no product available within specified category matches and complies with other specified requirements, comply with provisions in Part 2 "Product Substitutions" Article for proposal of product.
9. Visual Selection Specification: Where Specifications include phrase "as selected from manufacturer's colors, patterns, textures" or a similar phrase, select a product that complies with other specified requirements.
 - a. Standard Range: Where Specifications include phrase "standard range of colors, patterns, textures" or similar phrase, Contracting Officer will select color, pattern, density, or texture from manufacturer's product line that does not include premium items.
 - b. Full Range: Where Specifications include phrase "full range of colors, patterns, textures" or similar phrase, Contracting Officer will select color, pattern, density, or texture from manufacturer's product line that includes both standard and premium items.

2.2 APPROVED EQUAL PRODUCTS

- A. Conditions: Contracting Officer will consider Contractor's request when the following conditions are satisfied. If following conditions are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
 1. Evidence proposed product does not require revisions to Contract Documents, that it is consistent with Contract Documents and will produce indicated results and is compatible with other portions of Work.
 2. Detailed comparison of significant qualities of proposed product with those named in Specifications. Significant qualities include attributes such as performance, weight, size, durability, visual effect, and specific features and requirements indicated.
 3. Evidence proposed product provides specified warranty.
 4. List of similar installations for completed projects with project names and addresses and names and addresses of architects and owners, if requested.
 5. Samples, if requested.

PART 3 - EXECUTION

3.1 PROTECTION AFTER INSTALLATION

- A. Provide adequate coverings as necessary to protect installed materials from damage resulting from natural elements, traffic, and subsequent construction. Remove when no longer needed.

END OF SECTION 01 67 00

AFFIRMATIVE PROCUREMENT REPORTING FORM

Recycled Content Materials & Biobased Content Materials

Project Number: _____

License Number: _____

Contractor Address: _____

[illegible]

CERTIFICATION

I hereby certify the information provided herein is accurate and that the requisition/procurement of all materials listed on this form comply with current EPA standards for recycled/recovered materials content. The following exemptions may apply to the non-procurement of recycled/recovered content materials:

1. The product does not meet appropriate performance standards.
2. The product is not available within a reasonable time frame.
3. The product is not available competitively (from two or more sources).
4. The product is only available at an unreasonable price (compared with a comparable non-recycled content product.)

Date: _____

END OF
AFFIRMATIVE PROCUREMENT REPORTING FORM
Recycled Content Materials & Biobased Content Materials

SECTION 01 73 40 - EXECUTION

PART 1 - GENERAL

1.1 SUMMARY

- A. Section includes general procedural requirements governing execution of Work including:
 - 1. Construction layout
 - 2. Field engineering and surveying
 - 3. General installation of products
 - 4. Progress cleaning
 - 5. Starting and adjusting
 - 6. Protection of installed construction
 - 7. Correction of the Work

1.2 SUBMITTALS

- A. Certificates: Submit certificate signed by land surveyor certifying location and elevation of improvements comply with requirements.
- B. Landfill Receipts: Submit copy of receipts issued by landfill facility, licensed to accept hazardous materials, for hazardous waste disposal.
- C. Certified Surveys: Submit two copies signed by land surveyor.
- D. Quantity Surveys: Submit 2 copies showing quantities of work performed and actual construction completed in place.

1.3 QUALITY ASSURANCE

- A. Land Surveyor Qualifications: Professional land surveyor legally qualified to practice in jurisdiction where Project is located and-is experienced in providing land-surveying services of kind indicated.

PART 2 - PRODUCTS (Not Used)

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Existing Conditions: Existence and location of site improvements and other construction indicated as existing are not guaranteed.

1. Before beginning work, investigate and verify the existence and location of mechanical and electrical systems and other construction affecting the Work.
 2. Before construction, verify the location and points of connection of utility services.
- B. Existing Utilities: Existence and location of underground and other utilities and construction indicated as existing are not guaranteed. Before beginning sitework, investigate and verify existence and location of underground utilities and other construction affecting Work.
1. Before construction, verify location and invert elevation at points of connection of sanitary sewer, storm sewer, and water-service piping; and underground electrical services.
 2. Furnish location data for work related to Project that must be performed by public utilities serving Project site.
- C. Acceptance of Conditions: Examine substrates, areas, and conditions, with Installer or Applicator present where indicated, for compliance with requirements for installation tolerances and other conditions affecting performance. Record observations.
1. Verify compatibility with and suitability of substrates, including compatibility with existing finishes or primers.
 2. Proceed with installation only after unsatisfactory conditions have been corrected. Proceeding with the Work indicates acceptance of surfaces and conditions.

3.2 PREPARATION

- A. Field Measurements: Take field measurements as required to fit Work properly. Recheck measurements before installing each product. Where portions of Work are indicated fit to other construction, verify dimensions of other construction by field measurements before fabrication. Coordinate fabrication schedule with construction progress to avoid delaying Work.
- B. Space Requirements: Verify space requirements and dimensions of items shown diagrammatically on Drawings.
- C. Review of Contract Documents and Field Conditions: Immediately on discovery of need for clarification of the Contract Documents caused by differing field conditions outside control of Contractor, submit request for information to Contracting Officer in accordance with Section 01 31 00 "Project Management and Coordination."

3.3 CONSTRUCTION LAYOUT

- A. Verification: Verify layout information shown on Drawings, in relation to the existing benchmarks before proceeding to lay out Work. Notify Contracting Officer promptly if discrepancies are discovered.
- B. General: Engage a land surveyor to lay out Work using accepted surveying practices.
1. Establish benchmarks and control points to set lines of construction and elsewhere as needed to locate each element of Project.

2. Establish dimensions within tolerances indicated. Do not scale Drawings to obtain required dimensions.
 3. Inform installers of lines and levels to which they must comply.
 4. Check location, level and plumb, of every major element as Work progresses.
 5. Notify Contracting Officer when deviations from required lines and levels exceed allowable tolerances.
 6. Close site surveys with an error of closure equal to or less than the established standard.
- C. Site Improvements: Locate and lay out site improvements, including pavements, grading, fill and topsoil placement, utility slopes, and invert elevations.
- D. Building Lines and Levels: Locate and lay out control lines and levels for structures, building foundations, column grids, and floor levels, including those required for mechanical and electrical work. Transfer survey markings and elevations for use with control lines and levels. Level foundations and piers from two or more locations.
- E. Record Log: Maintain log of layout control work. Record deviations from required lines and levels. Include beginning and ending dates and times of surveys, weather conditions, name and duty of each survey party member, and types of instruments and tapes used. Make log available for review by National Park Service (NPS).

3.4 FIELD ENGINEERING

- A. Reference Points: Locate existing permanent benchmarks, control points, and similar reference points before beginning Work. Preserve and protect permanent benchmarks and control points during construction operations. Controls destroyed by Contractor will be replaced by Contractor at their expense.
1. Existing Monuments: All benchmarks, land corners, and triangulation points, established by other surveys, existing within construction area shall be preserved. If existing monuments interfere with Work, secure written permission before removing them.
- B. Certified Survey: On completion of foundation walls, major site improvements, and other work requiring field-engineering services, prepare a certified survey showing dimensions, locations, angles, and elevations of construction and sitework.

3.5 INSTALLATION

- A. General: Locate Work and components of Work accurately in correct alignment and elevation, as indicated.
1. Make vertical work plumb and make horizontal work level.
 2. Where space is limited, install components to maximize space available for maintenance and ease of removal for replacement.
 3. Conceal pipes, ducts, and wiring in finished areas, unless otherwise indicated.
- B. Comply with manufacturer's written instructions and recommendations for installing products in applications indicated.

- C. Install products at the time and under conditions for best possible results. Maintain conditions required for product performance until Substantial Completion.
- D. Conduct construction operations so no part of the Work is subjected to damaging operations or loading in excess of that expected during normal conditions of occupancy.
- E. Tools and Equipment: Do not use tools or equipment that produce harmful noise levels.
- F. Templates: Obtain and distribute to parties involved templates for work specified to be factory prepared and field installed. Check Shop Drawings of other work to confirm adequate provisions are made for locating and installing products to comply with indicated requirements.
- G. Anchors and Fasteners: Provide anchors and fasteners as required to anchor each component securely in place, accurately located and aligned with other portions of Work.
 - 1. Mounting Heights: Where mounting heights are not indicated, mount components at heights directed by Contracting Officer.
 - 2. Allow for building movement, thermal expansion, and contraction.
 - 3. Coordinate installation of anchorages. Furnish setting drawings, templates, and directions for installing anchorages, including sleeves, concrete inserts, anchor bolts, and items with integral anchors to be embedded in concrete or masonry. Deliver to Project site in time for installation.
- H. Joints: Make joints of uniform width. Where joint locations in exposed work are not indicated, arrange joints for best visual effect. Fit exposed connections together to form hairline joints.
- I. Hazardous Materials: Use products, cleaners, and installation materials not considered hazardous.
- J. Quantity surveys: Shall be conducted, and data derived from these surveys shall be used in computing quantities of work performed and actual construction completed and in place.
 - 1. Contractor shall conduct original and final surveys and surveys for any periods for which progress payments are requested. These surveys shall be conducted under direction of a representative of the Contracting Officer, unless Contracting Officer waives requirement in a specific instance. Government shall make such computations as are necessary to determine quantities of work performed or finally in place. Contractor shall make computations based on surveys for any periods for which progress payments are requested.
 - 2. Promptly upon completing a survey, Contractor shall furnish originals of field notes and other records relating to survey or layout of Work to Contracting Officer. Contractor shall retain copies of all such material furnished to Contracting Officer.

3.6 PROGRESS CLEANING

- A. General: Clean Project site, work areas, and common areas daily. Coordinate progress cleaning for joint-use areas where more than one Installer has worked. Enforce requirements strictly. Dispose of materials lawfully.
 - 1. Comply with requirements in National Fire Protection Association (NFPA) 241 for removal of combustible waste materials and debris at the end of every work session.

2. Do not hold materials more than 7 days during normal weather or 3 days if the temperature is expected to rise above 80 degrees Fahrenheit (27 degrees Celsius).
 3. Containerize hazardous and unsanitary waste materials separately from other waste. Mark containers appropriately and dispose of legally, according to regulations.
- B. Site: Maintain Project site free of waste materials and debris.
- C. Work Areas: Clean areas where work is in progress to level of cleanliness necessary for proper execution of Work.
1. Remove liquid spills promptly.
 2. Where dust would impair proper execution of Work, broom-clean or vacuum entire work area, as appropriate.
 3. Contractor shall provide progress cleaning that minimizes sources of food, water, and harborage available to pests.
- D. Installed Work: Keep installed work clean. Clean installed surfaces according to written instructions of manufacturer or fabricator of product installed, using only cleaning materials specifically recommended. If specific cleaning materials are not recommended, use cleaning materials not hazardous to health or property and will not damage exposed surfaces.
1. Utilize non-toxic cleaning materials and methods.
 - a. Comply with Green Seal Standard (GS) 37 for general purpose cleaning and bathroom cleaning.
 - b. Use natural cleaning materials where feasible. Natural cleaning materials include:
 - 1) Abrasive cleaners: substitute 1/2 lemon dipped in borax.
 - 2) Ammonia: substitute vinegar, salt and water mixture, or baking soda and water.
 - 3) Disinfectants: substitute 1/2 cup borax in gallon water.
 - 4) Drain cleaners: substitute 1/4 cup baking soda and 1/4 cup vinegar in boiling water.
- E. Concealed Spaces: Remove debris from concealed spaces before enclosing the space.
- F. Waste Disposal: Do not bury or burn waste materials on-site. Do not wash waste materials down sewers or into waterways.
- G. Clean and protect construction in progress and adjoining materials already in place during handling and installation. Apply protective covering where required to ensure protection from damage or deterioration at Substantial Completion.
- H. Limiting Exposures: Supervise construction operations so that no part of construction completed or in progress, is subject to harmful, dangerous, damaging, or deleterious exposure during construction period.
- I. Final Cleaning: At completion of Work, remove remaining waste materials, rubbish, tools, equipment, machinery and surplus materials. Clean exposed surfaces and leave Project clean.
1. Provide final cleaning in accordance with ASTM E1971.

3.7 PROTECTION OF INSTALLED CONSTRUCTION

- A. Provide final protection and maintain conditions that ensure installed Work is without damage or deterioration at time of Substantial Completion.

3.8 CORRECTION OF THE WORK

- A. Repair or remove and replace defective construction. Restore damaged substrates and finishes. Comply with requirements in Section 01 35 91 "Historic Preservation Treatment Procedures."
 - 1. Repairing includes replacing defective parts, refinishing damaged surfaces, and touching up with matching materials.
- B. Remove and replace damaged surfaces exposed to view if surfaces cannot be repaired without visible evidence of repair.

END OF SECTION 01 73 40

SECTION 01 74 19 - CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL

PART 1 - GENERAL

1.1 SUMMARY

- A. Section includes administrative and procedural requirements for:
 - 1. Salvaging nonhazardous demolition and construction waste.
 - 2. Recycling nonhazardous demolition and construction waste.
 - 3. Disposing of nonhazardous demolition and construction waste.

1.2 DEFINITIONS

- A. Construction Waste: Building and site improvement materials and other solid waste resulting from construction, remodeling, renovation, or repair operations. Construction waste includes packaging.
- B. Demolition Waste: Building and site improvement materials resulting from demolition or selective demolition operations.
- C. Solid Waste: Garbage, debris, sludge, or other discharged material (except hazardous waste) including solid, liquid, semisolid, or contained gaseous materials resulting from domestic, industrial, commercial, mining, or agricultural operations.
- D. Debris: Non-hazardous solid waste generated during construction, demolition, or renovation of a structure which exceeds 2.5 inch (60 millimeter) particle size that is: a manufactured object; plant or animal matter; or natural geologic material (e.g. cobbles and boulders). A mixture of debris and other material such as soil or sludge is also subject to regulation as debris if mixture is comprised primarily of debris by volume, based on visual inspection.
- E. Disposal: Removal off-site of demolition and construction waste and subsequent sale, recycling, reuse, or deposit in landfill or incinerator acceptable to authorities having jurisdiction.
- F. Environmental Pollution and Damage: Presence of chemical, physical, or biological elements or agents which adversely affect human health or welfare; unfavorably alter ecological balances; or degrade utility of environment for aesthetic, cultural, or historical purposes.
- G. Garbage: Refuse and scraps resulting from preparation, cooking, dispensing, and consumption of food.
- H. Hazardous Materials: Material regulated as a hazardous material in accordance with 49 CFR 173 (Code of Federal Regulations), requires a Material Safety Data Sheet (MSDS) in accordance with 29 CFR 1910.1200, or which during end use, treatment, handling, storage, transportation or disposal meets or has components which meet or have potential to meet the definition of Hazardous Waste in accordance with 40 CFR 261.

- I. Recycle: Recovery of demolition or construction waste for subsequent processing in preparation for reuse.
- J. Salvage: Recovery of demolition or construction waste and subsequent sale or reuse in another facility.
- K. Single-use plastic products: Plastic items intended to be disposed of immediately after use, including plastic and polystyrene food and beverage containers, bottles, straws, cups, cutlery, and disposable plastic bags.

1.3 PERFORMANCE REQUIREMENTS

- A. If waste materials encountered during deconstruction/demolition or construction phase are found to contain lead, asbestos, polychlorinated biphenyls (PCBs), (such as fluorescent lamp ballasts), or other harmful substances, they are to be handled and removed in accordance with local, state, and federal laws and requirements concerning hazardous waste.
- B. Existing items and material to be removed during deconstruction/demolition phase shall be reused in construction phase of the Project. Items that cannot be reused shall be recycled. Items considered for reuse must be in refurbishable condition and must meet quality standards set forth in these specifications. Contractor shall ensure quality of the item(s) in question will meet or exceed accepted industry or trade standards for first quality commercial grade application. During construction, deconstruction, or demolition Contracting Officer (CO) may designate other objects or materials for reuse.
- C. Salvage/Recycle Requirements: Government goal is to salvage and recycle as much nonhazardous demolition and construction waste as possible.

1.4 SUBMITTALS

- A. Waste Management Plan: After award of contract and prior to scheduled Pre-Construction Conference, Contractor shall submit a draft Waste Management Plan to Contracting Officer for approval. Submit 3 copies of plan. Revise and resubmit Plan as required by Contracting Officer. Approval of Contractor's Plan will not relieve Contractor of responsibility for compliance with applicable environmental regulations.
- B. Progress Documentation: Supplemental to Waste Management Plan, document solid waste disposal, diversion, and cost/revenue analysis and submit completed worksheet on a monthly basis. See Project Waste Management Plan Worksheet Sample, attached to the end of the Division 1 Specifications, and report totals to date for column headings.
- C. Landfill and Incinerator Disposal Records: Indicate receipt and acceptance of waste by landfills and incinerator facilities licensed to accept them. Include manifests, weight tickets, receipts, and invoices.
- D. Qualification Data: For Waste Management Coordinator
- E. Progress payment requirements:

1. With each Application for payment, submit an updated Project Waste Management Plan worksheet for solid waste disposal and diversion.
2. With each Application for Payment, submit manifests, weight tickets, receipts, and invoices specifically identifying the Project and waste material.

F. Closeout Submittals

1. With Closeout Submittals, submit a summary of a Project Waste Management Plan worksheet for solid waste disposal and diversion.

1.5 QUALITY ASSURANCE

- A. Regulatory Requirements: Comply with hauling and disposal regulations of authorities having jurisdiction.
- B. Waste Management Meeting: Conduct separate meeting or cover in Pre-Construction Conference and comply with requirements in Section 01 31 00 "Project Management and Coordination." Review methods and procedures related to waste management including:
 1. Review and discuss waste management plan including responsibilities of Waste Management Coordinator.
 2. Review requirements for documenting quantities of each type of waste and its disposition.
 3. Review and finalize procedures for materials separation and verify availability of containers and bins needed to avoid delays.
 4. Review procedures for periodic waste collection and transportation to recycling and disposal facilities.
 5. Review procedures for reduction of single-use plastic products on site.
 6. Review waste management requirements for each trade.

PART 2 - PRODUCTS

2.1 WASTE MANAGEMENT PLAN

- A. General: Develop plan consisting of waste identification and waste reduction work plan. quantities by weight or volume but use same units of measure throughout waste management plan.
- B. Waste Identification: Indicate anticipated types and quantities of demolition, site clearing, and construction waste generated by the Work. Include estimated quantities and assumptions for estimates.
- C. Waste Reduction Work Plan: List each type of waste and whether it will be salvaged, recycled, or disposed of in landfill or incinerator. Include points of waste generation, total quantity of each type of waste, quantity for each means of recovery, and handling and transportation procedures.
 1. Salvaged Materials for Reuse: For materials salvaged and reused in Project, describe methods for preparing salvaged materials before incorporation into the Work.

2. Recycled Materials: Include list of local receivers and processors and type of recycled materials each will accept. Include names, addresses, and telephone numbers.
 3. Disposed Materials: Indicate how and where materials will be disposed of. Include name, address, and telephone number of each landfill and incinerator facility.
 4. Handling and Transportation Procedures: Include method used for separating recyclable waste including sizes of containers, container labeling, and designated location on Project site where materials separation will be located.
 5. Reduction of single-use plastic products: Include methods for water distribution to include bottle refilling stations if available for personnel.
- D. Cost/Revenue Analysis: Indicate total cost of waste disposal as if there was no waste management plan and net additional cost or net savings resulting from implementing waste management plan. Include:
1. Landfill tip fees per ton.
 2. If diverted, tip fee savings from landfill diversion.
 3. Costs of recycling, salvage, or reuse.
 4. Revenue from recycling, salvage, or reuse.
 5. Total cost or savings from diversion. (Calculate by using tip fee savings and subtracting costs of recycling or adding revenue from recycling.)

PART 3 - EXECUTION

3.1 PLAN IMPLEMENTATION

- A. General: Implement waste management plan as approved by Contracting Officer. Provide handling, containers, storage, signage, transportation, and other items as required to implement waste management plan during entire duration of Contract.
- B. Waste Management Coordinator: Engage waste management coordinator responsible for implementing, monitoring, and reporting status of waste management work plan. Coordinator shall be present at Project site full time for duration of Project.
- C. Contractor shall establish contacts with local recycling and reuse companies to set up lines of responsibility. Contractor shall be responsible for coordination in terms of identifying materials, pickup schedules, and standard quality for recycled materials.
- D. Training: Train workers, subcontractors, and suppliers on proper waste management procedures, as appropriate for the Work occurring at Project site.
 1. Distribute waste management plan to everyone concerned within three days of submittal return.
 2. Distribute waste management plan to entities when they begin work on-site. Review plan procedures and locations established for salvage, recycling, and disposal.
- E. Site Access and Temporary Controls: Conduct waste management operations to ensure minimum interference with roads, streets, walks, walkways, and other adjacent occupied and used facilities.
- F. Separation facilities:

1. Contractor shall designate and Contracting Officer shall approve specific area or areas to facilitate separation of materials for potential reuse, salvage, recycling, and return.
 2. Place waste and recycling bins near each other, and close to point of waste generation but out of traffic pattern.
 3. Keep recycling and waste bin areas neat, clean, and clearly marked in order to avoid comingling of materials.
 4. Protect bins during non-working hours from off-site contamination.
 5. Check garbage dumpsters periodically for recyclables being thrown away and undocumented materials that could be recycled.
- G. Materials handling procedures: Material to be recycled shall be protected from contamination and shall be handled, stored, and transported in a manner that meets requirements set by designated facilities for acceptance. Establish defined area for operations of each trade, especially woodcutting so off-cuts are kept in one area and can be sorted by dimension for future reuse.

3.2 SALVAGING DEMOLITION WASTE

- A. Salvaged Items for Reuse in the Work:
1. Clean salvaged items.
 2. Pack or crate items after cleaning. Identify contents of containers.
 3. Store items in secure area until installation.
 4. Protect items from damage during transport and storage.
 5. Install salvaged items to comply with installation requirements for new materials and equipment. Provide connections, supports, and miscellaneous materials necessary to make items functional for use indicated.
- B. Salvaged Items for Sale and Donation: Not permitted on Project site.
- C. Salvaged Items for Governments Use:
1. Clean salvaged items.
 2. Pack or crate items after cleaning. Identify contents of containers.
 3. Allow for inspection if necessary.
 4. Store items in secure area until delivery to Government.
 5. Transport items to storage area designated by Government.
 6. Protect items from damage during transport and storage.

3.3 RECYCLING DEMOLITION AND CONSTRUCTION WASTE, GENERAL

- A. General: Recycle paper and beverage containers used by on-site workers.
- B. Recycling Incentives: Revenues, savings, rebates, tax credits, and other incentives received for recycling waste materials shall accrue to Contractor.
- C. Procedures: Separate recyclable waste from other waste materials, trash, and debris. Separate recyclable waste by type at Project site to maximum extent practical.

1. Provide appropriately marked containers or bins for controlling recyclable waste until removed from Project site. Include list of acceptable and unacceptable materials at each container and bin.
 - a. Inspect containers and bins for contamination and remove contaminated materials if found.
2. Stockpile processed materials on-site without intermixing with other materials. Place, grade, and shape stockpiles to drain surface water. Cover to prevent windblown dust.
3. Stockpile materials away from construction area. Do not store within drip line of remaining trees.
4. Store components off ground and protect from weather.
5. Remove recyclable waste off Government property and transport to recycling receiver or processor.

3.4 RECYCLING DEMOLITION WASTE

- A. Concrete: Remove reinforcement and other metals from concrete and sort with other metals.
 1. Pulverize concrete to maximum 1-1/2 inch (38 millimeter) size.
- B. Masonry: Remove metal reinforcement, anchors, and ties from masonry and sort with other metals.
 1. Pulverize masonry to maximum 1-1/2 inch (38 millimeter) size.
 2. Clean and stack undamaged, whole masonry units on wood pallets.
- C. Wood Materials: Sort and stack members according to size, type, and length. Separate lumber, engineered wood products, panel products, and treated wood materials.
- D. Metals: Separate metals by type.
 1. Remove and dispose of bolts, nuts, washers, and other rough hardware.
- E. Equipment: Drain tanks, piping, and fixtures. Seal openings with caps or plugs. Protect equipment from exposure to weather.
- F. Plumbing Fixtures: Separate by type and size.
- G. Piping: Reduce piping to straight lengths and store by type and size. Separate supports, hangers, and other components by type and size.
- H. Electrical Devices: Separate meters, panelboards, circuit breakers, and other devices by type.
- I. Conduit: Reduce conduit to straight lengths and store by type and size.

3.5 RECYCLING CONSTRUCTION WASTE

- A. Packaging:

1. Cardboard and Boxes: Break down packaging into flat sheets. Bundle and store in dry location.
 2. Polystyrene Packaging: Separate and bag materials.
 3. Pallets: As much as possible, require deliveries using pallets to remove pallets from Project site. For pallets that remain on-site, break down pallets into component wood pieces and comply with requirements for recycling wood.
 4. Crates: Break down crates into component wood pieces and comply with requirements for recycling wood.
- B. Site-Clearing Wastes: Chip brush, branches, and trees on-site.
- C. Wood Materials:
1. Clean Cut-Offs of Lumber: Grind or chip into small pieces.
 2. Clean Sawdust: Bag sawdust not containing painted or treated wood.

3.6 DISPOSAL OF WASTE

- A. General: Except for items or materials to be salvaged, recycled, or otherwise reused, remove waste materials from Project site and legally dispose in landfill or incinerator acceptable to authorities having jurisdiction.
1. Except as otherwise specified, do not allow waste materials to accumulate on-site.
 2. Remove and transport debris in manner preventing spillage on adjacent surfaces and areas.
- B. Burning: Do not burn waste materials.
- C. Burning: Burning of waste materials allowed only at designated areas on Government property, provided required permits are obtained. Provide full-time monitoring for burning materials until fires are extinguished.
- D. Disposal: Transport waste materials off Government property and legally dispose of them.

END OF SECTION 01 74 19

Project Waste Management Plan Worksheet Sample

National Park Service (NPS) - Denver Service Center (DSC) | 10-13-21

	A	B	C	D	E	F	G	H	I	J
Material	Quantity Recycled (in tons)	Quantity Salvaged or Reused (in tons)	A + B = Total Quantity Diverted from Landfill	Quantity to Landfill (in tons)	C + D = Total Quantity Generated (in tons)	Tip Fee/Ton at Landfill	C x F = Tip Fee Savings resulting from Landfill Diversion	Cost of Recycling (R), Salvage (S), or Reuse (Re) (Specify R, S, or Re)	Revenue from Recycling (R), Salvage (S), or Reuse (Re) (Specify R, S, or Re)	G - H + I = Total Cost (-) or Savings (+) from Diversion
Asphalt/Concrete										
Brick/Masonry/Tile										
Building Materials (doors, windows, fixtures, shingles, lumber, insulation, sheet goods, etc.)										
Carpet										
Carpet Padding, Foam Only										
Cardboard										
Ceiling Tile										
Drywall										
Glass										
Scrap Metal Aluminum										
Copper										
Steel										
Unpainted Wood & Pallets										
Yard Trimmings, Brush, Trees, Stumps, etc.										
Garbage/Trash										
Other										
Column Totals										
	Total Quantity Recycled	Total Quantity Reused or Salvaged	Total Quantity Diverted from Landfill	Total Quantity to Landfill	Total Quantity Generated		Tip Fee Savings from Diversion	Total Cost of Recycling, Salvage, or Reuse	Revenue from Recycling, Salvage, or Reuse	Total Cost (-) or Savings (+) from Diversion

Percentage Diverted = _____ (C divided by E from Column Totals). Should meet 60% diversion goal.

SECTION 01 77 00 - CLOSEOUT PROCEDURES

PART 1 - GENERAL

1.1 SUMMARY

- A. Section includes administrative and procedural requirements for contract closeout, including:
 - 1. Project Record Drawings
 - 2. Closeout Submittals
 - 3. Substantial Completion and Final Inspection
 - 4. Final Acceptance of the Work

1.2 PROJECT RECORD DRAWINGS

- A. Maintain one complete full-size set of contract drawings and one full-size set of vendor-supplied drawings. Clearly mark changes, deletions, and additions using National Park Service (NPS) drafting standards to show actual construction conditions. Show additions in red, deletions in green and special instructions in blue.
- B. Keep record drawings current. Make record drawings available to Contracting Officer (CO) for inspection at the time of monthly progress payment requests. If project record drawings are not current, Contracting Officer may retain an appropriate amount of progress payment.
- C. Submit complete record drawings on completion of total project. Include shop drawings, sketches, and additional drawings to be included in final set, with clear instructions showing the location of these drawings.

1.3 CLOSEOUT SUBMITTALS

- A. A list of closeout requirements has been attached at the end of the Division 1 Specifications for your convenience. The intent is to provide an overall summary of requirements and not a comprehensive list. Terms and conditions of the contract require satisfaction of requirements of individual specification sections regardless of what is shown on the list. Submit the following before requesting final inspection:
 - 1. Specific guarantees, workmanship bonds, final certifications, and similar documents.
 - 2. Project Record Documents, operation and maintenance manuals, final completion construction digital images recorded on CD-R (compact disc-recordable) or DVD-R (digital video disc-recordable) with index and descriptions, and similar final record information.
 - 3. Environmental Record Documents: As specified in Division 1.
 - 4. Terminate and remove temporary facilities, mockups, construction tools, and similar elements from Project site, complete final cleaning requirements, including touchup painting.
 - 5. Touch up and repair and restore marred exposed finishes to eliminate visual defects.

1.4 FINAL INSPECTION, SUBSTANTIAL COMPLETION AND ACCEPTANCE PROCEDURES

- A. Request final inspection in writing when project or designated portion of project is substantially complete. Contracting Officer will proceed with inspection within 10 days of receipt of written request or will advise Contractor of items that prevent project from being substantially complete.
- B. If work is determined substantially complete, following final inspection, Contracting Officer will prepare Punch List and issue a Letter of Substantial Completion.
- C. If work is not determined substantially complete following final inspection, Contracting Officer will notify Contractor in writing. Contractor shall request new final inspection after completing work. Re-inspection costs may be charged against Contractor in accordance with Inspection of Construction contract clause.
- D. Contractor shall complete Punch List within 30 calendar days, documented weather permitting.
- E. If Contractor completes items of work on Punch List and contractually required items, Contracting Officer will issue Letter of final acceptance of work.
- F. If Contractor fails to complete work within the time frame, Contracting Officer may correct work with an appropriate reduction in contract price or charge for re-inspection costs in accordance with Inspection of Construction contract clause.

PART 2 - PRODUCTS

2.1 MATERIALS

- A. See Division 1 Specification Section "Execution" for information on cleaning agents.

PART 3 - EXECUTION

3.1 FINAL CLEANING

- A. General: Conduct final cleaning and waste-removal operations to comply with local laws and ordinances and Federal and local environmental and antipollution regulations.
- B. Cleaning: Employ experienced workers or professional cleaners for final cleaning. Clean each surface or unit to condition expected in an average commercial building cleaning and maintenance program. Comply with manufacturer's written instructions.
 - 1. Complete following cleaning operations before requesting inspection for certification of Substantial Completion for entire Project or for a portion of Project:

- a. Clean Project site, yard, and grounds, in areas disturbed by construction activities, including landscape development areas, of rubbish, waste material, litter, and other foreign substances.
 - b. Sweep paved areas broom clean. Remove petrochemical spills, stains, and other foreign deposits.
 - c. Rake grounds that are neither planted nor paved to smooth, even-textured surface.
 - d. Remove tools, construction equipment, machinery, and surplus material from Project site.
 - e. Remove snow and ice to provide safe access to building.
 - f. Remove debris from limited access spaces.
- C. Waste Disposal: Comply with requirements of Section 01 74 19 "Construction Waste Management and Disposal."

END OF SECTION 01 77 00

SECTION 02 42 96 - HISTORIC REMOVAL AND DISMANTLING

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Historic treatment procedures for removal and dismantling work for designated historic spaces, areas, rooms, and surfaces and the following specific work:
 - a. Removal and dismantling of indicated portions of building or structure and debris hauling.
 - b. Removal and dismantling of indicated site elements and debris hauling.
 - c. Salvage of existing items to be reused or recycled.

B. Related Requirements:

1. Section 01 35 91 "Historic Preservation Treatment Procedures" for general historic treatment requirements.
2. Section 02 81 01 "Offsite Transportation and Disposal"
3. Section 02 83 33 "Removal and Disposal of Material Containing Lead"
4. Section 10 81 19 "Rodent and Bird Control"

1.2 DEFINITIONS

- A. Dismantle: To disassemble or detach a historic item from a surface, or a nonhistoric item from a historic surface, using gentle methods and equipment to prevent damage to historic items and surfaces; disposing of items unless indicated to be salvaged or reinstalled.
- B. Existing to Remain: Existing items that are not to be removed or dismantled, except to the degree indicated for performing required Work.
- C. Remove: To take down or detach a nonhistoric item located within a historic space, area, or room, using methods and equipment to prevent damage to historic items and surfaces; disposing of items unless indicated to be salvaged or reinstalled.
- D. Retain: To keep an element or detail secure and intact.
- E. Salvage: To protect removed or dismantled items and deliver them to Government.

1.3 PRECONSTRUCTION MEETINGS

A. Preconstruction Conference(s): Conduct conference(s) at Project site.

1. Review minutes of Preliminary Historic Treatment Conference that pertain to removal and dismantling procedures and protection of historic areas and surfaces.
2. Review list of items indicated to be salvaged.

3. Verify qualifications of personnel assigned to perform removal and dismantling.
4. Inspect and discuss condition of each construction type to be removed or dismantled.
5. Review requirements of other work that depends on condition of substrates exposed by removal and dismantling work.
6. Review methods and procedures related to removal and dismantling work, including, but not limited to, the following:
 - a. Historic removal and dismantling specialist's personnel, equipment, and facilities needed to make progress and avoid delays.
 - b. Materials, material application, sequencing, tolerances, and required clearances.
 - c. Fire prevention.
 - d. Coordination with building occupants.

1.4 SUBMITTALS

- A. Qualification Statements: For historic removal and dismantling specialist.
- B. Historic Preservation Treatment Plan according to Section 01 35 91.
- C. Schedule:
 1. Mandatory pre-construction meeting with Contracting Officer and park representatives to review all demolition work.
 2. Describe demolition, removal procedures, sequence and schedule.
 3. Include schedule for shutting off and capping utilities and re-establishing utility services.
- D. Shop Drawings:
 1. Plans and details of protection measures for building elements protected in situ. Must be reviewed and approved by the CO prior to installation of protection and start of demolition.
 2. Plans and details of proposed method of conducting new work under an historic building.
- E. Preconstruction Documentation: Show preexisting conditions of adjoining construction and site improvements, including finish surfaces, that might be misconstrued as damage caused by Contractor's removal and dismantling operations.
- F. Removal and Dismantling Historic Treatment Program: Submit 30 days before work begins.
- G. List of Items Indicated To Be Salvaged: Prepare a list of items indicated on Drawings to be salvaged for Government's use or for reinstallation. Submit 15 days before preconstruction conference.
- H. Inventory of Salvaged Items: After removal or dismantling Work is complete, submit a list of items that have been salvaged.
 1. Include item description, item condition, number of items if more than one of a type, and tag number. Include photo of item in original location.
 2. As work proceeds, include on the inventory items that were indicated to be salvaged and items of historic importance discovered during the work. Document reasons, if any, why

an item indicated to be salvaged was not salvaged.

1.5 QUALITY ASSURANCE

- A. Historic Removal and Dismantling Specialist Qualifications: A qualified historic treatment specialist with a minimum of 10 years successful experience in historic wood frame construction where careful removal and salvaging of wood was required. General selective demolition experience is insufficient experience for historic removal and dismantling work.
- B. Removal and Dismantling Historic Treatment Program: Prepare a written, detailed description of materials, methods, equipment, and sequence of operations to be used for each phase of removal and dismantling work, including protection of surrounding and substrate materials and Project site.
 - 1. Dust and Noise Control: Include locations of proposed temporary dust- and noise-control partitions and means of egress from occupied areas coordinated with continuing on-site operations and other known work in progress.
 - 2. Debris Hauling: Include plans clearly marked to show debris-hauling routes, turning radii, and locations and details of temporary protective barriers.

1.6 FIELD CONDITIONS

- A. Conditions existing at time of inspection for bidding purposes will be maintained by Government as long as practicable.
 - 1. Before removal and dismantling, Government will remove the following items:
 - a. Furniture and window treatments within the building.
- B. Notify Contracting Officer of discrepancies between existing conditions and Drawings before proceeding with removal and dismantling work.
- C. Hazardous Materials:
 - 1. Hazardous materials are present in construction affected by removal and dismantling work. A report on the presence of hazardous materials is on file for review and use. Examine report to become aware of locations where hazardous materials are present.
 - a. Hazardous material remediation is specified in Section 02 83 33 Removal and Disposal of Material Containing Lead.
 - b. Do not disturb hazardous materials or items suspected of containing hazardous materials, except under procedures specified in Section 02 83 33 Removal and Disposal of Material Containing Lead.
 - c. If unanticipated asbestos is suspected, stop work in the area of potential hazard, shut off fans and other air handlers ventilating the area, and rope off area until the questionable material is identified. Resume work in the area of concern after safe working conditions are verified.
- D. Storage or sale of removed or dismantled items on-site is not permitted unless otherwise indicated.

PART 2 - PRODUCTS - (Not Used)

PART 3 - EXECUTION

3.1 HISTORIC REMOVAL AND DISMANTLING EQUIPMENT

- A. Removal Equipment: Use manual, handheld tools. Handheld power tools may be permitted on a case-by-case basis with approval by Contracting Officer .
- B. Dismantling Equipment: Use manual, handheld tools, except as follows or otherwise approved by Contracting Officer on a case-by-case basis:
 - 1. Handheld power tools are permitted only as submitted in the historic treatment program. They must be adjustable so as to penetrate or cut only the thickness of material being removed.
 - 2. Pry bars more than 18 inches long and hammers weighing more than 2 lb are not permitted for dismantling work.

3.2 EXAMINATION

- A. Preparation for Removal and Dismantling: Examine construction to be removed or dismantled to determine best methods to safely and effectively perform removal and dismantling work. Examine adjacent work to determine what protective measures are necessary. Make explorations, probes, and inquiries as necessary to determine condition of construction to be removed or dismantled and location of utilities and services to remain that may be hidden by construction that is to be removed or dismantled.
 - 1. Verify that affected utilities are disconnected and capped.
 - 2. Inventory and record the condition of items to be removed and dismantled for reinstallation or salvage. Enter this information on the inventory of salvaged items.
 - 3. Before removal or dismantling of existing building elements that will be reproduced or duplicated in final Work, make permanent record of measurements, materials, and construction details required to make exact reproduction.
 - 4. Engineering Survey: Engage a professional engineer to survey condition of building to determine whether removing any element might result in structural deficiency or unplanned collapse of any portion of structure or adjacent structures as a result of removal and dismantling Work.
- B. Survey of Existing Conditions: Record existing conditions by use of preconstruction photographs.
 - 1. Comply with requirements specified in Section 01 32 33 "Photographic Documentation."
- C. Perform surveys as the Work progresses to detect hazards resulting from historic removal and dismantling procedures.

3.3 HISTORIC REMOVAL AND DISMANTLING

- A. General: Have removal and dismantling work performed by a qualified historic removal and dismantling specialist. Ensure that historic removal and dismantling specialist's field supervisors

are present when removal and dismantling work begins and during its progress.

- B. Comply with requirements in Section 01 35 91 "Historic Treatment Procedures" for identifying and storing historic items.
- C. Perform work according to the historic treatment program.
 - 1. Perform removal and dismantling to the limits indicated.
 - 2. Provide supports or reinforcement for existing construction that becomes temporarily weakened by removal and dismantling work, until the Project Work is completed unless otherwise indicated.
 - 3. Perform cutting by hand or with small power tools as permitted by Contracting Officer wherever possible. Cut holes and slots neatly to size required, with minimum disturbance of adjacent work.
 - 4. Do not operate air compressors inside building unless approved by Contracting Officer in each case.
 - 5. Do not drill or cut columns, beams, joints, girders, structural slabs, or other structural supporting elements, without having Contractor's professional engineer's written approval for each location before such work is begun.
 - 6. Dispose of removed and dismantled items off-site unless indicated to be salvaged or reinstalled.
- D. Water-Mist Sprinkling: Use water-mist sprinkling and other wet methods to control dust only with adequate, approved procedures and equipment according to the historic treatment program to ensure that such water does not create a hazard or adversely affect other building areas or materials.
- E. Unacceptable Equipment: Keep equipment that is not permitted for historic removal or dismantling work away from the vicinity where such work is being performed.
- F. Removing and Dismantling Items on or Near Historic Surfaces:
 - 1. Use only dismantling equipment and procedures within 12 inches of historic surface. Protect historic surface from contact with or damage by tools.
 - 2. Wherever possible, unfasten items in the opposite order from which they were installed.
 - 3. Support each item as it becomes loosened to prevent stress and damage to the historic surface.
 - 4. Dismantle anchorages.
- G. Anchorages:
 - 1. Remove anchorages associated with removed items.
 - 2. Dismantle anchorages associated with dismantled items.
 - 3. In nonhistoric surfaces, patch holes created by anchorage removal or dismantling according to the requirements for new work.
 - 4. In historic surfaces, patch or repair holes created by anchorage removal or dismantling according to Section that is specific to the historic surface being patched.

3.4 HISTORIC REMOVAL AND DISMANTLING SCHEDULE

A. The following non-historic items to be removed shall be disposed of according to Section 01 74 19 Construction Waste Management. Refer to drawings for additional information.

1. Conduit, electrical devices, meters, piping, hose bibbs, etc. that are required to be removed to facilitate new foundation work.
2. Interior plumbing fixtures required to be removed to facilitate new foundation work.
3. Concrete pad and foundation walls where indicated on drawings.

B. The following historic items to be removed shall be returned to park:

1. Interior wood doors, trim, threshold, and hardware where indicated on drawings.
2. 1x wood flooring where indicated on drawings.

C. The following non-historic items to be removed shall be returned to park:

1. Exterior wood door and hardware where indicated on drawings.

END OF SECTION 02 42 96

SECTION 02 81 01 – OFFSITE TRANSPORTATION AND DISPOSAL

PART 1 - GENERAL

1.1 SUMMARY

- A. This Section includes requirements for the offsite transportation and disposal of wastes generated during the planned historic rehabilitation of the Bear Valley School at Pinnacles National Park, CA (PINN), (Project Site). PMIS: 331043.
- B. The Contractor will be responsible for sampling, analyzing, and characterizing waste materials to be shipped offsite for final disposal or recycling as required by the Disposal Facility.
- C. If building materials coated with Lead-Containing Paint (LCP) and Lead-Based Paint (LBP) are demolished and will be disposed of, it will be the Contractor's sole responsibility to obtain temporary U.S. Environmental Protection Agency (EPA) ID number(s), as and if applicable, on behalf of the National Park Service (NPS) for any hazardous waste materials generated and removed from the Project Site during the execution of this Project. The Contracting Officer will identify appropriate NPS individuals to sign the required paperwork as the generator signatory for the Project.
- D. Offsite recycling and disposal facilities will be approved by the Contracting Officer prior to the offsite transportation of materials.
- E. Transportation of the materials will comply with the legal load restrictions of the U.S. Department of Transportation (DOT) and the California Department of Transportation (CalTrans).

1.2 RELATED SPECIFICATIONS

- A. Section 02 83 33: Removal and Disposal of Material Containing Lead
- B. Section 10 81 19: Rodent and Bird Control

1.3 SUBMITTALS

- A. The Contractor will include as a component of the Waste Management Plan (described in Section 01 74 19 "*Construction Waste Management and Disposal*") a description of planned means and methods for transporting and disposing of all the waste materials removed or generated during the Project.
- B. Permit for the selected Treatment Storage and/or Disposal Facility.

- C. The Contractor will provide an original and/or copies of all Bills of Lading and Manifests for all transported waste material loads from the Project Site, including all required generator, transporter, and waste disposal facility receipt/transfer signatures. These documents will be provided to the Contracting Officer at least weekly and will be accompanied by a detailed electronic waste tracking sheet. The waste tracking sheet will be cumulative and updated weekly. This document will provide the following:
1. Cumulative load numbers for each shipment.
 2. A description of the waste material being shipped for disposal.
 3. Company transporting the waste material.
 4. Estimated waste volumes for each shipment.
 5. Certified weight slips for each shipment.
 6. The disposal facility of record for each waste stream.
 7. The specific Manifest or Bill of Lading specific identifier number for the load being shipped for disposal.
- D. At the conclusion of the Project, all waste disposal information collected will be submitted to the Contracting Officer as an electronic deliverable package as part of the required Project closeout record documentation. Payments to the Contractor for waste transportation disposal fees will not be approved until completed disposal tracking documents, as described above, are received by the Contracting Officer.

1.4 WASTE CONTAINERS

- A. The Contractor will provide waste containers specific to the individual waste to be handled.

1.5 FIELD VERIFICATION

- A. Field verification of all new and existing dimensions affecting the work of this contract by the Contractor is required before ordering products.

1.6 TRANSPORTATION OF WASTES

- A. The Contractor will be responsible for transporting all wastes generated during the Project that cannot be salvaged, reused, or recycled. This includes materials generated by final Project Site cleanup activities and the dismantling of the temporary facilities and controls.
- B. The Contractor will be responsible for coordinating the number and schedule of vehicles required for the offsite transportation of waste materials generated during the execution of the Project.
- C. The Contractor will be responsible for inspecting transportation vehicles before and after loading to ensure compliance with all local, state, and federal regulations for the safe transport of wastes from the Project Site to the receiving facility. The Contractor will provide the necessary labor and materials to ensure all trucks and containers are covered prior to departure.

- D. The Contractor will ensure that transporters arriving at the Project Site for loading do not cause undue congestion to local roadways and will stage trucks either within the perimeter of the Project Site or at an offsite staging area approved by the Contracting Officer.
- E. The Contractor's transporters will proceed directly from the Project Site to a designated staging area or the designated offsite receiving facility. Temporary staging or storage of material will only be allowed at designated staging areas approved by the Contracting Officer.
- F. The Contractor will originate, maintain, and provide the Contracting Officer with a copy of each executed Bill of Lading for all loads shipped offsite. In addition, the Contractor will provide the Contracting Officer with documentation and records verifying the receipt of each truckload by the receiving facility. Such documentation will indicate the actual weight of each load shipped.
- G. Transporters will proceed from the Project Site along traffic routes established by the Contractor and approved by the Contracting Officer. Transporters will call back weights after each load and modify the loads accordingly. The Contractor will ensure that trucks leaving the Project Site are within appropriate weight limitations for the local roads along the designated route.
- H. The weight measurements for the materials transported to recycling and disposal facilities will be determined by scales certified in accordance with local, state, and federal regulations and requirements.

1.7 DISPOSAL OF WASTES

- A. The Contractor will be responsible for the proper disposal of all solid and liquid wastes generated during the execution of this Project in conformance with all local, state, and federal regulations and requirements. Proper disposal requires that the facility accepting the waste be a state-licensed disposal/recycling facility approved for acceptance of the waste based on the results of the characterization testing and analysis.
- B. The disposal facilities will be approved by the Contracting Officer prior to transporting waste. The Contractor will not change facilities without prior written consent of the Contracting Officer.
- C. If applicable, for wastes determined to have lead toxicity characteristics as a result of LCP or LBP, the Contractor shall obtain temporary EPA ID number(s) on behalf of the NPS, as directed by the Contracting Officer, for hazardous waste generated during the execution of this Project.

PART 2 - PRODUCTS (Not Used)

PART 3 - EXECUTION (Not Used)

END OF SECTION 02 81 01

SECTION 02 83 33 – REMOVAL AND DISPOSAL OF MATERIAL CONTAINING LEAD

PART 1 - GENERAL

1.1 DESCRIPTION

This Section consists of tasks associated with the removal and disposal of material containing lead during the planned historic rehabilitation of the Bear Valley School at Pinnacles National Park, CA (PINN), (Project Site). This structure was found to have Lead-Containing Paint (LCP) and Lead-Based Paint (LBP) on interior surfaces. This specification establishes the required environmental controls to be used during impacts to LBP or LCP during the planned rehabilitation work.

- A. Testing and analysis at the Project Site identified LCP/LBP, including the following five different types of LCP, including:
 - The green wood floor near the west entryway is 0.056%.
 - The green wood window screen area is 0.011%.
 - The white metal pipe on the northwest exterior is 0.088%.
 - The white wood wall on the west exterior is 0.022%.
 - The off-white window glazing on the window sash in the kitchen is 0.2%.
- B. Testing and analysis at the Project Site identified six different types of LBP, including:
 - The off-white wood window in the kitchen/pantry area is 17.0%.
 - The off-white wood wall in the kitchen area is 0.9%.
 - The blue wood countertop in the kitchen is 1.4%.
 - The off-white wood door in the classroom area is 0.5%.
 - The off-white wood door frame in the classroom area is 2.4%.
 - The white wood wall on the northeast exterior is 0.6%.
- A. The Contractor will be responsible for the collection and analysis of personnel exposure samples for lead in accordance with the U.S. Occupational Safety and Health Administration (OSHA) regulations.

- B. Appropriate training will be provided by the Contractor prior to their personnel performing work impacting identified LCP and potentially discovered LBP as required by applicable regulations, California OSHA's Lead Standards (California Code of Regulations (CCR), Title 8 (T8), Sections 1532.1 (for the Construction Industry) and 5198 (for General Industry).
- C. Two hours of lead awareness training will be provided to applicable National Park Service (NPS) personnel and their subcontractors working within the areas that contain LBP/LCP who can reasonably be expected to be impacted as part of this work.
- D. Work shall be conducted so that no visible emissions are generated by construction that impacts LBP/LCP building components. Manual removal methods that do not generate dust or debris must be utilized. The Contractor's Competent Person (CP) shall determine if the methods used present a risk of exposure above the Cal/OSHA's Action Level (AL) of 2 micrograms per cubic meter over an 8-hour Time-Weighted Average (TWA) as required by the Cal/OSHA lead standard CCR, T8, Section 1532.1.
- E. Any employee who may be exposed to lead levels in excess of the OSHA AL will have a minimum of two hours of lead awareness training.
- F. The Contractor is responsible for providing an independent Industrial Hygienist (IH). This IH will coordinate with the Contracting Officer to provide waste profile sampling.
- G. The Contractor must carefully collect, containerize, characterize for waste disposal, transport, and dispose of any regulated or hazardous wastes generated during this project, including LCP dust and debris and other potentially lead-contaminated wastes. If LCP components or debris are disposed of offsite, a representative waste profile sample shall be collected and analyzed as required by the receiving landfill for waste profiling purposes. The Contractor shall provide temporary staging areas for waste and conduct waste profile sampling; analyses of the sample shall be by Toxicity Characteristic Leaching Procedure (TCLP).
- H. The Contractor will be responsible for the cleanup and testing of any area(s) contaminated due to Contractor negligence at no additional cost to the NPS.

1.2 REFERENCES AND RELATED SPECIFICATIONS

- A. All work under this contract will be conducted in strict accordance with all applicable federal, state, and local regulations, standards, and codes governing lead remediation and other concurrent trade work. The most recent edition of any relevant regulation, standard, document, or code will be in effect. Where conflict among the requirements or within these specifications exists, the most stringent requirements will be utilized.
- B. Section 02 81 01: Offsite Transportation and Disposal.

- C. Landmark Environmental, Inc., *Hazardous Materials Assessment Letter for Bear Valley School Rehabilitation Phases I and 2: Title II Documents, Pinnacles National Park, CA. Landmark Project No. 25097.001*. October 17, 2025.
- D. American Society of Safety Engineers (ASSE), *ASSE/SAFE Z9.2 Fundamentals Governing the Design and Operation of Local Exhaust Ventilation Systems*, 2012.
- E. Underwriters Laboratories (UL), *UL Standard for Safety High-Efficiency Particulate, Air Filter Units, UL 586*, 2009; reprint 2017.
- F. Iowa Department of Natural Resources (IDNR) lead resources page at <https://www.iowadnr.gov/environmental-protection/land-quality/contaminated-sites/brownfields/lead-based-paint#Proper-Disposal-and-Testing-327>.
- G. The U.S. Occupational Safety and Health Administration (OSHA), 29 CFR 1910.1025 - General Industry Standards for Lead.
- H. OSHA, 29 CFR 1926.62 – Construction Industry Standards for Lead.
- I. OSHA, 20 CFR 1910.120 – Hazardous Waste Operations and Emergency Response.
- J. OSHA, 29 CFR 1910.134 – Respiratory Protection.
- K. OSHA, 29 CFR 1910.145 – Accident Prevention.
- L. OSHA, 29 CFR 1910.2 – Medical Records.
- M. OSHA, 29 CFR 1910.38 – Emergency/Fire Prevention Programs.
- N. OSHA, 29 CFR 1910.1200 – Hazard Communication.
- O. California OSHA (Cal/OSHA) California Code of Regulations (CCR), Title 8 (T8), 1532.1 – Lead Standard for the Construction Industry.
- P. Cal/OSHA CCR T8, Section 5198 – Lead Standard for the General Industry.
- Q. The U.S. Environmental Protection Agency (EPA), 40 CFR 260 – Hazardous Waste Management System: General.
- R. EPA, 40 CFR 261 – Identification and Listing of Hazardous Waste.
- S. EPA, 40 CFR 262 – Standards Applicable to Generators of Hazardous Waste.
- T. EPA, 40 CFR 263 – Standards Applicable to Transporters of Hazardous Waste.
- U. The U.S. Department of Transportation (US DOT), 49 CFR 171 and 172 – Hazardous Materials Regulations.

- V. National Institute for Occupational Safety and Health (NIOSH), *Certified Equipment List*.
- W. NIOSH, *Occupational Exposure Sampling Strategy*, 77-173.
- X. NIOSH, *Guide to Industrial Respiratory Protection*, 87-116.
- Y. ASTM International, *Collection of Settled Dust Samples Using Wipe Sampling Methods for Subsequent Lead Determination E1792*, 2003, reprinted in 2016.

1.3 DEFINITIONS

- A. Atomic Absorption Spectrometry (AAS): The laboratory method used to determine lead in bulk and air samples.
- B. Authorized Person: Any person(s) authorized by the employer and required by work duties to be present in regulated areas.
- C. Critical Barrier: One or more layers of plastic sealed over all openings into a work area(s) or any other similarly placed physical barrier sufficient to prevent airborne dust in a work area(s) from migrating into an adjacent area.
- D. Competent Person (CP): The Contractor's employee who identifies lead hazards in the workplace, has the authority to promptly implement corrective measures to eliminate them, and is trained and accredited as defined in 29 CFR 1926.62. The duties of the Competent Person include but are not limited to the following activities: 1) controlling entry to and exit from the lead work area; 2) supervising any employee exposure monitoring required by the standards; 3) ensuring that all employees working within a work area wear the appropriate Personal Protective Equipment (PPE) and are trained in the use of appropriate methods of exposure control and decontamination, and 4) ensuring that engineering controls in use are functioning properly.
- E. Disposal Bag: A properly labeled 6- mil thick, leak-tight polyethylene bag used for containerizing and/or transporting lead waste to the disposal site.
- F. Disturbance: Activities that disturb LCP or LBP within the facility.
- G. Generator: Any person or site whose processes and actions create hazardous waste per 40 CFR 260.10.
- H. HEPA Filter: A High-Efficiency Particulate Air (HEPA) filter capable of trapping and retaining 99.97% of all mono-dispersed particles (e.g., asbestos fibers) of 0.3 microns in diameter.
- I. Industrial Hygiene: An area of specialization in the field of industrial safety and health concerned with predicting, recognizing, assessing, controlling, and preventing environmental stressors in the workplace that can lead to sickness, disease, or other forms of impaired health.

- J. Industrial Hygienist: A person having a college or university degree(s) in engineering, chemistry, physics, medicine, or related physical or biological sciences who, by virtue of special studies and training, has acquired competence in industrial hygiene.
- K. Lead: Metallic lead, inorganic lead compounds, and organic lead soaps. Excludes other forms of organic lead compounds.
- L. Lead-Based Paint (LBP): Paint or other coatings containing at least 0.5% lead by weight or 1.0 milligrams (mg) per square centimeter.
- M. Lead-Containing Paint (LCP): Paint or other coatings containing concentrations of lead that are quantifiable through laboratory analysis.
- N. Negative Initial Exposure Assessment (IEA): A demonstration by the employer that demonstrates employee exposure during an operation is expected to be consistently below the Permissible Exposure Limit (PEL) for representative 8-hour TWA samples.
- O. Personal Monitoring: Sampling of the dust/fiber concentrations within the breathing zone of an employee.
- P. Regulated Area: An area established to demarcate the areas where lead-containing dust in the air exceeds, or there is a reasonable possibility it may exceed, the PEL or other applicable industry standards for occupational exposure limits.
- Q. Removal: All work operations where lead is taken out of or stripped from structures or substrates, including demolition operations.
- R. Renovation: Modifying any existing structure or portion thereof.
- S. Time-Weighted Average (TWA): The employee's average airborne exposure in any 8-hour work shift of a 40-hour work week which shall not be exceeded. The 8-hour TWA PEL is the level of exposure established as the highest level of exposure an employee may be exposed to without incurring the risk of adverse health effects.
- T. Toxicity: The ability of a substance to cause harmful effects. Toxic waste is harmful or fatal when ingested or absorbed (e.g., containing mercury, lead, etc.). When toxic waste is land-disposed, contaminated liquid may leach from this waste and pollute groundwater.
- U. Toxicity Characteristic Leaching Procedure (TCLP): Analytical test designed to determine the mobility of both organic and inorganic analytes present in liquid, solid, and multi-phase wastes. TCLP is used to determine if a waste will be a hazardous waste as defined by 40 CFR Part 261.
- V. Transporter: Hazardous waste transporters are individuals or entities that move hazardous waste from one site to another by highway, railway, water, or air per 40 CFR 260.10. This includes transporting hazardous waste from a generator's site to a facility that can recycle, treat, store, or

dispose of the waste. It can also include transporting treated hazardous waste to a site for further treatment or disposal.

- W. Treatment: Treatment is defined as any method, technique, or process designed to change the physical, chemical, or biological character or composition of any hazardous waste to neutralize such waste, recover energy or material resources from the waste, or render such waste non-hazardous, or less hazardous; safer to transport, store or dispose of; or amenable for recovery, amenable for storage, or reduced in volume.
- X. Visible Emissions: Any emissions potentially containing particulate and/or lead-containing dust material that is visually detectable without the aid of instruments.
- Y. Work Area: The area(s) where lead-related work or removal operations are performed, which is defined and/or isolated to prevent the spread of lead-containing dust or debris and entry by unauthorized personnel. The work area includes the waste load-out area.

1.4 FIELD VERIFICATION

- A. The Contractor shall field-verify all new and existing dimensions affecting the work of this contract before ordering products.

1.5 CONSTRUCTION MATERIALS

- A. The Government will not furnish any materials for this project.

1.6 SALVAGED MATERIALS

- A. Excess salvaged materials not reinstalled in the project remain the property of the Government.
- B. Stockpile material within the rooms where the materials are recovered for reuse. If more space is required, the Contractor shall coordinate storage location with the Contracting Officer.

1.7 PROJECT SUBMITTAL REQUIREMENTS

- A. Health and Safety-Related Submittals:
 - 1. The Contractor's IH will prepare submittals on behalf of the Contractor for submission to the Contracting Officer for approval. These submittals will include a Health and Safety Plan that includes a Respiratory Protection Program, Medical Surveillance Program, Accident Prevention Program, and Lead Waste Management and Disposal Plan.
 - 2. The Contractor will submit a resume and copies of professional registrations/certifications for the Contractor's employee, who will act as the CP for the project. The CP must meet qualifications and have the training and certifications defined by OSHA.

3. The Contractor will submit copies of attendance records and certificates for each employee for all lead hazard awareness training required by this Section.
4. The Contractor will submit proof that all employees have passed the appropriate medical examinations required by OSHA for respirator use and work around lead where the AL may be exceeded.
5. The Contractor will submit copies of current physicians' written medical opinions for all employees subject to the medical surveillance requirements defined in OSHA Respiratory Protection Standards.
6. If the CP determines that respiratory protection is required, the Contractor shall submit certification that each employee was properly fitted with a personally assigned respirator.
7. The Contractor will submit certification that HEPA vacuums, HEPA-equipped negative pressure air machines, and other equipment required to contain airborne lead dust or other particulates meets American National Standards Institute (ANSI) Z9.2 79.
8. For HEPA vacuums used for LBP and LCP debris cleanup or HEPA-equipped negative pressure air machines, the Contractor shall maintain and submit copies of HEPA-filter change logs for each piece of HEPA-equipped equipment at the project site to verify timely filter changes (filters should be changed every 750 hours of use).

B. Occupational Exposure Sampling:

1. After initial exposure monitoring, the Contractor shall submit sampling results to the NPS's Contracting Officer within three working days of sample collection, signed by the testing laboratory employee performing the analysis, the employee that performed the sampling, and the CP.
2. The Contractor shall provide documentation that:
 - a. The initial air monitoring is representative of each job classification or previously collected exposure data and represents the worker's job tasks for this project.
 - b. The worker exposure data, including a complete description of the job tasks (such as manual demolition and manual scraping, etc.) where lead-containing coatings are present, is submitted.

C. Lead Waste Management and Disposal Plan:

1. The Contractor shall submit a Lead Waste Management and Disposal Plan that complies with applicable requirements of federal, state, and local hazardous waste regulations and shall include:
 - a. Identification and classification of wastes associated with the work.
 - b. Estimated quantities of wastes to be generated and disposed of.
 - c. Names and qualifications of each Contractor who will be transporting, storing, treating, and disposing of the wastes.
 - d. Description of the waste containment, removal, and disposal procedures. Proper temporary containment of the waste includes using acceptable leak-tight waste containers such as 55-gallon drums or other secure and covered storage areas that prevent spillage and exposure to stormwater, as well as proper marking/labeling of the containers. Waste will be cleaned up and containerized daily.

- e. Description of any site construction process, such as the use of chemical strippers or caustic agents that may alter or treat waste, rendering hazardous waste non-hazardous.
- D. The Contractor is responsible for the waste profile sampling and analysis for disposal of waste streams that include building components with LCP or LBP.

1.8 POST-PROJECT SUBMITTAL REQUIREMENTS

- A. After every disposal operation, provide copies of transport manifests and disposal receipts for all lead waste materials removed from the site. The Chain-Of-Custody (COC) form will include the date, address of the pickup site, name and address of the Contractor, names of persons responsible for pickup, name, and address of the disposal site, quantity and type of hazardous waste, and the type of containers used. The form will be signed by the Contractor, Contracting Officer, disposal site operator, and hauler if a private hauler is employed.
- B. Closeout Report: Prepare a closeout report with the following items:
 - 1. Description of removal and stabilization activities performed during the project.
 - 2. Description and results of occupational and environmental sampling.
 - 3. Description of cleanup activities.
 - 4. Description and results of confirmatory sampling.
 - 5. Results of waste characterization sampling.
 - 6. COC forms for confirmatory sampling and waste characterization sampling.
 - 7. Description of completed work, including the onsite processing of waste (if any), handling and transportation procedures, and disposal site.
 - 8. Certificates of disposal.
 - 9. Transport manifests.

1.9 QUALITY ASSURANCE

- A. Qualifications:
 - 1. Competent Person: The Contractor shall submit the name, address, and telephone number of the CP selected to perform the responsibilities specified in Section 1.3 Definitions, Item D, "Competent Person." Provide documentation of construction project-related experience with implementation of OSHA's Lead in Construction standard (29 CFR 1926.62), which shows the ability to assess occupational and environmental exposure to lead, experience with the use of respirators, PPE, and other exposure reduction methods to protect employee health. Submit proper documentation that the CP is trained and certified in accordance with federal, state, and local laws.
 - 2. Workers/Supervisors: The Contractor shall ensure that all workers will be fully qualified and experienced in handling and disposing of lead-painted materials. Submit training certifications and proper documentation for each worker, supervisor, and CP performing LBP/LCP removal activities in accordance with federal, state, and local laws.

3. Analytical Laboratory: The Contractor and IH shall use a laboratory participating in the EPA National Lead Laboratory Accreditation Program (NLLAP) that is accredited by either the American Association for Laboratory Accreditation (AALA) or the American Industrial Hygiene Association (AIHA) and is also a successful participant in the Environmental Lead Proficiency Analytical Testing (ELPAT) program to perform sample analysis. Laboratories selected to perform blood lead analysis will be OSHA-approved.
 4. The Contractor's Third-Party Industrial Hygiene Consultant: The Contractor shall submit the name, address, and telephone number of a third-party industrial hygiene consultant selected to assist the Contractor with any project submittals, etc. Submit the proper documentation that the consultant is trained and certified as an inspector technician or inspector/risk assessor by the EPA-authorized state (or local) certification and accreditation program.
- B. Regulatory Requirements: Develop and implement a written Environmental Control Program covering the employees, visitors, and government personnel who can potentially be exposed to hazardous materials on the project. The program must include, but not be limited to, the following elements: a list of chemicals, labels, and other forms of warning, material Safety Data Sheets (SDS), all employee information and training, procedures, chemical safety and spill response information, and subcontractor notifications.
- C. All work under this contract will be performed in strict accordance with all applicable federal, state, and local regulations, standards, and codes governing lead removal and other concurrently performed trade work. The most recent edition of any relevant regulation, standard, document, or code will be in effect. Where conflict among the requirements or with these specifications exists, the most stringent requirements will be utilized. Copies of applicable standards, regulations, and codes shall be maintained in the Contractor's project records and be available for access at the worksite during the work.

PART 2 - PRODUCTS

2.1 PERSONAL PROTECTIVE EQUIPMENT

- A. Respiratory Protection:
1. If required by the CP, workers shall be provided with personally issued, individually identified respirators that meet the required level of protection. If construction activities generate any airborne dust emissions, then the Contractor will utilize appropriate respirators with HEPA filter cartridges until the initial air sampling exposure assessment demonstrates that the use of the respirators is not required to protect personnel from air concentrations of lead dust. The Contractor will provide workers with and require the use of respirators approved by NIOSH for lead, in accordance with the OSHA Standard 29 CFR 1926.62. Disposable single-use respirators or dust masks are not permitted.
 2. Workers must perform positive and negative air pressure fit tests each time a respirator is worn.

3. If respiratory protection is required, workers will be given a qualitative fit test in accordance with procedures detailed in OSHA 29 CFR 1926.62 for all respirators to be used. An appropriately administered quantitative fit test may be substituted for a qualitative fit test. Documentation of adequate respirator fit must be provided to the workers. Fit tests will be administered at the start of the project.
4. Persons having facial hair that may interfere with the seal of a respirator will not be allowed to use respirators.

B. Personnel Protection for Lead-Containing and Lead-Based Paints:

1. If the work generates dust and debris from LBP/LCP, disposable clothing, including head, foot, and full-body protection, shall be provided in sufficient quantities and adequate sizes for all workers and any authorized visitors. All personnel engaged in LCP and LBP demolition work that may potentially exceed the OSHA AL will wear approved disposable protective clothing constructed of spun-bonded olefin or polypropylene fabrics or other material of equivalent resistance to penetration. Disposable whole-body clothing, including head covers, gloves, and shoe coverings, will be provided to and worn by all personnel in the removal work areas. If elastic sleeve closures are not provided, sleeves will be secured with duct tape to the gloves. Washable footwear having a non-skid tracking surface will be provided and used by all personnel within the lead work area.
2. Contaminated clothing will be treated as lead waste and undergo the same disposal procedures utilized for lead waste.
3. Protective eyewear, gloves, rubber boots and/or other footwear will be provided as required for workers and authorized visitors. Safety shoes may be required for some activities.
4. Eating, drinking, smoking, chewing gum, or tobacco will not be allowed in the LCP/LBP work areas.

2.2 EQUIPMENT

A. HEPA-equipped filtration equipment:

1. Vacuum Equipment: HEPA-filtered vacuuming equipment with a filter system capable of collecting and retaining lead particulates shall be used for housekeeping during work that creates dust and debris from LBP/LCP. The filters must be 99.97% efficient for retaining particles 0.3 microns or larger.
2. Use of HEPA filters will not exceed the manufacturers' recommended machine hours (typically 600 to 900 hours). Verification of proper/timely filter changes for each piece of equipment shall be submitted to NPS in the form of a filter change log. The Contractor will use only those brands manufactured and tested for commercial purposes. No "homemade" units will be permitted. The HEPA filters will be approved by UL Method 586.

- B. If utilizing rental equipment that will be impacted by or used to dispose of LCP and LBP or other lead-containing materials, notify the rental agency in writing concerning the intended use of the equipment.

2.3 PLASTIC SHEETING

- A. Six mil in thickness and sized to minimize the frequency of joints.
- B. Polyethylene sheeting utilized for the worker decontamination enclosure, if required by the CP, will be opaque white or black.

2.4 TAPE

- A. For use in dry and wet conditions, and capable of being cleaned off surfaces without permanent marks or damage to the substratum.

2.5 CONTAINERS

- A. Air and watertight, 55-gallon metal or fiberglass drums with tightly fitting lids will be used to package the waste. The Contractor will line the drum(s) with 6-mil plastic bags and label containers in accordance with the OSHA requirements and the EPA Resource Conservation and Recovery Act (RCRA) requirements. All containers will carry appropriate labels as required by OSHA, EPA, and U.S. DOT, and the Arizona Department of Transportation (ADOT). No container on the Project Site will be unlabeled as to its contents (empty or otherwise).

PART 3 - EXECUTION

3.1 DECONTAMINATION ENCLOSURE SYSTEMS

- A. Worker Decontamination
 - 1. A hygiene facility, as described in OSHA 1926.62, must be utilized during the lead removal activities.

3.2 AIR MONITORING

- A. Air samples for lead exposure will be collected by the Contractor for a minimum of the first three days that LCP/LBP work occurs. Lead air samples will be analyzed by an independent accredited laboratory on a rush basis. The purpose of air monitoring is to confirm the protection of workers at the point of operation and to confirm the effectiveness of emission controls established by the Contractor. The Contractor will prevent the contamination of areas/outside of the regulated areas with airborne lead-containing dust. Should any of the above occur, the Contractor shall:
 - 1. Immediately cease removal activities within the regulated areas until the fault is corrected and any contamination outside the regulated area(s) is completely remediated. Do not commence work within the original regulated area until authorized by the Contracting Officer or the Contractor's Industrial Hygienist.

2. Complete the corrective work at no additional cost to the Government if high airborne lead-containing dust levels are identified during the Contractor's activities.

3.3 WORK PRACTICES

The following sections describe protocols for the various LCP and LBP removal tasks associated with this project.

- A. Manual or power sanding or grinding of lead-coated surfaces or materials is not permitted unless the tools are equipped with HEPA attachments or wet methods. The dry sanding or grinding of surfaces that contain lead is prohibited.
- B. Select lead removal processes to minimize contamination of work areas outside the control area with lead-contaminated dust or other debris or waste and to ensure that unprotected personnel are not exposed to hazardous concentrations of lead.
- C. Cleanup
 1. Ensure surfaces in the lead control area are free of accumulations of dust and debris.
 2. Restrict the spread of dust and debris; keep the waste from being distributed over the work areas.
 3. Do not dry sweep or use pressurized air to clean up the area. At the end of each shift and when the lead operations have been completed, clean the controlled area of all visible contamination by vacuuming with a HEPA-filtered vacuum cleaner and wet mopping and wet wiping the areas.
 4. Re-clean areas that show dust or debris.
 5. After removing visible dust and debris, wet wipe and HEPA vacuum all surfaces in the controlled area, as applicable.
 6. If adjacent areas become contaminated at any time during the work, clean and visually inspect all contaminated areas.
 7. The CP will certify in writing that the areas have been cleaned of lead contamination.

3.4 DISPOSAL

- A. Additional material will not be added to bags, and bags will not be reopened after they have been sealed. The lead wastes will be transported by a properly licensed transporter and disposed of at a landfill approved for lead.
- B. Labels must be affixed by the Contractor to all bags, drums, or other containers containing lead waste. The label will be prominently displayed on the waste containers. All warning label print will be readily visible in large bold letter-size printed on a contrasting background. This applies only to waste that is considered hazardous waste per the waste profile.

3.5 AIR AND WATER POLLUTION CONTROL

- A. The Contractor must take all necessary reasonable measures to reduce air and water pollution by any material or equipment used during construction.
- B. Contractor personnel and visitors must be protected from exposure to airborne lead-containing dust.
- C. Lead air or dust samples will be analyzed by AAS using a laboratory accredited by the Environmental Lead Laboratory Accreditation Program and American Industrial Hygiene Association.
- D. The Contractor will not dispose of lead, coating wastes, or oils in storm or sanitary drains.
- E. The Contractor will not allow lead or other waste materials to be washed into streams or bodies of water.

3.6 REPORTS

- A. Prepare and submit a final closeout report at the completion of the work.

END OF SECTION 02 83 33

SECTION 03 20 00 - CONCRETE REINFORCING

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Steel reinforcement bars.

1.2 ACTION SUBMITTALS

A. Product Data: For the following:

1. Each type of steel reinforcement.
2. Bar supports.

B. Shop Drawings: Comply with ACI SP-066:

1. Include placing drawings that detail fabrication, bending, and placement.
2. Include bar sizes, lengths, materials, grades, bar schedules, stirrup spacing, bent bar diagrams, bar arrangement, location of splices, lengths of lap splices, tie spacing, hoop spacing, and supports for concrete reinforcement.

1.3 INFORMATIONAL SUBMITTALS

A. Qualification Statements: For testing and inspection agency.

B. Material Test Reports: For the following, from a qualified testing agency:

1. Steel Reinforcement:

C. Field quality-control reports.

1.4 QUALITY ASSURANCE

- A. Testing Agency Qualifications: An independent agency, acceptable to authorities having jurisdiction, qualified in accordance with ASTM C1077 and ASTM E329 for testing indicated.

1.5 DELIVERY, STORAGE, AND HANDLING

- A. Steel Reinforcement: Deliver, store, and handle steel reinforcement to prevent bending and damage and to avoid damaging coatings on steel reinforcement.

1. Store reinforcement to avoid contact with earth.

PART 2 - PRODUCTS

2.1 STEEL REINFORCEMENT

- A. Reinforcing Bars: ASTM A615/A615M, Grade 60, deformed.

2.2 REINFORCEMENT ACCESSORIES

- A. Bar Supports: Bolsters, chairs, spacers, and other devices for spacing, supporting, and fastening reinforcing bars and welded-wire reinforcement in place.
 1. Manufacture bar supports from steel wire, plastic, or precast concrete in accordance with CRSI's "Manual of Standard Practice," of greater compressive strength than concrete and as follows:
 - a. For concrete surfaces exposed to view, where legs of wire bar supports contact forms, use CRSI Class 1 plastic-protected steel wire, all-plastic bar supports, or CRSI Class 2 stainless steel bar supports.
- B. Steel Tie Wire: ASTM A1064/A1064M, annealed steel, not less than 0.0508 inch in diameter.
 1. Finish: Plain.

2.3 FABRICATING REINFORCEMENT

- A. Fabricate steel reinforcement according to CRSI's "Manual of Standard Practice."

PART 3 - EXECUTION

3.1 PREPARATION

- A. Clean reinforcement of loose rust and mill scale, earth, ice, and other foreign materials that reduce bond to concrete.

3.2 INSTALLATION OF STEEL REINFORCEMENT

- A. Comply with CRSI's "Manual of Standard Practice" for placing and supporting reinforcement.
- B. Accurately position, support, and secure reinforcement against displacement.
 1. Locate and support reinforcement with bar supports to maintain minimum concrete

cover.

2. Do not tack weld crossing reinforcing bars.
- C. Preserve clearance between bars of not less than 1 inch, not less than one bar diameter, or not less than 1-1/3 times size of large aggregate, whichever is greater.
- D. Provide concrete coverage in accordance with ACI 318.
- E. Set wire ties with ends directed into concrete, not toward exposed concrete surfaces.
- F. Splices: Lap splices as indicated on Drawings.
 1. Bars indicated to be continuous, and all vertical bars to be lapped not less than 36 bar diameters at splices, or 24 inches, whichever is greater.
 2. Stagger splices in accordance with ACI 318.

3.3 INSTALLATION TOLERANCES

- A. Comply with ACI 117.

3.4 FIELD QUALITY CONTROL

- A. Testing Agency: Engage a qualified testing and inspecting agency to perform tests and inspections and to submit reports.
- B. Inspections:
 1. Steel-reinforcement placement.

END OF SECTION 03 20 00

SECTION 03 30 00 - CAST-IN-PLACE CONCRETE

PART 1 - GENERAL

1.1 SUMMARY

1. Concrete standards.
2. Concrete materials.
3. Repair materials.
4. Concrete mixture materials.
5. Concrete mixture class types.
6. Concrete mixing.

B. Related Requirements:

1. Section 03 20 00 "Concrete Reinforcing" for steel reinforcing bars and welded-wire reinforcement.

1.2 DEFINITIONS

A. Cementitious Materials: Portland cement or blended hydraulic cement alone or in combination with one or more of the following:

1. Fly ash, slag cement, other pozzolans, and silica fume; materials subject to compliance with requirements.

B. Water/Cementitious Materials (w/cm) Ratio: The ratio by weight of mixing water to cementitious materials.

1.3 ACTION SUBMITTALS

A. Product Data:

1. Portland cement.
2. Fly ash.
3. Slag cement.
4. Silica fume.
5. Natural or other pozzolans.
6. Aggregates.
7. Admixtures:
 - a. Include limitations of use. Admixtures that do not comply with reference ASTM International requirements must be submitted with test data for approval.
8. Repair materials.

- B. Design Mixtures: For each concrete mixture, include the following:
1. Mixture identification.
 2. Compressive strength at 28 days or other age as specified.
 3. Compressive strength required at stages of construction.
 4. Durability exposure classes for Exposure Categories F, S, W, and C.
 5. Maximum w/cm ratio.
 6. Slump or slump flow limit.
 7. Air content.
 8. Nominal maximum aggregate size.
 9. Intended placement method.
 10. Submit adjustments to design mixtures when characteristics of materials, Project conditions, weather, test results, or other circumstances warrant changes.
- C. Concrete Schedule: For each location of each class of concrete indicated in "Concrete Mixture Class Types" Article, including the following:
1. Concrete class designation.
 2. Location within Project.
 3. Exposure class designation.
 4. Formed surface finish designation and final finish.

1.4 INFORMATIONAL SUBMITTALS

- A. Qualification Data: For the following:
1. Installer: Include copies of applicable ACI certificates.
 2. Testing Agency: Include documentation indicating compliance with ASTM E329 or ASTM C1077 and copies of applicable ACI certificates for testing technicians or ACI Concrete Construction Special Inspector - MH, ASCC.
- B. Material Certificates: For each of the following:
1. Cementitious materials.
 2. Admixtures.
 3. Curing compounds.
 4. Bonding agents.
 5. Adhesives.
 6. Repair materials.
- C. Material Test Reports: For the following:
1. Portland cement.
 2. Fly ash.
 3. Slag cement.
 4. Silica fume.
 5. Natural or other pozzolans.
 6. Aggregates.
 7. Admixtures.

D. Research Reports:

1. For concrete admixtures in accordance with ICC's Acceptance Criteria AC198.

E. Field quality-control reports.

1.5 QUALITY ASSURANCE

A. Installer Qualifications: A qualified Installer who employs Project personnel qualified as an ACI-certified Concrete Flatwork Associate and Concrete Flatwork Finisher and a supervisor who is a certified ACI Advanced Concrete Flatwork Finisher/Technician or an ACI Concrete Flatwork Finisher with experience installing and finishing concrete.

1. Post-Installed Concrete Anchors Installers: ACI-certified Adhesive Anchor Installer.

B. Ready-Mixed Concrete Manufacturer Qualifications: A firm experienced in manufacturing ready-mixed concrete products and that complies with ASTM C94/C94M requirements for production facilities and equipment.

C. Laboratory Testing Agency Qualifications: A testing agency qualified in accordance with ASTM C1077 and ASTM E329 for testing that performs duties on behalf of the Contractor.

1. Personnel performing laboratory tests to be an ACI-certified Concrete Strength Testing Technician and Concrete Laboratory Testing Technician, Level 1. Testing agency laboratory supervisor tests to be an ACI-certified Concrete Laboratory Testing Technician, Level 2.

D. Field Quality-Control Testing Agency Qualifications: An independent agency, acceptable to authorities having jurisdiction, qualified in accordance with ASTM C1077 and ASTM E329 for testing indicated.

1. Personnel conducting field tests on plastic concrete properties are to be qualified as an ACI Concrete Field Testing Technician, Grade 1, in accordance with policies from ACI CPP 610.1 or an equivalent certification program.

1.6 DELIVERY, STORAGE, AND HANDLING

A. Comply with ASTM C94/C94M and ACI 301.

1.7 FIELD CONDITIONS

A. Cold-Weather Placement: Comply with ACI 301 as follows:

1. Protect concrete work from physical damage or reduced strength that could be caused by frost, freezing actions, or low temperatures.
2. When air temperature has fallen to, or is expected to fall below 40 deg F during the protection period, maintain delivered concrete mixture temperature within the

- temperature range required by ACI 301.
3. Do not use frozen materials or materials containing ice or snow.
4. Do not place concrete in contact with surfaces less than 35 deg F, other than reinforcing steel.

B. Hot-Weather Placement: Comply with ACI 301 and ACI 305.1, and as follows:

1. Maintain concrete temperature at time of discharge to not exceed 95 deg F.
2. Fog-spray forms, steel reinforcement, and subgrade just before placing concrete. Keep subgrade uniformly moist without standing water, soft spots, or dry areas.

PART 2 - PRODUCTS

2.1 CONCRETE STANDARDS

A. ACI Publications: Comply with ACI 301 unless modified by requirements in the Contract Documents.

2.2 CONCRETE MATERIALS

A. Source Limitations:

1. Obtain all concrete mixtures from a single ready-mixed concrete manufacturer for entire Project.
2. Obtain each type of admixture from single source from single manufacturer.

B. Cementitious Materials:

1. Portland Cement: ASTM C150/C150M, Type I Type II, gray.
2. Pozzolans: ASTM C618, Class C, F, or N.
3. Slag Cement: ASTM C989/C989M, Grade 100 or 120.
4. Ground Glass Pozzolan: ASTM C1866/C1866M, Type GS or GE.

C. Normal-Weight Aggregates:

1. Coarse Aggregate: ASTM C33/C33M, Class 3M
2. Maximum Coarse-Aggregate Size: 1 inch nominal.
3. Fine Aggregate: ASTM C33/C33M.
4. Recycled Aggregate: Provide documentation of characteristics of recycled aggregate and mechanical properties and durability of proposed concrete, which incorporates recycled aggregate to conform to applicable requirements for the class of concrete.

2.3 ADMIXTURES

A. Air-Entraining Admixture: ASTM C260/C260M.

B. Chemical Admixtures: Do not use calcium chloride or admixtures containing calcium

chloride in steel-reinforced concrete.

1. Water-Reducing Admixture: ASTM C494/C494M, Type A.
 2. Retarding Admixture: ASTM C494/C494M, Type B.
 3. Water-Reducing and -Retarding Admixture: ASTM C494/C494M, Type D.
 4. High-Range, Water-Reducing Admixture: ASTM C494/C494M, Type F.
 5. High-Range, Water-Reducing and -Retarding Admixture: ASTM C494/C494M, Type G.
 6. Admixtures with special properties, with documentation of claimed performance enhancement, ASTM C494/C494M, Type S.
- C. Mixing Water for Concrete Mixtures and Water Used to Make Ice: ASTM C1602/C1602M. Include documentation of compliance with limits for alkalis, sulfates, chlorides, or solids content of mixing water from Table 2 in ASTM C1602/C1602M.

2.4 CURING MATERIALS

- A. Absorptive Cover: AASHTO M 182, Class 2, burlap cloth made from jute or kenaf, weighing approximately 9 oz./sq. yd. when dry.
- B. Moisture-Retaining Cover: ASTM C171, polyethylene film burlap-polyethylene sheet.
- C. Water: Potable water that does not cause staining of the surface.

2.5 ACCESSORIES

- A. Bonding Agent: ASTM C1059/C1059M, Type II, nonredispersible, acrylic emulsion or styrene butadiene.
- B. Epoxy Bonding Adhesive: ASTM C881/C881M, two-component epoxy resin, capable of humid curing and bonding to damp surfaces, of class suitable for application temperature and of grade and class to suit requirements, and as follows:

2.6 REPAIR MATERIALS

- A. Repair Overlayment: Cement-based, polymer-modified, self-leveling product that can be applied in thicknesses from 1/4 inch and that can be filled in over a scarified surface to match adjacent floor elevations.
 1. Cement Binder: ASTM C150/C150M portland cement or hydraulic or blended hydraulic cement, as defined in ASTM C219.
 2. Primer: Product of topping manufacturer recommended for substrate, conditions, and application.
 3. Aggregate: Well-graded, washed gravel, 1/8 to 1/4 inch or coarse sand as recommended by topping manufacturer.
 4. Compressive Strength: Not less than 5000 psi at 28 days when tested in accordance with ASTM C109/C109M.

2.7 CONCRETE MIXTURE MATERIALS

- A. Prepare design mixtures for each type and strength of concrete, proportioned on the basis of laboratory trial mixture or field test data, or both, in accordance with ACI 301.
 - 1. Use a qualified testing agency for preparing and reporting proposed mixture designs, based on laboratory trial mixtures.
- B. Cementitious Materials: Limit percentage, by weight, of cementitious materials other than portland or hydraulic cement in concrete assigned to Exposure Class F3 as follows:
 - 1. Fly Ash or Other Pozzolans: 25 percent by mass.
 - 2. Slag Cement: 50 percent by mass.
 - 3. Silica Fume: 10 percent by mass.
 - 4. Total of Fly Ash or Other Pozzolans, Slag Cement, and Silica Fume: 50 percent by mass, with fly ash or pozzolans not exceeding 25 percent by mass and silica fume not exceeding 10 percent by mass.
 - 5. Total of Fly Ash or Other Pozzolans and Silica Fume: 35 percent by mass with fly ash or pozzolans not exceeding 25 percent by mass and silica fume not exceeding 10 percent by mass.
- C. Admixtures: Use admixtures in accordance with manufacturer's written instructions.

2.8 CONCRETE MIXTURE CLASS TYPES

- A. Class A: Normal-weight concrete used for shallow foundation footings and grade beams.
 - 1. Exposure Class: ACI 318 Class F2 Class S0 Class W1 Class C1.
 - 2. Minimum Compressive Strength: 4500 psi at 56 days.
 - 3. Maximum w/cm Ratio: 0.45.
 - 4. Slump Limit: 4 inches, plus or minus 1 inch for concrete.
 - 5. Air Content:
 - a. Exposure Classes F2: 6.0 percent, plus or minus 1.5 percent at point of delivery for concrete containing 1-inch nominal maximum aggregate size.
 - 6. Limit water-soluble, chloride-ion content in hardened concrete to 0.30 percent by weight of cement.
- B. Class B: Normal-weight concrete used for exterior slab on grade.
 - 1. Exposure Class: ACI 318 Class F3 Class S0 Class W1 Class C0.
 - 2. Minimum Compressive Strength: 5000 psi at 28 days.
 - 3. Maximum w/cm Ratio : 0.40.
 - 4. Slump Limit: 4 inches, plus or minus 1 inch for concrete.
 - 5. Air Content:
 - a. Exposure Class F3: 6.0 percent, plus or minus 1.5 percent at point of delivery for concrete containing 1-inch nominal maximum aggregate size.

6. Limit water-soluble, chloride-ion content in hardened concrete to 1.00 percent by weight of cement.

2.9 CONCRETE MIXING

- A. Ready-Mixed Concrete: Measure, batch, mix, and deliver concrete in accordance with ASTM C94/C94M and furnish delivery ticket.
- B. Project-Site Mixing: Measure, batch, and mix concrete materials and concrete in accordance with ASTM C94/C94M. Mix concrete materials in appropriate drum-type batch machine mixer.
 1. For mixer capacity of 1 cu. yd. or smaller, continue mixing at least 1-1/2 minutes, but not more than five minutes after ingredients are in mixer, before any part of batch is released.
 2. For mixer capacity larger than 1 cu. yd., increase mixing time by 15 seconds for each additional 1 cu. yd..
 3. Provide batch ticket for each batch discharged and used in the Work, indicating Project identification name and number, date, mixture type, mixture time, quantity, and amount of water added. Record approximate location of final deposit in structure.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Verification of Conditions:
 1. Before placing concrete, verify that installation of concrete forms, accessories, reinforcement, and embedded items is complete and that required inspections have been performed.
 2. Do not proceed until unsatisfactory conditions have been corrected.

3.2 PREPARATION

- A. Provide reasonable auxiliary services to accommodate field testing and inspections, acceptable to testing agency, including the following:
 1. Daily access to the Work.
 2. Incidental labor and facilities necessary to facilitate tests and inspections.
 3. Secure space for storage, initial curing, and field curing of test samples, including source of water and continuous electrical power at Project site during site curing period for test samples.
 4. Security and protection for test samples and for testing and inspection equipment at Project site.

3.3 TOLERANCES

- A. Comply with ACI 117.

3.4 INSTALLATION OF EMBEDDED ITEMS

- A. Place and secure anchorage devices and other embedded items required for adjoining Work that is attached to or supported by cast-in-place concrete.
 - 1. Use setting drawings, templates, diagrams, instructions, and directions furnished with items to be embedded.

3.5 INSTALLATION OF CAST-IN-PLACE CONCRETE

- A. Before placing concrete, verify that installation of formwork, reinforcement, and embedded items is complete and that required inspections are completed.
- B. Notify Contracting Officer and testing and inspection agencies 24 hours prior to commencement of concrete placement.
- C. Water addition in transit or at the Project site must be in accordance with ASTM C94/C94M and must not exceed the permitted amount indicated on the concrete delivery ticket.
- D. Deposit concrete continuously in one layer or in horizontal layers of such thickness that no new concrete is placed on concrete that has hardened enough to cause seams or planes of weakness.
 - 1. If a section cannot be placed continuously, provide construction joints as indicated.
 - 2. Deposit concrete to avoid segregation.
 - 3. Deposit concrete in horizontal layers of depth not to exceed formwork design pressures and in a manner to avoid inclined construction joints.
 - 4. Consolidate placed concrete with mechanical vibrating equipment in accordance with ACI 301.
 - a. Do not use vibrators to transport concrete inside forms.
 - b. Insert and withdraw vibrators vertically at uniformly spaced locations to rapidly penetrate placed layer and at least 6 inches into preceding layer.
 - c. Do not insert vibrators into lower layers of concrete that have begun to lose plasticity.
 - d. At each insertion, limit duration of vibration to time necessary to consolidate concrete, and complete embedment of reinforcement and other embedded items without causing mixture constituents to segregate.
- E. Deposit and consolidate concrete for slabs in a continuous operation, within limits of construction joints, until placement of a panel or section is complete.
 - 1. Consolidate concrete during placement operations, so concrete is thoroughly worked around reinforcement and other embedded items and into corners.

2. Maintain reinforcement in position on chairs during concrete placement.
3. Screed slab surfaces with a straightedge and strike off to correct elevations.
4. Level concrete, cut high areas, and fill low areas.
5. Begin initial floating using bull floats or darbies to form a uniform and open-textured surface plane, before excess bleedwater appears on the surface.
6. Do not further disturb slab surfaces before starting finishing operations.

3.6 APPLICATION OF FINISHING FLOORS, SLABS, AND EXPOSED FOUNDATION TOPS

A. Float Finish:

1. When bleedwater sheen has disappeared and concrete surface has stiffened sufficiently to permit operation of specific float apparatus, consolidate concrete surface with power-driven floats or by hand floating if area is small or inaccessible to power-driven floats.
2. Repeat float passes and restraighening until surface is left with a uniform, smooth, granular texture and complies with ACI 117 tolerances for conventional concrete.
3. Apply float finish to surfaces to receive trowel finish.

B. Trowel Finish:

1. After applying float finish, apply first troweling and consolidate concrete by hand or power-driven trowel.
2. Continue troweling passes and restraighen until surface is free of trowel marks and uniform in texture and appearance.
3. Grind smooth any surface defects that would telegraph through applied coatings or floor coverings.
4. Do not add water to concrete surface. Use of an approved finishing aid is acceptable.
5. Do not apply troweled finish to concrete, which has a total air content greater than 3 percent.
6. Apply a trowel finish to surfaces exposed to view.

3.7 APPLICATION OF FINISHING FORMED SURFACES

A. As-Cast Surface Finishes:

1. ACI 301 (ACI 301M) Surface Finish SF-1.0: As-cast concrete texture imparted by form-facing material.
 - a. Patch voids larger than 1-1/2 inches wide or 1/2 inch deep.
 - b. Remove projections larger than 1 inch.
 - c. Tie holes do not require patching.
 - d. Surface Tolerance: ACI 117, Class D.
 - e. Apply to concrete surfaces for metal lap pan deck formed surfaces and those surfaces that are buried or covered with subsequent installed surfaces.

3.8 APPLICATION OF CONCRETE CURING

- A. Protect freshly placed concrete from premature drying and excessive cold or hot temperatures.
 - 1. Comply with ACI 301 for cold weather protection during curing.
 - 2. Comply with ACI 301 and ACI 305.1 for hot-weather protection during curing.
 - 3. Maintain moisture loss no more than 0.2 lb/sq. ft. x h, calculated in accordance with ACI 305R, before and during finishing operations.
- B. Curing Formed Surfaces: Comply with ACI 308.1 as follows:
 - 1. Cure formed concrete surfaces, including underside of beams, supported slabs, and other similar surfaces.
 - 2. Cure concrete containing color pigments in accordance with color pigment manufacturer's instructions.
 - 3. If forms remain during curing period, moist cure after loosening forms.
 - 4. If removing forms before end of curing period, continue curing for remainder of curing period as follows:
 - a. Continuous Fogging: Maintain standing water on concrete surface until final setting of concrete.
 - b. Continuous Sprinkling: Maintain concrete surface continuously wet.
 - c. Absorptive Cover: Pre-dampen absorptive material before application; apply additional water to absorptive material to maintain concrete surface continuously wet.
 - d. Water-Retention Sheeting Materials: Cover exposed concrete surfaces with sheeting material, taping, or lapping seams.
 - e. Membrane-Forming Curing Compound: Apply uniformly in continuous operation by power spray or roller in accordance with manufacturer's written instructions.
 - 1) Recoat areas subject to heavy rainfall within three hours after initial application.
 - 2) Maintain continuity of coating and repair damage during curing period.
- C. Curing Unformed Surfaces: Comply with ACI 308.1 as follows:
 - 1. Begin curing after finishing concrete.

3.9 INSTALLATION OF CONCRETE SURFACE REPAIRS

- A. Defective Concrete:
 - 1. Repair and patch defective areas when approved by Contracting Officer.
 - 2. Remove and replace concrete that cannot be repaired and patched to meet specification requirements.
- B. Patching Mortar: Mix dry-pack patching mortar, consisting of 1 part portland cement to

2-1/2 parts fine aggregate passing a No. 16 sieve, using only enough water for handling and placing.

- C. Repairing Formed Surfaces: Surface defects include color and texture irregularities, cracks in excess of 0.01 inch spalls, air bubbles exceeding surface finish limits, honeycombs, rock pockets, fins and other projections on the surface exceeding surface finish limits, and stains and other discolorations that cannot be removed by cleaning.

1. Immediately after form removal, cut out honeycombs, rock pockets, and voids more than 1/2 inch in any dimension to solid concrete.
 - a. Limit cut depth to 3/4 inch.
 - b. Make edges of cuts perpendicular to concrete surface.
 - c. Clean, dampen with water, and brush-coat holes and voids with bonding agent.
 - d. Fill and compact with patching mortar before bonding agent has dried.
 - e. Fill form-tie voids with patching mortar or cone plugs secured in place with bonding agent.
2. Repair defects on surfaces exposed to view by blending white portland cement and standard portland cement, so that, when dry, patching mortar matches surrounding color.
 - a. Patch a test area at inconspicuous locations to verify mixture and color match before proceeding with patching.
 - b. Compact mortar in place and match surrounding surface.
3. Repair defects on concealed formed surfaces that will affect concrete's durability and structural performance, as determined by Contracting Officer.

- D. Repairing Unformed Surfaces:

1. Repair finished surfaces containing surface defects, including spalls, popouts, honeycombs, rock pockets, crazing, and cracks in excess of 0.01 inch wide or that penetrate to reinforcement or completely through unreinforced sections regardless of width.
2. Repair defective areas, except random cracks and single holes 1 inch or less in diameter, by cutting out and replacing with fresh concrete.
 - a. Remove defective areas with clean, square cuts, and expose steel reinforcement with at least a 3/4-inch clearance all around.
 - b. Dampen concrete surfaces in contact with patching concrete and apply bonding agent.
 - c. Mix patching concrete of same materials and mixture as original concrete, except without coarse aggregate.
 - d. Place, compact, and finish to blend with adjacent finished concrete.
 - e. Cure in same manner as adjacent concrete.
3. Repair random cracks and single holes 1 inch or less in diameter with patching mortar.

- a. Groove top of cracks and cut out holes to sound concrete, and clean off dust, dirt, and loose particles.
 - b. Dampen cleaned concrete surfaces and apply bonding agent.
 - c. Place patching mortar before bonding agent has dried.
 - d. Compact patching mortar and finish to match adjacent concrete.
 - e. Keep patched area continuously moist for at least 72 hours.
- E. Perform structural repairs of concrete, subject to Contracting Officer's approval, using epoxy adhesive and patching mortar.
- F. Repair materials and installation not specified above may be used, subject to Contracting Officer's approval.

3.10 FIELD QUALITY CONTROL

- A. Testing Agency: Engage a qualified testing and inspecting agency to perform tests and inspections and to submit reports.
 - 1. Testing agency to be responsible for providing curing facility for initial curing of strength test specimens on-site and verifying that test specimens are cured in accordance with standard curing requirements in ASTM C31/C31M.
 - 2. Testing agency to immediately report to Contracting Officer, Contractor, and concrete manufacturer any failure of Work to comply with Contract Documents.
 - 3. Testing agency to report results of tests and inspections, in writing, to Contracting Officer, Contractor, and concrete manufacturer within 48 hours of inspections and tests.
 - a. Test reports to include reporting requirements of ASTM C31/C31M, ASTM C39/C39M, and ACI 301, including the following as applicable to each test and inspection:
 - 1) Project name.
 - 2) Name of testing agency.
 - 3) Names and certification numbers of field and laboratory technicians performing inspections and testing.
 - 4) Name of concrete manufacturer.
 - 5) Date and time of inspection, sampling, and field testing.
 - 6) Date and time of concrete placement.
 - 7) Location in Work of concrete represented by samples.
 - 8) Date and time sample was obtained.
 - 9) Truck and batch ticket numbers.
 - 10) Design compressive strength at 28 days.
 - 11) Concrete mixture designation, proportions, and materials.
 - 12) Field test results of fresh concrete, including slump or slump flow, air content, temperature and density.
 - 13) Information on storage and curing of samples at the Project site, including curing method and maximum and minimum temperatures during initial curing period.
 - 14) Type of fracture and compressive break strengths at seven days and 28 days.

4. Provide a space and source of power or other resources for curing and access to test specimens by the testing agency.
- B. Delivery Tickets: comply with ASTM C94/C94M.
- C. Inspections:
1. Verification of use of required design mixture.
 2. Concrete placement, including conveying and depositing.
 3. Curing procedures and maintenance of curing temperature.
 4. Batch Plant Inspections: On a random basis, as determined by Contracting Officer.
- D. Concrete Tests: Testing of composite samples of fresh concrete obtained in accordance with ASTM C 172/C 172M to be performed in accordance with the following requirements:
1. Testing Frequency: Obtain one composite sample for each day's pour of each concrete mixture exceeding 5 cu. yd., but less than 25 cu. yd., plus one set for each additional 150 cu. yd. or fraction thereof.
 - a. When frequency of testing provides fewer than five compressive-strength tests for each concrete mixture, testing is to be conducted from at least five randomly selected batches or from each batch if fewer than five are used.
 2. Slump: ASTM C143/C143M:
 - a. One test at point of delivery for each composite sample, but not less than one test for each day's pour of each concrete mixture.
 - b. Perform additional tests as needed.
 3. Slump Flow: ASTM C1611/C1611M:
 - a. One test at point of delivery for each composite sample when strength test specimens are cast, but not less than one test for each day's pour of each concrete mixture.
 - b. Perform additional tests as needed.
 4. Air Content: ASTM C231/C231M pressure method, for normal-weight concrete.
 - a. One test for each composite sample when strength test specimens are cast, but not less than one test for each day's pour of each concrete mixture.
 5. Concrete Temperature: ASTM C1064/C1064M:
 - a. One test hourly when air temperature is 40 deg F and below or 80 deg F and above, and one test for each composite sample when strength test specimens are cast.
 6. Concrete Density: ASTM C138/C138M:
 - a. One test for each composite sample when strength test specimens are cast.

7. Compression Test Specimens: ASTM C31/C31M:
 - a. Cast and standard cure two sets of two 6 inches by 12-inches or 4-inch by 8-inch cylindrical specimens for each composite sample.
 - b. Cast, and field cure two sets of two standard cylindrical specimens for each composite sample.
8. Compressive-Strength Tests: ASTM C39/C39M.
 - a. Test one set of two standard cured specimens at seven days and one set of two specimens at 28 days.
 - b. Test one set of two field-cured specimens at seven days and one set of two specimens at 28 days.
 - c. A compressive-strength test to be the average compressive strength from a set of two specimens obtained from same composite sample and tested at age indicated.
9. When strength of field-cured cylinders is less than 85 percent of companion laboratory-cured cylinders, Contractor to evaluate operations and provide corrective procedures for protecting and curing in-place concrete.
10. Strength of each concrete mixture will be satisfactory if every average of any three consecutive compressive-strength tests of standard cured cylinders equals or exceeds specified compressive strength, and no compressive-strength test value falls below specified compressive strength by more than 500 psi if specified compressive strength is 5000 psi, or no compressive strength test value is less than 10 percent of specified compressive strength if specified compressive strength is greater than 5000 psi.
11. Additional testing and inspecting, at Contractor's expense, will be performed to determine compliance of replaced or additional work with specified requirements.
12. Correct deficiencies in the Work that test reports and inspections indicate do not comply with the Contract Documents.

3.4 INSPECTION

- A. Contractor will commence accessibility inspection immediately after finishing work. Observations, measurements and other notes will be recorded using DSC-CQC Accessibility Inspection Report as basis. Report can include digital images, product literature, drawings and sketches, etc. to communicate conditions, especially items not in compliance with ABAAS. Submit report to CO within 24 hours of commencing inspection.

END OF SECTION 03 30 00

SECTION 06 10 00 - ROUGH CARPENTRY

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Wood products.
2. Wood-preservative-treated lumber.
3. Dimension lumber framing.
4. Miscellaneous lumber.

B. Related Requirements:

1. Section 06 16 00 "Sheathing" for sheathing and subflooring.

1.2 DEFINITIONS

- A. Boards or Strips: Lumber of less than 2 inches nominal size in least dimension.
- B. Dimension Lumber: Lumber of 2 inches nominal size or greater but less than 5 inches nominal size in least dimension.
- C. Exposed Framing: Framing not concealed by other construction.
- D. Lumber grading agencies, and abbreviations used to reference them, include the following:
1. SPIB: The Southern Pine Inspection Bureau.
 2. WCLIB: West Coast Lumber Inspection Bureau.
 3. WWPA: Western Wood Products Association.

1.3 ACTION SUBMITTALS

- A. Product Data: For each type of process and factory-fabricated product. Indicate component materials and dimensions and include construction and application details.
1. Include data for wood-preservative treatment from chemical treatment manufacturer and certification by treating plant that treated materials comply with requirements. Indicate type of preservative used and net amount of preservative retained.

1.4 INFORMATIONAL SUBMITTALS

- A. Material Certificates:

1. For preservative-treated wood products. Indicate type of preservative used and net amount of preservative retained.

1.5 DELIVERY, STORAGE, AND HANDLING

- A. Stack wood products flat with spacers beneath and between each bundle to provide air circulation. Protect wood products from weather by covering with waterproof sheeting, securely anchored. Provide for air circulation around stacks and under coverings.

PART 2 - PRODUCTS

2.1 WOOD PRODUCTS

- A. Lumber: Comply with DOC PS 20 and applicable rules of grading agencies indicated. If no grading agency is indicated, comply with the applicable rules of any rules-writing agency certified by the ALSC Board of Review. Grade lumber by an agency certified by the ALSC Board of Review to inspect and grade lumber under the rules indicated.
 1. Factory mark each piece of lumber with grade stamp of grading agency.
 2. Dress lumber, S4S, unless otherwise indicated.
- B. Maximum Moisture Content:
 1. Boards: 15 percent.
 2. Dimension Lumber: 19 percent unless otherwise indicated.

2.2 WOOD-PRESERVATIVE-TREATED LUMBER

- A. Preservative Treatment by Pressure Process: AWP A U1, Use categories as follows:
 1. UC2: Interior construction not in contact with ground but may be subject to moisture. Include the following items:
 - a. Wood sills, sleepers, blocking, furring, and similar concealed members in contact with masonry or concrete.
 2. UC3B (Commodity Specification A): Uncoated sawn products in exterior construction not in contact with ground, exposed to all weather cycles including intermittent wetting but with sufficient air circulation for wood to dry. Excludes sawn products not in contact with ground but with ground contact-type hazards. Include the following items:
 - a. Wood cants, nailers, curbs, equipment support bases, blocking, stripping, and similar members in connection with roofing, flashing, vapor barriers, and waterproofing.
 - b. Wood framing members that are less than 18 inches above the ground in crawlspaces or unexcavated areas.
 - c. Wood decking, railings, and joists and beams for decks that are not critical

to the performance and safety of the entire system/construction and that are in locations easily accessible for maintenance, repair, or replacement.

3. Preservative Chemicals: Acceptable to authorities having jurisdiction and containing no arsenic or chromium.
 4. After treatment, redry dimension lumber to 19 percent maximum moisture content.
- B. Kiln-dry lumber after treatment to a maximum moisture content of 19 percent. Do not use material that is warped or that does not comply with requirements for untreated material.
- C. Mark lumber with treatment quality mark of an inspection agency approved by the ALSC Board of Review.
- D. Application: Treat items indicated on Drawings, and the following:
1. Wood sills, sleepers, blocking, furring, and similar concealed members in contact with masonry or concrete.
 2. Wood framing members that are less than 18 inches above the ground in crawlspaces or unexcavated areas.

2.3 DIMENSION LUMBER FRAMING

- A. Load-Bearing Partitions by Grade: No. 2 grade.
1. Application: Exterior walls.
 2. Species:
 - a. Douglas fir-larch; WCLIB or WWPA.
- B. Ceiling Joists: Construction or No. 2 grade.
1. Species:
 - a. Douglas fir-larch; WCLIB or WWPA.
- C. Joists, Rafters, and Other Framing by Grade: No. 2 grade.
1. Species:
 - a. Southern pine; SPIB.
 - b. Douglas fir-larch; WCLIB or WWPA.

2.4 MISCELLANEOUS LUMBER

- A. Provide miscellaneous lumber indicated and lumber for support or attachment of other construction, including the following:
1. Blocking.

- B. Dimension Lumber Items: Construction or No. 2 grade lumber of the following species:
 - 1. Douglas fir-larch; WCLIB or WWP.
- C. Concealed Boards: 15 percent maximum moisture content and any of the following species and grades:
 - 1. Mixed southern pine or southern pine; No. 2 grade; SPIB.
 - 2. Hem-fir or hem-fir (north); Construction or No. 2 Common grade; NLGA, WCLIB, or WWP.
 - 3. Spruce-pine-fir (south) or spruce-pine-fir; Construction or No. 2 Common grade; NeLMA, NLGA, WCLIB, or WWP.

2.5 FASTENERS

- A. General: Fasteners are to be of size and type indicated and comply with requirements specified in this article for material and manufacture. Provide nails or screws, in sufficient length, to penetrate not less than 1-1/2 inches into wood substrate.
 - 1. Where rough carpentry is exposed to weather, in ground contact, pressure-preservative treated, or in area of high relative humidity, provide fasteners with hot-dip zinc coating complying with ASTM A153/A153M or ASTM F2329.
- B. Nails, Brads, and Staples: ASTM F1667.
- C. Power-Driven Fasteners: Fastener systems with an evaluation report acceptable to authorities having jurisdiction, based on ICC-ES AC70.
- D. Post-Installed Anchors: Fastener systems with an evaluation report acceptable to authorities having jurisdiction, based on ICC-ES AC01 ICC-ES AC58 ICC-ES AC193 or ICC-ES AC308 as appropriate for the substrate.

2.6 METAL FRAMING CONNECTORS

- A. Allowable design loads, as published by manufacturer, shall meet or exceed those of basis-of-design products. Manufacturer's published values shall be determined from empirical data or by rational engineering analysis and demonstrated by comprehensive testing performed by a qualified independent testing agency. Framing anchors shall be punched for fasteners adequate to withstand same loads as framing anchors.
- B. Galvanized-Steel Sheet: Hot-dip, zinc-coated steel sheet complying with ASTM A653/A653M, G60 (Z180) coating designation.
 - 1. Use for interior locations unless otherwise indicated.
- C. Hot-Dip, Heavy-Galvanized Steel Sheet: ASTM A653/A653M; structural steel (SS), high-strength low-alloy steel Type A (HSLAS Type A), or high-strength low-alloy steel Type B (HSLAS Type B); G185 (Z550) coating designation; and not less than 0.036 inch (0.9 mm) thick.

1. Use for wood-preservative-treated lumber and where indicated.
- D. Salient characteristics of Simpson Strong-Tie products specified in the drawings, to be matched with alternate products:
1. Framing Angles:
 - a. A34, 18ga x 1 7/16" wide legged nailed angle, 545 lbs allowable shear
 - b. A35, 18ga x 1 7/16" wide legged nailed angle, 650 lbs allowable shear
 2. Joist Hangers:
 - a. LUS28, 20ga U-shaped nailed hanger with 1 3/4" long seat x 1" flanges, allowable shear of 945 lbs
 3. Holdowns:
 - a. DTT2Z, 14ga U-shaped screwed tie x 3 1/4" wide x 1/2"Ø thread rod attachment, 2,145 lbs allowable tension
 - b. HDU4, 14ga U-shaped screwed tie x 3" wide x 5/8"Ø thread rod attachment, 4,565 lbs allowable tension

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Framing Standard: Comply with AF&PA's WCD 1, "Details for Conventional Wood Frame Construction," unless otherwise indicated.
- B. Set work to required levels and lines, with members plumb, true to line, cut, and fitted. Fit rough carpentry accurately to other construction. Locate furring, nailers, blocking, and similar supports to comply with requirements for attaching other construction.
- C. Install metal framing anchors to comply with manufacturer's written instructions. Install fasteners through each fastener hole.
- D. Do not splice structural members between supports unless otherwise indicated.
- E. Provide blocking and framing as indicated and as required to support facing materials, fixtures, specialty items, and trim.
- F. Sort and select lumber so that natural characteristics do not interfere with installation or with fastening other materials to lumber. Do not use materials with defects that interfere with function of member or pieces that are too small to use with minimum number of joints or optimum joint arrangement.
- G. Comply with AWWA M4 for applying field treatment to cut surfaces of preservative-treated lumber.
 1. Use inorganic boron for items that are continuously protected from liquid water.

2. Use copper naphthenate for items not continuously protected from liquid water.
- H. Securely attach rough carpentry work to substrate by anchoring and fastening as indicated, complying with the following:
 1. Table 2304.10.2, "Fastening Schedule," in ICC's International Building Code (IBC).
 2. ICC-ES evaluation report for fastener.
- I. Use steel common nails unless otherwise indicated. Select fasteners of size that will not fully penetrate members where opposite side will be exposed to view or will receive finish materials. Make tight connections between members. Install fasteners without splitting wood. Drive nails snug but do not countersink nail heads unless otherwise indicated.

3.2 INSTALLATION OF WOOD BLOCKING AND NAILERS

- A. Install where indicated and where required for screeding or attaching other work. Form to shapes indicated and cut as required for true line and level of attached work. Coordinate locations with other work involved.
- B. Attach wood blocking to substrates to support applied loading. Recess bolts and nuts flush with surfaces unless otherwise indicated.

3.3 INSTALLATION OF FLOOR JOIST FRAMING

- A. General: Install floor joists with crown edge up and support ends of each member with not less than 1-1/2 inches of bearing on wood or metal, or 3 inches on masonry. Attach floor joists as follows:
 1. Where supported on wood members, by toe nailing or by using metal framing anchors.
 2. Where framed into wood supporting members, by using wood ledgers as indicated or, if not indicated, by using metal joist hangers.
- B. Do not notch in middle third of joists; limit notches to one-sixth depth of joist, one-third at ends. Do not bore holes larger than one-third depth of joist; do not locate closer than 2 inches from top or bottom.
- C. Provide solid blocking of 2-inch nominal thickness by depth of joist at ends of joists unless nailed to header or band.

3.4 PROTECTION

- A. Protect wood that has been treated with inorganic boron (SBX) from weather. If, despite protection, inorganic boron-treated wood becomes wet, apply EPA-registered borate treatment. Apply borate solution by spraying to comply with EPA-registered label.

3.7 INSPECTION

- A. Contractor will commence accessibility inspection immediately after finishing work. Observations, measurements and other notes will be recorded using DSC-CQC Accessibility Inspection Report as basis. Report can include digital images, product literature, drawings and sketches, etc. to communicate conditions, especially items not in compliance with ABAAS. Submit report to CO within 24 hours of commencing inspection.

END OF SECTION 06 10 00

SECTION 06 16 00 - SHEATHING

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Floor sheathing.

1.2 ACTION SUBMITTALS

A. Product Data:

1. Floor sheathing.

B. Sustainable Design Submittals:

1. Chain-of-Custody Certificates: For certified wood products. Include statement of costs.

1.3 DELIVERY, STORAGE, AND HANDLING

- ##### A.
- Stack panels flat with spacers beneath and between each bundle to provide air circulation. Protect sheathing from weather by covering with waterproof sheeting, securely anchored. Provide for air circulation around stacks and under coverings.

PART 2 - PRODUCTS

2.1 WOOD PANEL PRODUCTS

- ##### A.
- Thickness: As needed to comply with requirements specified, but not less than thickness indicated.
- ##### B.
- Factory mark panels to indicate compliance with applicable standard.

2.2 FLOOR SHEATHING

- ##### A.
- Plywood Sheathing, Floors: Either DOC PS 1 or DOC PS 2, Exposure 1 sheathing.
1. Span Rating: Not less than 48/24.
 2. Nominal Thickness: Not less than 3/4 inch.
- ##### B.
- Oriented-Strand-Board Sheathing, Floors: DOC PS 2, Exposure 1 sheathing.

1. Span Rating: Not less than 48/24.
2. Nominal Thickness: Not less than 3/4 inch.

2.3 FASTENERS

- A. Nails, Brads, and Staples: ASTM F1667.

PART 3 - EXECUTION

3.1 INSTALLATION, GENERAL

- A. Do not use materials with defects that impair quality of sheathing or pieces that are too small to use with minimum number of joints or optimum joint arrangement. Arrange joints so that pieces do not span between fewer than three support members.
- B. Cut panels at penetrations, edges, and other obstructions of work; fit tightly against abutting construction unless otherwise indicated.
- C. Securely attach to substrate by fastening as indicated, complying with the following:
 1. Table 2304.10.1, "Fastening Schedule," in the ICC's International Building Code.
 2. ICC-ES evaluation report for fastener.
- D. Use common wire nails unless otherwise indicated. Select fasteners of size that will not fully penetrate members where opposite side will be exposed to view or will receive finish materials. Make tight connections. Install fasteners without splitting wood.

3.2 INSTALLATION OF WOOD STRUCTURAL PANEL

- A. General: Comply with applicable recommendations in APA Form No. E30, "Engineered Wood Construction Guide," for types of structural-use panels and applications indicated.
- B. Fastening Methods: Fasten panels as indicated below:
 1. Floor Sheathing:
 - a. Nail to wood framing. Apply a continuous bead of glue to framing members.
 - b. Space panels 1/8 inch apart at edges and ends.

END OF SECTION 06 16 00

SECTION 07 16 00 - CEMENTITIOUS DAMPPROOFING

PART 1 - GENERAL

1.1 SUMMARY

- A. Section Includes:
 - 1. Polymer-modified cementitious dampproofing coating, suitable for exterior above-grade exposure, gray in color, at exterior of new concrete foundation walls.
- B. Related Requirements:
 - 1. Section 03 30 00 "Cast-in-Place Concrete"

1.2 ACTION SUBMITTALS

- A. Product Data: For each type of product.

1.3 FIELD CONDITIONS

- A. Weather Limitations: Proceed with application only when existing and forecasted weather conditions permit dampproofing to be performed in accordance with to manufacturers' written instructions.
- B. Ventilation: Provide adequate ventilation during application of dampproofing in enclosed spaces. Maintain ventilation until dampproofing has cured.

PART 2 - PRODUCTS

2.1 SOURCE LIMITATIONS

- A. Obtain primary dampproofing materials and primers from single source from single manufacturer. Provide protection course auxiliary materials recommended in writing by manufacturer of primary materials.

2.2 Materials

- A. Polymer-modified portland cement coating:
 - 1. Component "A" shall be a liquid polymer emulsion of an acrylic co-polymer base and additives.
 - 2. Component "B" shall be a blend of selected Portland cements, specially graded aggregates, and admixtures to control setting time and workability.
 - 3. The ratio of Component A: Component B shall be: Mortar 1:1.647 by weight
 - 4. The material shall be non-combustible, either before or after cure.

2.3 PERFORMANCE REQUIREMENTS

- A. VOC Content: Products are to comply with VOC content limits of authorities having jurisdiction unless otherwise indicated.
- B. Properties of the mixed polymer-modified portland cement coating:
 - 1. Pot Life: Approx. 4 hours
 - 2. Color: gray or white
 - 3. Recoat Time: 2 hours minimum between coats
 - 4. Application thickness: 8-16 mils/ coat
- C. Properties of the cured polymer-modified portland cement coating:
 - 1. Bond Strength (ACI 503R-30 Modified): Pull-off test 7 days - concrete substrate failure 2
 - 2. Moisture Vapor permeability (ASTM E96) 7 days 1 coat 27 grains/hr./ft² 2 coats 24 grains/hr./ft²
 - 3. Abrasion Resistance (ASTM D-968 Modified) 7 days 55liters/mil
 - 4. The material shall not produce a vapor barrier.
 - 5. The material shall be thermally compatible with portland cement mortar and concrete.
 - 6. The material shall be approved by the United Staes Department of Agriculture.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine substrates, areas, and conditions, with Applicator present, for compliance with requirements for surface smoothness, maximum surface moisture content, and other conditions affecting performance of the Work.
- B. Proceed with application only after substrate construction and penetrating work have been completed and unsatisfactory conditions have been corrected.

3.2 PREPARATION

- A. Substrate must be clean, sound, and free of surface contaminants. Remove dust, laitance, grease, oils, curing compounds, form release agents and all foreign particles by mechanical means. An open-textured, sandpaper-like substrate is ideal. Substrate shall be in accordance with ICRI Guideline No. 03732 for coatings and fall within CSP4. All surfaces must be saturated surface dry (SSD), with no standing water at time of application.

3.3 Mixing and Application

- A. Mixing: Under normal circumstances, full quantities of both components are mixed together, a slurry consistency will result. Mix in a clean container by slowly adding the powder component to the liquid component and mixing with a slow speed (400-600rpm) drill and mixing paddle.
- B. Coating Application: Apply with stiff bristle brush, roller, or spray equipment. Work material into the prepared substrates, filling all pores and voids.

1. For brush grade: Apply first coat, with horizontal brush strokes and leave to harden (2 to 4 hours). Apply second coat with vertical brush strokes.
 2. For spray application: Use a hopper gun spray equipment, textured sprayer (e.g. Texspray E110c by Graco), or a rotor/stator pump equipment. Allow the first coat to harden (2 to 4 hours) prior to the application of the second coat. As soon as the second coat is applied an optional back roller to get desired appearance.
- C. When applying the coating, never stop the application until the entire surface has been coated. Always stop application at an edge, corner, or joint. Never let a previously coated film dry; always coat into a wet film. Always apply the coating at a 45o angle to an edge, corner, or joint.
- D. Adhere to all limitations and cautions for the polymer-modified cement coating in the manufacturer's printed literature.

3.4 Cleaning

- A. The uncured polymer-modified portland cement coating can be cleaned from tools with water. The cured polymermodified portland cement coating can only be removed mechanically.
- B. Leave finished work and work area in a neat, clean condition without evidence of spillovers onto adjacent areas.

END OF SECTION 07 16 00

SECTION 07 21 00 - THERMAL INSULATION

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Mineral-wool blanket insulation.

1.2 ACTION SUBMITTALS

- A. Product Data: For each type of product.
- B. Installer qualifications for over 10 years on commercial installations.
- C. Post installation inspection report and data.

1.3 INFORMATIONAL SUBMITTALS

- A. Installer's Certification: Listing type, manufacturer, and R-value of insulation installed in each element of the building thermal envelope.
 1. Sign, date, and post the certification in a conspicuous location on Project site.
- B. Product Test Reports: For each product, for tests performed by a qualified testing agency.
- C. Research Reports: For foam-plastic insulation, from ICC-ES.

1.4 DELIVERY, STORAGE, AND HANDLING

- A. Protect insulation materials from physical damage and from deterioration due to moisture, soiling, and other sources. Store inside and in a dry location. Comply with manufacturer's written instructions for handling, storing, and protecting during installation.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

- A. Surface-Burning Characteristics Maximum flame-spread and smoke-developed indexes less than Class A, 25 and 450 when tested in accordance with ASTM E84.
- B. Fire-Resistance Ratings: Comply with ASTM E119 or UL 263; testing by a qualified testing agency. Identify products with appropriate markings of applicable testing agency.

1. Indicate design designations from UL's "Fire Resistance Directory" or from listings of another qualified testing agency.
- C. Fire Propagation Characteristics: Passes NFPA 285 testing as part of an approved assembly.
- D. Labeling: Provide identification of mark indicating R-value of each piece of insulation 12 inches and wider in width.
- E. Recycled Content: Postconsumer recycled content plus one-half of preconsumer recycled content not less than 10 percent.

2.2 MINERAL-WOOL BLANKET INSULATION

- A. Mineral-Wool Blanket Insulation, Unfaced: ASTM C665, Type I (blankets without membrane facing); consisting of fibers; passing ASTM E136 for combustion characteristics.
 1. Flame-Spread Index: 0 when tested in accordance with ASTM E84.
 2. Smoke Developed Index: 0 when tested in accordance with ASTM E84.
 3. Density: >2lbs/cu. ft. when tested in accordance to ASTM C167.
 4. Thickness: As required to meet minimum R-values indicated on drawings and to fully fill cavities.

2.3 ACCESSORIES

- A. Miscellaneous Application Accessories:
 1. Detailing Foam Insulation for Voids: Urethane foam complying with AAMA 812, low expansion pressure suitable for filling insulation gaps and voids adjacent to openings to protect against water, air, and sound intrusion.

PART 3 - EXECUTION

3.1 PREPARATION

- A. Clean substrates of substances that are harmful to insulation, including removing projections capable of puncturing insulation or vapor retarders, or those that interfere with insulation attachment.

3.2 INSTALLATION, GENERAL

- A. Comply with insulation manufacturer's written instructions applicable to products, applications and applicable codes.
- B. Install insulation that is undamaged, dry, and unsoiled and that has not been left exposed to ice, rain, or snow at any time.
- C. Extend insulation to envelop entire area to be insulated. Fit tightly around obstructions and fill

voids with insulation. Remove projections that interfere with placement.

- D. Provide sizes to fit applications and selected from manufacturer's standard thicknesses, widths, and lengths. Apply single layer of insulation units unless multiple layers are otherwise shown.

3.3 INSTALLATION OF INSULATION IN FRAMED CONSTRUCTION

- A. Blanket Insulation: Install in cavities formed by framing members in accordance with the following requirements:
 - 1. Use insulation widths and lengths that fill the cavities formed by framing members. If more than one length is required to fill the cavities, provide lengths that will produce a snug fit between ends.
 - 2. Place insulation in cavities formed by framing members to produce a friction fit between edges of insulation and adjoining framing members.
 - 3. For wood-framed construction, install blankets in accordance with ASTM C1320 and as follows:
 - a. With faced blankets having stapling flanges, lap blanket flange over flange of adjacent blanket to maintain continuity of vapor retarder once finish material is installed over it.
- B. Miscellaneous Voids: Install insulation in miscellaneous voids and cavity spaces where required to prevent gaps in insulation using the following materials:
 - 1. Glass-Fiber Insulation: Compact to approximately 40 percent of normal maximum volume equaling a density of approximately 2.5 lb/cu. ft.
 - 2. Detailing Foam Insulation for Voids: Apply in accordance with manufacturer's written instructions.

3.4 PROTECTION

- A. Protect installed insulation from damage due to harmful weather exposures, physical abuse, and other causes.
- B. Provide temporary coverings or enclosures where insulation is subject to abuse and cannot be concealed and protected by permanent construction immediately after installation.

END OF SECTION 07 21 00

SECTION 07 26 00 - VAPOR RETARDERS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Fire-retardant, reinforced-polyethylene vapor retarders.

B. Related Requirements:

1. Section 03 30 00 "Cast-in-Place Concrete"

1.2 ACTION SUBMITTALS

A. Product Data:

1. Fire-retardant, reinforced-polyethylene vapor retarders.

1.3 INFORMATIONAL SUBMITTALS

- ##### A. Product Test Reports:
- For each product, for tests performed by a qualified testing agency.

PART 2 - PRODUCTS

2.1 POLYETHYLENE VAPOR RETARDERS

- ##### A. Polyethylene Vapor Retarders:
- ASTM D4397, 10-mil- thick sheet, with maximum permeance rating of 0.1 perm.

2.2 ACCESSORIES

- ##### A. Vapor-Retarder Tape:
- Pressure-sensitive tape of type recommended by vapor-retarder manufacturer for sealing joints and penetrations in vapor retarder.
- ##### B. Adhesive for Vapor Retarders:
- Product recommended by vapor-retarder manufacturer and has demonstrated capability to bond vapor retarders securely to substrates indicated.
- ##### C. Vapor-Retarder Fasteners:
- Pancake-head, self-tapping steel drill screws; with fender washers.

PART 3 - EXECUTION

3.1 PREPARATION

- A. Clean substrates of substances that are harmful to vapor retarders, including removing projections capable of puncturing vapor retarders.

3.2 INSTALLATION OF VAPOR RETARDERS IN CRAWL SPACES

- A. Install vapor retarders over prepared grade. Lap joints a minimum of 12 inches and seal with manufacturer's recommended tape. Install second layer over pathways to equipment.
- B. Extend vapor retarder over footings and seal to foundation wall or grade beam with manufacturer's recommended tape.
 - 1. Extend vapor retarder vertically minimum 16 inches above top of footing.
- C. Seal around penetrations such as utilities and columns in order to create a monolithic, airtight membrane at grade surface, perimeter, and all vertical penetrations.

3.3 PROTECTION

- A. Protect vapor retarders from damage until concealed by permanent construction.

END OF SECTION 07 26 00

SECTION 07 62 00 - SHEET METAL FLASHING AND TRIM

PART 1 - GENERAL

1.1 SUMMARY

- A. Section Includes: Custom flashing and trim fabrications, made from the following:
 - 1. Sheet metal materials.
- B. Related Requirements:
 - 1. Section 06 10 00 "Rough Carpentry" for wood nailers, curbs, and blocking.

1.2 ACTION SUBMITTALS

- A. Product Data: For each type of product.
- B. Shop Drawings: For sheet metal flashing and trim.
 - 1. Plans, elevations, sections, and attachment details.
 - 2. Fabrication and installation layouts, expansion-joint locations, and keyed details. Distinguish between shop- and field-assembled Work.
 - 3. Identification of material, thickness, weight, and finish for each item and location in Project.
 - 4. Details for forming, including profiles, shapes, seams, and dimensions.
 - 5. Details for joining, supporting, and securing, including layout and spacing of fasteners, cleats, clips, and other attachments. Include pattern of seams.
 - 6. Details of termination points and assemblies.
 - 7. Details of special conditions.
 - 8. Details of connections to adjoining work.
 - 9. Formed flashing and trim at scale of not less than 1-1/2 inches per 12 inches.
- C. Samples: For each exposed product and for each color and texture specified, 12 inches long by actual width.
- D. Samples for Initial Selection: Manufacturer's standard color sheets, showing full range of available colors for each type of exposed finish.
- E. Samples for Verification: Actual sample of finished products for each type of exposed finish for sheet metal and other metal accessories.
 - 1. Sheet Metal Flashing and Trim: Manufacturers' standard size . Include finished seam with required profile. Include fasteners, cleats, clips, closures, and other attachments.
 - 2. Anodized Aluminum Samples: Samples to show full range to be expected for each color required.

1.3 INFORMATIONAL SUBMITTALS

- A. Product Test Reports: For each product, for tests performed by a qualified testing agency.
- B. Qualification Statements: For fabricator.
- C. Sample warranties.

1.4 CLOSEOUT SUBMITTALS

- A. Maintenance Data: For sheet metal flashing and trim, and its accessories.

1.5 QUALITY ASSURANCE

- A. Fabricator Qualifications: Entity that employs skilled workers who custom fabricate sheet metal flashing and trim similar to that required for this Project and whose products have a record of successful in-service performance.
- B. Installer Qualifications: Entity that employs a supervisor who is an NRCA ProCertified Foreman or installers who are NRCA ProCertified Architectural Metal Flashings and Accessories Installers.

1.6 DELIVERY, STORAGE, AND HANDLING

- A. Do not store sheet metal flashing and trim materials in contact with other materials that might cause staining, denting, or other surface damage.
 - 1. Store sheet metal flashing and trim materials away from uncured concrete and masonry.
 - 2. Protect stored sheet metal flashing and trim from contact with water.
- B. Protect strippable protective covering on sheet metal flashing and trim from exposure to sunlight and high humidity, except to extent necessary for period of sheet metal flashing and trim installation.

1.7 COORDINATION

- A. Coordinate sheet metal flashing and trim layout and seams with sizes and locations of penetrations to be flashed, and joints and seams in adjacent materials.
- B. Coordinate sheet metal flashing and trim installation with adjoining wall materials, joints, and seams to provide leakproof, secure, and noncorrosive installation.

1.8 WARRANTY

- A. Special Warranty on Finishes: Manufacturer agrees to repair finish or replace sheet metal flashing and trim that shows evidence of deterioration of factory-applied finishes within specified warranty period.

1. Exposed Panel Finish: Deterioration includes, but is not limited to, the following:
 - a. Color fading more than 5 Delta E units when tested in accordance with ASTM D2244.
 - b. Chalking in excess of a No. 8 rating when tested in accordance with ASTM D4214.
 - c. Cracking, checking, peeling, or failure of paint to adhere to bare metal.
2. Finish Warranty Period: 20 years from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

- A. Sheet metal flashing and trim assemblies, including cleats, anchors, and fasteners, are to withstand wind loads, structural movement, thermally induced movement, and exposure to weather without failure due to defective manufacture, fabrication, installation, or other defects in construction. Completed sheet metal flashing and trim are not to rattle, leak, or loosen, and are to remain watertight.
- B. Sheet Metal Standard for Flashing and Trim: Comply with SMACNA's "Architectural Sheet Metal Manual" requirements for dimensions and profiles shown unless more stringent requirements are indicated.
- C. Thermal Movements: Allow for thermal movements from ambient and surface temperature changes.
 1. Temperature Change: 120 deg F, ambient; 180 deg F, material surfaces .

2.2 SHEET METAL MATERIALS

- A. Protect mechanical and other finishes on exposed surfaces from damage by applying strippable, temporary protective film before shipping.
- B. Metallic-Coated Steel Sheet: Zinc-coated (galvanized) steel sheet complying with minimum ASTM A653/A653M, G90 coating designation, or aluminum-zinc alloy-coated steel sheet complying with minimum ASTM A792/A792M, Class AZ50 coating designation; structural quality. Prepainted by the coil-coating process to comply with ASTM A755/A755M.
 1. Nominal Thickness: 0.022 inch.
 2. Surface: Smooth, flat.
 3. Color: As selected by Contracting Officer from manufacturer's full range.
 - a. Basis of Design Color: to match color of metal roofing
 4. Concealed Finish: Pretreat with manufacturer's standard white or light-colored acrylic or polyester backer finish, consisting of prime coat and wash coat with minimum total dry film thickness of 0.5 mil.

2.3 MISCELLANEOUS MATERIALS

- A. Provide materials and types of fasteners , protective coatings, sealants, and other miscellaneous items as required for complete sheet metal flashing and trim installation and as recommended by manufacturer of primary sheet metal or manufactured item unless otherwise indicated.
- B. Fasteners: Wood screws, annular threaded nails, self-tapping screws, self-locking rivets and bolts, and other suitable fasteners designed to withstand design loads and recommended by manufacturer of primary sheet metal or manufactured item .
 - 1. General: Blind fasteners or self-drilling screws, gasketed, with hex-washer head.
 - a. Blind Fasteners: High-strength aluminum or stainless steel rivets suitable for metal being fastened.
 - b. Spikes and Ferrules: Same material as gutter; with spike with ferrule matching internal gutter width.
 - 2. Fasteners for Zinc-Coated (Galvanized) or Aluminum-Zinc Alloy-Coated Steel Sheet: Series 300 stainless steel or hot-dip galvanized steel in accordance with ASTM A153/A153M or ASTM F2329/F2329M.
- C. Sealant Tape: Pressure-sensitive, 100 percent solids, polyisobutylene compound sealant tape with release-paper backing. Provide permanently elastic, nonsag, nontoxic, nonstaining tape 1/2 inch wide and 1/8 inch thick.
- D. Elastomeric Sealant: ASTM C920, elastomeric silicone polymer sealant; of type, grade, class, and use classifications required to seal joints in sheet metal flashing and trim and remain watertight.
- E. Butyl Sealant: ASTM C1311, single-component, solvent-release butyl rubber sealant; polyisobutylene plasticized; heavy bodied for hooked-type expansion joints with limited movement.
- F. Epoxy Seam Sealer: Two-part, noncorrosive, aluminum seam-cementing compound, recommended by aluminum manufacturer for exterior nonmoving joints, including riveted joints.

2.4 FABRICATION, GENERAL

- A. Custom fabricate sheet metal flashing and trim to comply with details indicated and recommendations in cited sheet metal standard that apply to design, dimensions, geometry, metal thickness, and other characteristics of item required.
 - 1. Fabricate sheet metal flashing and trim in shop to greatest extent possible.
 - 2. Fabricate sheet metal flashing and trim in thickness or weight needed to comply with performance requirements, but not less than that specified for each application and metal.
 - 3. Verify shapes and dimensions of surfaces to be covered and obtain field measurements for accurate fit before shop fabrication.
 - 4. Form sheet metal flashing and trim to fit substrates without excessive oil-canning, buckling, and tool marks; true to line, levels, and slopes; and with exposed edges folded

- back to form hems.
5. Conceal fasteners and expansion provisions where possible. Do not use exposed fasteners on faces exposed to view.
- B. Fabrication Tolerances:
1. Fabricate sheet metal flashing and trim that is capable of installation to a tolerance of 1/4 inch in 20 ft. on slope and location lines indicated on Drawings and within 1/8-inch offset of adjoining faces and of alignment of matching profiles.
 2. Fabricate sheet metal flashing and trim that is capable of installation to tolerances specified.
- C. Expansion Provisions: Form metal for thermal expansion of exposed flashing and trim.
1. Form expansion joints of intermeshing hooked flanges, not less than 1 inch deep, filled with butyl sealant concealed within joints.
 2. Use lapped expansion joints only where indicated on Drawings.
- D. Sealant Joints: Where movable, nonexpansion-type joints are required, form metal in accordance with cited sheet metal standard to provide for proper installation of elastomeric sealant.
- E. Fabricate cleats and attachment devices from same material as accessory being anchored or from compatible, noncorrosive metal.
- F. Fabricate cleats and attachment devices of sizes as recommended by cited sheet metal standard for application, but not less than thickness of metal being secured.
- G. Seams:
1. Fabricate nonmoving seams with flat-lock seams. Tin edges to be seamed, form seams, and solder.
 2. Fabricate nonmoving seams with flat-lock seams. Form seams and seal with elastomeric sealant unless otherwise recommended by sealant manufacturer for intended use.
- H. Do not use graphite pencils to mark metal surfaces.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine substrates, areas, and conditions, with Installer present, for compliance with requirements for installation tolerances, substrates, and other conditions affecting performance of the Work.
1. Verify compliance with requirements for installation tolerances of substrates.
 2. Verify that substrate is sound, dry, smooth, clean, sloped for drainage, and securely anchored.
 3. Verify that air- or water-resistant barriers have been installed over substrate to prevent air infiltration or water penetration.

- B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 INSTALLATION OF SHEET METAL FLASHING AND TRIM, GENERAL

- A. Install sheet metal flashing and trim to comply with details indicated and recommendations of cited sheet metal standard that apply to installation characteristics required unless otherwise indicated on Drawings.
 - 1. Install fasteners , protective coatings, separators, sealants, and other miscellaneous items as required to complete sheet metal flashing and trim system.
 - 2. Install sheet metal flashing and trim true to line, levels, and slopes. Provide uniform, neat seams with minimum exposure of sealant .
 - 3. Anchor sheet metal flashing and trim and other components of the Work securely in place, with provisions for thermal and structural movement.
 - 4. Install sheet metal flashing and trim to fit substrates and to result in watertight performance.
 - 5. Install continuous cleats with fasteners spaced not more than 12 inches o.c.
 - 6. Space individual cleats not more than 12 inches apart. Attach each cleat with at least two fasteners. Bend tabs over fasteners.
 - 7. Install exposed sheet metal flashing and trim with limited oil-canning, and free of buckling and tool marks.
 - 8. Do not field cut sheet metal flashing and trim by torch.
 - 9. Do not use graphite pencils to mark metal surfaces.
- B. Metal Protection: Where dissimilar metals contact each other, or where metal contacts pressure-treated wood or other corrosive substrates, protect against galvanic action or corrosion by painting contact surfaces with bituminous coating or by other permanent separation as recommended by sheet metal manufacturer or cited sheet metal standard.
 - 1. Coat concealed side of stainless steel sheet metal flashing and trim with bituminous coating where flashing and trim contact wood, ferrous metal, or cementitious construction.
- C. Expansion Provisions: Provide for thermal expansion of exposed flashing and trim.
 - 1. Space movement joints at maximum of 10 ft. with no joints within 24 inches of corner or intersection.
 - 2. Form expansion joints of intermeshing hooked flanges, not less than 1 inch deep, filled with sealant concealed within joints.
 - 3. Use lapped expansion joints only where indicated on Drawings.
- D. Fasteners: Use fastener sizes that penetrate substrate not less than recommended by fastener manufacturer to achieve maximum pull-out resistance .
- E. Conceal fasteners and expansion provisions where possible in exposed work and locate to minimize possibility of leakage. Cover and seal fasteners and anchors as required for a tight installation.
- F. Seal joints as required for watertight construction.

1. Use sealant-filled joints unless otherwise indicated.
 - a. Embed hooked flanges of joint members not less than 1 inch into sealant.
 - b. Form joints to completely conceal sealant.
 - c. When ambient temperature at time of installation is between 40 and 70 deg F, set joint members for 50 percent movement each way.
 - d. Adjust setting proportionately for installation at higher ambient temperatures.
 - 1) Do not install sealant-type joints at temperatures below 40 deg F.

3.3 INSTALLATION TOLERANCES

- A. Shim and align sheet metal flashing and trim within installed tolerance of 1/4 inch in 20 ft. on slope and location lines indicated on Drawings and within 1/8-inch offset of adjoining faces and of alignment of matching profiles.

3.4 CLEANING

- A. Clean and neutralize flux materials. Clean off excess solder.
- B. Clean off excess sealants.

3.5 PROTECTION

- A. Remove temporary protective coverings and strippable films as sheet metal flashing and trim are installed unless otherwise indicated in manufacturer's written installation instructions.
- B. On completion of sheet metal flashing and trim installation, remove unused materials and clean finished surfaces as recommended in writing by sheet metal flashing and trim manufacturer.
- C. Maintain sheet metal flashing and trim in clean condition during construction.
- D. Replace sheet metal flashing and trim that have been damaged or that have deteriorated beyond successful repair by finish touchup or similar minor repair procedures, as determined by Contracting Officer.

END OF SECTION 07 62 00

SECTION 08 95 16 - WALL VENTS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Air vents in concrete foundation walls with screening.
2. Custom air vent in wood framed wall for attic ventilation.
3. Metal screening at concrete foundation.

1.2 ACTION SUBMITTALS

- A. Product Data: For each type of product.
- B. Samples: For each type of metal finish required.

1.3 INFORMATIONAL SUBMITTALS

- A. Evaluation Reports: For flood vents, from ICC-ES.
- B. Sample Warranties: For manufacturer's special warranties.

1.4 WARRANTY

- A. Special Finish Warranty, Factory-Applied Finishes: Standard form in which manufacturer agrees to repair finishes or replace aluminum that shows evidence of deterioration of baked enamel, powder coat, or organic finishes within specified warranty period.
1. Deterioration includes, but is not limited to, the following:
 - a. Color fading more than 5 Delta E units when tested in accordance with ASTM D2244.
 - b. Chalking in excess of a No. 8 rating when tested in accordance with ASTM D4214.
 - c. Cracking, checking, peeling, or failure of paint to adhere to bare metal.
 2. Warranty Period: 20 years from date of Substantial Completion.
- B. Special Finish Warranty, Anodized Finishes: Standard form in which manufacturer agrees to repair finishes or replace aluminum that shows evidence of deterioration of anodized finishes within specified warranty period.
1. Deterioration includes, but is not limited to, the following:

- a. Color fading more than 5 Delta E units when tested in accordance with ASTM D2244.
 - b. Chalking in excess of a No. 8 rating when tested in accordance with ASTM D4214.
 - c. Cracking, peeling, or chipping.
2. Warranty Period: 10 years from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 SOURCE LIMITATIONS

- A. Obtain vents from single source from single manufacturer approved for Wildland Urban Interface.

2.2 FOUNDATION VENT

- A. Frame and Flange: G90 Galvanized Steel, 26 gauge.
 1. Free area required: 93 sq. in.
 2. Size and location as indicated on drawings.
 3. Flange front
- B. Intumescent Core
 1. 5mm hexagonal matrix
 2. Aluminum with intumescent coating
- C. Ember Mesh
 1. 304 Stainless steel, faces interior
 2. Self-closing mesh, effectively blocking fire and ember intrusion.
 3. Passes ASTM E2886 to block ember entry.
- D. Insect and Rodent Screen
 1. 1/8" galvanized screen, faces exterior

2.3 CUSTOM ATTIC VENT

- A. Custom-fabricated attic ventilation assembly, field-fabricated by Contractor, consisting of the following:
 1. Frame:
 - a. Wood frame sized to opening indicated on drawings.
 - b. Species and grade suitable for exterior exposure or protected by exterior cladding.
 - c. Frame depth coordinated with wall construction.
 2. Hinged Access Panel:

- a. Continuous stainless steel or galvanized steel piano hinge.
 - b. Hinge sized for full width of vent opening.
 - c. Hinges concealed where possible.
- 3. Screening / Ember Protection:
 - a. WUI-compliant ember-resistant mesh.
 - b. Maximum opening size as required to pass ASTM E2886 for ember intrusion resistance.
 - c. Noncombustible material, corrosion resistant.
 - d. Mesh securely fastened to frame to prevent displacement under wind or pressure.
- 4. Operation:
 - a. Panel shall be operable for inspection and maintenance.
 - b. Provide secure closure method to prevent unintended opening.
- 5. Finish:
 - a. Exposed metal components to be corrosion resistant.
 - b. Wood components to receive paint finish to match adjacent construction.

2.4 CONCRETE FOUNDATION SCREEN

- A. Non-proprietary screening assembly for concrete foundation wall opening, consisting of the following:
 - 1. Screening:
 - a. WUI-compliant ember-resistant metal mesh.
 - b. Corrosion-resistant material.
 - c. Mesh size compliant with ember intrusion resistance requirements.
 - 2. Support and Attachment:
 - a. Continuous metal angle frame at perimeter of opening.
 - b. Galvanized steel or stainless steel angles sized to securely retain mesh.
 - c. Angles mechanically fastened to concrete substrate.
 - 3. Performance Requirements:
 - a. Assembly shall resist displacement due to wind, debris impact, or thermal movement.
 - b. Assembly shall maintain required free area indicated on drawings.

PART 3 - EXECUTION

3.1 INSTALLATION

A. FOUNDATION VENTS

- 1. Install according to manufacturer's written instructions.
- 2. Locate and place vents level, plumb, and at indicated alignment with adjacent work.
- 3. Prior to installing the vent, run a bead of fire caulking behind the nailing flange to ensure an air tight seal where the vent meets the building material.
- 4. Attach vents securely in place using fasteners supplied or approved by manufacturer.
- 5. Protect unpainted surfaces that are in contact with concrete, masonry, or dissimilar metals from corrosion and galvanic action by applying a heavy coating of bituminous paint or by separating surfaces with waterproof gaskets or nonmetallic flashing.

B. CUSTOM ATTIC VENT

1. Install vent assemblies plumb, level, and square.
2. Coordinate fabrication dimensions with field-verified opening sizes.
3. Secure wood frame to framing substrate with corrosion-resistant fasteners.
4. Install WUI mesh tightly and continuously within frame; no gaps permitted.
5. Install piano hinges true and aligned to allow smooth operation.
6. Seal perimeter between vent frame and wall construction with fire-resistant sealant to provide an airtight assembly.
7. Verify operable panel opens and closes freely after installation.

C. CONCRETE FOUNDATION SCREEN

1. Install mesh centered and taut within metal angle frame.
2. Fasten metal angles securely to concrete using corrosion-resistant anchors.
3. Provide continuous bearing of mesh at all edges.
4. Seal perimeter of angle frame to concrete as required to prevent air and ember infiltration.
5. Protect dissimilar metals from galvanic action.

3.2 ADJUSTING AND CLEANING

- A. Before final inspection, clean exposed surfaces with water and a mild soap or detergent not harmful to finishes. Thoroughly rinse surfaces and dry.
- B. Restore vents damaged during installation and construction, so no evidence remains of corrective work. If results of restoration are unsuccessful, as determined by Contracting Officer, remove damaged units and replace with new units.

END OF SECTION 08 95 16

SECTION 10 81 19 – RODENT AND BIRD CONTROL

PART 1 - GENERAL

1.1 DESCRIPTION

- A. This Section consists of abating health hazards associated with rodents and birds and furnishing, installing, and maintaining barriers to protect from rodent/bird entry or intrusions during planned historic rehabilitation of the Bear Valley School at Pinnacles National Park, CA (PINN), (Project Site). PMIS: 331043
- B. The Project Site had significant evidence of rodent, bird, and insect infestation, including observations of rodent droppings, bird nests, and insect nests, per Landmark's October 17, 2025, Hazardous Materials Assessment Summary Letter for the Project Site. Any rodent or bird intrusion removal and exclusion work should be performed by a licensed and insured pest control contractor.
- C. All removal of rodent feces, urine, and nesting materials must be conducted in accordance with the National Park Service's (NPS) *Hantavirus Risk Reduction Worker Protection* guidelines and requirements.
- D. All removal of bird feces, urine, and nesting materials must be conducted in accordance with the *Histoplasmosis Protecting Workers from Risk* guidance document produced by the National Institute for Occupational Safety and Health (NIOSH).
- E. Locations on the exterior building envelope that are open and allow rodent and bird species to enter the interior of the Project Site will require treatment to exclude these animals. This project will also require disinfecting surfaces contaminated with rodent feces and bird nesting areas before work that disturbs these areas begins.
- F. Items to be protected from animal entry include, but are not limited to, interior ceilings, base trim, window and door trim, walls, floors, and other materials installed or replaced according to project requirements.
- G. Neoprene seals, spray-in-place foam, and similar products commonly used to close openings are not rodent-proof. Even in new buildings, utility pipes, electrical conduit (often at meters or circuit breaker panels), water and gas lines, and communication cables may have large openings that permit the entry of rodents and birds. Once animals have entered walls, they generally have ready access to much of a building via holes for utility pipes and wires in the framing, overhead ceilings, beneath raised floors, or other types of construction adjacent to utility enclosures.
- H. The Contractor will maintain a clean and debris-free Project Site. If the areas are known to have serious rodent infiltration problems, the Contractor will use exterior surface materials that have a

hard, smooth surface on at least the lower levels that will be subject to contact with the ground surface, ornamental plantings, fences, and other potential areas of rodent harborage.

- I. The Contractor will tightly rodent and bird-proof any electrical penetrations associated with solar panels that may be installed for this building and openings associated with other potential planned utility alterations.
- J. During construction, the Contractor will implement measures to keep the job site free of rubbish, foodstuff, and other items as necessary to avoid attracting animals and will trap/bait rodents as necessary to control rodent intrusion.

1.2 REFERENCE DOCUMENTATION

- A. All work will also comply with the following:
 - 1. National Park Service, *Rodent Exclusion Manual*, Published by the U.S. Department of the Interior, National Park Service, Natural Resource Stewardship and Science: Fort Collins, CO, September 2014.
 - 2. National Park Service, *Hantavirus Risk Reduction Worker Protection* – Revised Edition, December 2013.
 - 3. National Park Service, *Rodent Exclusion Manual*, September 2014.
 - 4. National Institute for Occupational Safety and Health, *Histoplasmosis: Protecting Workers at Risk* – Revised Edition, December 2004.

1.3 FIELD VERIFICATION

- A. Field-verify all new and existing dimensions affecting the work of this contract before ordering products.

1.4 CONSTRUCTION MATERIALS

- A. The Government will not furnish any materials for this project.

1.5 PROJECT CONDITIONS

- A. Do not damage any existing building materials not associated with the work or work area. Any damage to building materials not protected in or outside the work areas and not associated with the work will be repaired or replaced by the Contractor at no cost to the Government.
- B. The Government will supply electricity and water service to the Project Site. The Contractor will provide the necessary temporary utility connections and the water source needed to complete the scope of work. If any generators are provided by the Contractor for temporary electric service, a written Spill Prevention, Control, and Countermeasure (SPCC) plan and equipment will also be

required. The Contractor will be required to operate generators at noise levels that do not exceed applicable regulatory limits.

PART 2 - PRODUCTS

2.1 PERSONAL PROTECTIVE EQUIPMENT

A. Respiratory Protection:

1. Workers who abate rodent droppings and/or bird nesting material wastes will be provided with personally issued and individually identified respirators that meet the required level of protection. During the pre-cleaning, installation of critical barriers, and High-Efficiency Particulate Air (HEPA) vacuuming activities, the Contractor will utilize (at a minimum) half-face air-purifying respirators with approved HEPA filter cartridges. At a minimum, the Contractor will also utilize the half-face air-purifying respirators with approved HEPA-filter cartridges for the removal and cleanup work. The respiratory cartridges used will be, at a minimum, N-100 or P-100 filters.
2. The Contractor will provide workers with, and require the use of, respirators approved by the NIOSH in accordance with the OSHA Standard 29 Code of Federal Regulations (CFR) 1926.1101. Minimum respiratory protection will be an approved half-mask air-purifying respirator with HEPA cartridges. Disposable single-use respirators will not be permitted. The initial respiratory protection provided will be chosen prior to the commencement of removal operations based on data from past exposure assessment monitoring or the initial monitoring performed to accurately determine the airborne concentration of dust/fibers to which employees may be exposed.
3. Workers must perform positive and negative air pressure fit tests each time a respirator is donned and whenever the respirator design permits. Powered air-purifying respirators will be tested for adequate flow as specified by the manufacturer.
4. Workers will be given a qualitative fit test in accordance with procedures detailed in the OSHA 29 CFR 1926.1101 requirements for all respirators to be used. An appropriately administered quantitative fit test may be substituted for a qualitative fit test. Documentation of adequate respirator fit must be provided to the Contractor. Fit tests will be administered annually.

B. Personal Protection Required for Rodent Removal/Rodent-Contaminated Material Handling:

1. Prior to the commencement of cleaning of animal-infested areas, all personnel who will be required to enter the work areas for handling animal-infested and contaminated materials will have hazard awareness training. Special onsite training on equipment and procedures unique to this job site will be performed as required. Training in emergency response and evacuation procedures will also be provided.
2. All respiratory protection will be provided to the workers through a written respiratory protection program. This program will be available at the Project Site.
3. Disposable clothing, including head, foot, and full body protection, will be provided in sufficient quantities and in adequate sizes for all workers and authorized visitors. All

personnel engaged in the animal removal work will wear approved disposable protective clothing constructed of spun-bonded olefin, polypropylene fabrics, or other material of equivalent resistance to penetration. A full-body suit is recommended instead of a separate set of coveralls, head covers, and shoe covers. Disposable whole-body clothing, including head covers, gloves, and shoe coverings, will be provided to and worn by all personnel in the abatement work area. If elastic sleeve closures are not provided, sleeves will be secured with duct tape to the gloves. Washable footwear having a non-skid tracking surface will be provided and used by all personnel within the abatement work area.

4. Persons with facial hair that may interfere with the respirator's seal will not be allowed to enter the work area.
5. Respirators will be worn at all times if animal-contaminated materials are impacted to the extent that airborne emissions are created.
6. Protective eyewear, gloves, rubber boots, and/or other footwear will be provided as required for workers and authorized visitors. Safety shoes may be required for some activities.
7. Eating, drinking, smoking, and chewing gum or tobacco will not be allowed in the work areas.

2.2 EQUIPMENT

- A. The materials utilized may be new or used, but will be suitable for the intended purpose. Rodent exclusion barriers will be structurally adequate, neat in appearance, and consistent with historical preservation or the historical fabric on or within the structure.

2.3 SCREENS AND BARRIERS

- A. Use only metal window screening materials where windows or doors are accessible to animals. Avoid unnecessary ledges outside of the windows. When necessary, screen ventilation openings and windows with a woven/welded galvanized hardware cloth. Such screening work is critical in a building where high animal pressures are found. For large openings or where the screen may be subject to abuse, add crossbars to support the hardware cloth. If the opening is an access route, install the screen on a hinged frame.

2.4 DISINFECTING SOLUTIONS

- A. Two types of disinfecting solutions are recommended for the cleanup of animal-contaminated materials:
 1. General-Purpose Household Disinfectant: Prepare according to the label if not prediluted. Almost any agent commercially available in the U.S. is sufficient as long as the label states that it is a disinfectant product. Effective agents include those based on phenols, quaternary ammonium compounds, and hypochlorite.

2. Hypochlorite Solution: A chlorine solution that is freshly prepared by mixing 1½ cups of household bleach in one gallon of water (or a 1:10 solution) can be used in place of a commercial disinfectant product. When using a chlorine solution, avoid spilling the mixture on clothing or other items that might be damaged by bleach. Wear rubber, latex, vinyl, or nitrile gloves when preparing and using chlorine solutions. Chlorine solutions should be prepared fresh daily.

PART 3 - EXECUTION

3.1 PROTECTION OF STRUCTURES

- A. Rodents and birds seek shelter behind, under, or in appliances, sinks, cabinets, drawers, stored goods, wall voids, false ceilings, and other undisturbed areas.
- B. Rodents can climb the outside of vertical pipes and conduits up to 3 inches (7.6 cm) in diameter and the outsides of larger pipes attached to buildings by bracing themselves between the wall and pipe.
- C. Rodents can also climb the inside of vertical pipes, wall voids, or structural/expansion seams and joints between 1.5 and 4 inches (3.8 and 10.2 cm) in diameter. All building installations must include systems that prevent these intrusions.
- D. Junctures, where utilities (i.e., pipes, cables, etc.) enter the structure, require special consideration in preventing rodent entry.
- E. Exclusion must include no exterior openings higher or wider than one-quarter inch.
- F. Close openings around pipes and electric cables where they enter structures with Portland cement mortar, Concrete Patch®, masonry, or metal collars.

3.2 RODENT REMOVAL

- A. Completely remove rodents when approved by the Contracting Officer. Do not use poisons due to the potential for negative environmental or health effects.
- B. The Federal Migratory Bird Act from the U.S. Fish & Wildlife Service forbids Contractors from moving or destroying a nest that has eggs, chicks, or young birds in it. If live birds and eggs are present at the Project Site, the Contracting Officer needs to be informed, and the Contractor must seek approval from the NPS to cautiously work around any live animals and their offspring/eggs.

3.3 CLEANUP OF ANIMAL URINE, DROPPINGS, AND CONTAMINATED SURFACES

- A. Prior to cleaning areas such as sheds, outbuildings, or attics, ventilate the area by opening doors or windows to the outdoors for at least 30 minutes. Provide cross ventilation where possible.
- B. Utilize appropriate Personal Protective Equipment (PPE).
- C. Spray all animal urine and the droppings with a disinfectant or chlorine solution until thoroughly soaked.
- D. To avoid generating potentially infectious aerosols, do not vacuum or sweep the animal urine, droppings, or contaminated surfaces until they have been disinfected.
- E. Use a paper towel to absorb the animal urine and pick up the droppings. Place the paper towel in the garbage.
- F. After animal droppings and urine have been removed, disinfect items that might have been contaminated by animals or their urine and droppings.
 - 1. Mop floors with a disinfectant or chlorine solution.
 - 2. Disinfect countertops, cabinets, drawers, and other durable surfaces with a disinfectant or chlorine solution.
 - 3. Disinfect carpets with a disinfectant or commercial-grade steam cleaner or shampoo.
 - 4. Steam-clean or shampoo rugs and upholstered furniture.
 - 5. Leave books, papers, and other items that cannot be thrown away or cleaned with a liquid disinfectant outdoors in the sunlight for several hours or in an indoor area free of rodents for approximately one week before cleanup. After that time, the virus should no longer be infectious. However, to further reduce risk, wear rubber, latex, vinyl, or nitrile gloves and wipe the items with a cloth moistened with disinfectant.

3.4 CLEAN UP OF DEAD ANIMALS AND NESTS

- A. Utilize appropriate PPE.
- B. Use insect repellent (containing DEET) on clothing, socks, and arms to reduce the risk of fleabites that might transmit the plague.
- C. Spray dead animals and nests with a disinfectant or a chlorine solution, soaking them thoroughly. Wait 10 minutes before disturbing to ensure the inactivation of the virus.
- D. Place the dead animal or nest in a plastic bag or remove the dead rodent from the trap and place it within a plastic bag. When the cleanup is complete (or when the bag is full), seal the bag, place it into a second plastic bag, and seal the second bag. Dispose of the material in the double bag by burning it or discarding it in a covered trash can that is regularly emptied.

- E. Clean up the surrounding area(s) as described in Section 3.3 - Cleanup of Animal Urine and Droppings and Contaminated Surfaces above.

3.5 VERIFICATION OF RODENT EXCLUSION ACTIVITIES

- A. The Contractor will procure the services of a third-party rodent and bird exclusion specialist to verify the animal exclusion activities.
- B. The selected third-party specialist will document the measures taken to prevent animals from entering the Project Site.

END OF SECTION 10 81 19

SECTION 32 40 00 – TREE AND VEGETATION RETENTION AND PROTECTION

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Work under this section includes protecting existing trees and vegetation to remain, furnishing, installing, and maintaining tree protection fencing in a vertical and effective position to protect existing trees and vegetation for the entire duration of construction, and practices to protect existing trees and vegetation within the tree protection zones.

1.2 DEFINITIONS

- A. **Drip Line:** The outermost edge of the tree's canopy or branch spread. The area within a tree's dripline is all the ground under the total branch spread.
- B. **Critical Root Zone:** Generally, all of the ground area included in the dripline and approximately one foot per inch of tree at a minimum of a 6' diameter from trunk, whichever is greater.
- C. **Diameter (Caliper):** The size (in inches) of a tree's trunk is measured at:
1. Six (6) inches above grade for trunk diameters up to and including four (4) inches.
 2. Twelve (12) inches above grade for trunk diameters from four (4) inches up to and including eight (8) inches.
 3. Four and a half (4½) feet above grade for trunk diameters greater than eight (8) inches.
- D. **Exploratory Excavation:** A method approved by the CO to determine the amount and size of tree roots under the surface. Methods may include hand excavation, a pneumatic air excavation tool, or water excavation.
- E. **Water Excavation:** Hydrovac excavation that utilizes high-pressure water to cut and liquefy the soil, while simultaneously using a vacuum to remove the soil from the excavation and will require that import fill is brought to the site to fill trenches after excavation.
- F. **Pneumatic Air Excavation Tool:** A tool specifically designed to excavate soil from around tree roots. Air Spade or Air Knife are examples of pneumatic air excavation tools.
- G. **Tree Protection Work Plan:** The tree protection plan is a plan based on the contract drawings that includes the Contractor's approach to working within and around the Tree Protection Zones.
- H. **Tree and Vegetation Protection Zone (TPZ):** The Tree and Vegetation Protection Zone is the area above and below grade around each tree where construction activities are limited or restricted to preserve tree health and structural integrity of the protected trees, and areas where existing vegetation is to be protected per park and CO direction.
1. The diameter of the tree shall be measured at four and one-half feet (4.5') above grade (referred to as diameter breast height). For every-one inch (1") of tree diameter the Tree

Protection Zone shall extend one foot (1.0') radially from the base of the trunk or be placed at the dripline, whichever is **greater**.

2. For areas with groups or groupings of trees, if the distance between trees is less than thirty feet (30'), the Tree Protection Zone may be combined and treated as one contiguous area to create a more clearly defined and manageable Tree Protection Zone.
3. Fencing for trees not immediately adjacent (within two feet of) to construction activities, but for which construction or staging activities will occur within the TPZ, orange construction fencing will be installed. Fencing will only be installed around parts of the tree trunk that face construction activities and are vulnerable to damage. Construction fencing will be placed at the greatest possible distance of the following from trees a) TPZ edge, b) TPZ edge, or c) [least desirable] construction/staging zone limits (>2 ft from trunk).
4. Trunk protection: For trees immediately adjacent to the work zone (< 2 feet) and or which fencing is not practicable, trunk protection may be used consisting lumber, loosely banded onto the trees with straps, and angled so that the trunk flare and buttress roots are also protected. Closed cell foam padding minimum shall be placed between the trunk and the lumber with straps or wire to bind the planks in place, avoiding any fastener contact with bark or limbs, and do not drive fasteners into the tree. Lumber-covered sections of trees will be wrapped with orange construction fencing to provide a visual cue to heavy equipment operators.
5. Soil protection: Where surface access is unavoidable through either root protection zone, but no subsurface construction actions are being implemented, soil protection consisting of a steel plate or plywood placed on wood straw (or other NPS approved mulch), may be used to bridge over the root protection zone.

1.3 QUALITY ASSURANCE

- A. Project Consulting Arborist: Consulting arborist shall be field experienced in tree and vegetation preservation or on-site monitoring of public works or construction projects involving tree and vegetation retention and protection.
 1. Consulting Arborist shall provide an analysis/ condition assessment trees within the limits of construction and provide this documentation to the National Park Service.
 2. Consulting arborist shall be an independent consultant with a degree in forestry, horticulture, arboriculture, plant pathology, entomology or plant biology; an American Society of Consulting Arborists (ASCA) registered consulting arborist, or an International Society of Arboriculture (ISA) Board Certified Arborist, with at least five (5) years of field experience, including tree and vegetation preservation or on-site monitoring of public works or construction projects involving tree retention and protection. The Project Consulting Arborist shall be approved by the Contracting Officer.
- A. If it appears that the completion of the construction may cause damage to the branches, roots or trunk of any tree or existing vegetation to remain, the Contractor shall contact CO prior to work commencement.
- B. To prevent or minimize soil compaction and to minimize disruption to existing trees and vegetation, designated routes for equipment and foot traffic by work crews shall be determined prior to commencing construction activities. These routes shall be marked at the site, prior to commencement of construction, with tree and vegetation protection fencing as specified.
- C. Motorized equipment and trailers, including tractors, bobcats, bulldozers, trackhoes, trucks, cars, and carts shall not be allowed access within tree and vegetation protection areas.

- D. Materials and supplies shall not be stockpiled or stored within the tree and vegetation protection area.
- E. Under no circumstances shall any objects or materials be leaned against or supported by a tree's trunk, branches, or exposed roots. The attachment or installation to trees of any sign, cable, wire, nail, swing, or any other material that is not needed to help support the natural structure of the tree is prohibited.
- F. Tree and Vegetation Protection Work Plan: Contractor shall submit a tree and vegetation protection plan by Project Consulting Arborist based on the contract drawings for approval by Contracting Officer. The Contractor shall minimize disturbance to native vegetation, particularly in landscape / soil stabilization areas where ground disturbance is not needed.
 - 1. Proposed methods and schedule for implementing tree and vegetation protection shall be submitted for approval.
 - 2. Proposed methods, materials, and schedule for root pruning, branch pruning, and other tree maintenance and native vegetation retention practices shall be submitted for approval.
- G. Construction Schedule: Contractor shall submit a construction schedule that includes a time frame for work near existing trees and existing areas of native vegetation. Approval of such shall be obtained from Project Consulting Arborist prior to commencement of construction near Tree and Vegetation Protection Zones.
- H. Maintenance Schedule: Submit maintenance schedule to Contracting Officer.

1.4 REPLACEMENT COSTS

- A. Replacement costs for damage of trees and vegetation and park resources outside the designated construction limits or outside designated areas set aside for the contractor's use is as defined by the park and CO.
- A. Replacement costs for damaged and/or removed trees and vegetation noted to remain within the limit of construction will incur fines and penalties as applied defined by the park and CO.
- B. Pruning or removal of vegetation when authorized and supervised by the Contracting Officer is exempted from these penalties.

1.5 SUBMITTALS

- A. All Submittals shall be approved by the Contracting Officer prior to mobilization and any Work being conducted on site.
- B. Tree and Vegetation Protection Work Plan: Submit a tree and vegetation protection work plan based on the Contract Drawings for approval. Work shall be accomplished to minimize impacts and disturbance to site vegetation, which is a character-defining element of Ohanapecosh Campground. The plan shall include the following:
 - 1. Existing conditions including existing trees and vegetation within the limits of construction.
 - 2. Limit of work and Limit of Construction.
 - 3. Tree caliper / diameter size, location of dripline, and species.
 - 4. Areas of existing vegetation to remain and areas of existing vegetation to be removed, salvaged and reinstalled.

5. Tree and Vegetation Protection Zones with Tree and Vegetation Protection Zone sign locations.
 6. Tree and vegetation protection fencing (chain link) with moveable base shall correspond to the level of construction activity adjacent to protected tree(s). Including but not limited to areas of high traffic, material drop off/pickup, high intensity activity, and earthwork of greater intensity than grubbing.
 7. Plan for the Work occurring within or adjacent to Tree and Vegetation Protection Zones:
 - a. Access routes, including designated routes for equipment and foot traffic by work crews to minimize soil compaction and to minimize disturbance to existing vegetation.
 - b. Proposed exploratory excavation locations and methods to remove soil from around tree roots within Tree and Vegetation Protection Zones.
 8. Maintenance of Tree and Vegetation Protection Zones and any trees and existing vegetation affected by Construction during the Work.
 9. Trees identified for regular soil moisture readings.
 10. During construction, Tree and Vegetation Protection Plans may need to change per field conditions and contractor work. The Contractor shall continue to update the Tree and Vegetation Protection Work Plan in the field as a redlined or revised plan and submit as often as changes are needed for approval from the Contracting Officer.
- C. Tree and Vegetation Protection Work Plan Updates: The Contractor shall submit a highlighted / annotated section of their Tree and Vegetation Protection Work Plan at the Construction Progress Meetings to discuss where they intend to work and what Tree and Vegetation Protection Zones may be impacted for the upcoming week.
- D. Pruning: Proposed methods, materials, the Licensed and Insured Project Consulting Arborist, and schedule for root pruning, branch pruning, and other tree maintenance shall be submitted for approval.
- E. Pneumatic Air Excavation: Type of pneumatic air excavation tool, the Licensed and Insured Project Consulting Arborist, and schedule for excavation shall be submitted for approval.
- F. Water Excavation: Type of hydrovac excavation tool, the Licensed and Insured Project Consulting Arborist, and the schedule for excavation shall be submitted for approval.
- G. Watering Plan and Schedule: The Contractor shall submit a watering plan and schedule prior to the start of work that details watering of all trees affected by the Project for review and approval by the park and CO. Watering Plan and Schedule shall be developed in consultation with the park and CO. When watering is required the following information shall be included:
1. Area of the project site to be watered and how watering will be phased based on construction.
 2. Location, identification and caliper inch of each tree or area of existing vegetation to be watered.
 3. Total number of trees to be watered and total caliper inches, and total square footage of vegetation to be watered. Identify the amount of water to be applied based on total caliper inches.
 4. Schedule for watering during the duration of the project.
 5. How trees and will be watered:
 - a. Temporary system
 - b. Hydrant.
 - c. Water truck.
 - d. Combination of on-site, hydrant, and water truck.

- H. Soil Moisture and Watering Log: The Contractor shall fill out and submit to the Contracting Officer and the park the watering log, attached as an Exhibit. Information on the watering log shall include:
1. Tree(s) watered, identified by the Contractor's Tree and Vegetation Protection Plan.
 2. Number of gallons of water applied to each tree during every watering period or irrigation schedule.
 3. Soil moisture level readings, on a scale of one to ten (1 – 10). Refer to Part 3 - Execution for soil moisture reading requirements.
 4. Dates of each watering.
 5. The Soil Moisture and Watering Log shall be provided monthly via email to the Contracting Officer and reviewed at Construction Progress Meetings. If there appears to be an issue with on-site watering, Soil Moisture and Watering Logs may be requested at any time by the Contracting Officer.

1.6 QUALITY CONTROL

- A. The Contractor and sub-contractors shall attend a Preconstruction Meeting as outlined in Division 01 "Project Meetings." When the Project requires new sub-contractors to be on-site to complete Work, the Contractor shall include the sub-contractors in the Construction Progress Meetings to review the requirements of this specification.
- B. Contractor shall comply with applicable requirements and recommendations of the most current versions of the following standards and guidelines. Where these conflict with other specified requirements, the more restrictive requirements shall govern.
1. ANSI Z133.1: American National Standard for Tree Care Operations.
 2. ANSI A300: Tree, Shrub, and Other Woody Plant Management – Standard Practices – Parts - 1,2,5,6 and 8.
 3. Guide for Plant Appraisal – Current Edition: Authored by the Council of Tree and Landscape Appraisers; published by the International Society of Arboriculture.
 4. Requirements as set forth in these Contract Documents, particularly related to Laminated Root Rot and Douglas Fir protection best practices, and as identified by park staff.
- C. Motorized equipment and trailers, including but not limited to tractors, skid loaders, bulldozers, rubber-tired excavators, tracked excavators, trucks, cars, and carts shall not be allowed access within Tree and Vegetation Protection Zones with the exception of paved surfaces. Motorized equipment and trailers shall not be allowed access outside of paved areas or unvegetated areas except as explicitly shown on the plans or as approved by the CO or Project Consulting Arborist. Should access be necessary within designated Tree and Vegetation Protection Zones the Contracting Officer and Project Consulting Arborist shall be notified and CO shall approve of the access route and driving surface prior to use.
- D. Materials, supplies, tools, and construction facilities shall not be stockpiled or stored within the Tree and Vegetation Protection Zones unless otherwise approved by the Contracting Officer. Should temporary storage be approved by the Contracting Officer within designated Tree and Vegetation Protection Zone(s), the existing grade shall be covered as identified in Part 2 – Materials and Part 3 – Execution to help distribute the weight of equipment and to minimize soil compaction and rutting. Plywood and/or mulch are not acceptable bridging materials for driving over exposed tree roots.
- E. Under no circumstances shall any objects or materials be leaned against or supported by a tree's trunk, branches, or exposed roots or lain on existing vegetation to remain. The attachment or installation of any sign, cable, wire, nail, swing, or any other material to trees that is not needed to help support the natural structure of the tree is prohibited. Standard arboricultural techniques

such as bracing or cabling that are performed by professional arborists and recommended by the Project Consulting Arborist and are acceptable upon approval of the Contracting Officer.

The Contractor shall notify the Contracting Officer prior to mobilization on site with their schedule for installation of tree and vegetation protection. Tree protection shall be in place and approved prior to any work being conducted on-site, this includes erosion control.

PART 2 – PRODUCTS

2.1 TREE AND VEGETATION PROTECTION FENCING

- A. For trees not immediately adjacent (within two feet of) to construction activities, but for which construction or staging activities will occur within the TPZ, fencing shall be installed. Fencing shall only be installed around parts of the tree trunk that face construction activities and are vulnerable to damage. Construction fencing shall be placed at the greatest possible distance of the following from trees: a) TPZ edge, b) TPZ edge, or c) [least desirable] construction/staging zone limits (>2 ft from trunk).
- B. Black Vinyl-Coated Steel Chain-link fencing – Six feet (6') in height. Black vinyl-coated steel chain link fence panels with above ground base units to hold fence in a vertical position with no staking into ground. Fencing as per tree and vegetation protection detail on plans.
- C. Tree and Vegetation Protection Zone (11x17") signs printed and laminated by the Contractor.
- D. Refer to Drawings for locations for tree and vegetation retention and protection. Additional areas for protection may be needed to minimize disturbance to trees and native vegetation. and for additional locations as indicated by Contracting Officer.
- E. Contractor shall facilitate an on-site pre-construction meeting to include the park and CO to review areas of tree and vegetation retention and protection. Contractor shall adjust the limits of tree and vegetation retention and protection as directed by the CO and park.

1.7 TRUNK PROTECTION: For trees immediately adjacent to the work zone (< 2 feet) and which fencing is not practicable.

- A. Lumber - 2 x 4 lumber, minimum 4-foot in height..
 - 1. Lumber shall be loosely banded onto the trees with straps, and angled so that the trunk flare and buttress roots are also protected.
 - 2. Closed cell foam padding minimum of 1/4 in thick shall be placed between the trunk and the lumber.
 - 3. Use straps or wire to bind the planks in place, avoiding any fastener contact with bark or limbs, and do not drive fasteners into the tree. Additionally, lumber-covered sections of trees will be wrapped with orange construction fencing to provide a visual cue to heavy equipment operators.

1.8 SOIL PROTECTION: Where surface access is unavoidable through either root protection zone, but no subsurface construction actions are being implemented.

- A. Steel Plate - A minimum 1/2 in steel plate placed on top of a bed of minimum 2 in of wood straw (or other NPS approved mulch).

- B. Plywood - A minimum 3/4 in plywood over minimum of 4 in of woodstraw (or other NPS approved mulch), may be used to bridge over the root protection zone.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. The Contractor shall follow the approved Tree Protection Work Plans and access routes.
- B. The Contractor shall report any injured or dead trees within the limits of work to the CO and Project Consulting Arborist.
- C. Work shall be completed to ensure no damage to existing trees and native vegetation to remain. If it appears any Work may cause damage to a tree or native vegetation to remain, the Contractor shall contact the Contracting Officer and Project Consulting Arborist, who will make the determination as to whether such damage is likely, and the next steps.
- D. Methods for working within a Tree and Vegetation Protection Zone shall be approved prior to the start of work and may include the following depending on project site conditions:
 - 1. Pneumatic air excavation tool. Maximum pressure applied shall be an amount not to damage roots and shall not exceed ninety pounds per square inch (90 PSI). Pneumatic air excavation may not be allowed when in known asbestos areas.
 - 2. Water excavation. Maximum pressure applied shall be an amount not to damage roots and shall not exceed sixty pounds per square inch (60 PSI). Water excavation may be required when in known asbestos areas.
 - 3. Directional boring.
 - 4. Hand digging.
 - 5. Other methods recommended by Project Consulting Arborist and approved by the Contracting Officer.

3.2 TREE AND VEGETATION PROTECTION FENCING

- A. Tree and vegetation protection fencing shall be installed and approved by the Contracting Officer prior to any site activity, including installation of erosion control, and shall remain in place and maintained in the condition in which it was approved until removal is authorized by the Contracting Officer.
- B. Tree and vegetation protection fencing shall be constructed as follows:
 - 1. Fencing shall be installed to protect trees and native vegetation within the Limit(s) of Construction as indicated on the plans.
 - 2. Fencing is required as delineated on the plans.
 - a. Fence shall have posts no less than six feet 6) utilizing above-ground base units and sandbags as shown in the Tree and Vegetation Protection Detail.
 - b. Fencing shall not be fully enclosed to prevent entrapment of animals.
 - c. Fencing shall be self-supported in an effective vertical position for the duration of construction. Leaning the fence on trees or other site elements shall not be permitted.
 - d. Installation of the fencing shall not result in injury to tree above or below ground or to native vegetation.

3.3 TREE AND VEGETATION PROTECTION SIGNS

- A. A tree and vegetation protection sign shall be placed one (1) per each Tree and Vegetation Protection Zone, minimum, or per the direction of the Contracting Officer.
 - 1. The Contractor shall print and laminate (11"x17") size tree and vegetation protection signs.
 - a. The Contractor shall use zip ties on all four (4) corners of the signs to attach to the tree and vegetation protection fencing.
 - 2. The tree and vegetation protection signs shall be replaced within twenty-four (24) hours if they become unreadable, destroyed, missing, or per the direction of the Contracting Officer.

3.4 GENERAL

- B. Provide temporary barriers (e.g., tree and vegetation protection fence) to protect plants, trees, and critical root zones that are designated to remain but are within the construction limits and to protect those areas as indicated by the CO. Barriers shall be in place and approved by the Contracting Officer before any construction equipment is allowed on site. Coordinate with the Contracting Officer to allow adequate time to conduct salvaging or transplanting of plant materials into or out of the project construction limits.
- A. Unless areas are specifically designated for the purpose by the Contracting Officer, the following activities are prohibited outside the limits of construction and within the fenced areas surrounding trees and native vegetation designated to remain or as directed to remain by the Contracting Officer and as described above:
 - 1. Parking vehicles.
 - 2. Storing materials or supplies.
 - 3. Disposing of materials including, but not limited to paint, paint thinners, fiberglass materials, fiberglass thinners, concrete or cement residue, washings from cement or concrete mixers or any other material considered harmful by the Contracting Officer.
- B. Adhere to the park's pruning guidelines when pruning of trees or shrubs is necessary for construction activities. Avoid damage to trees or roots in or adjacent to the project during construction. All workers on the project involved in pruning operations must receive training by park staff to review methods and requirements prior to the commencement of pruning activities. Coordinate training with the Contracting Officer.
- C. Trimming or pruning of branches as approved by the CO.
- D. Provide additional water to trees during construction activities and the growing season after construction is complete as approved by Contracting Officer. Do not provide additional fertilizer as this may further stress the trees.

3.4 CONSTRUCTION PROCEDURES IN TREE AND VEGETATION PROTECTION ZONES

- A. Construction and excavation work within tree and vegetation protection zones shall be done with hand tools to reduce impacts to root systems.
- B. The Contractor shall protect all trees and native vegetation from injury due to construction related work. All injuries to trees and native vegetation shall be mitigated to the satisfaction of the Contracting Officer, and, if appropriate in accordance with penalty guidelines established by

the park and NPS. All costs of such mitigating shall be charged to and paid by the Contractor or offending sub-contractor.

- C. If access within designated Tree and Vegetation Protection Zones is approved by the Contracting Officer, the existing grade shall be covered with twelve inches (12") of wood mulch and overtopped by three-quarter inch (3/4") plywood with overlapping sheets, six-inches (6") minimum, or the plywood may be approved for use on the bottom of the mulch, to help distribute the weight of equipment and to minimize soil compaction and rutting. Mulch must be maintained at twelve inches (12") and plywood must be replaced when it becomes damaged and cracked to the point of being ineffective as determined by the Contracting Officer.
 - 1. Ground tracking mats or steel plates may be used as an alternative to plywood and may be required in high-traffic areas. Mulch is still required in combination with ground tracking mats or steel plates. Mats or plates must be appropriately sized for the type of equipment and shall be approved by the Contracting Officer prior to installation.
 - 2. Mulch shall be kiln dried and approved by the park prior to purchasing and prior to installation.
- D. The following procedures shall be used when constructing any hardscape or drainage improvements:
 - 1. All materials and equipment shall be kept on existing hardscape or within previously approved, compacted sub-base areas.
 - 2. Protect exposed roots from damage and contamination by stabilization materials and concrete.
 - 3. Locate concrete washouts outside of and away from Tree and Vegetation Protection Zones. Washout runoff shall be strictly contained within the washout area and shall not flow into Tree and Vegetation Protection Zones.
 - 4. When excavating, excavated soil should be retained for use on site and covered when temporary stockpiled. NPS must approve movement of excavated soil.
 - 5. Excavated soil shall be deposited in trucks and hauled off or deposited temporarily outside the Tree and Vegetation Protection Zones. Excavated and fill soil shall not be deposited, even temporarily, in Tree and Vegetation Protection Zones.
 - 6. If root pruning is needed, refer to section "Pruning."
- E. The compaction of sub soil within Tree and Vegetation Protection Zones shall not exceed eighty percent (80%) proctor density for areas of stabilization, unless the Contractor can confirm by testing that the existing compaction within the Tree and Vegetation Protection Zones already exceeds eighty percent (80%).
- F. If part of an approved plan, grading within the Tree and Vegetation Protection Zones shall be performed by hand or a method approved by the Contracting Officer. Any fill material that needs to be placed in the Tree and Vegetation Protection Zone in non-paved areas shall be limited to a maximum of four inches (4") of fill material over the area. Fill should consist of sandy loam topsoil. Clay soils shall not be used as fill. When using fill soil, the existing surface to receive fill should be scarified by hand, avoiding root damage and only to a depth to ensure proper incorporation of fill material to meet finished grade. Any filling operation shall not occur during water saturated soil conditions.
- G. Concrete or chemicals spilled within Tree and Vegetation Protection Zones shall be completely removed at the time of the spill. Contaminated soil shall be completely removed by hand and/or approved methods to minimize disturbance to root systems. Approved soil shall be added as necessary to restore the grade and the Contractor shall provide remedial tree maintenance as outlined in section "Tree and Vegetation Maintenance During Construction." Contact the Contracting Officer immediately in the event of a spill within a Tree and Vegetation Protection Zone

3.5 TRENCHING

- A. When trenching is approved within the Tree and Vegetation Protection Zones by the Contracting Officer, the Work shall be dug by hand, pneumatic air excavation, water excavation, or be done by directional boring in a manner to minimize root damage. Contractor shall receive approval of CO prior to use of pneumatic air excavation or use of water excavation.
- B. Whenever trenching exposes roots, as called out in section "Pruning," extending through the trench wall, the Contractor shall contact the Contracting Officer or the Project Consulting Arborist immediately for inspection and evaluation. Refer to section "Pruning."

3.6 UTILITY INSTALLATION

- A. All Excavation or Utility installation occurring within the Tree and Vegetation Protection Zones as indicated on the Contract Documents shall be marked out in the field and reviewed in the field with the Contracting Officer and Project Consulting Arborist prior to installation.

3.7 PRUNING

- A. Work Within Tree Disease Zones: Tree or soil work within known laminated root rot (LRR) pockets or in any area of downed green Douglas Fir trees shall be in consultation with NPS vegetation staff.
 - 1. Contractor shall follow BMPs to avoid spread disease in consultation with NPS vegetation staff.
 - 2. Contractor shall avoid damage and exposure of woody tissue on trees adjacent to LRR zones as a critical measure to prevent disturbance scars that act as pathways for fungal establishment and spread of disease.
 - 3. All trenching and falling equipment shall be cleaned of visible soil and woody root material when moving from a location affected by LRR location to others. Chemical sanitation of equipment is not required.
 - 4. LRR infected portions of trees shall be left in place (i.e. butt/roots/stumps). Any signs of LRR rot shall be removed prior to moving any aboveground portion of the tree. Unaffected tree boles may be used for walkways, LWD for stream habitat, or campsite delineation (on-site logs), etc.
 - 5. Contractor shall notify NPS vegetation staff or CO approved qualified staff to review and survey freshly cut stumps in LRR zones for LRR staining, to be done at time of cutting or shortly after.
 - 6. Contractor shall consult with park staff and CO on disease protection measures to prevent Douglas-fir beetle infestation.
 - 7. Under no circumstances should diseased or infected wood be left or taken for firewood, mulch or taken to a wood processing mill.
- B. Branch Pruning: Shall be identified in the pre-construction meeting and documented by the Contractor's Tree and Vegetation Protection Plan.
 - 1. Branch pruning shall be the responsibility of the Contractor for construction equipment access, specialty pruning required for the Project, and any damage caused by the Contractor during the Project and shall be done by a Licensed, and Insured Arborist.
 - 2. Pruning shall only be done with approval and at the direction of the Contracting Officer or Project Consulting Arborist.
 - 3. Tools used for branch pruning shall be disinfected between use on individual trees.
 - 4. Information on storm damage is found in section "Tree Maintenance During Construction".

- C. Root Pruning: Shall be done in accordance with ANSI A300 Root Management Standard latest edition and in accordance with Best Management Practices and be performed by a Licensed, and Insured Arborist.
1. The Contracting Officer or Project Consulting Arborist shall be notified prior to any Work operation known or suspected to involve cutting of roots within the Tree Protection Zone. Tree roots shall not be pruned or cut unless their removal is unavoidable and shall be authorized by the Contracting Officer.
 - a. When contacting the Contracting Officer, provide the following information:
 - 1) Tree type and/or identification number.
 - 2) Condition of the tree.
 - 3) Percent of disturbance anticipated within the Tree Protection Zone.
 - 4) Size of roots.
 2. All roots needing to be pruned or removed shall be cut cleanly with sharp hand tools, with oversight by the Contracting Officer or the Project Consulting Arborist. No wound dressings shall be used. Backfill shall promptly be installed over the exposed roots, filling all voids.
 3. Recommended root pruning tools shall be free of defects and have sharp cutting edges
 - a. Scissor-type lopper.
 - b. Scissor-type pruner.
 - c. Pruning saws designed specifically for tree work.
 4. After root-pruning, cover exposed roots within thirty (30) minutes to minimize desiccation. Roots may be covered with soil, mulch, or moistened burlap, and shall be kept moist until the final grade is established or roots are buried in soil.
 5. Where appropriate, and under the direction of the Contracting Officer or the Project Consulting Arborist, root restricting barriers can be installed in accordance with manufacturer's recommendations.
 6. If root pruning is authorized within the Tree Protection Zones, the Contractor shall provide remedial tree maintenance as identified in section "Tree Maintenance During Construction."
- D. Water Exposed root zones (from ground disturbance) or cover those exposed roots to avoid drying. If construction occurs on hot days, any root zones exposed by trenching or grubbing shall be covered with loose, wet matting to avoid drying. Exposure shall be limited.

3.8 SURFACE AND EXPOSED ROOTS

- A. When encountered, exposed roots shall be worked around in a manner that does not damage the outer layer of the root surface or bark
1. The Contracting Officer shall be notified should tree roots become exposed.
 2. Exposed roots must be wrapped in burlap and kept moist. At or below forty degrees (40°) F, the excavation area shall be covered in concrete blankets or as directed by the Contracting Officer.
 3. Exposed tree roots shall not be driven over. Plywood and/or mulch are not acceptable bridging materials for driving over exposed roots.

3.9 TREE MAINTENANCE DURING CONSTRUCTION

- A. Common Tree Maintenance: Practices that maintain the health and vigor of the tree, such as: watering, mulching, remedial pruning (if needed, and approved by the Contracting Officer). The Contractor shall be responsible for all trees impacted by Construction throughout the Project.
- B. Remedial Tree Maintenance: Practices such as soil remediation and soil/tissue sampling, and these practices are on an "as needed" basis, or as directed by the Contracting Officer. The

Contractor shall be responsible for all trees and native vegetation impacted by Construction throughout the Project.

- C. Tree and vegetation watering during Construction shall consist of the following minimum requirements for all trees within a Tree and Vegetation Protection Zone or otherwise impacted by Construction:
1. Watering Plan shall be submitted to the park and CO for approval prior to implementing watering.
 2. Watering of trees shall occur throughout the year during Construction, including winter if construction activities impact root zones, exposed roots or otherwise impacted the tree or their access to water.
 3. Watering shall occur when daytime temperatures are at or above forty degrees (40°) F.
 4. Watering frequency shall be based on visual indications, on the average soil moisture level and as directed by the park and CO.
 - a. The Contractor shall take soil moisture readings throughout the Project.
 - b. Soil moisture readings shall be taken every week, at a minimum, during the Construction period in areas of active construction to depth of twelve inches (12").
 - c. Readings shall be taken and recorded for twenty percent (20%) of the trees within the Limit of Construction and impacted by Construction and shall be indicated on the Tree and Vegetation Protection Work Plan.
 - d. All readings shall be recorded and submitted as outlined in Part 1 – Submittals.
 5. Depending on weather conditions, the Contracting Officer or Project Consulting Arborist may approve less frequent watering.
 6. The Contracting Officer will look for visual indications to determine if watering is adequate during Construction. If it appears that the trees are suffering based on visual indications, the Contracting Officer will check the soil moisture levels and determine the corrective action. Criteria includes:
 - a. The surrounding soil and/or landscape is dry.
 - b. The surrounding vegetation is dying or dead.
 - c. The leaves on the trees are wilted.
 - d. Color.
 7. At the Contractor's expense, they may install a temporary irrigation system to water with the approval of the Contracting Officer.
- D. Branch and Root Pruning: Refer to Section "Pruning."
- E. Maintenance for Storm Damage or Emergencies: The Contractor shall coordinate trees affected by storm damage with the Contracting Officer. Guidelines for responsibilities are identified below:
1. Non-imminent threat or immediate threat that is not a danger to the public: Considered low priority clearing and will be removed as time allows by the Contracting Officer. If the Contractor needs access to the Limit of Work, the Contractor shall remove the damage at their expense.
 2. Immediate threat that affects the public: Considered high priority clearing for the Contracting Officer and will be removed by the Contracting Officer inside the Limit of Work.
- F. The timing duration and frequency of necessary maintenance practices shall be determined and approved by the Contracting Officer or Project Consulting Arborist, based on factors associated with the site and affected trees.

3.5 DAMAGE ASSESSMENT

- A. If the Contractor destroys or seriously injures plant material designated for protection or outside of the work limits, damages will be assessed prior to final progress payment. Existing vegetation shall not be used to support temporary construction or permanent fencing.
- A. A damage assessment process will be triggered by any of the following types of damage to vegetation outside of clearing or construction limits or inside fenced vegetation within the construction and clearing limits:
 - 1. Removal of any undesignated plant material.
 - 2. Pruning or removal of more than 30% of a tree or shrub canopy.
 - 3. Removal or fracture of a limb or trunk that is one of the major structural entities of the plant.
 - 4. Removal or fracture of any limb greater than 12 inches in diameter.
 - 5. Bark damage or removal around more than 30% of the trunk circumference.
 - 6. Trenching or soil disturbance within the drip line that is deeper than 6 inches.

END OF SECTION 32 40 00

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October 17, 2025

Chelsea Rooney
Anderson Hallas Architects, P.C.
1317 Washington Ave.
Golden, CO 80401
chelsearooney@andarch.com

Subject: Hazardous Materials Assessment Summary Letter for Bear Valley School Rehabilitation Phases I and 2: Title II Documents, Pinnacles National Park, CA. Landmark Project No. 25097.001. National Park Service Project: PINN – 331043.

Dear Ms. Rooney:

Landmark Environmental, Inc. (Landmark) is pleased to provide this hazardous materials assessment report to Anderson Hallas Architects, P.C. (Client) to support the Bear Valley School Rehabilitation Phases I and 2, Pinnacles National Park, CA, PMIS –331043 (Project Site). The purpose of the hazardous materials assessment is to identify Asbestos-Containing Materials (ACMs), Lead-Based Paint (LBP), Lead-Containing Paint (LCP), Regulated Building Materials (RBMs), and evidence of animal infestation, if any, before the rehabilitation of the historic Bear Valley School. Landmark conducted the assessment of the Project Site on September 25, 2025. The project scope includes the Bear Valley School, a wood-framed, one-room schoolhouse constructed in 1902. The structure is plagued with rodent and bird infestations and is in poor condition, requiring major rehabilitation.

On September 25, 2025, Landmark mobilized an AHERA-accredited and CAL/OSHA-certified asbestos inspector to the Project Site to perform the ACM inspection, LBP/LCP survey, and RBM inventory. Landmark's inspection of the Project Site was in support of the Design Documentation (DD) and Construction Document (CD) phases being prepared by the Client for the National Park Service (NPS) for PINN 331043 Rehabilitation Phase I and 2: Title II Documents. Accordingly, Landmark's screening-level environmental assessments are designed to provide the appropriate information to support the DD and CD documentation and cost estimation for the future rehabilitation of the Bear Valley School at the Project Site.

Hazardous Materials Assessments and Laboratory Analysis

Landmark conducted the hazardous materials assessments below at the Project Site.

Asbestos

Inspection activities were conducted according to the U.S. Environmental Protection Agency (EPA) Title 40 Code of Federal Regulations (CFR) Part 763, Asbestos Hazard and Emergency Response Act (AHERA), the U.S. Occupational Safety and Health Administration (OSHA) Title 29 CFR 1926.1101, Asbestos in Construction Standard requirements, and California OSHA (Cal/OSHA) requirements, as

well as generally accepted industry standards. Asbestos assessment activities were conducted by AHERA-accredited and Cal/OSHA-certified personnel. Bulk samples of accessible suspect ACMs were collected in a representative manner as determined by the inspector. Bulk sampling was limited to materials that were visible and readily accessible to the inspector.

Sampling of suspect ACM at the Project Site was performed in general accordance with EPA, AHERA, OSHA Chapter 29 CFR 1926.1101, CAL/OSHA, and generally accepted industry standards. Bulk samples of suspect materials were submitted to SGS in Hayward, California. SGS is a laboratory accredited by the National Voluntary Laboratory Accreditation Program. A copy of the laboratory results is found in Attachment A.

The bulk samples were analyzed using Polarized Light Microscopy (PLM) via EPA method 600/R-93/116 to determine the asbestos type and percentage using progressive analysis. Applicable asbestos regulations define ACM as material containing greater than 1% asbestos by weight, volume, or surface area and distinguishable between friable and non-friable forms of ACM. Friable ACM can be crumbled or reduced to powder by hand pressure when dry, while non-friable material cannot. Regulations stipulate that if any one sample of a homogenous material contains greater than 1% asbestos, the entire homogenous area is considered asbestos-containing. The ACM information is summarized in Attachment A.

Lead

A certified lead paint inspector performed the LCP/LBP assessment concurrently with the ACM survey. The LCP portion of this survey was performed by a trained LBP Inspector utilizing paint chip sampling in the field. Chip samples of suspect lead paint were collected in a representative manner as determined by the inspector based on identified testing combinations where surface color, substrate, or building components were visibly different. The sampling was limited to materials that were visible and readily accessible to Landmark's inspector. LBP is defined as a paint material containing 0.5% lead by weight or 1.0 milligram per square centimeter (mg/cm²) by both the EPA and the Department of Housing and Urban Development (HUD). LCP is paint containing any detectable concentrations of lead. The lead screening results are summarized in Attachment B.

Regulated Building Materials

The RBM inventory included the visual identification and quantification of building components potentially requiring special handling or management during removal, including refrigerants Chlorofluorocarbons (CFCs); waste oils or chemicals; fluorescent bulbs and ballasts potentially containing mercury and Polychlorinated Biphenyls (PCBs), respectively; transformers that potentially contain PCBs; utility switches and thermostats potentially containing mercury; high-intensity discharge exterior lighting potentially containing mercury; and fixtures that may contain batteries or radiation sources. This inventory was limited to the RBMs that were visible and readily accessible to the inspector. The RBMs inventory is summarized in Attachment C.

Rodent and Bird Infestation

Landmark inspected the Project Site and interviewed the building occupants regarding evidence of animal infestation, specifically evidence and health hazards related to rodents and birds.

Findings

Asbestos

Nine bulk samples were collected from four suspect homogenous ACMs at the Project Site. Of the nine samples collected and analyzed, none were positive for asbestos.

Attachment A provides photographs of the homogeneous areas and laboratory analytical reports. Figure 1 presents sample locations, and Attachment D includes inspector and company credentials.

Lead

The LCP portion of this survey was performed utilizing paint chip sampling in the field. Eleven different paint chip samples were collected in a representative manner as determined by the inspector based on identified testing combinations where the surface color, substrate, or building component were visibly different. The LBP summary sheet is included in Attachment B.

The following paints were found to be LCP:

- The green wood floor near the west entryway is 0.056%.
- The green wood window screen area is 0.011%.
- The white metal pipe on the northwest exterior is 0.088%.
- The white wood wall on the west exterior is 0.022%.
- The off-white window glazing on the window sash in the kitchen is 0.2%.

The following paints are found to be LBP:

- The off-white wood window in the kitchen/pantry area is 17.0%.
- The off-white wood wall in the kitchen area is 0.9%.
- The blue wood countertop in the kitchen is 1.4%.
- The off-white wood door in the classroom area is 0.5%.
- The off-white wood door frame in the classroom area is 2.4%.

- The white wood wall on the northeast exterior is 0.6%.

Regulated Building Materials

Identified RBMs included one fire extinguisher and two 1-gallon paint cans in the classroom area.

Rodent and Bird Infestation

Significant areas of rodent, bird, and insect infestation, including observations of rodent droppings, bird nests, and insect nests, were observed throughout the structure. Landmark did not observe ground squirrels or burrows. No other evidence of animals or animal infestation was observed or reported.

Conclusions and Recommendations

Asbestos

No ACMs are identified at the Project Site. If any future ACMs are discovered during renovations or demolition that were not identified during this assessment, the ACMs should be abated and/or managed according to applicable federal and state regulations.

Any ACM that will be disturbed during the renovation or demolition work must be removed in accordance with all applicable EPA, OSHA, and CAL/OSHA regulations. The California Air Resource Board (CARB) requires permits for all asbestos removal projects of 100 square feet or more. This varies from the Federal trigger level amounts. Cal/OSHA also considers any material with more than 0.1% asbestos an ACM. Trace materials require 1,000-point counts to determine if they are less than 0.1%. Materials with more than 0.1% asbestos, which will be removed, still require CARB permits. Materials with less than 1% asbestos can still be discarded as construction debris.

Lead

Six painted testing combinations were identified as LBP, and five painted testing combinations were identified as LCP. The EPA, the U.S. Department of Housing and Urban Development (HUD), and the California Department of Public Health (CDPH) regulate LBP abatement work activities in target housing and child-occupied facilities. Target housing is defined as any housing constructed before 1978 occupied by a child under six. A child-occupied facility is a building constructed before 1978 that is regularly visited by the same child under six.

The EPA, HUD, and CDPH regulations related to LBP abatement do not apply to the Project Site unless abatement activities occur (specifically designed to permanently remove the LBP/activity designed to last more than 20 years). LBP is defined by the EPA and HUD as paint containing 0.5% by weight or 1.0 (mg/cm²).

The EPA also developed a program for renovation activities in child-occupied/target housing and child-care facilities called the Lead Renovation, Repair, and Paint Rule (RRP Rule). This rule requires that firms performing renovation, repair, and painting projects that disturb LBP in homes, child-care

facilities, and pre-schools built before 1978 have their firm certified by EPA (or an EPA-authorized state), use certified renovators who are trained by EPA-approved training providers, and follow lead-safe work practices.

The EPA and HUD regulate LBP abatement activities in target housing and child-occupied facilities. EPA and HUD regulations related to LBP abatement do not apply to the buildings at the Project Site, as they are not considered target housing or child-occupied facilities.

The Cal/OSHA lead in construction standard 8 *CCR 1532.1* addresses the requirements for sites where the employer has reason to believe that any employee may be exposed to lead. In addition to the basic requirements for evaluating worker exposure, additional requirements are required when workers are exposed above the Cal/OSHA Action Level (AL) of $2 \mu\text{g}/\text{m}^3$ and the Permissible Exposure Limit (PEL) of $10 \mu\text{g}/\text{m}^3$. This Cal/OSHA standard applies to all construction activities that may impact LCP (any detectable lead).

Materials reused or recycled, such as metal components, are not subject to waste characterization provisions under the EPA. Still, proper disclosure of LCPs should be provided to the recycling facility for hazard communication purposes.

The types and locations of LBP and LCP and regulatory requirements should be disclosed to the General Contractor and the building owner's staff to avoid accidental disturbance and ensure contractor compliance with applicable regulations (to ensure proper worker protection and material disposal).

Regulated Building Materials

Identified RBMs included one fire extinguisher and two 1-gallon paint cans in the classroom area. These RBMS will require proper disposal and management if they are to be impacted by renovations. These items must be disposed of and handled in accordance with applicable regulations. If discovered, any fluorescent light ballasts without the "No PCBs" label are considered an RBM and must be managed in accordance with applicable regulations.

Rodent and Bird Infestation

Significant areas of rodent, bird, and insect infestation, including observations of rodent droppings, bird nests, and insect nests, were observed throughout the structure. Landmark did not observe ground squirrels or burrows. No other evidence of animals or animal infestation was observed or reported. Any rodent removal and exclusion work should be performed by a licensed and insured pest control contractor. Personal Protective Equipment (PPE) is recommended when working in this building, and workers should be trained on applicable hazards and exposure prevention.

Additionally, renovation activities that occur on or near materials contaminated with rodent feces or nesting will require disinfection before work begins. It is recommended that the NPS Hantavirus Guidelines (Attachment E) be followed prior to working in areas with rodent contamination.

Limitations

Landmark performed this hazardous materials assessment consistent with the level of care and expertise exercised by industrial hygiene inspection and assessment professionals. Landmark does not imply or guarantee that every suspect ACM, LCP, RBM, or biological hazard on or in the Project Site has been identified or sampled. A limited intrusive investigation was conducted. In the event of a more extensive renovation or complete demolition of the Project Site, ACM could be discovered that was not identified during this inspection.

No subsurface investigation was conducted. It should be noted that in the event of demolition or a more extensive renovation of the Project Site, asbestos that was not identified during this assessment could be discovered. If additional suspect materials are discovered, they should be tested for the presence of asbestos before disturbance.

Landmark appreciates the opportunity to provide our services on this important project. Please feel free to contact us if you have any questions or if we can provide additional assistance.

Respectfully,

Landmark Environmental, Inc.



Nicole Christ
Senior Project Manager

Attachments:

Figure 1: Asbestos Bulk Sampling Locations

Attachment A: Asbestos Information

Attachment B: Lead Information

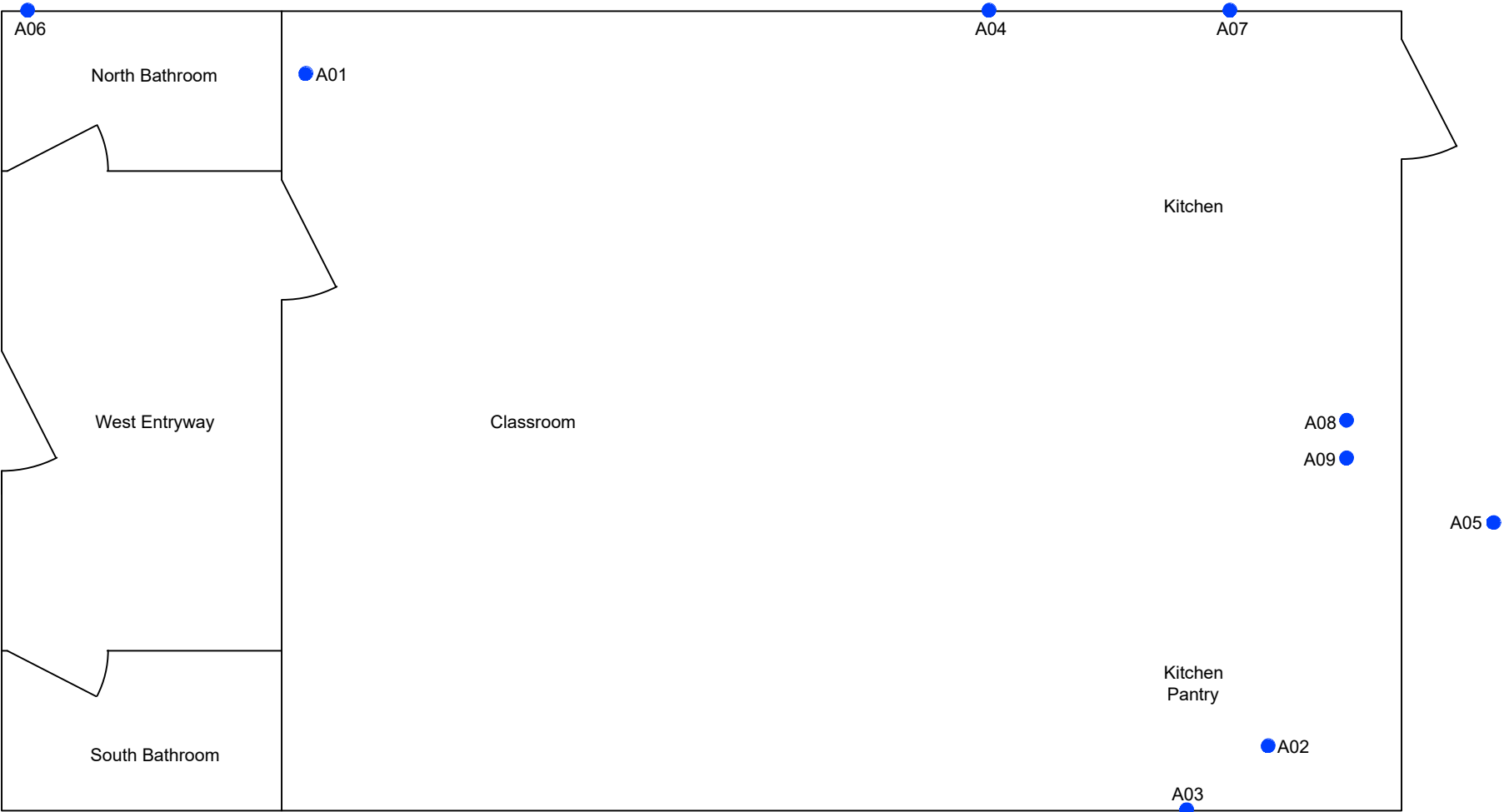
Attachment C: RBMs Inventory

Attachment D: Certifications

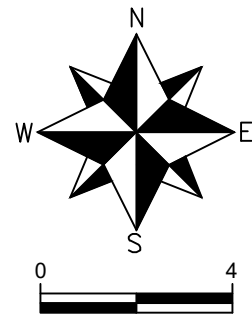
Attachment E: NPS Hantavirus Guidelines

Figure 1

ASBESTOS BULK SAMPLING LOCATIONS



Landmark Environmental, Inc.
7961 Shaffer Parkway Suite 4, Littleton, CO 80127



LEGEND
● Non-Asbestos Containing Building Material Sample Location

Figure 1
Asbestos Bulk Sample Locations

Pinnacles National Park
Bear Valley School
California 95043

Project #:	25097.001	Date:	10/13/2025
Drafted By:	J.G.M.		

Attachment A

ASBESTOS INFORMATION

Homogeneous Area Building Inspection Form

Client: Anderson Hallas	Landmark Project No.: 25097.001	Date: 09/25/25
Inspector: Jim Sevilla	Project Site: Bear Valley School	Page: 1 of 2

Homogeneous Material Description:			Photo
Tan RSF over black mastic			
Material Code:	Material Sequence:	Size:	
1			
Color:	Pattern / Texture:	Associated Materials:	
Tan			
Quantity (SF/LF/EA):	Material Type:	Friability:	
3,200 SF	M	Friable (F)	
Condition:	Disturbance Potential:	Assessment Category:	
Damaged	High		
Material Notes:			
Quantities are approximations.			
Sample Number:	Sample Location Description:	Analytical Results:	
A01	Classroom/NW area/floor	None-detected	
A02	Kitchen Pantry/ SE area/ floor	None-detected	

Homogeneous Material Description:			Photo
Off white window glazing			
Material Code:	Material Sequence:	Size:	
2			
Color:	Pattern / Texture:	Associated Materials:	
Off-white			
Quantity (SF/LF/EA):	Material Type:	Friability:	
		Type II Non-Friable	
Condition:	Disturbance Potential:	Assessment Category:	
Significantly Damaged	Moderate		
Material Notes:			
Quantities are approximations			
Sample Number:	Sample Location Description:	Analytical Results:	
A03	Kitchen pantry/South wall/window	None-detected	
A04	Kitchen/ North wall/Window	None-detected	

Material Type:
Surfacing (S)
Thermal System Insulation (TSI)
Miscellaneous (M)

Friability:
Friable (F)
Type I Non-Friable (I)
Type II Non-Friable (II)

Condition:
Good (G)
Damaged (D)
Significantly Damaged (S)

Disturbance Potential:
Low (L)
Moderate (M)
High (H)

Assessment Categories:
1. Damaged or significantly damaged TSI ACM
2. Damaged friable surfacing ACM
3. Significantly damaged friable surfacing ACM
4. Damaged or significantly damaged friable miscellaneous ACM
5. ACBM with potential for damage
6. ACBM with potential for significant damage
7. Any remaining friable ACBM of friable suspected ACBM
N/A – Not Applicable

Homogeneous Area Building Inspection Form

Client: Anderson Hallas	Landmark Project No.: 25097.001	Date: 09/25/25
Inspector: Jim Sevilla	Project Site: Bear Valley School	Page: 2 of 2

Homogeneous Material Description:			Photo
Concrete Slab			
Material Code:	Material Sequence:	Size:	
3			
Color:	Pattern / Texture:	Associated Materials:	
Gray			
Quantity (SF/LF/EA):	Material Type:	Friability:	
3,400 SF		Type II Non-Friable	
Condition:	Disturbance Potential:	Assessment Category:	
Damaged	Low		
Material Notes:			
Additional samples were taken due to concrete being poured at different times. Quantities are approximations.			
Sample Number:	Sample Location Description:	Analytical Results:	
A05	Ext/ East area	None-detected	
A06	Ext/ West area	None-detected	
A07	Ext/ North area	None-detected	

Homogeneous Material Description:			Photo
Brick and Mortar			
Material Code:	Material Sequence:	Size:	
4			
Color:	Pattern / Texture:	Associated Materials:	
Red/gray			
Quantity (SF/LF/EA):	Material Type:	Friability:	
150 SF		Type II Non-Friable	
Condition:	Disturbance Potential:	Assessment Category:	
Damaged	Low		
Material Notes:			
Only accessible through the attic space. Quantities are approximations.			
Sample Number:	Sample Location Description:	Analytical Results:	
A08	Int/ Attic space/ East area/Chimney	None-detected	
A09	Int/ Attic space/ East area/Chimney	None-detected	

--	--	--

Material Type:
Surfacing (S)
Thermal System Insulation (TSI)
Miscellaneous (M)

Friability:
Friable (F)
Type I Non-Friable (I)
Type II Non-Friable (II)

Condition:
Good (G)
Damaged (D)
Significantly Damaged (S)

Disturbance Potential:
Low (L)
Moderate (M)
High (H)

Assessment Categories:
1. Damaged or significantly damaged TSI ACM
2. Damaged friable surfacing ACM
3. Significantly damaged friable surfacing ACM
4. Damaged or significantly damaged friable miscellaneous ACM
5. ACBM with potential for damage
6. ACBM with potential for significant damage
7. Any remaining friable ACBM of friable suspected ACBM
N/A – Not Applicable

Bulk Asbestos Analysis

(EPA Method 40CFR, Part 763, Appendix E to Subpart E and EPA 600/R-93-116, Visual Area Estimation)
 NVLAP Lab Code: 101459-0

Forensic Analytical Consulting Svcs
 Wilson Wong
 21228 Cabot Blvd.

 Hayward, CA 94545

Client ID: HAY01
Report Number: B376721
Date Received: 09/29/25
Date Analyzed: 10/06/25
Date Printed: 10/06/25
First Reported: 10/06/25

Job ID/Site: PJ88077; Landmark Environmental Inc Bear Valley School Pinnacles National
 Park CA 95043
Date(s) Collected: 09/26/2025

SGSFL Job ID: HAY01
Total Samples Submitted: 9
Total Samples Analyzed: 9

Sample ID	Lab Number	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer
01-A01	12830003						
Layer: Tan Sheet Flooring			ND				
Layer: Fibrous Backing			ND				
Layer: Black Mastic			ND				
Layer: Wood			ND				
Total Percentage Values of Non-Asbestos Fibrous Components:							
Cellulose (30 %)	Fibrous Glass (5 %)	Synthetic (10 %)					
01-A02	12830004						
Layer: Tan Sheet Flooring			ND				
Layer: Fibrous Backing			ND				
Layer: Black Mastic			ND				
Layer: Wood			ND				
Total Percentage Values of Non-Asbestos Fibrous Components:							
Cellulose (30 %)	Fibrous Glass (5 %)	Synthetic (10 %)					
02-A03	12830005						
Layer: Off-White Non-Fibrous Material			ND				
Total Percentage Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							
02-A04	12830006						
Layer: Off-White Non-Fibrous Material			ND				
Total Percentage Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							
04-A05	12830007						
Layer: Grey Cementitious Material			ND				
Total Percentage Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							
04-A06	12830008						
Layer: Grey Cementitious Material			ND				
Total Percentage Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							
04-A07	12830009						
Layer: Grey Cementitious Material			ND				
Total Percentage Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							

Client Name: Forensic Analytical Consulting Svcs

Report Number: B376721

Date Printed: 10/06/25

Sample ID	Lab Number	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer
05-A08	12830010						
Layer: Red Cementitious Material			ND				
Total Percentage Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							
05-A09	12830011						
Layer: Red Cementitious Material			ND				
Layer: Off-White Cementitious Material			ND				
Total Percentage Values of Non-Asbestos Fibrous Components:							
Cellulose (Trace)							

Maria E. Casper

Maria Casper, Laboratory Supervisor, Hayward Laboratory

Note: Limit of Quantification ('LOQ') = 1%. 'Trace' denotes the presence of asbestos below the LOQ. 'ND' = 'None Detected'.

Inhomogeneous samples are separated into homogenous subsamples and analyzed individually. Analytical results and reports are generated by SGS at the request of and for the exclusive use of the person or entity (client) named on such report. Results, reports or copies of same will not be released by SGS to any third party without prior written request from client. This report applies only to the sample(s) tested. Supporting laboratory documentation is available upon request. This report must not be reproduced except in full, unless approved by SGS. The client is solely responsible for the use and interpretation of test results and reports requested from SGS. This report must not be used by the client to claim product endorsement by NVLAP or any other agency of the U.S. Government. SGS is not able to assess the degree of hazard resulting from materials analyzed. SGS reserves the right to dispose of all samples after a period of thirty (30) days, according to all state and federal guidelines, unless otherwise specified. All samples were received in acceptable condition unless otherwise noted.

Attachment B

LEAD INFORMATION

LEAD INSPECTION SAMPLE LOG

Project Name:	Pinnacles National Park			Inspector:	Jim Sevilla			
Location:	Bear Valley School			Date:	9/25/2025			
Client:	Anderson Hallas			Laboratory Report No.	M272683			
Sample No.	Location	Component	Substrate	Feature	Condition	Color	Result (weight %)	Screen Result
Pb01	Kitchen pantry/South wall	Window	Wood		Poor	Off white	17	LBP
Pb02	Kitchen East area	Wall	Wood		Poor	Off white	0.9	LBP
Pb03	Kitchen/Center area	Counter top	Wood		Poor	Blue	1.4	LBP
Pb04	Classroom/NW area	Door	Wood		Poor	Off white	0.5	LBP
Pb05	Classroom/NW area	Door frame	Wood		Poor	Off white	2.4	LBP
Pb06	West Entryway/NW area	Floor	Wood		Poor	Green	0.056	LCP
Pb07	Classroom/West area	Window screen	Wood		Poor	Green	0.011	LCP
Pb08	Ext/NW area	Pipe	Metal		Poor	White	0.088	LCP
Pb09	Ext/West area	Wall	Wood		Poor	White	0.022	LCP
Pb10	Ext/NE area	Wall	Wood		Poor	White	0.6	LBP
Pb11	Kitchen pantry/South wall	Window sash	Window glazing		Poor	Off white	0.2	LCP

Notes:

NL - Not Lead

LCP - Lead-Containing Paint

LBP - Lead-Based Paint





Landmark Project No. 25097.001	Description	Off white paint on the window	Photo 1
	Site Name	Bear Valley School	Photo Date 09/25/25



Landmark Project No. 25097.001	Description	Off white paint on the wall	Photo 2
	Site Name	Bear Valley School	Photo Date 09/25/25



Landmark Project No. 25097.001	Description	Blue paint on the countertop	Photo 3
	Site Name	Bear Valley School	Photo Date 09/25/25



Landmark Project No. 25097.001	Description	Off white paint on the door	Photo 4
	Site Name	Bear Valley School	Photo Date 09/25/25



Landmark Project No. 25097.001	Description	Off white paint on the door frame	Photo 5
	Site Name	Bear Valley School	Photo Date 09/25/25



Landmark Project No. 25097.001	Description	Green paint on the floor	Photo 6
	Site Name	Bear Valley School	Photo Date 09/25/25



Landmark Project No. 25097.001	Description	Green paint on the window screen	Photo 7
	Site Name	Bear Valley School	Photo Date 09/25/25



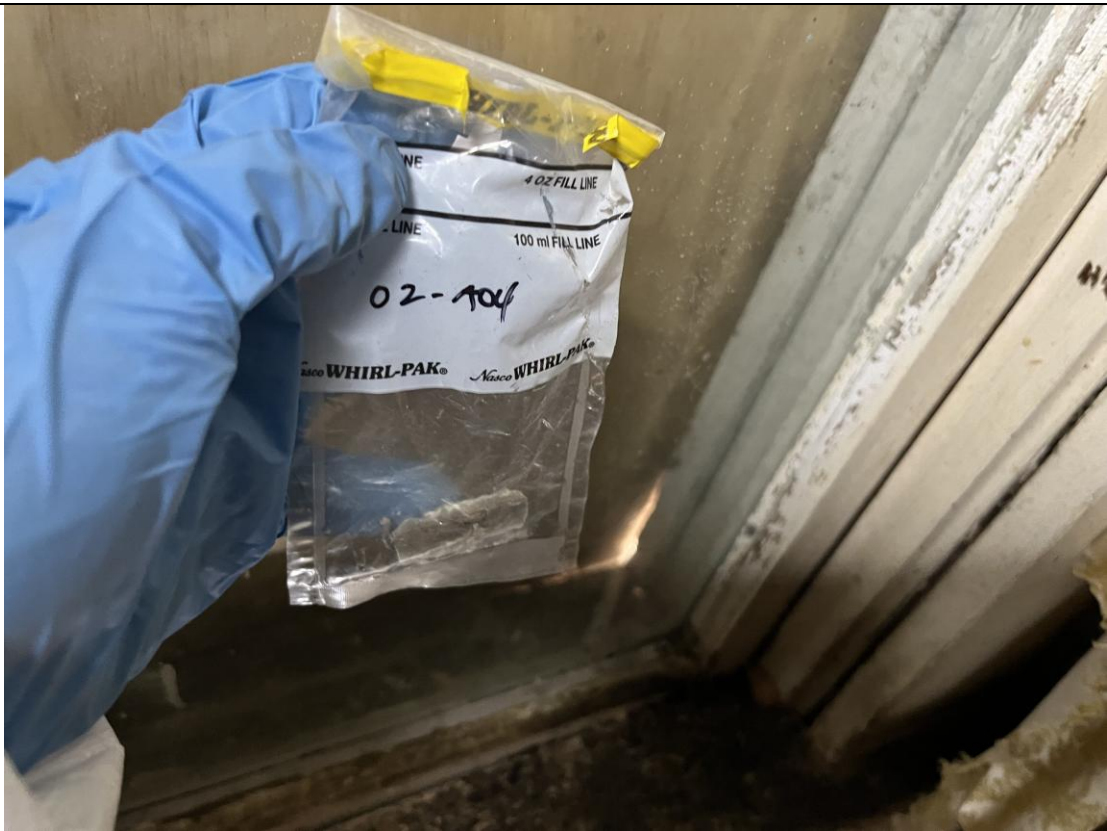
Landmark Project No. 25097.001	Description	White paint on the pipe	Photo 8
	Site Name	Bear Valley School	Photo Date 09/25/25



Landmark Project No. 25097.001	Description	White paint on the exterior wall	Photo 9
	Site Name	Bear Valley School	Photo Date 09/25/25



Landmark Project No. 25097.001	Description	White paint on the exterior wall	Photo 10
	Site Name	Bear Valley School	Photo Date 09/25/25



Landmark Project No. 25097.001	Description	Off white window glazing	Photo 11
	Site Name	Bear Valley School	Photo Date 09/25/25



Metals Analysis of Paints

(AIHA-LAP, LLC Accreditation, Lab ID #101762)

Forensic Analytical Consulting Svcs
Wilson Wong
21228 Cabot Blvd.

Hayward, CA 94545

Client ID: HAY01
Report Number: M272683
Date Received: 09/29/25
Date Analyzed: 10/02/25
Date Printed: 10/03/25
First Reported: 10/03/25

Job ID / Site: PJ88077; Landmark Environmental Inc Bear Valley School Pinnacles National
Park CA 95043

SGSFL Job ID: HAY01

Date(s) Collected: 9/25

Total Samples Submitted: 11

Total Samples Analyzed: 11

Sample Number	Lab Number	Analyte	Result	Result Units	Reporting Limit*	Method Reference
PB01	30961680	Pb	17	wt%	3	EPA 3050B/7000B
PB02	30961681	Pb	0.9	wt%	0.2	EPA 3050B/7000B
PB03	30961682	Pb	1.4	wt%	0.2	EPA 3050B/7000B
PB04	30961683	Pb	0.5	wt%	0.2	EPA 3050B/7000B
PB05	30961684	Pb	2.4	wt%	0.2	EPA 3050B/7000B
PB06	30961685	Pb	0.056	wt%	0.007	EPA 3050B/7000B
PB07	30961686	Pb	0.011	wt%	0.006	EPA 3050B/7000B
PB09	30961687	Pb	0.022	wt%	0.006	EPA 3050B/7000B
PB10	30961688	Pb	0.6	wt%	0.2	EPA 3050B/7000B
PB11	30961689	Pb	0.20	wt%	0.02	EPA 3050B/7000B
PB08	30961690	Pb	0.088	wt%	0.007	EPA 3050B/7000B

* The Reporting Limit represents the lowest amount of analyte that the laboratory can confidently detect in the sample, and is not a regulatory level. The Units for the Reporting Limit are the same as the Units for the Final Results.

Kevin Poon, Laboratory Supervisor, Hayward Laboratory

Analytical results and reports are generated by SGS at the request of and for the exclusive use of the person or entity (client) named on such report. Results, reports or copies of same will not be released by SGS to any third party without prior written request from client. This report applies only to the sample(s) tested. Supporting laboratory documentation is available upon request. This report must not be reproduced except in full, unless approved by SGS. The client is solely responsible for the use and interpretation of test results and reports requested from SGS. SGS is not able to assess the degree of hazard resulting from materials analyzed. SGS reserves the right to dispose of all samples after a period of thirty (30) days, according to all state and federal guidelines, unless otherwise specified. Any modifications that have been made to referenced test methods are documented in SGS Standard Operating Procedures Manual. Sample results have not been blank corrected. Quality control and sample receipt condition were acceptable unless otherwise noted.

Note* Sampling data used in this report was provided by the client as noted on the associated chain of custody form.

Attachment C

RBMS INVENTORY

Other RBMs Presumptive Inventory List

Room #	CE	CFC	EL	EX	FE	CL	FL-2'	FL-4'	FL-8'	FL-0	LB	FS	HID	LS	MEF	MTG	RB	SD	TF	OT	XX	Comments	
Classroom					1															2		2 paint cans observed in the classroom	
TOTAL					1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2		
CE = Comp. equip.; CFC = Refrigerants; EL = Emergency lighting; EX = Exit signs; FE = Fire extinguishers (hand held); CL = Compact light bulb; FL-2' = 2' Fluorescent lamp; FL-4' = 4' Fluorescent lamp; FL-8' = 8' Fluor. lamp; FL-0 = Fluorescent lamp other; FS = Fire suppression system; HID = High intensity lights; LB = Light ballast; LS = Lead sheeting; MEF = Mech equip. fluids; MTG = Mercury switch/gauge/etc.; RB = Rechargeable battery; SD = Smoke detectors; TF = Transformer; OT = Other; XX = No RBMs																							

Attachment D
CERTIFICATIONS



STATE OF CALIFORNIA
DEPARTMENT OF PUBLIC HEALTH



LEAD-RELATED CONSTRUCTION CERTIFICATE

INDIVIDUAL:



Virgilito Sevilla

CERTIFICATE TYPE:

Lead Inspector/Assessor

NUMBER:

LRC-00010379

EXPIRATION DATE:

2/20/2026

Disclaimer: This document alone should not be relied upon to confirm certification status. Compare the individual's photo and name to another valid form of government issued photo identification. Verify the individual's certification status by searching for Lead-Related Construction Professionals at www.cdph.ca.gov/programs/clppb or calling (800) 597-LEAD

STATE OF CALIFORNIA

DEPARTMENT OF INDUSTRIAL RELATIONS

Division of Occupational Safety and Health-Asbestos & Carcinogen Unit

1750 Howe Avenue, Suite 460

Sacramento, CA 95825

(916) 574-2993 Office <http://www.dir.ca.gov/dosh/asbestos.html> actu@dir.ca.gov

Gavin Newsom, Governor



911256720T

9/9/2025

Forensic Analytical Consulting Services, Inc.
Virgilito Sevilla
21228 Cabot Boulevard
Hayward CA 94545

Dear Certified Asbestos Consultant or Technician:

Enclosed is your certification card. **To maintain your certification, you must abide by the rules printed on the back of the certification card.**

Your certification is valid for a period of one year. If you wish to renew your certification, you must apply for renewal at least 60 days before the expiration date shown on your card. [8 CCR 341.15(h)(1)].

Please hold and do not send copies of your required AHERA refresher renewal certificates to our office until you apply for renewal of your certification.

Certificates must be kept current if you are actively working as a CAC or CSST. The grace period is only for those who are not actively working as an asbestos consultant or site surveillance technician.

Please contact our office at the above address or email w any changes in your contact/mailling information within 15 days of the change.

Sincerely,

Dean Mochrie, CAC
Senior Safety Engineer

Attachment: Certification Card

cc: File

Renewal – Card Attached (08/24)

State of California
Division of Occupational Safety and Health
Certified Site Surveillance Technician
Virgilito C. Sevilla

Name

Certification No. **19-6720**

Expires on **10/06/2026**

This certification was issued by the Division of Occupational Safety and Health as authorized by Sections 7180 et seq of the Business and Professions Code.



Attachment E

NPS HANTAVIRUS GUIDANCE



Hantavirus Risk Reduction

Worker Protection

Updated December 2013

Hantavirus Risk Reduction

Worker Protection

Updated December 2013

Office of Public Health
1201 Eye Street NW
Room 52
Washington, DC 20005

Wildlife Health Branch, Biological Resource Management Division
1201 Oakridge Drive
Suite 200
Fort Collins, CO 80525

Integrated Pest Management Program, Biological Resource Management Division
1201 Oakridge Drive
Suite 200
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The following National Park Service contributors helped revise this document: Bruce Badzik, Integrated Pest Management Coordinator & Biologist, Golden Gate National Recreation Area; Dr. Danielle Buttke, DVM, PhD, MPH, DACVPH, One Health Coordinator, Biological Resource Management Division/Wildlife Health Branch and Office of Public Health; Myron Chase, Integrated Pest Management Coordinator –Biologist, Intermountain Regional Office IPM Coordinator; Carol DiSalvo, Servicewide Integrated Pest Management Coordinator, Biological Resource Management Division; and Ciro Monaco, Biological Technician, Servicewide Integrated Pest Management Program.

December 2013

U.S. Department of the Interior
National Park Service
Office of Public Health
Washington, DC
Natural Resource Stewardship and Science
Fort Collins, Colorado

This report received formal peer review by subject-matter experts who were not directly involved in the collection, analysis, or reporting of the data, and whose background and expertise put them on par technically and scientifically with the authors of the information.

Views, statements, findings, conclusions, recommendations, and data in this report do not necessarily reflect views and policies of the National Park Service, U.S. Department of the Interior. Mention of trade names or commercial products does not constitute endorsement or recommendation for use by the U.S. Government.

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Introduction

This document summarizes the updated recommendations from the Centers for Disease Control and Prevention (CDC) for hantavirus risk reduction for workers. The information is adapted from the Morbidity and Mortality Weekly Report, July 26, 2002; Vol. 51; No. RR09.

Precautions

Precautions for Workers Frequently Exposed to Rodents

Persons who frequently handle or are exposed to wild rodents are probably at higher risk for hantavirus infection than the general public because of the frequency of their exposures. Such persons include, but are not limited to: wildlife specialists; maintenance workers; employees involved in rodent management; concessions managers; some custodial staff; and building and fire inspectors. Therefore, enhanced precautions are warranted to protect them against hantavirus infection, as described below.

- Workers in potentially high-risk settings should be informed by their employers about hantavirus transmission and symptoms of infection, and be given detailed guidance and training on prevention measures. Determining the level of risk for HPS in each work setting is the responsibility of the park. The Regional Public Health Consultant and Safety Officer may be contacted for assistance, if necessary.
- Workers who develop a febrile or respiratory illness within 7 weeks of the last potential exposure should immediately seek medical attention and inform the attending physician of the potential occupational risk of hantavirus infection.
- When removing rodents from traps, handling rodents, or cleaning heavily infested areas, workers should wear either a NIOSH-approved half-face or full-face, tight-seal, negative-pressure respirator or a positive pressure PAPR (powered air-purifying respirator), both options must be equipped with P-100 or N-100 filters. Employees must be in compliance with NPS Director's Order #50B and Reference Manual #50B for respiratory protection. Requirements include medical clearance and annual training and fit testing for each approved respirator type. Any individual wearing a respirator must be clean shaven.
- Workers should wear rubber, latex, vinyl, or nitrile gloves when cleaning or working in rodent infested areas, handling rodents or handling traps containing rodents. Before removing the gloves, wash gloved hands in a disinfectant or chlorine solution and then wash bare hands in soap and water.
- Mammalogists or wildlife biologists who handle wild rodents for research or management purposes should refer to the published safety guidelines available on CDC's website, All About Hantavirus (<http://www.cdc.gov/ncidod/dvrd/spb/mnpages/rodentmanual.htm>).

Precautions for Workers Having Potential Contact with Rodents

Persons who work in occupations with unpredictable or incidental contact with rodents or their nesting sites should follow general risk reduction recommendations and seek guidance from their safety manager or the Office of Public Health. Examples of such occupations include: archaeologists; natural resource specialists; utility operators; curators; and certain construction workers. Workers in these jobs may have to enter buildings and crawl spaces, or might otherwise be exposed to sites or materials that are potentially rodent-infested. Recommendations for such circumstances must be made on a case-by-case basis after the specific working environment has been assessed. The Regional Public Health Consultant or the Safety Officer may be consulted as needed to assist in the assessment. Determining the level of risk present and implementing appropriate protective measures is the responsibility of the park.

Areas with evidence of rodent activity (e.g., dead rodents, nests, and droppings) should be thoroughly cleaned to reduce the likelihood of exposure to hantavirus-infected materials. Cleanup procedures must be performed in a manner that limits the potential for dirt or dust from contaminated surfaces to become airborne. Recommendations are listed in this report for cleaning up (1) rodent urine and droppings, and surfaces potentially contaminated by rodents; and (2) dead rodents and rodent nests.

Cleanup

Cleanup of Rodent Urine, Droppings, and Contaminated Surfaces

- During cleaning, wear rubber, latex, vinyl, or nitrile gloves.
- Spray rodent urine and droppings with an EPA registered disinfectant or chlorine solution until thoroughly soaked. (See Cleanup of Dead Rodents and Rodent Nests below.) Allow disinfectant-soaked area to sit for at least 10 minutes before proceeding.
- To avoid generating potentially infectious aerosols, do not sweep rodent urine, droppings, or contaminated surfaces until they have been disinfected (soaked with disinfectant for at least 10 minutes).
- Use a paper towel to absorb the urine and disinfectant and pick up the droppings. Place the paper towel in the garbage.
- After the rodent droppings and urine have been removed, disinfect items and underlying surfaces that might have been contaminated by rodents or their urine and droppings.
 - o Mop floors with a disinfectant or chlorine solution. Allow to sit for 10 minutes before rinsing.

- o Disinfect countertops, cabinets, drawers, and other durable surfaces with a disinfectant or chlorine solution. Allow disinfectant to sit on surface for 10 minutes before wiping down.
- o Spray dirt floors with a disinfectant or chlorine solution.
- o Disinfect carpets with a disinfectant or with a commercial-grade steam cleaner or shampoo.
- o Steam-clean or shampoo rugs and upholstered furniture.
- o Launder potentially contaminated bedding and clothing with hot water and detergent. Use rubber, latex, vinyl, or nitrile gloves when handling contaminated laundry. Machine-dry laundry on a high setting or hang it to air dry in the sun.
- o Leave books, papers, and other items that cannot be cleaned with a liquid disinfectant or thrown away, outdoors in the sunlight for several hours, or in an indoor area free of rodents for approximately 3 weeks before cleanup. After that time, the virus should no longer be infectious. However, to further reduce risk, wear rubber, latex, vinyl, or nitrile gloves and wipe the items with a cloth moistened with disinfectant.
- o Before removing the gloves, wash gloved hands in a disinfectant or chlorine solution and then wash bare hands in soap and water.

Cleanup of Dead Rodents and Rodent Nests

- Wear rubber, latex, vinyl, or nitrile gloves.
- In the western United States, use insect repellent (containing DEET) on clothing, socks, and arms to reduce the risk of fleabites that might transmit plague, tularemia, or other diseases.
- Spray dead rodents and rodent nests with a disinfectant or a 10% chlorine solution, soaking them thoroughly. Wait 10 minutes before disturbing to ensure inactivation of the virus.
- Place the dead rodent or nest in a plastic bag, or remove the dead rodent from the trap and place it in a plastic bag. When cleanup is complete (or when the bag is full), seal the bag, place it into a second plastic bag, and seal the second bag. Dispose of the material in the double bag discarding it in a covered trash can that is regularly emptied.
- Clean up the surrounding area and area that was underneath the rodent as described in “Cleanup of Rodent Urine and Droppings and Contaminated Surfaces.”

Disinfecting Solutions

Two types of disinfecting solutions are recommended to clean up rodent materials.

1. General-Purpose Household Disinfectant: These can be used for light infestations (ie, rodent droppings present, evidence of chewing, but no extensive nesting or droppings). Prepare according to the label, if not prediluted. Almost any agent commercially available in the United States is sufficient as long as the label states that it is a “disinfectant” and it has an EPA registration number on the label. Effective agents include those based on phenols, quaternary ammonium compounds, and hypochlorite solutions at a 1:100 or greater concentration.
2. Hypochlorite Solution: A 10% chlorine solution, freshly prepared by mixing 1½ cups of household bleach in 1 gallon of water (or a 1:10 solution) can be used in place of a commercial disinfectant and should be used for heavily infested areas (ie, several rodent nests with extensive droppings present). When using chlorine solution, avoid spilling the mixture on clothing or other items that might be damaged by bleach. Wear rubber, latex, vinyl, or nitrile gloves when preparing and using chlorine solutions. Chlorine solutions should be prepared fresh daily.

Cleaning Shed and Other Buildings

Before cleaning closed sheds and other outbuildings, ventilate the building by opening doors and windows for at least 30 minutes. Use cross ventilation if possible. Be sure that you do not stir up any dust when entering to open windows and leave the area during the airing-out period. This airing helps to remove infectious primary aerosols that might be created by hantavirus-infected rodents. In substantially dirty or dusty environments, additional protective clothing or equipment may be worn. Such equipment includes coveralls (disposable when possible) and safety glasses or goggles, in addition to rubber, latex, vinyl, or nitrile gloves. For recommendations regarding precautions for cleanup of outbuildings with heavy rodent infestations, see below.

Recommendations for Cleaning Homes or Buildings with Heavy Rodent Infestations

Special precautions are indicated for cleaning homes or buildings with heavy rodent infestations. A rodent infestation is considered heavy if piles of feces or numerous nests or dead rodents are observed. Persons cleaning these homes or buildings should contact their Safety Officer or Public Health Consultant. These precautions also can apply to vacant dwellings that have attracted rodents while unoccupied and to dwellings and other structures that have been occupied by persons with

confirmed hantavirus infection. Workers who are either hired specifically to perform the cleanup or asked to do so as part of their work activities should receive a thorough orientation about hantavirus transmission and disease symptoms and should be trained to perform the required activities safely.

- If the building has been closed and unoccupied for a long period (weeks or months), ventilate the building by opening doors and windows for at least 30 minutes before beginning any work. The ventilation helps to remove aerosolized virus inside the structure. Use cross ventilation if possible. Leave the area during the airing-out period.
- Persons involved in the cleanup should wear coveralls (disposable, if possible); rubber boots or disposable shoe covers; rubber, latex, vinyl, or nitrile gloves; protective goggles; and a respirator with appropriate clearance, annual fit-testing, and approvals as detailed in “Precautions for Workers Frequently Exposed to Rodents.”
- Personal protective gear should be decontaminated or safely disposed of upon removal at the end of the day. If the coveralls are not disposable, they should be laundered on site. If no laundry facilities are available, the coveralls should be immersed in liquid disinfectant until they can be washed.
- Wash hands thoroughly after personal protective equipment is removed.
- Unless burned on site, all potentially infectious waste material from cleanup operations should be double-bagged in durable plastic bags and then discarded in a covered trash can that is regularly emptied. Contact the local or state health department concerning other appropriate disposal methods.
- Persons involved in the cleanup who develop a febrile or respiratory illness within seven weeks of the last potential exposure should immediately seek medical attention and inform the attending physician of the potential occupational risk of hantavirus infection.

Special Considerations for Historic Structures or Structures with Dirt Floors

- Consult cultural resources staff before beginning any work in historic structures.
- Some disinfectants, such as a 10% or greater bleach solutions or repeat use of other disinfectants, can change the color of an organic surface (such as wood, cloth, etc). If this is a concern for a historic structure, disinfectants such as household disinfectants or isopropyl/ethyl alcohol might be more appropriate.
- Dirt floors should be treated as other hard surfaces, with extra care taken not to stir up dust. Thoroughly wet the area and adjacent area with disinfectant, allow the disinfectant to sit for at least 10 minutes, and use a wetted paper towel to clean up the droppings or material.

Contacts

Integrated Pest Management (IPM) – 202-513-7183

Public Health – 202-513-7217

Risk Management – 202-513-7214

Wildlife Health – 970-267-2118

The Department of the Interior protects and manages the nation's natural resources and cultural heritage; provides scientific and other information about those resources; and honors its special responsibilities to American Indians, Alaska Natives, and affiliated Island Communities.

NPS, December 2013

National Park Service
U.S. Department of the Interior



Office of Public Health
1201 Eye Street NW, Room 52
Washington, DC 20005

Natural Resource Stewardship and Science
1201 Oakridge Drive, Suite 150
Fort Collins, CO 80525

www.nature.nps.gov

Geotechnical Evaluation
Bear Gulch Nature Center and Bear Valley School
Pinnacles National Park
San Benito County, California

Anderson Hallas Architects, PC
715 Fourteenth Street | Golden, Colorado 80401

January 27, 2025 | Project No. 502075008



Geotechnical | Environmental | Construction Inspection & Testing | Forensic Engineering & Expert Witness

Geophysics | Engineering Geology | Laboratory Testing | Industrial Hygiene | Occupational Safety | Air Quality | GIS

Ninyo & Moore
Geotechnical & Environmental Sciences Consultants

Geotechnical Evaluation
Bear Gulch Nature Center and Bear Valley
School
Pinnacles National Park
San Benito, California

Ms. Elizabeth Hallas, AIA, LEED AP BD+C
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715 Fourteenth Street | Golden, Colorado 80401

January 27, 2025 | Project No. 502075008



Forrest K. Wright, PE
Senior Engineer

FKW/ARD/rk



Anthony R. Dover, PE, GE
Principal Engineer



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2B – Boring and Test Pit Locations for Bear Valley School

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B – Laboratory Testing by Ninyo & Moore
C – Laboratory Testing by Others
D – Photographs from Field Exploration

1 INTRODUCTION

In accordance with your request and authorization, we have performed a geotechnical evaluation for the proposed rehabilitation of Bear Gulch Nature Center and Bear Valley School project located in Pinnacles National Park and San Benito County, California. The approximate location of the site is depicted on Figure 1.

The purpose of our study was to evaluate the subsurface conditions and to provide design and construction recommendations regarding geotechnical aspects of the proposed project. This report presents the findings of our subsurface exploration, results of our laboratory testing, conclusions regarding the subsurface conditions at the site, and geotechnical recommendations for design and construction of this proposed improvements.

2 SCOPE OF SERVICES

The scope of our services for the project generally included:

- Review of referenced background information, including aerial imagery, published geologic maps, in-house geotechnical data, and available topographical information pertaining to the project site and vicinity.
- Notification of Utility Notification Center of California of the boring locations prior to drilling.
- Subcontracting the services of a private utility locating firm to mark traceable privately-owned utilities.
- Drilling, logging, and sampling of two small-diameter exploratory borings, one outside each building, to depths of approximately 11 to 35 feet below the ground surface (bgs). The boring logs are presented in Appendix A. Boring locations are presented on Figures 2A (Bear Gulch Nature Center) and 2B (Bear Valley School).
- Performance of laboratory tests on selected samples obtained from the borings to evaluate engineering properties including in-situ moisture content, Atterberg limits, percent materials passing the No. 200 sieve, particle size analysis, and soil corrosivity characteristics (including pH, resistivity, water soluble sulfates, and chlorides). The results of the laboratory testing performed by Ninyo & Moore are presented on the boring logs and in Appendix B. The results of the laboratory testing for corrosivity performed by others are presented in Appendix C.
- Excavating four test pits to expose the exterior foundations of the buildings up to a depth of approximately 4 feet bgs. The test pit locations are presented in Figures 2A (Bear Gulch Nature Center) and 2B (Bear Valley School). Test pit logs are provided in Appendix A. Photographs from the test pit excavations are provided in Appendix D.
- Compilation and analysis of the data obtained.
- Preparation of this report presenting our findings, conclusions, and geotechnical recommendations regarding design and construction of the project.

3 SITE DESCRIPTION AND BACKGROUND REVIEW

The project consists of the rehabilitation of two buildings within Pinnacles National Park. The first site is the Bear Gulch Nature Center located near the center of Pinnacle National Park at the terminus of a two-lane asphalt access road approximately 0.75 miles west of Highway 146. The site is located in Bear Gulch, which is surrounded on three sides by steep mountain slopes of volcanic rock. The second site is Bear Valley School, which is located at the perimeter of Pinnacles National Park approximately 160 feet east of Highway 25. Bear Valley is located along a seasonal creek bisecting steeply sloped mountains to the west and moderately sloped mountains to the east. Highway 25 runs along the base of the steeply sloped volcanic rock mountains to the west. The seasonal creek is approximately 100 feet to the east of the Bear Valley School building.

The Bear Gulch Nature Center is an approximately 700 square-foot (sf) single-story building. The building was built in the 1937 by the Civilian Conservation Corps (CCC) using Pumice Lapilli Tuff. Based on observations provided by the project structural engineer and design team, we understand the building was constructed with grouted boulder foundations and has a wooden structural floor. The building has experienced distress due to differential movement. The most significant distress are separations in the grouted boulders and framing, which bisects the building at the chimney locations. Ninyo & Moore understands that these separations were previously repaired in the 1970's but the building continues to exhibit distress at these locations.

Based on our observations of the site photographs the rear of the building is cut approximately 3 feet into the rock slope. However, the front of the building provides utility access for water and is supported on fill material placed during site construction. There are below grade utilities to include water and stormwater in asphalt pavement in front of the building. Additionally, a corrugated metal culvert bridge, up to a depth of approximately 6 feet, is located approximately 25 feet southwest of the building.

The Bear Valley School was originally constructed in 1903 as a one-room, single-story, wood-framed structure with an approximately footprint of 1,100 sf. We understand the building is supported on a concrete slab-on-grade floor with turned down concrete edge foundations. It appears the building foundation was reinforced in 1986 by doweling in additional concrete footings. The site is serviced with a water line connecting to a well on the north side of the building along with a sanitary line running to the east towards a septic field. Additionally, an unknown line was identified going from the building to the east (towards the seasonal creek bed).

4 PROPOSED CONSTRUCTION

The purpose of this project is to consider potential repair and renovation of the existing buildings. The Bear Gulch Nature Center will continue to be used as a visitor contact and educational center for public usage. Several proposed uses for the Bear Valley school have been provided by Anderson Hallas (Anderson Hallas, 2022) to include educational center, event venue, or concessioner. No significant additions or foundation loading increases are currently anticipated as part of this renovation and repair process.

5 FIELD EXPLORATION AND LABORATORY TESTING

On November 4, 2024, Ninyo & Moore conducted a subsurface exploration at the site to evaluate the existing subsurface conditions and to collect soil samples for laboratory testing. The evaluation consisted of the drilling and logging two boring using 6-inch inner diameter hollow-stem augers. Boring B-1 was drilled near the Bear Gulch Nature Center and Boring B-2 was drilled near Bear Valley School building. Boring B-1 was advanced to a depth of approximately 11 feet bgs prior to being terminated due to auger refusal. Boring B-2 was advanced to a boring depth of approximately 35 feet bgs due to auger refusal. The approximate locations of the borings for the Bear Gulch Nature Center and Bear Valley School are presented on Figures 2A and 2B, respectively. Split-spoon soil and modified California samples were collected at selected intervals. The sampling methods used during the subsurface evaluation are presented in Appendix A.

Additionally, two test pits were advanced around the exterior foundation systems for both the Bear Gulch Nature Center (TP-1 and TP-2) and the Bear Valley School (TP-3 and TP-4). Test pits for the Bear Gulch Nature Center were extended to the foundation depths of approximately 3.6 feet bgs on the north side of the structure and 1.8 feet bgs on the south side of the structure. Test pits for the Bear Valley School were extended to depth of approximately 2.5 to 3 feet bgs on the east and west corner of the building. The approximate locations of test pits at the Bear Gulch Nature Center and Bear Valley School are presented on Figure 2A and 2B, respectively.

Soil samples collected during the subsurface exploration were transported to the Ninyo & Moore laboratory and a third-party laboratory (CERCO Analytical) for geotechnical laboratory analyses. Selected samples were analyzed to evaluate engineering properties including in-situ moisture content, dry density, Atterberg limits, percent materials finer than the No. 200 sieve, particle size analysis, and soil corrosivity characteristics (including resistivity, pH, water soluble sulfates and chlorides). The results of the in-situ moisture content tests are presented on the boring logs in Appendix A. Descriptions of the laboratory test methods and the remainder of the test results performed by Ninyo & Moore are presented in Appendix B. Additional laboratory test results performed by CERCO Analytical are provided in Appendix C.

6 GEOLOGY AND SUBSURFACE CONDITIONS

The geology and subsurface conditions at the site are described in the following sections.

6.1 Geologic Setting

The sites are located in the southern portion of the Gabilan Range in central and eastern portion of Pinnacles National Park in San Benito County, California. This region is heavily impacted by the both volcanic and faulting forces in their geologic past. The Pinnacles portion of the Park were formed approximately 23 million years ago by the Neenach Volcano. This extinct Miocene volcano is located near Lancaster, California (approximately 195 miles to the northwest) but is split by the San Andres Fault.

Geologic mapping by Dibblee and Minch (2007a) indicates the Bear Gulch Nature site is underlain by Miocene-age volcanic rock intrusive into the granitic basement. During our site reconnaissance we generally observed exposed bedrock around the back of the building which was cut into the hillside. Geologic mapping by Dibblee and Minch (2007b) indicates the Bear Valley School is underlain by alluvium consisting of sands and gravels from major stream channels. The underlying bedrock material is mapped as Miocene aged sandstone and conglomerate sedimentary deposits (Wagner, 2002).

6.2 Subsurface Conditions

Our understanding of the subsurface conditions at the project site is based on our field exploration and laboratory testing, review of published geologic maps, and historic aerial imagery. The following sections provide a generalized description of the subsurface materials encountered. More detailed descriptions are presented on the boring logs in Appendix A.

6.2.1 Bear Gulch Nature Center

Surficial pavement material was encountered at the ground surface at Boring B-1 near Bear Gulch Nature Center consisting of 1 inch of asphalt underlain by 2 inches of aggregate base. Underlying the surficial pavement section, fill material was encountered extended to a depth of approximately 10 feet bgs. Fill material within the boring generally consisted of light brown to reddish brown, moist to wet, well-graded sand with fines. Fill materials encountered in our borings is considered undocumented. Based on the results of the laboratory testing, the fill material had moisture contents between approximately 3.1 to 4.8 percent. The fill material was underlain by hard rhyolite bedrock which exhibited auger refusal at a depth of approximately 11 feet.

6.2.2 Bear Valley School

Alluvium was encountered at the ground surface in Boring B-2 near the Bear Valley School and extended to the boring's termination depth of approximately 35 feet bgs. The alluvium generally consisted of dark to light brown with some gray, moist to wet, loose to medium dense, clayey sand, medium dense, poorly graded sand, and stiff to very stiff, lean clay. Based on the results of the laboratory testing, selected samples of the alluvium generally had in-place moisture contents ranging from approximately 4.9 to 21.6 percent and dry densities 96.6 to 112.1 pounds per cubic foot (pcf).

6.3 Groundwater

Groundwater was encountered in Borings B-1 and B-2 at depths of approximately 8.5 to 18.5 feet bgs, respectively, at time of drilling. Stabilized groundwater measurements were not included in the scope of this investigation. Groundwater levels can fluctuate due to seasonal variations, precipitation, flash flooding, groundwater withdrawal or injection, and other factors. Depending on the time of year construction occurs, groundwater could be encountered.

7 GEOLOGIC HAZARDS

The following sections describe potential geologic hazards at the site including faulting and seismicity, expansive soils, compressible/collapsible soils, liquefaction potential, and landslides.

7.1 Seismic Hazards

The seismic hazards considered in this study include the potential for ground rupture due to faulting, seismic ground shaking, liquefaction, dynamic settlement, sand-boil-induced ground subsidence, lateral spreading, compressible/collapsible soils, and landslides. These potential hazards are discussed in the following subsections.

7.1.1 Faulting and Ground Surface Rupture

Based on our review of seismic hazard maps, geologic literature, and geologic maps, the site is not located within a State of California Earthquake Fault Zone (formerly known as Alquist-Priolo Special Studies Zone), and no active faults are known to cross the subject site. Therefore, the probability of damage from surface fault rupture is considered to be low. However, localized lurching or cracking of the ground surface as a result of nearby seismic events is possible.

7.1.2 Strong Ground Motion

The sites are within a seismically active region and the potential for future seismic ground shaking is considered high. The 2022 California Building Code (CBC) specifies that the potential for liquefaction and soil strength loss be evaluated, where applicable, for the Maximum Considered Earthquake Geometric Mean (MCE_G) peak ground acceleration with adjustment for site class effects in accordance with the American Society of Civil Engineers (ASCE) 7-22 Standard. The PGA associated with the MCE_G was calculated in accordance with the American Society of Civil Engineers (ASCE) 7-22 Standard and the 2022 CBC. The MCE_G peak ground acceleration with adjustment for site class effects (PGA_M) was calculated as 0.51g for the Bear Gulch Nature Center and 0.67g for the Bear Valley School using the ASCE Hazard Tool seismic design map (ASCE, 2024) for Site Class C and CD, respectively.

Table 2 presents the Risk-Targeted, Maximum Considered Earthquake (MCER) spectral response accelerations consistent with the 2022 CBC and corresponding site-adjusted and design level spectral response accelerations based on the ASCE Seismic Hazard Tool (ASCE, 2024). Using the referenced USGS seismic web application (ASCE, 2024), estimated MCE_R spectral response accelerations for short (0.2 second) and long (1.0 second) periods were obtained for the project sites. Based on the information provided in Table 1 and Table 2 the site classifies as Seismic Design Category D for Bear Gulch Nature Center and E for Bear Valley School.

Table 1 – Bear Gulch Nature Center Seismic Design Criteria	
Seismic Design Factors	Value
Site Class	C
Mapped Spectral Acceleration at 0.2-second Period, S_s	1.5 g
Mapped Spectral Acceleration at 1.0-second Period, S_1	0.6 g
Spectral Acceleration at 0.2-second Period Adjusted for Site Class, S_{MS}	1.51 g
Spectral Acceleration at 1.0-second Period Adjusted for Site Class, S_{M1}	2.69 g
Design Spectral Response Acceleration at 0.2-second Period, S_{DS}	1.08 g
Design Spectral Response Acceleration at 1.0-second Period, S_{D1}	0.57 g

Table 2 – Bear Valley School Seismic Design Criteria

Seismic Design Factors	Value
Site Class	CD
Mapped Spectral Acceleration at 0.2-second Period, S_s	2.06 g
Mapped Spectral Acceleration at 1.0-second Period, S_1	0.77 g
Spectral Acceleration at 0.2-second Period Adjusted for Site Class, S_{MS}	1.58 g
Spectral Acceleration at 1.0-second Period Adjusted for Site Class, S_{M1}	3.37 g
Design Spectral Response Acceleration at 0.2-second Period, S_{DS}	1.45 g
Design Spectral Response Acceleration at 1.0-second Period, S_{D1}	1.19 g

7.1.3 Liquefaction and Strain Softening

The strong vibratory motions generated by earthquakes can trigger a rapid loss of shear strength in saturated, loose, granular soils of low plasticity (liquefaction) or in wet, sensitive, cohesive soils (strain softening). Liquefaction and strain softening can result in a loss of foundation bearing capacity or lateral spreading of sloping or unconfined ground. Liquefaction can also generate sand boils leading to subsidence at the ground surface. Liquefaction (or strain softening) is generally not a concern at depths more than 50 feet below ground surface.

During our subsurface exploration, we encountered layers of loose to medium dense and relatively clean sand and gravel in Boring B-2 below historic high groundwater levels that could be susceptible to liquefaction. We evaluated the liquefaction susceptibility of these deposits in the vicinity of the existing buildings in accordance with the method presented by Idriss & Boulanger (2008) using the penetration resistance data collected during our subsurface exploration and considering a PGA of 0.548g corresponding to a Magnitude 6.85 earthquake on the San Andreas Fault Zone (Caltrans, 2012) and OSHPD produced PGA max of 0.634. For the liquefaction analysis, we assumed a groundwater depth of 18.5 feet. The results of our analysis indicate that clayey sand layers in Borings B-2 between depths of approximately 18 to 25 feet below the existing ground surface will liquefy under the considered ground motion based on a factor of safety against liquefaction of less than one. The anticipated ground surface settlement is approximately 1.5 inches; however, this liquefaction will not result in a loss of foundation bearing capacity.

7.1.4 Dynamic Settlement

The strong vibratory motion associated with earthquakes can also dynamically compact loose granular soil leading to surficial settlements. Dynamic settlement is not limited to the near

surface environment and may occur in both dry and saturated sand and silt. Cohesive soil is not typically susceptible to dynamic settlement.

During our subsurface exploration, near surface loose to medium dense clayey soils were encountered continuing to boring termination. Due to these subsurface conditions, this site does meet the parameters of dynamic settlement. Dynamic settlement is anticipated to be approximately ½-inch. This settlement should be included with the potential liquefaction settlement discussed in Section 7.1.3., for a total settlement of about 2 inches. Differential settlement from a design seismic event is estimated be about 1 inch over a distance of about 40 feet.

Based on our experience the proposed structural improvements may tolerate such differential dynamic settlement, but if such settlement is unacceptable to the structural engineer, mitigation measures should be considered. Mitigation for the dynamic settlement hazard may include structural mitigation and/or geotechnical mitigation such as ground improvement or underpinning. We do not anticipate that reconstructing the building from the ground up or underpinning the footprint area of the proposed restroom building would be in the scope of the proposed rehabilitation, and as such we are not providing recommendations for such scopes of work. However, we can provide more detailed geotechnical recommendations to mitigate the dynamic settlement hazard upon request.

7.2 Landslides

Landslides may be triggered by several mechanisms including strong vibratory motion produced by earthquakes. Research and historical data indicate that seismically induced landslides tend to occur in weak soil and rock on sloping terrain. The subject sites and immediate vicinities are at the base of slopes and as such, we do not regard landsliding as a design consideration for the proposed project. It should be noted that on the relatively steeper slopes ascending from the subject sites, we observed cobbles and small boulders which may descend into the site if dislodged. We did not observe indications of slope instability on the subject slopes during our site reconnaissance. A detailed rock fall hazard analysis or slope stability analysis of the slopes surrounding the sites are out of the scope of our geotechnical evaluation.

7.3 Expansive Soils

Moisture changes to bedrock or surficial deposits containing swelling clays can result in volumetric expansion and collapse of those units. Changes in soil moisture content can result from rainfall, irrigation, pipeline leakage, surface drainage, perched groundwater, drought, or other factors. Volumetric change of expansive soil may cause excessive cracking and heaving of structures with

shallow foundations, concrete slabs-on-grade, or pavements supported on these materials. Construction on soils known to be potentially expansive could have a significant impact to the project.

Based on the results of our subsurface exploration and laboratory testing, the alluvium deposits generally exhibited a very low potential for expansion on a scale ranging from very low to high. The alluvium had a relatively low percentage of particles passing the No. 200 sieve, and Atterberg limits testing indicated that the alluvium materials are have low plasticity index.

7.4 Compressible/Collapsible Soils

Compressible soils are generally comprised of soils that undergo consolidation when exposed to new loadings, such as fill or foundation loads. Soil collapse (or hydrocollapse) is a phenomenon where soils undergo a significant decrease in volume upon an increase in moisture content, with or without an increase in external loads. Buildings, structures, and other improvements may be subject to excessive settlement-related distress when compressible soils or collapsible soils are present.

Based on the nature along with the depositional environment of the alluvium encountered during our subsurface exploration, it is our opinion that the potential for post-construction elastic and consolidation settlement of the alluvium is low, ranging from about 0.25 to 0.5 inches of differential settlement over a span of 30 feet due to site grade increases of no more than 2 feet. New foundations with loading similar to the existing structure loads would be expected to settle in the same range. Laboratory testing was not performed to characterize the consolidation characteristics of the fill materials, but due to the age of the fill material, the relatively high blow counts, and granular nature of the fill it is our opinion that the existing fill material has a low consolidation potential. This fill should be further evaluated in the field by a Licensed Professional Engineer prior to placement of any new foundation system.

8 CONCLUSIONS

Based on the results of the subsurface evaluation, laboratory testing, and data analyses, it is our opinion that the proposed project is feasible from a geotechnical standpoint, provided the recommendations presented herein are implemented and appropriate construction practices are followed. Geotechnical design and construction considerations for the proposed project include the following:

- Surficial pavement material was encountered at the ground surface at Boring B-1 near Bear Gulch Nature Center consisting of 1 inch of asphalt underlain by 2 inches of aggregate base. Underlying the surficial pavement section, fill material was encountered extended to a depth of approximately 10 feet bgs.

- Alluvium was encountered at the ground surface in Boring B-2 near the Bear Valley School and extended to the boring's termination depth of approximately 35 feet bgs. The alluvium generally consisted of dark to light brown with some gray, moist to wet, loose to medium dense clayey sand, medium dense poorly graded sand, and stiff to very stiff lean clay.
- The on-site fill materials and alluvium may generally be excavated with moderate- to heavy-duty earthmoving or excavation equipment in good operating condition. Excavations around Bear Valley Nature Center are not anticipated to extend into the rhyolite bedrock, however, if they do more aggressive excavation techniques will be required.
- Groundwater was encountered in Boring B-2 at a depth of approximately 18.5 feet bgs at time of drilling. Stabilized groundwater measurements were not included in the scope of this investigation. Groundwater was encountered at a depth of approximately 8.5 feet bgs in the Boring B-1. Groundwater levels can fluctuate due to seasonal variations, precipitation, flash flooding, groundwater withdrawal or injection, and other factors. Depending on the time of year construction occurs, groundwater could be encountered.
- Based on our laboratory data, the sulfate content of the tested fill material and alluvial soils presents a negligible risk of sulfate attack to concrete. We recommend the use of Type I/II cement for construction of concrete structures at this site.
- Based on our laboratory data, the subgrade soils at the Bear Gulch Nature Center are highly corrosive and at the Bear Valley School are moderately corrosive to ferrous metals.
- The primary geologic hazard affecting the proposed development at the site is seismicity and associated ground motion generated from the San Andres Fault. Known active faults are not mapped at the proposed locations for the improvements, therefore the probability of damage from surface fault rupture is considered to be low. Based on our analysis, strong ground motion are design considerations for the proposed improvements.

9 RECOMMENDATIONS

Based on our understanding of the project, the following sections present our geotechnical recommendations for design and renovation of the existing buildings and other site improvements.

9.1 Demolition

The subject project may include demolition of existing pavements and other site improvements. In addition, it is unknown whether previous iterations of construction at the locations of the proposed improvements have occurred. As a result, remnants from the previous demolition activities may be encountered during construction. Remnants from previous or current demolition activities should be removed from the site. Demolition of the existing improvements should include rerouting, removal, or in-place abandonment of underground utilities. Utilities should be adequately capped or rerouted at the project perimeter in accordance with the requirements of the governing authorities and the recommendations of the geotechnical consultant at the time of demolition. Abandoned underground utility pipes under proposed building limits should be removed from the site, or, if the pipes are left in place, they should be filled with flowable fill, such as grout or controlled low strength material (CLSM). The contractor should take adequate

precautions when conducting excavations associated with the demolition activities to reduce the potential for damage to existing utilities that are to remain in service.

If structural components of existing underground utility pipes will be left in place under the proposed building footprints, these should be discussed on a case-by-case basis with the project Geotechnical Engineer and the design team. Additional site-specific evaluations and data analysis may be needed to address these conditions.

9.2 Earthwork

The following sections provide our earthwork recommendations for this project. We anticipate the site grading may consist of material cuts and fills of approximately 2 feet or less. Deeper cuts and fills may be needed to install buried utilities.

9.2.1 Site Grading

Prior to grading, the ground surface in proposed improvement areas should be cleared of any surface obstructions, existing pavements, debris, topsoil, organics (including vegetation), and other deleterious material. Based on our site observations, stripping depths of 6 inches or less should be anticipated. Material generated from site stripping should not be incorporated into engineered fill, but may be stockpiled for re-use as landscaping material or other non-structural material. Topsoil-contaminated fill materials should not be used during site grading.

Materials generated from clearing operations should be removed from the project sites for disposal (e.g., at a legal landfill site). Obstructions that extend below finish grade, if present, should be removed and resulting voids filled with compacted, engineered fill or CLSM.

Undocumented fill was encountered at the boring location B-1 near Bear Gulch Nature Center at the sites to a depth up to approximately 10 feet bgs. This fill is considered undocumented and will need to be further evaluated during construction for suitability to support the site improvements.

The exposed alluvium materials at Bear Valley School should be firm and unyielding prior to fill placement. Subgrades should be evaluated by a geotechnical representative during the excavation work. Additional recommendations specific to the site conditions encountered may be provided at the time of construction. The project budget should include additional cost associated with the removal and replacement of unsuitable subgrade material. Subgrade materials that are disturbed during grading should be moisture-conditioned and re-compacted according to the recommendations provided in this report.

The proposed building improvements may be supported on a shallow foundation system consisting of spread footings bearing on 12 or more inches of engineered fill extending to firm alluvium or approved fill. Deepening of the footing excavation may be needed to penetrate through isolated, softer site soils/rock. The engineered fill limits should extend laterally 1-foot for every 1-foot of overexcavation depth beyond the footing limits.

The proposed slab-on-grade floor systems may be supported on 12 or more inches of engineered fill extending to firm alluvium or approved fill. The engineered fill limits should extend laterally 1-foot for every 1-foot of overexcavation depth beyond the equipment pad limits.

New pavements and exterior flatwork should be placed on 12 or more inches of moisture-conditioned and compacted engineered fill extending to firm alluvium or approved fill.

9.2.2 Excavations

Our evaluation of the excavation characteristics of the on-site materials is based on the results of the subsurface exploration, our site observations, and our experience with similar materials. The on-site surface and near-surface soils (fill and alluvium) may generally be excavated with moderate to heavy-duty earthmoving or excavation equipment in good operating condition. Excavations may slow down in zones of the fill and alluvium with cobbles. Excavations around Bear Valley Nature Center are not anticipated to extend into the rhyolite bedrock, however, if they do more aggressive excavation techniques will be required. These techniques may require the use of excavators equipped with rock teeth, hoe rams, or blasting.

Equipment and procedures that do not cause significant disturbance to the excavation bottoms should be used. Excavators and backhoes with buckets having large claws to loosen the soil should be avoided when excavating the bottom 6 to 12 inches of excavations as such equipment may disturb the excavation bases.

The fill materials and alluvium are susceptible to variations in moisture content. As a result, and depending on the time of year construction occurs, wet or saturated soils may be encountered after periods of heavy or prolonged precipitation. These materials may soften under the action of light equipment and foot traffic. Where encountered, drying or overexcavation of these materials is recommended. If the subgrade becomes disturbed, it should be compacted or removed and replaced before placing additional backfill material.

Groundwater was encountered in Boring B-1 at a depth of approximately 8.5 feet bgs and Boring B-2 at depth of approximately 18.5 feet bgs. Groundwater levels can fluctuate due to

seasonal variations in precipitation, flash flooding, snowmelt, irrigation, groundwater withdrawal or injection, and other factors.

Depending on the depth of foundation associated excavations and buried utility lines, groundwater may be encountered. Excavations within 5 feet of the groundwater table may encounter soft and/or wet conditions; therefore, implementation of various stabilization dewatering techniques and associated permitting may be needed for construction of the proposed improvements.

Stabilization methods should be provided by the grading contractor, as needed, and may include the use of geogrids, geotextiles, pushing oversized rock into the subgrade, and/or chemical stabilization. The subgrade stabilization methods proposed should be discussed with the Geotechnical Engineer prior to implementation. The stabilization method selected should consider the effects of such stabilization on future utility installation and/or repair work.

The contractor should provide safely sloped excavations or an adequately constructed and braced shoring system, in compliance with Occupational Safety and Health Administration (OSHA, 2005) guidelines, for employees working in an excavation that may expose employees to the danger of moving ground. If material is stored or equipment is operated near an excavation, stronger shoring should be used to resist the extra pressure due to superimposed loads.

9.2.3 Temporary Excavations

Temporary excavations will be needed for this project to construct utilities. Based on the subsurface information obtained from our exploratory borings and our experience with similar projects, we anticipate that the soil conditions and stability of the excavation sidewalls may vary with depth. Soils with higher fines content may stand vertically for a short time (less than 12 hours) with little sloughing. However, as the soil dries after excavation or as the excavations are exposed to rainfall, sloughing may occur. Soils with low cohesion (e.g., predominately sandy or gravelly material), may slough or cave during excavation, especially if wet or saturated.

The contractor should provide safely sloped excavations or an adequately constructed and braced shoring system, in compliance with OSHA regulations as mentioned in Section 9.2.2.

In our opinion, the fill and alluvium materials on site should generally be considered Type C soil when applying the OSHA regulations. For Type C soil conditions, OSHA recommends a temporary slope inclination of 1.5:1 (horizontal to vertical) or flatter for excavations 20 feet or

less in depth. Appropriate slope inclinations should be evaluated in the field by an OSHA-qualified “Competent Person” based on the conditions encountered.

9.2.4 Re-Use of Site Soils

The on-site soils encountered during the subsurface exploration in the fill and alluvium were generally granular. The granular fill and alluvium are anticipated to be suitable to be re-used as engineered fill, provided they meet the specifications of imported fill in Section 9.2.5. However, additional testing should be performed at the time of fill placement to check the suitability of the material. Due to presence of cobbles encountered in the fill and alluvium during the subsurface exploration, we anticipate that screens will be needed to process these materials to be re-used as engineered fill.

Fragments of rock, cobbles, and inert construction debris (e.g., concrete or asphalt) larger than 3 inches in diameter may be incorporated into the project fills in non-structural areas and below the anticipated utility installation depths. A Geotechnical Engineer should be consulted regarding appropriate recommendations for usage of such materials on a case-by-case basis when such materials have been observed during earthwork. Care should be taken to avoid nesting of oversized materials during placement.

9.2.5 Imported Soil

Imported soil for use as engineered fill should have 35 or less percent passing the No. 200 sieve, a low swell potential (approximately 1-percent or less when wetted against a surcharge pressure of 200 psf), and a low plasticity index (approximately 15 or less). Imported soil should not contain organic matter, clay lumps, bedrock (shale, claystone, sandstone, etc.) fragments, debris, other deleterious matter, or particles larger than approximately 3 inches nominal diameter.

Imported soil for use as engineered fill should exhibit low corrosion potential. Imported soil placed in contact with ferrous materials should have a saturated soil resistivity of 2,000 ohm-cm or more and a chloride content of 25 parts per million or less. Soils in contact with concrete should exhibit soluble sulfate content less than 0.1 percent.

We further recommend that proposed import material be evaluated by the project’s geotechnical consultant at the borrow source for its suitability prior to importation to the project site.

9.2.6 Fill Placement and Compaction

Granular soils (on-site soils that classify as sands/gravels, or import soils) used as engineered fill should be moisture-conditioned to moisture contents within 2 percent of optimum moisture content. Engineered fill should be compacted to a relative compaction of 90 percent, or more, as evaluated by American Society for Testing and Materials (ASTM) D1557.

Lift thickness for fill will be dependent upon the type of compaction equipment utilized, but should generally be placed in lifts not exceeding 8 inches in loose thickness. Fill materials should not be placed, worked, rolled while they are frozen, thawing, or during poor/inclement weather conditions. Compaction areas should be kept separate, and no lift should be covered by another until relative compaction and moisture content within the recommended ranges are obtained.

Use of a CLSM should be considered in lieu of compacted fill for areas with low tolerances for surface settlements and vibrations, and in areas with difficult access for compaction equipment.

9.2.7 Controlled Low Strength Material (CLSM)

Use of CLSM should be considered in lieu of compacted fill for areas with low tolerances for surface settlements, for excavations that extend below the groundwater table and in areas with difficult access for compaction equipment. CLSM consists of a fluid, workable mixture of aggregate, Portland cement, and water. CLSM should be placed in lifts of 5 feet or less with a 24-hour or more curing period between each lift.

The use of CLSM has several advantages:

- A narrower excavation can be used where shoring is present, thereby minimizing the quantity of soil to be excavated and possibly reducing disturbance to the near-by traffic;
- Compaction requirements do not apply;
- There is less risk of damage to improvements, since little compaction is needed to place CLSM;
- CLSM can be batched to flow into irregularities in excavation bottoms and walls; and
- The number of workers needed inside the trench excavation is reduced.

The CLSM mix design should be submitted for review prior to placement. The 28-day strength of the material should be no less than 50 pounds per square inch (psi) and no more than 150 psi. CLSM should be observed and tested by the geotechnical consultant.

9.2.8 Utility Installation

The contractor should provide adequate mechanical compaction in utility trench backfills. The contractor should take particular care to achieve and maintain adequate compaction of the backfill soils around manholes, valve risers and other vertical pipeline elements where settlements are commonly observed. Use of CLSM should be considered in lieu of compacted soil backfill for areas with low tolerances for surface settlements in deep utility excavations and in areas with difficult access for compaction equipment. Use of imported soil, blending the site soils with high moisture contents with drier material, use of imported soil may be needed when backfilling utility trenches that extend deeper than 4 feet bgs (see Section 9.2.5).

Pipe bedding materials, placement and compaction should meet the specifications of the pipe manufacturer and applicable local and state standards. Materials proposed for use as pipe bedding should be tested for suitability prior to use.

Special care should be exercised to avoid damaging the pipe or other structures during the compaction of the backfill. In addition, the underside (or haunches) of the buried pipe should be supported on bedding material that is compacted as described above. This may need to be performed with placement by hand or small-scale compaction equipment.

Positive surface drainage should be maintained over utility trench alignments. Where topography, site constraints or other factors limit or preclude adequate surface drainage, the granular bedding materials should be surrounded by a non-woven geotextile fabric having a maximum average Apparent Opening Size less than 0.43 mm (US Sieve #40) by ASTM D4751 and a minimum grab tensile strength of 120 pounds by ASTM D4632 to reduce the potential for migration of fines into the bedding and subsequent, associated settlement.

Sandy pipe bedding materials can function as conduits for re-distribution of natural and applied waters in the subsurface. Development of site grading plans should consider the subsurface transfer of water in utility trenches and the pipe bedding in areas where the utility trenches penetrate the building footprint. Cut-off walls in utility trenches or other water-stopping measures should be implemented to reduce the rates and volumes of water transmitted along utility alignments and transmitted toward the proposed building, or other structures, where wetting of the underlying soils increases the potential for soil movements, material degradation, or structural distresses. Incorporation of water cut-offs and/or outlet mechanisms for saturated bedding materials into development plans should be considered for the project. Cutoffs and outlet mechanisms will help reduce the potential for migration of fines into the bedding and subsequent, associated settlement.

9.3 Foundation Systems

The following foundation design parameters may be used for the design of foundation systems, if required, for any new buildings.

9.3.1 Spread-Footing Foundations

We anticipate that new spread footings may be constructed for the existing building along with additional loading of the existing shallow foundations. Perimeter and interior spread footings should extend to 12 or more inches below the lowest exterior finished grade. Continuous footings and isolated pad footings should have widths of 18 and 24 inches or more, respectively. Footings should be reinforced in accordance with the recommendations of the Structural Engineer.

New footings or existing shallow foundations bearing on 12 or more inches of compacted engineered fill or approved existing fill/alluvium, as recommended in Section 9.2.1, may be designed using a net allowable soil bearing pressure of 2,500 pounds per square foot (psf) for static conditions. The bearing capacity may be increased by one-third when considering loads of short duration such as wind or seismic forces. The foundations should preferably be proportioned such that the resultant force from design loads, including lateral loads, falls within the kern (i.e., middle one-third of the footing base). Total and differential settlements for footings designed and constructed in accordance with the above recommendations are estimated to be less than approximately 1-inch and ½-inch over a horizontal span of 40 feet, respectively.

Footings bearing on 12 or more inches of compacted engineered fill, as recommended in Section 9.2.1, may be designed using a coefficient of friction of 0.40, where the total frictional resistance equals the coefficient of friction times the dead load. Footings may be designed using a allowable passive resistance value of 300 psf per foot of depth for a level ground condition up to a value of 3,000 psf per foot. This value assumes that the ground is horizontal for a distance of 10 feet, or three times the height generating the passive pressure, whichever is greater. We recommend that the upper 24 inches of soil not protected by pavement or a concrete slab be neglected when calculating passive resistance.

The allowable lateral resistance can be taken as the sum of the frictional resistance and passive resistance provided the passive resistance does not exceed one-half of the total allowable resistance. The passive resistance (including the maximum value) may be increased by one-third when considering loads of short duration such as wind or seismic forces.

The bottom surface of foundation excavations should be compacted with hand-held dynamic compaction equipment (i.e., jumping jack, flat-plate vibrator) prior to placement of forms and reinforcing steel. The base of foundation excavations should be free of ice, snow, water, frost, and loose soil prior to placing concrete. Concrete should be placed soon after subgrade compaction to reduce bearing soil disturbance. Should the soils at bearing level become excessively dry, disturbed, or saturated, the affected soil should be moisture-conditioned and recompacted. It is recommended that Ninyo & Moore be retained to observe, test, and evaluate the foundation bearing materials.

9.4 Slab-On-Grade Floors

Slab-on-grade floors should bear on 12 or more inches of engineered fill. The remedial grading recommendations provided in Section 9.2.1 should be followed.

For slab design, a design modulus of subgrade reaction (K) of 75 pci (pounds per square inch per inch of deflection) may also be used for the subgrade soils in evaluating such deflections.

The design of the floor slabs (including jointing and reinforcement) is the responsibility of the Structural Engineer. Joints should be constructed at intervals designed by the Structural Engineer to help reduce random cracking of the slab. Floor slabs should be adequately reinforced. Recommendations based on structural considerations for slab thickness, jointing, and steel reinforcement should be developed by the Structural Engineer in accordance with American Concrete Institute recommendations. Proper placement of reinforcement in the slab is vital for satisfactory performance.

Floor slabs should be separated from bearing walls and columns with expansion joints, which allow unrestrained vertical movement. Joints should be observed periodically, particularly during the first several years after construction. Slab movement can cause previously free-slipping joints to bind. Measures should be taken so that slab isolation is maintained in order to reduce the likelihood of damage to walls and other interior improvements.

Where floor slabs are tied to perimeter walls or turn-down slabs to meet structural or other construction objectives, our experience indicates that any differential movement between the walls and slabs will probably be observed in adjacent slab expansion joints or floor slab cracks that occur beyond the length of the structural dowels. The Structural Engineer should account for this potential differential settlement through use of sufficient control joints, appropriate reinforcing or other means.

9.5 Pavements

We anticipate project pavements will be maintained by the NPS. Pavement section alternatives are included herein for various pavements at the sites.

The current subgrade soils at Bear Gulch Nature Center encountered in our borings typically consisted of relatively clean to clayey sand and gravel that classify as A-2-6 soils in accordance with the American Association of State Highway and Transportation Officials (AASHTO) classification system. The current subgrade soils at Bear Valley School encountered in our borings typically consisted of relatively clean to clayey sand and gravel that classify as A-1 and A-2 soils in accordance with the American Association of State Highway and Transportation Officials (AASHTO) classification system. It is anticipated that fill imported to the site will classify as A-2 or better. We utilized a design R-Value of 20 for the pavement subgrade soils for the project.

The pavement sections recommended below were developed in general accordance with the guidelines and procedures of the AASHTO, (AASHTO, 1993). Table 3 summarizes the minimum pavement sections for asphaltic concrete pavements placed over aggregate base course (ABC), and Portland cement concrete (PCC) pavements placed over ABC.

9.5.1 Pavement Design

Specific traffic loadings for the project were not available at the time of this report preparation. Based on our experience with similar facilities, an equivalent 18-kip daily load application (EDLA) value of 5 was assumed for the standard duty, automobile parking areas and driveways and 50 was assumed for the heavy-duty drive lanes and fire lanes. The EDLA values of 5 and 50 were converted to an equivalent 18-kip single axle load value of 36,500 and 365,000 for 20-year design lives, respectively. These values should be confirmed as a part of the traffic study to be performed as a part of the final pavement design. If design traffic loadings differ significantly from this assumed value, we should be notified to re-evaluate the pavement recommendations below.

The design of flexible pavements was based on the following input parameters:

Initial Serviceability:	4.5
Terminal Serviceability:	2.0
Reliability	90%
Overall Standard Deviation:	0.44
Resilient Modulus:	4,941 psi (R-value of 20)
Stage Construction:	1.0

The design of rigid pavements was based on the following input parameters:

Initial Serviceability:	4.5
Terminal Serviceability:	2.0
Reliability	90%
28-Day Mean PCC Modulus Rupture:	600 psi
28-Day Mean Modulus of Elasticity:	3.6×10^6 psi
Mean Effective k value:	150 psi/in
Overall Standard Deviation:	0.35
Load Transfer Coefficient:	3.9
Overall Drainage Coefficient:	1.0

Based on the above-mentioned guidelines, procedures, and input parameters, Table 3 provides our recommended pavement section thicknesses.

Table	3 – Recommended Pavement Thickness		
Traffic Type	Composite AC / ABC (inches)	Full-Depth AC (inches)	PCCP (inches)
Standard-Duty Areas	3.0 / 4.0	5.0	5.0
Heavy Duty Areas	4.0 / 6.0	6.0	6.0
Notes: AC = Asphalt Concrete, ABC = Aggregate Base Course, PCCP = Portland Cement Concrete Pavement			

We recommend PCCP be utilized in entrance and exit sections, dumpster pads, loading areas, or other areas where extensive wheel maneuvering are expected. The dumpster pad should be large enough to support the wheels of the truck, which will bear the load of the dumpster.

We recommend the placement of 4 or more inches of ABC below PCCP pavements which will develop a more stable subgrade for concrete truck traffic associated with the pavement construction and help reduce potential slab curl, shrinkage cracking, and subgrade “pumping” through joints. Adequate joint spacing and reinforcement is recommended to prevent loss of load transfer across saw-cut crack control joints. Joints should be sealed to reduce water infiltration. The design guidelines provided in the referenced ACI 330R-01 guide should be followed for joint spacing and reinforcing.

Where practical, we recommend “early-entry” cutting of crack-control joints in PCCP. Cutting of PCCP in its “green” state typically reduces the potential for micro-cracking of the pavements prior to the crack control joints being formed, compared to cutting the joints after

the concrete has fully set. Micro-cracking of pavements may lead to crack formation in locations other than the sawed joints, and/or reduction of fatigue life of the pavement.

9.5.2 Pavement Subgrade Preparation

The contractor should be prepared either to dry the subgrade materials or moisten them, as needed, prior to compaction. New pavements should be placed on 12 or more inches of compacted engineered fill per the recommendations presented in Section 9.2.1. Some site soils may pump or deflect during compaction if moisture levels are not carefully monitored. The contractor should be prepared to process and compact such soils to establish a stable platform for paving, including use of chemical stabilization or geotextiles, where needed.

The prepared subgrade should be protected from the elements prior to pavement placement. Subgrades that are exposed to the elements may need additional moisture conditioning and compaction, prior to pavement placements.

Immediately prior to paving, the subgrade should be proof rolled with a heavily loaded, pneumatic tired vehicle and checked for moisture. Areas that show excessive deflection during proof rolling should be excavated and replaced and/or stabilized. Areas allowed to pond prior to paving may need to be re-worked prior to proof rolling.

9.5.3 Pavement Materials

The AC pavement shall consist of a bituminous plant mix composed of a mixture of high-quality aggregate and bituminous material, which meets the requirements of a job-mix formula established by a qualified engineer. Pavement layer thickness should be between 2 and 4 inches for the lower lifts and 2 inches for the top lift. The geotechnical engineer should be consulted to review the proposed pavement mix designs, grading, and lift thicknesses prior to construction.

PCC should consist of a plant mix composed of a mixture of aggregate, Portland cement and appropriate admixtures meeting applicable local requirements. Concrete should have a modulus of rupture of third point loading of 600 psi or more. The concrete should be air-entrained with approximately 6 percent air and should have a cement content of six or more sacks per cubic yard. Allowable slump should be approximately 4 inches.

Thickened edges should be used along outside edges of PCC. The edge thickness should be 2 inches or more than the recommended PCC thickness and taper to the recommended PCC thickness 36 inches inward from the edge. Integral curbs may be used in lieu of thickened edges.

PCC pavement should have longitudinal and transverse joints that meet applicable local requirements.

9.5.4 Pavement Maintenance

The collection and diversion of surface drainage away from paved areas is vital to satisfactory performance of the pavements. The subsurface and surface drainage systems should be carefully designed to facilitate removal of the water from paved areas and subgrade soils. Allowing surface waters to pond on pavements will cause premature pavement deterioration. Where topography, site constraints or other factors limit or preclude adequate surface drainage, pavements should be provided with edge drains to reduce loss of subgrade support. The long-term performance of the pavement also can be improved greatly by backfilling and compaction behind curbs, gutters, and sidewalks so that ponding is not permitted and water infiltration is reduced.

The standard care of practice in pavement design describes the recommended flexible pavement section as a “20-year” design pavement; however, many pavements will not remain in satisfactory condition without routine, preventive maintenance and rehabilitation procedures performed during the life of the pavement. Preventive pavement treatments are surface rehabilitation and operations applied to improve or extend the functional life of a pavement. These treatments preserve, rather than improve, the structural capacity of the pavement structure. In the event the existing pavement is not structurally sound, the preventive maintenance will have no long-lasting effect. Therefore, a routine maintenance program to seal joints and cracks, and repair distressed areas is recommended.

9.6 Exterior Concrete Flatwork

Ground-supported exterior concrete flatwork, such as walkways, will be subject to soil-related movements resulting from settlement, etc. Thus, where these types of elements abut rigid building foundations or isolated/suspended structures, differential movements should be anticipated. We recommend that flexible joints be provided where such elements abut the main structure to allow for differential movement at these locations. In no case should exterior flatwork extend to under any portion of the building where there is less than 2 inches of clearance between the flatwork and any element of the building. Exterior flatwork in contact with brick, rock facades, or any other element of the building can cause damage to the structure if the flatwork experiences movements.

Concrete walkways should be in accordance with local requirements, 4 inches or more thick, and be supported on improved subgrade as described in Section 9.2.1 of this report. Positive drainage

should be established and maintained adjacent to flatwork. Water should not be allowed to pond on flatwork.

Exterior concrete flatwork, including curbs and gutters, should be constructed in accordance with the recommendations of the project's civil or structural engineer. Recommendations regarding concrete utilized in construction of proposed improvements are provided in Section 9.7.1.

9.7 Corrosion Considerations

The corrosion potential of on-site soils to concrete and buried metal was evaluated in the laboratory using representative samples obtained from the exploratory borings. Laboratory testing was performed to assess the effects of sulfate on concrete and the effects of soil resistivity on buried metal. Results of these tests are presented in Appendix B. Recommendations regarding concrete to be utilized in construction of proposed improvements and for buried metal pipes are provided in the following sections.

9.7.1 Concrete

The test for water-soluble sulfate content of the soils was performed and laboratory test results are presented in Appendix C. The percentage of water-soluble sulfates in water measured was negligible. Based on the laboratory results, the on-site soils represent a Class 0 severity of sulfate exposure to concrete on a scale that ranges between Class 0 and Class 3. Therefore, we recommend that the concrete used for this project should have a maximum water to cementitious material ratio of 0.45 and the cementitious materials should meet one of the below outlined requirements.

- ASTM C 150 Type I, II or V
- ASTM C 595 Type IP, IP(MS) or IP(HS)
- ASTM C 1157 Type GU, MS or HS
- ASTM C 150 Type III cement if it is allowed, as in Class E concrete

The Structural Engineer should ultimately select the concrete design strength based on the project specific loading conditions. However, higher strength concrete may be selected for increased durability, resistance to slab curling and shrinkage cracking. We recommend the use of concrete with a design 28-day compressive strength of 4,000 psi or more, for concrete slabs at this site. Concrete exposed to the elements should be air-entrained.

9.7.2 Buried Metal Structures

The corrosion potential of on-site materials was analyzed to evaluate potential effects on foundations and structures. Corrosion potential was evaluated using the results of laboratory testing of samples obtained during the subsurface evaluation that were considered representative of soils at the subject site.

The results of the laboratory testing indicate the on-site materials could be mildly corrosive at the Bear Gulch Nature Center and moderately corrosive at the Bear Valley School to ferrous metals. Therefore, special consideration should be given to the use of heavy gauge, corrosion-protected, underground steel pipe or culverts, if any are planned. As an alternative, plastic pipe or reinforced concrete pipe could be considered. A corrosion specialist should be consulted for further recommendations.

9.8 Site Drainage

Infiltration of water into subsurface soils can lead to soil movement and associated distress, and chemically and physically related deterioration of concrete structures. To reduce the potential for infiltration of moisture into subsurface soils at the site, we recommend the following:

- Positive drainage should be established and maintained away from the proposed buildings. Positive drainage may be established by providing a surface gradient for paved areas of 2 to 5 percent or more for a distance of 10 feet or more away from structures. Where concrete flatwork is placed adjacent to structures and other considerations are required by law, such as ADA requirements, slopes of 1 percent or more are considered acceptable. For unpaved areas, positive drainage may be established by a slope of 5 to 10 percent for 10 feet or more away from structures, where possible.
- Adequate surface drainage should be provided to channel surface water away from on-site structures and off paved surfaces to a suitable outlet such as a storm drain. Adequate surface drainage may be enhanced by utilization of graded swales, area drains, and other drainage devices. Surface run-off should not be allowed to pond near structures or structure footings.
- Building roof drains should have downspouts tightlined to an appropriate outlet, such as a storm drain or the street, away from structures, pavements, and flatwork. If tightlining of the downspouts is not practicable, they should discharge 5 feet or more away from structures and onto surfaces that slope away from the structure. Downspouts should not be allowed to discharge onto the ground surface adjacent to building foundations.
- The possibility of moisture infiltration beneath a structure, in the event of plumbing leaks, should be considered in the design and construction of underground water and sewer conduits. Permitting increases in moisture to the building supporting soils may result in a decrease in bearing capacity and an increase in settlement, heave, and/or differential movement. Incorporating a perimeter drainage system around the building foundations that will aid in reduction of the moisture infiltration of subsurface soils may be considered.
- Utility trenches should be backfilled with compacted, low permeability fill (i.e. permeability of 10-5 cm/s or less) within 5 feet of the building. Planters, if any, should be maintained 5 feet or more from the building and constructed with closed bottoms or with drainage systems to drain

excess irrigation away from the building.

9.9 Construction Observation and Testing

A qualified geotechnical consultant should perform appropriate observation and testing services during grading and construction operations. These services should include observation of any soft, loose, or otherwise unsuitable soils, evaluation of subgrade conditions where soil removals are performed, evaluation of the suitability of proposed borrow materials for use as fill, evaluation of the stability of open temporary excavations, evaluation of the results of any subgrade stabilization or dewatering activities, and performance of observation and testing services during placement and compaction of engineered fill and structural fill soils.

The geotechnical consultant should also perform observation and testing services during placement of concrete, mortar, grout, asphalt concrete, and steel reinforcement. If another geotechnical consultant is selected to perform foundation inspection, and construction observation and materials testing services for the project, we request that the selected consultant provide a letter to the owner, with a copy to Ninyo & Moore, indicating that they fully understand our geotechnical recommendations and that they are in full agreement with the recommendations contained in this report. Qualified subcontractors utilizing appropriate techniques and construction materials should perform construction of the proposed improvements.

9.10 Plan Review

The recommendations presented in this report are based on preliminary and conceptual design information for the proposed project and on the findings of our geotechnical evaluation. When finished, project plans and specifications should be reviewed by the geotechnical consultant prior to submitting the plans and specifications for bid. Additional field exploration, laboratory testing and geotechnical recommendations may be needed upon review of the project design plans.

9.11 Pre-Construction Meeting

We recommend a pre-construction meeting be held. The owner or the owner's representative, the architect, the contractor, and the geotechnical consultant should be in attendance to discuss the plans and the project.

10 LIMITATIONS

The field evaluation, laboratory testing, and geotechnical analyses presented in this geotechnical report have been conducted in general accordance with current practice and the standard of care exercised by geotechnical consultants performing similar tasks in the project area. No warranty, expressed or implied, is made regarding the conclusions, recommendations, and opinions

presented in this report. There is no evaluation detailed enough to reveal every subsurface condition. Variations may exist and conditions not observed or described in this report may be encountered during construction. Uncertainties relative to subsurface conditions can be reduced through additional subsurface exploration. Additional subsurface evaluation will be performed upon request. Please also note that our evaluation was limited to assessment of the geotechnical aspects of the project, and did not include evaluation of structural issues, environmental concerns, or the presence of hazardous materials.

This document is intended to be used only in its entirety. No portion of the document, by itself, is designed to completely represent any aspect of the project described herein. Ninyo & Moore should be contacted if the reader requires additional information or has questions regarding the content, interpretations presented, or completeness of this document.

This report is intended for design purposes only. It does not provide sufficient data to prepare an accurate bid by contractors. It is suggested that the bidders and their geotechnical consultant perform an independent evaluation of the subsurface conditions in the project areas. The independent evaluations may include, but not be limited to, review of other geotechnical reports prepared for the adjacent areas, site reconnaissance, and additional exploration and laboratory testing.

Our conclusions, recommendations, and opinions are based on an analysis of the observed site conditions. If geotechnical conditions different from those described in this report are encountered, our office should be notified and additional recommendations, if warranted, will be provided upon request. It should be understood that the conditions of a site could change with time as a result of natural processes or the activities of man at the subject site or nearby sites. In addition, changes to the applicable laws, regulations, codes, and standards of practice may occur due to government action or the broadening of knowledge. The findings of this report may, therefore, be invalidated over time, in part or in whole, by changes over which Ninyo & Moore has no control.

This report is intended exclusively for use by the client. Any use or reuse of the findings, conclusions, and/or recommendations of this report by parties other than the client is undertaken at said parties' sole risk.

11 REFERENCES

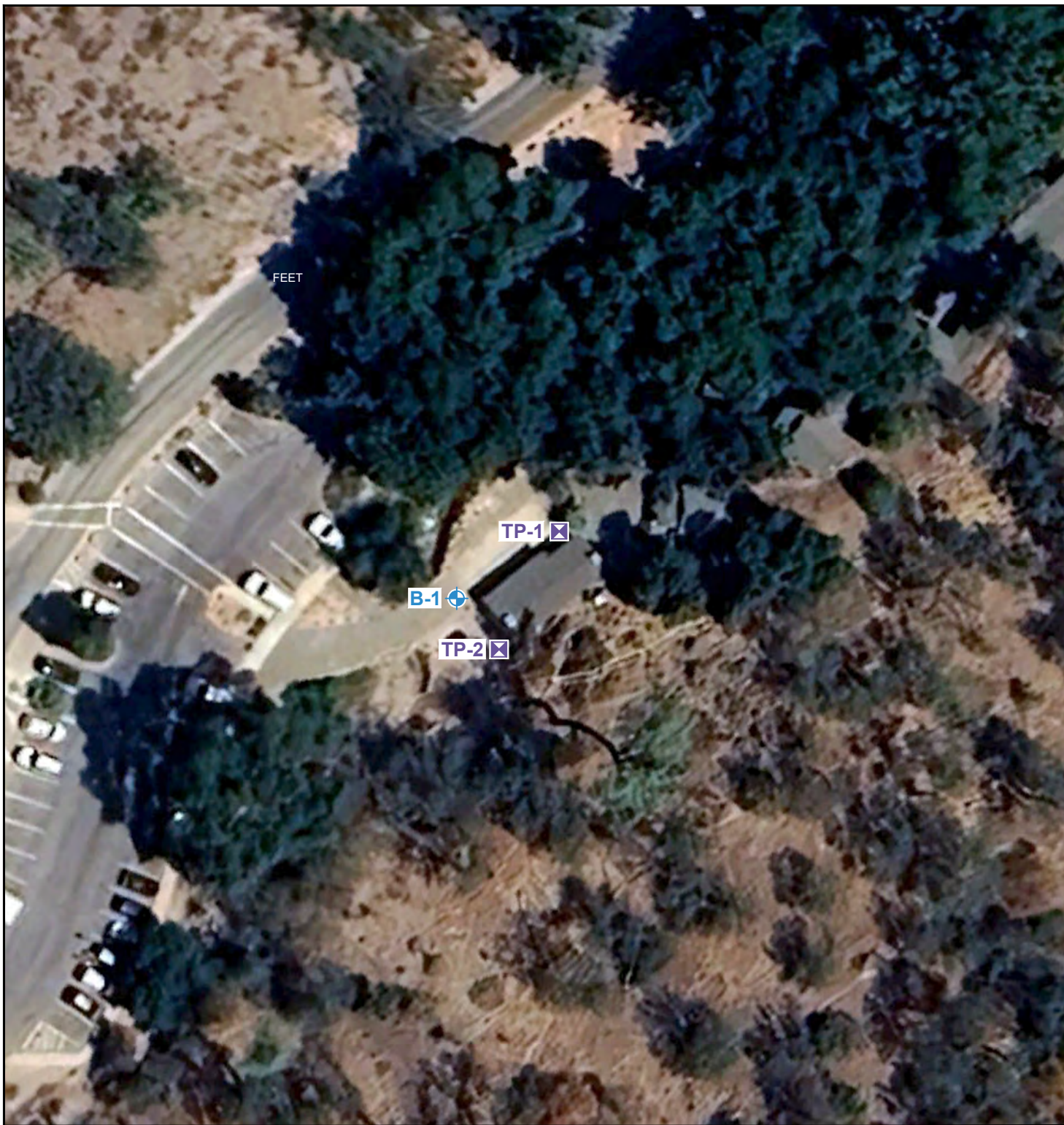
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FIGURES

502075008.dwg_BTPL1 12/26/2024 JDP



LEGEND

- B-1**  BORING
- TP-2**  TEST PIT

NOTE: DIMENSIONS, DIRECTIONS AND LOCATIONS ARE APPROXIMATE. | REFERENCE: GOOGLE EARTH, 2024.

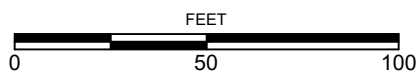


FIGURE 2A

502075008.dwg_BTPL2 12/26/2024 JDP



LEGEND

- B-2**  BORING
- TP-4**  TEST PIT

NOTE: DIMENSIONS, DIRECTIONS AND LOCATIONS ARE APPROXIMATE. | REFERENCE: GOOGLE EARTH, 2024.

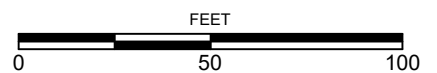


FIGURE 2B



APPENDIX A

Boring and Test Pit Logs

APPENDIX A

BORING LOGS

Field Procedure for the Collection of Disturbed Samples

Disturbed soil samples were obtained in the field using the following methods.

The Standard Penetration Test (SPT) Sampler

Disturbed drive samples of earth materials were obtained by means of a Standard Penetration Test sampler. The sampler is composed of a split barrel with an external diameter of 2 inches and an unlined internal diameter of 1-3/8 inches. The sampler was driven into the ground 12 to 18 inches with a 140-pound hammer falling freely from a height of 30 inches in general accordance with ASTM D 1586. The blow counts were recorded for every 6 inches of penetration; the blow counts reported on the logs are those for the last 12 inches of penetration. Soil samples were observed and removed from the sampler, bagged, sealed and transported to the laboratory for testing.

Field Procedure for the Collection of Ring-lined Samples

Ring-lined soil samples were obtained in the field using the following methods.

The Modified California Split-Barrel Drive Sampler

The sampler, with an external diameter of 3.0 inches, was lined with thin brass rings with inside diameters of approximately 2.4 inches. The sample barrel was driven into the ground with the weight of a hammer in general accordance with ASTM D 3550. The driving weight was permitted to fall freely. The approximate length of the fall, the weight of the hammer or bar, and the number of blows per foot of driving are presented on the boring logs as an index to the relative resistance of the materials sampled. The samples were removed from the sample barrel in the brass rings, sealed, and transported to the laboratory for testing.

BORING LOG EXPLANATION SHEET

DEPTH (feet)	Bulk Samples Driven	BLOWS/FOOT	MOISTURE (%)	DRY DENSITY (PCF)	SYMBOL	CLASSIFICATION U.S.C.S.	
0							Bulk sample.
							Modified split-barrel drive sampler.
							No recovery with modified split-barrel drive sampler.
							Sample retained by others.
							Standard Penetration Test (SPT).
5							No recovery with a SPT.
	XX/XX						Shelby tube sample. Distance pushed in inches/length of sample recovered in inches.
							No recovery with Shelby tube sampler.
							Continuous Push Sample.
10							Seepage.
							Groundwater encountered during drilling.
							Groundwater measured after drilling.
						SM	MAJOR MATERIAL TYPE (SOIL):
							Solid line denotes unit change.
						CL	Dashed line denotes material change.
15							Attitudes: Strike/Dip b: Bedding c: Contact j: Joint f: Fracture F: Fault cs: Clay Seam s: Shear bss: Basal Slide Surface sf: Shear Fracture sz: Shear Zone sbs: Shear Bedding Surface
20							The total depth line is a solid line that is drawn at the bottom of the boring.

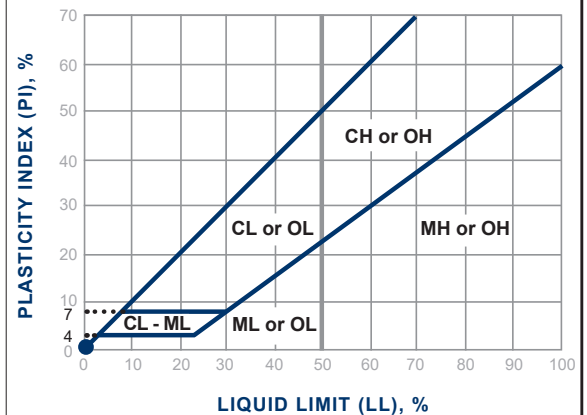
Soil Classification Chart Per ASTM D 2488

Primary Divisions			Secondary Divisions	
			Group Symbol	Group Name
COARSE-GRAINED SOILS more than 50% retained on No. 200 sieve	GRAVEL more than 50% of coarse fraction retained on No. 4 sieve	CLEAN GRAVEL less than 5% fines	 GW	well-graded GRAVEL
			 GP	poorly graded GRAVEL
		GRAVEL with DUAL CLASSIFICATIONS 5% to 12% fines	 GW-GM	well-graded GRAVEL with silt
			 GP-GM	poorly graded GRAVEL with silt
			 GW-GC	well-graded GRAVEL with clay
			 GP-GC	poorly graded GRAVEL with
		GRAVEL with FINES more than 12% fines	 GM	silty GRAVEL
			 GC	clayey GRAVEL
			 GC-GM	silty, clayey GRAVEL
	SAND 50% or more of coarse fraction passes No. 4 sieve	CLEAN SAND less than 5% fines	 SW	well-graded SAND
			 SP	poorly graded SAND
		SAND with DUAL CLASSIFICATIONS 5% to 12% fines	 SW-SM	well-graded SAND with silt
			 SP-SM	poorly graded SAND with silt
			 SW-SC	well-graded SAND with clay
			 SP-SC	poorly graded SAND with clay
		SAND with FINES more than 12% fines	 SM	silty SAND
			 SC	clayey SAND
			 SC-SM	silty, clayey SAND
	SILT and CLAY liquid limit less than 50%	INORGANIC	 CL	lean CLAY
			 ML	SILT
			 CL-ML	silty CLAY
		ORGANIC	 OL (PI > 4)	organic CLAY
			 OL (PI < 4)	organic SILT
	SILT and CLAY liquid limit 50% or more	INORGANIC	 CH	fat CLAY
			 MH	elastic SILT
		ORGANIC	 OH (plots on or above "A"-line)	organic CLAY
			 OH (plots below "A"-line)	organic SILT
		Highly Organic Soils	 PT	Peat

Grain Size

Description		Sieve Size	Grain Size	Approximate Size
Boulders		> 12"	> 12"	Larger than basketball-sized
Cobbles		3 - 12"	3 - 12"	Fist-sized to basketball-sized
Gravel	Coarse	3/4 - 3"	3/4 - 3"	Thumb-sized to fist-sized
	Fine	#4 - 3/4"	0.19 - 0.75"	Pea-sized to thumb-sized
Sand	Coarse	#10 - #4	0.075 - 0.19"	Rock-salt-sized to pea-sized
	Medium	#40 - #10	0.017 - 0.075"	Sugar-sized to rock-salt-sized
	Fine	#200 - #40	0.0029 - 0.017"	Flour-sized to sugar-sized
Fines		Passing #200	< 0.0029"	Flour-sized and smaller

Plasticity Chart

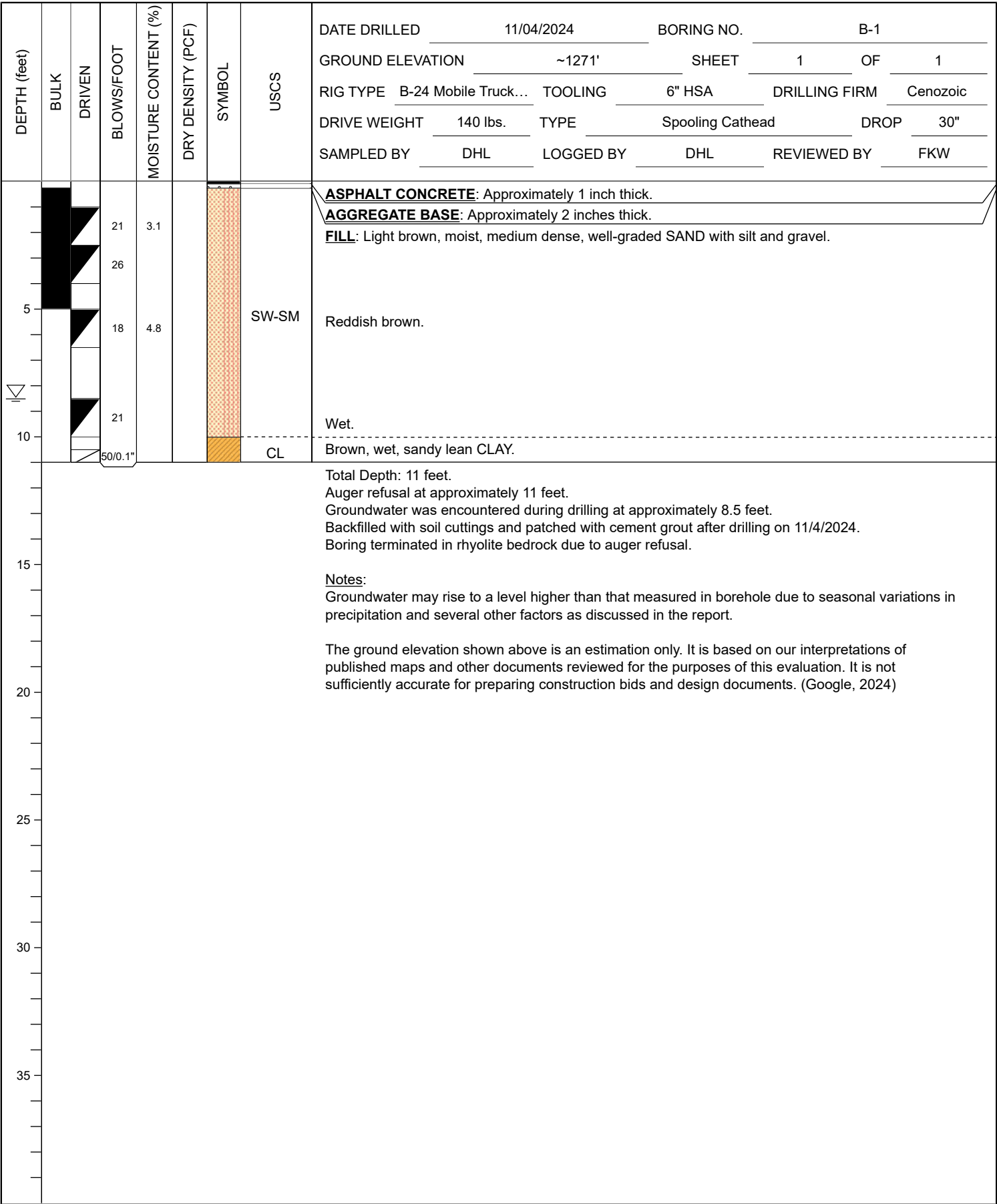


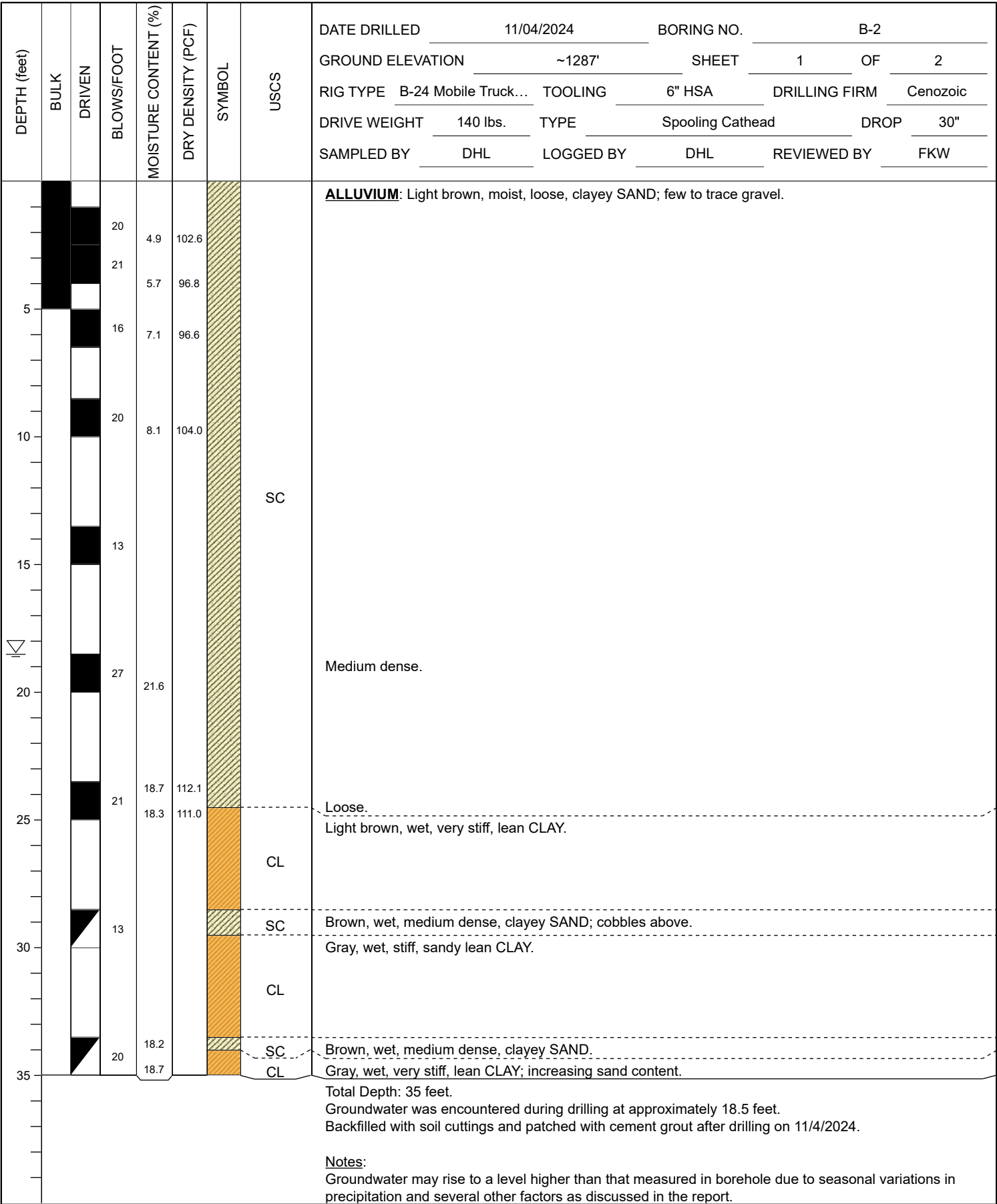
Apparent Density - Coarse-Grained Soil

Apparent Density	Spooling Cable or Cathead		Automatic Trip Hammer	
	SPT (blows/foot)	Modified Split Barrel (blows/foot)	SPT (blows/foot)	Modified Split Barrel (blows/foot)
Very Loose	≤ 4	≤ 8	≤ 3	≤ 5
Loose	5 - 10	9 - 21	4 - 7	6 - 14
Medium Dense	11 - 30	22 - 63	8 - 20	15 - 42
Dense	31 - 50	64 - 105	21 - 33	43 - 70
Very Dense	> 50	> 105	> 33	> 70

Consistency - Fine-Grained Soil

Consistency	Spooling Cable or Cathead		Automatic Trip Hammer	
	SPT (blows/foot)	Modified Split Barrel (blows/foot)	SPT (blows/foot)	Modified Split Barrel (blows/foot)
Very Soft	< 2	< 3	< 1	< 2
Soft	2 - 4	3 - 5	1 - 3	2 - 3
Firm	5 - 8	6 - 10	4 - 5	4 - 6
Stiff	9 - 15	11 - 20	6 - 10	7 - 13
Very Stiff	16 - 30	21 - 39	11 - 20	14 - 26
Hard	> 30	> 39	> 20	> 26





DEPTH (feet)	BULK	DRIVEN	BLOWS/FOOT	MOISTURE CONTENT (%)	DRY DENSITY (PCF)	SYMBOL	USCS	DATE DRILLED	11/04/2024	BORING NO.	B-2		
								GROUND ELEVATION	~1287'	SHEET	2	OF	2
								RIG TYPE	B-24 Mobile Truck...	TOOLING	6" HSA	DRILLING FIRM	Cenozoic
								DRIVE WEIGHT	140 lbs.	TYPE	Spooling Cathead	DROP	30"
								SAMPLED BY	DHL	LOGGED BY	DHL	REVIEWED BY	FKW
45								The ground elevation shown above is an estimation only. It is based on our interpretations of published maps and other documents reviewed for the purposes of this evaluation. It is not sufficiently accurate for preparing construction bids and design documents. (Google, 2024)					



Test Pit LOG

PINNACLE NATIONAL PARK

PROJECT NO.

502075008

DATE

1/25

DEPTH (FEET)

Bulk

Driven

Sand Cone

SAMPLES

MOISTURE (%)

DRY DENSITY (PCF)

CLASSIFICATION
U.S.C.S.

DATE EXCAVATED 11/4/2024 TEST PIT NO. TP-1

GROUND ELEVATION -- LOGGED BY DHL

METHOD OF EXCAVATION Backhoe Excavator (Hawkins)

LOCATION Bear Gulch Visitor Center

DESCRIPTION

GW

Brown, moist, well-graded GRAVEL; few sand.

SW

Brown, moist, well-graded SAND; few cobbles and gravel.

Total Depth: 4 feet
Groundwater was not encountered during drilling.
Backfilled with on-site soil after drilling on 11/4/2024.

Notes:

Groundwater, though not encountered at the time of test pit, may rise to a higher level due to seasonal variations in precipitation and several other factors as discussed in the report.

GREEN
ROCK

CONCRETE

2" PIPE

6

5

4

3

2

1

0

SCALE = 1 in./1 ft.

FIGURE



Test Pit LOG

PINNACLE NATIONAL PARK

PROJECT NO.

502075008

DATE

1/25

DEPTH (FEET)

Bulk

Driven

Sand Cone

SAMPLES

MOISTURE (%)

DRY DENSITY (PCF)

CLASSIFICATION
U.S.C.S.

DATE EXCAVATED 11/4/2024 TEST PIT NO. TP-2

GROUND ELEVATION -- LOGGED BY DHL

METHOD OF EXCAVATION Backhoe Excavator (Hawkins)

LOCATION Bear Gulch Visitor Center

DESCRIPTION

Gray, blue, and white, COBBLES with sand.

SW

Brown, moist, well-graded SAND; few cobbles and gravel.

Total Depth: 1.8 feet.

Groundwater was not encountered during drilling.

Backfilled with on-site soil after drilling on 11/4/2024.

Test Pit Terminated in weathered bedrock.

Notes:

Groundwater, though not encountered at the time of test pit, may rise to a higher level due to seasonal variations in precipitation and several other factors as discussed in the report.

GREEN
ROCK

CONCRETE

4" PIPE

FIGURE

SCALE = 1 in./1 ft.



Test Pit LOG

PINNACLE NATIONAL PARK

PROJECT NO.

502075008

DATE

1/25

DEPTH (FEET)

Bulk

Driven

Sand Cone

SAMPLES

MOISTURE (%)

DRY DENSITY (PCF)

CLASSIFICATION
U.S.C.S.

DATE EXCAVATED 11/4/2024 TEST PIT NO. TP-3

GROUND ELEVATION -- LOGGED BY DHL

METHOD OF EXCAVATION Backhoe Excavator (Hawkins)

LOCATION Bear Valley School

DESCRIPTION

6" METAL SEWER PIPE

CL

TOPSOIL: Approximately 2 inches thick.
Light brown, moist, lean CLAY.

CL

Brown, moist, lean CLAY; scattered roots.

Total Depth: 3 feet.
Groundwater was not encountered during drilling.
Backfilled with on-site soil after drilling on 11/4/2024.

Notes:

Groundwater, though not encountered at the time of test pit, may rise to a higher level due to seasonal variations in precipitation and several other factors as discussed in the report.

SCALE = 1 in./1 ft.

FIGURE



Test Pit LOG

PINNACLE NATIONAL PARK

PROJECT NO.

502075008

DATE

1/25

DEPTH (FEET)

Bulk

Driven

Sand Cone

SAMPLES

MOISTURE (%)

DRY DENSITY (PCF)

CLASSIFICATION
U.S.C.S.

DATE EXCAVATED 11/4/2024 TEST PIT NO. TP-4

GROUND ELEVATION -- LOGGED BY DHL

METHOD OF EXCAVATION Backhoe Excavator (Hawkins)

LOCATION Bear Valley School

DESCRIPTION

Brown, moist, sandy lean CLAY.

Total Depth: 2.5 feet.
Groundwater was not encountered during drilling.
Backfilled with on-site soil after drilling on 11/4/2024.

Notes:

Groundwater, though not encountered at the time of test pit, may rise to a higher level due to seasonal variations in precipitation and several other factors as discussed in the report.

CONCRETE

0

1

2

3

4

5

6

SCALE = 1 in./1 ft.

FIGURE



APPENDIX B

Laboratory Testing by Ninyo & Moore

APPENDIX B

LABORATORY TESTING

Classification

Soils were visually and texturally classified in accordance with the Unified Soil Classifications System (USCS) in general accordance with ASTM D 2488. Soil classifications are indicated on the logs of the exploratory excavations in Appendix A.

In-Place Moisture and Density Tests

The moisture content and dry density of ring-lined samples obtained from the exploratory borings were evaluated in general accordance with ASTM D 2837. These test results are presented on the logs of the exploratory borings in Appendix A.

Atterberg Limits

Tests were performed on selected representative fine-grained soil samples to evaluate the liquid limit, plastic limit, and plasticity index in general accordance with ASTM D 4318. These test results were utilized to evaluate the soil classification in accordance with the Unified Soil Classification System. The test results and classifications are shown on Figure B-1.

200 Wash

An evaluation of the percentage of particles finer than the No. 200 sieve in selected soil samples was performed in general accordance with ASTM D 1140. The results of the tests are presented on Figure B-2.

Gradation Analysis

Gradation analysis tests were performed on selected representative soil samples in general accordance with ASTM D6913. The grain size distribution curves are shown on Figures B-3 through B-5. These test results were utilized in evaluating the soil classifications in accordance with the USCS.

Gradation and Hydrometer Analysis

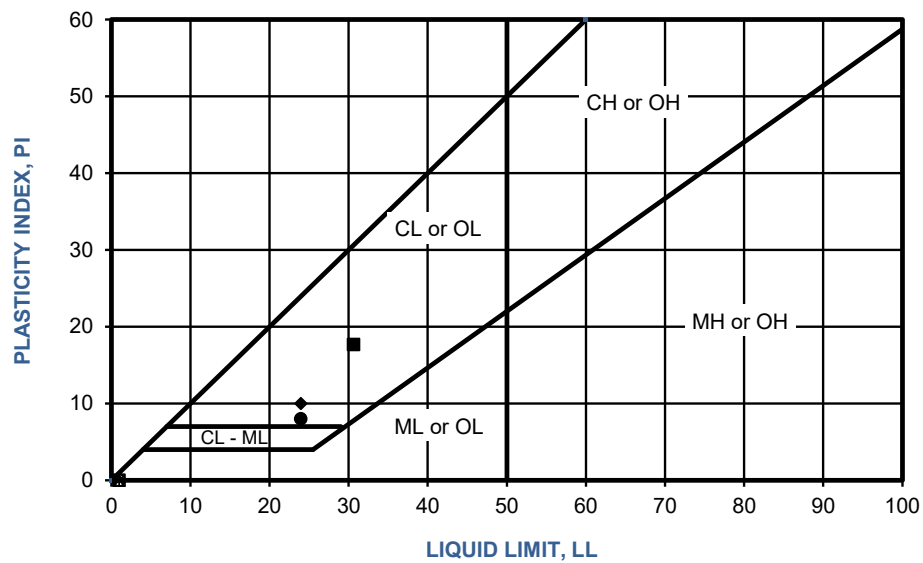
Gradation analysis and hydrometer analysis tests were performed on select representative soil samples in general accordance with ASTM D422. The test results were utilized in evaluating the soil classifications in accordance with the Unified Soil Classification System. The grain-size distribution curves are shown on Figures B-6 through B-9.

Soil Corrosivity Tests

A soil pH test was performed on a representative sample in general accordance with ASTM Test Method D 4972. A soil minimum resistivity test was performed on a representative sample in general accordance with AASHTO T288. The test results are presented in Appendix C.

SYMBOL	LOCATION	DEPTH (ft)	LIQUID LIMIT	PLASTIC LIMIT	PLASTICITY INDEX	USCS CLASSIFICATION (Fraction Finer Than No. 40 Sieve)	USCS
●	B-2	2.0-2.5	24	16	8	CL	SC
■	B-2	9.5-10.0	31	13	18	CL	SC
◆	B-2	24.5-25.0	24	14	10	CL	SC
○							
□							
△							
X							
+							

NP - INDICATES NON-PLASTIC



PERFORMED IN ACCORDANCE WITH ASTM D 4318

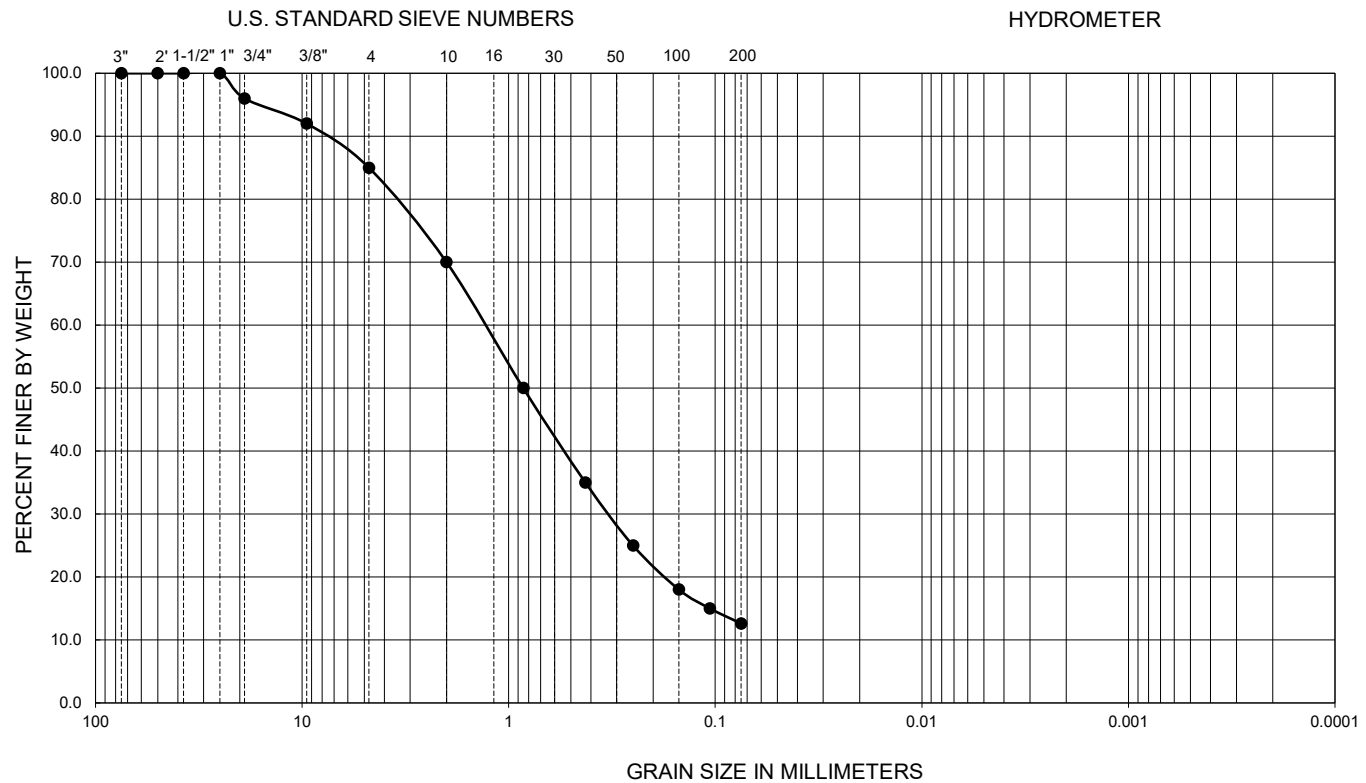
FIGURE B-1

SAMPLE LOCATION	SAMPLE DEPTH (ft)	DESCRIPTION	PERCENT PASSING NO. 4	PERCENT PASSING NO. 200	USCS (TOTAL SAMPLE)
B-2	2.0-2.5	Light Brown Clayey SAND Few Gravel	91.0	28.63	SC
B-2	9.5-10.0	Light Brown Clayey SAND Few Gravel	94.6	29.37	SC
B-2	24.5-25.0	Light Brown SAND with Clay and Trace Gravel	98.7	10.50	SW-SC

PERFORMED IN ACCORDANCE WITH ASTM D 1140

FIGURE B-2

GRAVEL		SAND			FINES	
Coarse	Fine	Coarse	Medium	Fine	SILT	CLAY



Symbol	Sample Location	Depth (ft)	Liquid Limit	Plastic Limit	Plasticity Index	D ₁₀	D ₃₀	D ₆₀	C _u	C _c	Passing No. 200 (percent)	USCS
●	B-1	1.0-2.5	--	--	--	--	0.34	1.43	--	--	12.61	SW-SM

PERFORMED IN ACCORDANCE WITH ASTM D6913

Group Name:

Soak Time: 2.00

% Gravel	15.0
% Sand	72.4
% Fines	12.61

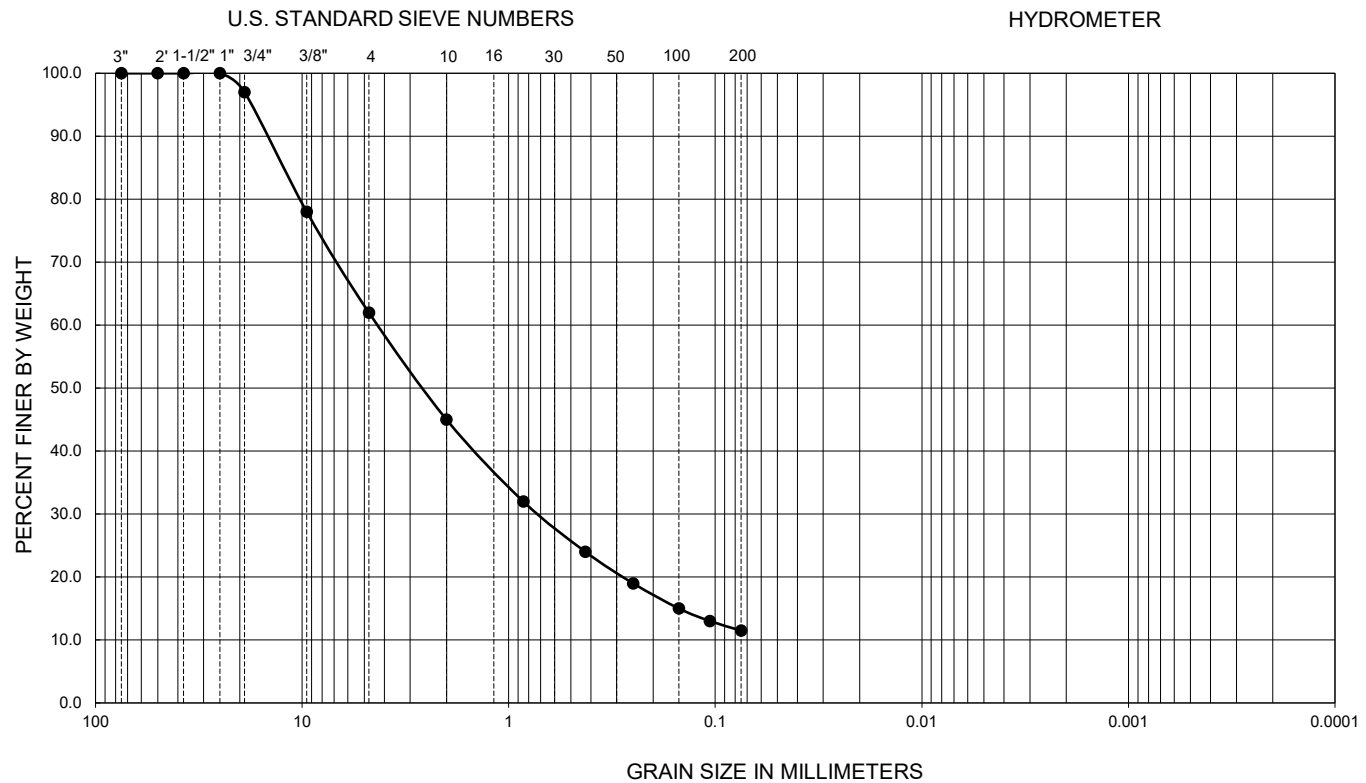


Geotechnical & Environmental Sciences Consultants

FIGURE B-3

GRADATION TEST RESULTS
 ANDERSON HALLAS/PINN BEAR GULCH/GEO
 PINNACLE NATIONAL PARK, PAICINES, CA
 502075008 | JANUARY 2025

GRAVEL		SAND			FINES	
Coarse	Fine	Coarse	Medium	Fine	SILT	CLAY



Symbol	Sample Location	Depth (ft)	Liquid Limit	Plastic Limit	Plasticity Index	D ₁₀	D ₃₀	D ₆₀	C _u	C _c	Passing No. 200 (percent)	USCS
●	B-1	5.0-6.5	--	--	--	--	0.74	4.43	--	--	11.49	SW-SM

PERFORMED IN ACCORDANCE WITH ASTM D6913

Group Name:

Soak Time: 2.10

% Gravel	38.0
% Sand	50.5
% Fines	11.49

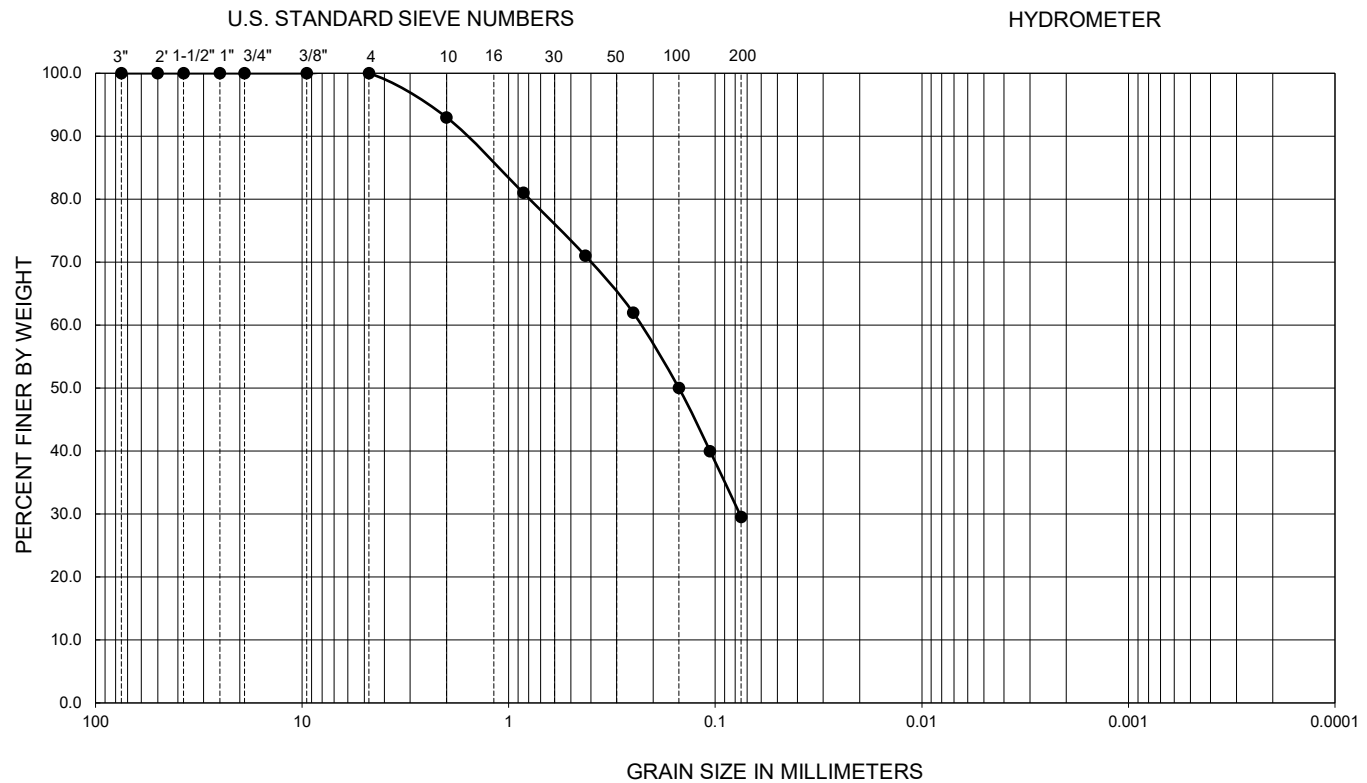


Geotechnical & Environmental Sciences Consultants

FIGURE B-4

GRADATION TEST RESULTS
 ANDERSON HALLAS/PINN BEAR GULCH/GEO
 PINNACLE NATIONAL PARK, PAICINES, CA
 502075008 | JANUARY 2025

GRAVEL		SAND			FINES	
Coarse	Fine	Coarse	Medium	Fine	SILT	CLAY



Symbol	Sample Location	Depth (ft)	Liquid Limit	Plastic Limit	Plasticity Index	D ₁₀	D ₃₀	D ₆₀	C _u	C _c	Passing No. 200 (percent)	USCS
●	B-2	34.5-35.0	--	--	--	--	0.08	0.23	--	--	29.55	SC

PERFORMED IN ACCORDANCE WITH ASTM D6913

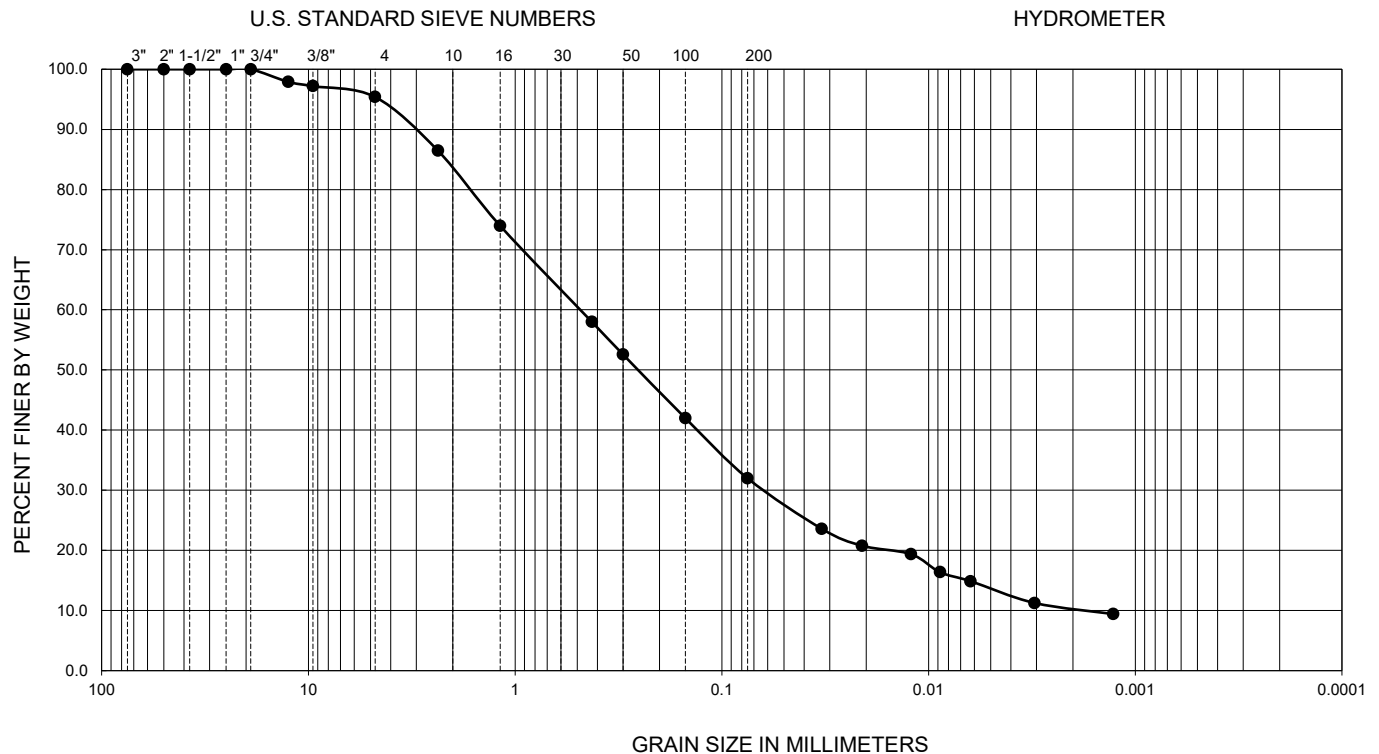
Group Name:

Soak Time: 2.20

% Gravel 0.0
 % Sand 70.5
 % Fines 29.55

FIGURE B-5

GRAVEL		SAND			FINES	
Coarse	Fine	Coarse	Medium	Fine	SILT	CLAY

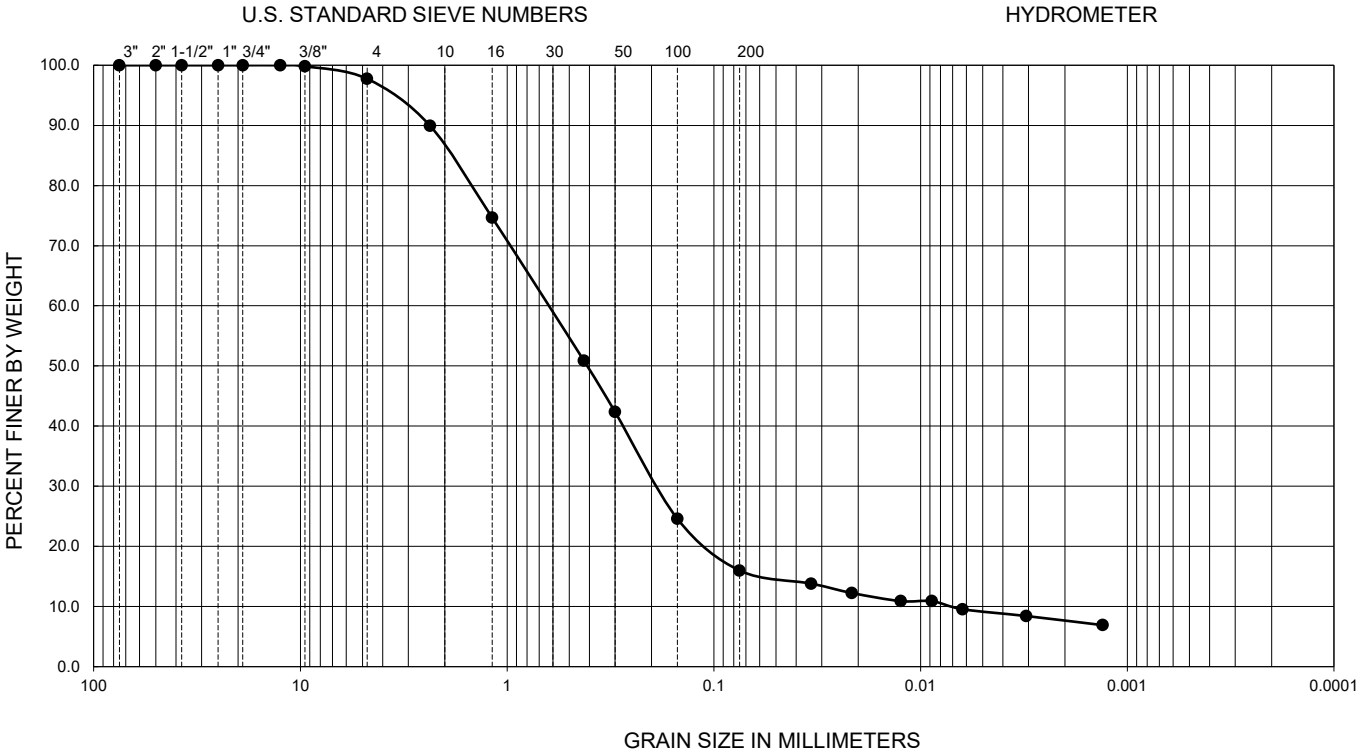


Symbol	Sample Location	Depth (ft)	Liquid Limit	Plastic Limit	Plasticity Index	D ₁₀	D ₃₀	D ₆₀	C _u	C _c	Passing No. 200 (%)	USCS
●	B-2	5.5-6.0	--	--	--	0.002	0.065	0.520	282.8	4.4	32.0	SC

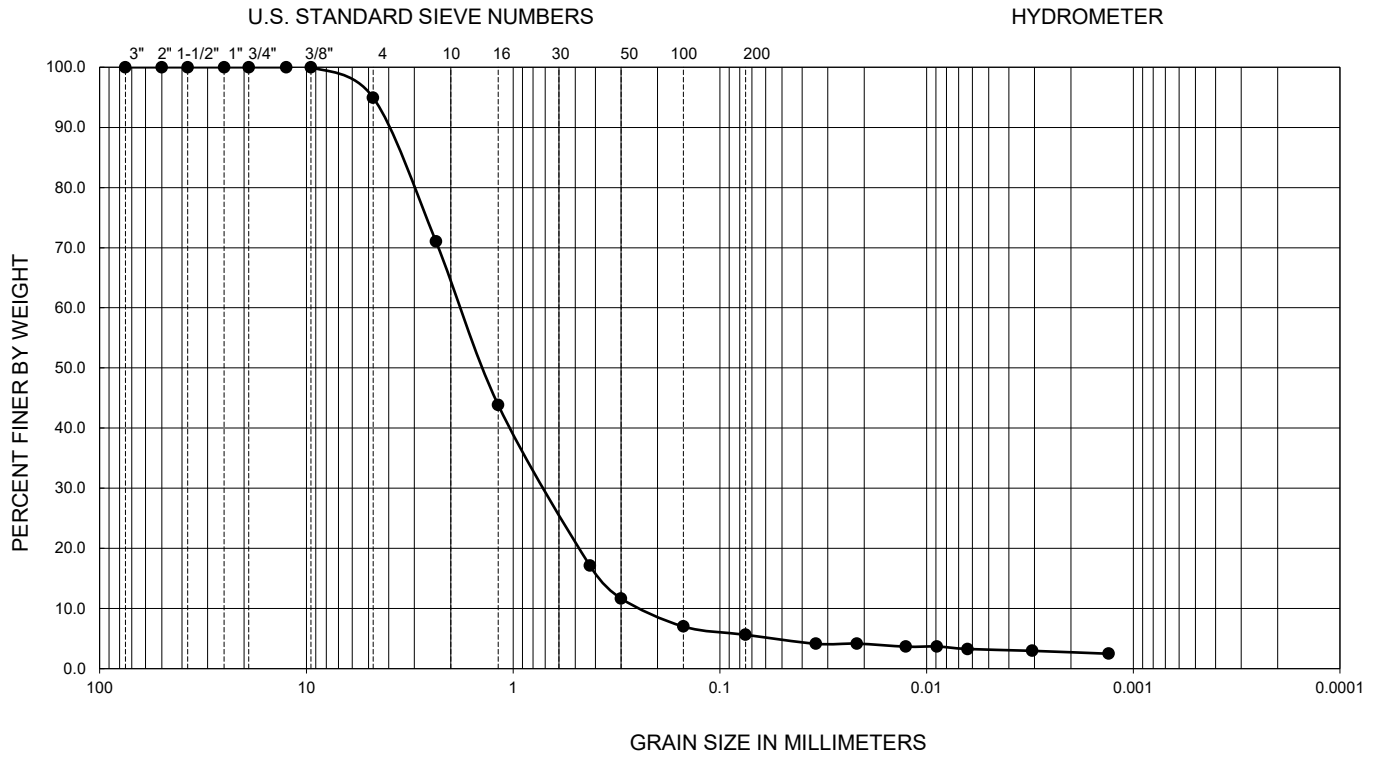
PERFORMED IN ACCORDANCE WITH ASTM D 422

FIGURE B-6

GRAVEL		SAND			FINES	
Coarse	Fine	Coarse	Medium	Fine	SILT	CLAY



GRAVEL		SAND			FINES	
Coarse	Fine	Coarse	Medium	Fine	SILT	CLAY

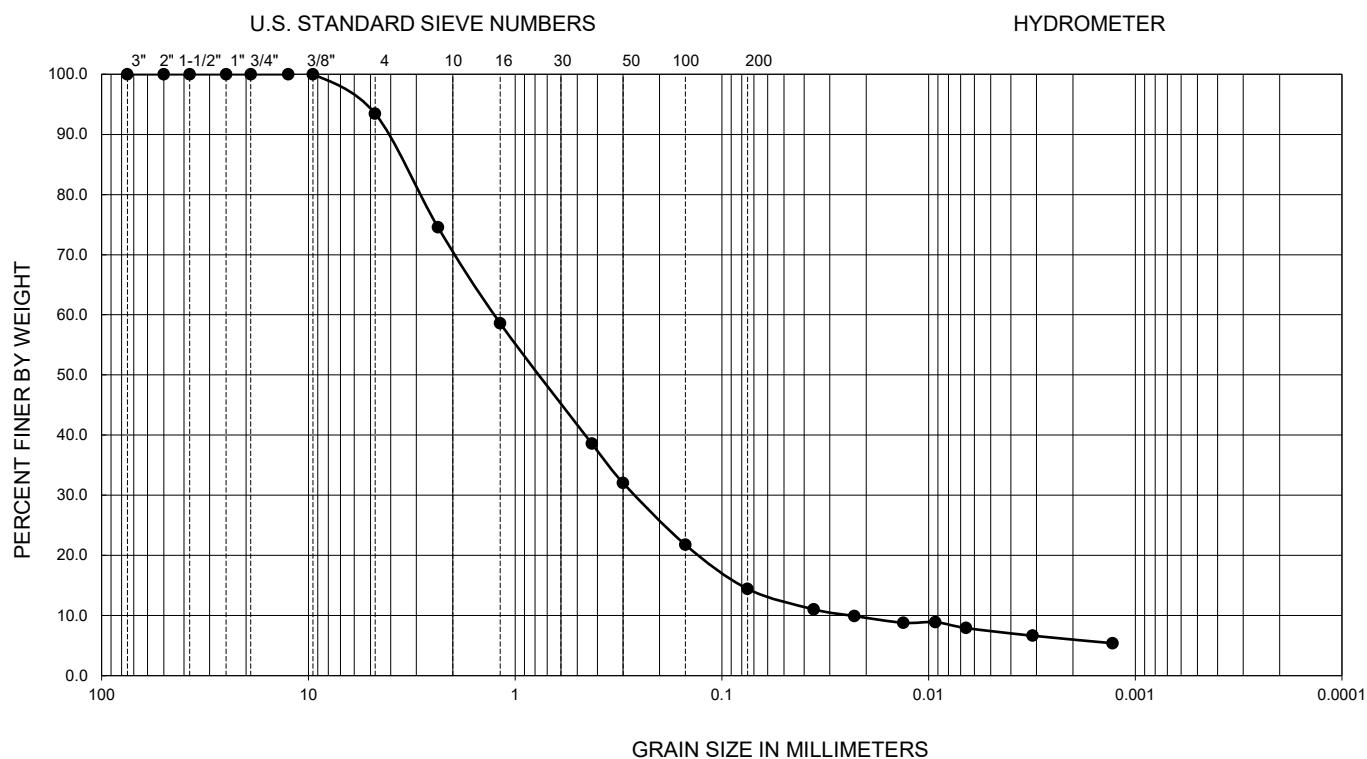


Symbol	Sample Location	Depth (ft)	Liquid Limit	Plastic Limit	Plasticity Index	D ₁₀	D ₃₀	D ₆₀	C _u	C _c	Passing No. 200 (%)	USCS
●	B-2	23.5-24.0	--	--	--	0.246	0.789	1.881	7.6	1.3	5.6	SW-SC

PERFORMED IN ACCORDANCE WITH ASTM D 422

FIGURE B-8

GRAVEL		SAND			FINES	
Coarse	Fine	Coarse	Medium	Fine	SILT	CLAY



Symbol	Sample Location	Depth (ft)	Liquid Limit	Plastic Limit	Plasticity Index	D ₁₀	D ₃₀	D ₆₀	C _u	C _c	Passing No. 200 (%)	USCS
●	B-2	24.5-25.0	--	--	--	0.024	0.270	1.284	54.0	2.4	14.4	SC

PERFORMED IN ACCORDANCE WITH ASTM D 422

FIGURE B-9



APPENDIX C

Laboratory Testing by Others

19 November, 2024

Job No. 2411032

Cust. No. 13270

Mr. David Lin
Ninyo & Moore
2149 O'Toole Avenue, Suite 30
San Jose, CA 95131

Subject: Project No.: 502075008
Project Name: Old Pinnacle Trailhead, Pinnacle National Park, Paicines, CA
Corrosivity Analysis – ASTM Test Methods

Dear Mr. Lin:

Pursuant to your request, CERCO Analytical has analyzed the soil samples submitted on November 14, 2024. Based on the analytical results, this brief corrosivity evaluation is enclosed for your consideration.

Based upon the resistivity measurement, Sample No. 001 is classified as "moderately corrosive" and Sample No. 002 is classified as "mildly corrosive". All buried iron, steel, cast iron, ductile iron, galvanized steel and dielectric coated steel or iron should be properly protected against corrosion depending upon the critical nature of the structure. All buried metallic pressure piping such as ductile iron firewater pipelines should be protected against corrosion.

The chloride ion concentrations reflect none detected with a reporting limit of 15 mg/kg.

The sulfate ion concentrations are none detected and 24 mg/kg and are determined to be insufficient to damage reinforced concrete structures and cement mortar-coated steel at these locations.

The pH of the soils are 7.26 and 8.09, which does not present corrosion problems for buried iron, steel, mortar-coated steel and reinforced concrete structures.

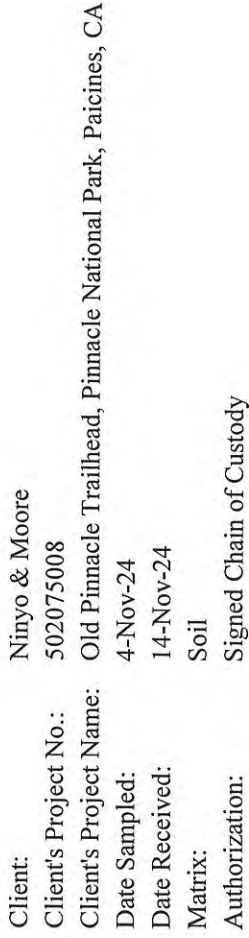
This corrosivity evaluation is based on general corrosion engineering standards and is non-specific in nature. For specific long-term corrosion control design recommendations or consultation, please call *JDH Corrosion Consultants, Inc.* at (925) 927-6630.

We appreciate the opportunity of working with you on this project. If you have any questions, or if you require further information, please do not hesitate to contact us.

Very truly yours,
CERCO ANALYTICAL, INC.


J. Darby Howard, Jr., P.E.
President

JDH/jdl
Enclosure



Sulfate



APPENDIX D

Field Exploration Photographs



Photograph 1:



Photograph 2:

FIGURE A-1



Photograph 3:



Photograph 4:

FIGURE A-2



Photograph 5:



Photograph 6:

FIGURE A-3



Photograph 7:



Photograph 8:

FIGURE A-4



Photograph 9:



Photograph 10:

FIGURE A-5



Photograph 11:



Photograph 12:

FIGURE A-6



Photograph 13:



Photograph 14:

FIGURE A-7



Photograph 15:



Photograph 16:

FIGURE A-8



Photograph 17:



Photograph 18:

FIGURE A-9



Photograph 19:



Photograph 20:

FIGURE A-10

PHOTOGRAPHS

PROJECT
ADDRESS, CITY, STATE
XXXXXXX | MONTH YEAR



Photograph 21:



Photograph 22:

FIGURE A-11



Photograph 23:



Photograph 24:

FIGURE A-12



Photograph 25:



Photograph 26:

FIGURE A-13



Photograph 27:



Photograph 28:

FIGURE A-14



Photograph 29:



Photograph 30:

FIGURE A-15

PHOTOGRAPHS

PROJECT
ADDRESS, CITY, STATE
XXXXXXX | MONTH YEAR



Photograph 31:



Photograph 32:

FIGURE A-16



Photograph 33:



Photograph 34:

FIGURE A-17



Photograph 35:



Photograph 36:

FIGURE A-18



Photograph 37:



Photograph 38:

FIGURE A-19



Photograph 39:



Photograph 40:

FIGURE A-20



Photograph 41:



Photograph 42:

FIGURE A-21



Photograph 43:



Photograph 44:

FIGURE A-22



Photograph 45:

FIGURE A-23



9707 East Easter Lane | Centennial, Colorado 80112 | p. 303.629.6000

ARIZONA | CALIFORNIA | COLORADO | NEVADA | TEXAS | UTAH

ninyoandmoore.com

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Geotechnical & Environmental Sciences Consultants